

Story-Based Mathematics

for Grades 1–6

A Comprehensive Collection of 750+ Story Problems
with Detailed Solutions and Explanations

Topics Include:

- Arithmetic Operations
- Fractions and Decimals
 - Geometry Basics
- Ratios and Proportions
 - Word Problems
 - Measurement
 - Data and Graphs
- Mathematical Vocabulary

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Chapter 1: Arithmetic Operations

This chapter focuses on fundamental arithmetic operations including addition, subtraction, multiplication, and division. Students will practice these operations through engaging story problems that relate to real-world situations.

Grade 1 Problems

Problem 1: Olivia has 6 stickers. They give 1 stickers to a friend. How many stickers does Olivia have left?

Answer: 5

Explanation: Olivia started with 6 stickers. After giving away 1: $6 - 1 = 5$ stickers.

Problem 2: Jack has 7 apples. Their friend gives them 3 more apples. How many apples does Jack have now?

Answer: 10

Explanation: Jack started with 7 apples. Adding 3 more: $7 + 3 = 10$ apples.

Problem 3: Emily has 6 apples. Their friend gives them 4 more apples. How many apples does Emily have now?

Answer: 10

Explanation: Emily started with 6 apples. Adding 4 more: $6 + 4 = 10$ apples.

Problem 4: Benjamin has 12 stickers. They give 5 stickers to a friend. How many stickers does Benjamin have left?

Answer: 7

Explanation: Benjamin started with 12 stickers. After giving away 5: $12 - 5 = 7$ stickers.

Problem 5: Aiden has 14 stickers. They give 3 stickers to a friend. How many stickers does Aiden have left?

Answer: 11

Explanation: Aiden started with 14 stickers. After giving away 3: $14 - 3 = 11$ stickers.

Problem 6: Samuel has 12 stickers. They give 1 stickers to a friend. How many stickers does Samuel have left?

Answer: 11

Explanation: Samuel started with 12 stickers. After giving away 1: $12 - 1 = 11$ stickers.

Problem 7: Sebastian has 9 apples. Their friend gives them 8 more apples. How many apples does Sebastian have now?

Answer: 17

Explanation: Sebastian started with 9 apples. Adding 8 more: $9 + 8 = 17$ apples.

Problem 8: Olivia has 15 stickers. They give 2 stickers to a friend. How many stickers does Olivia have left?

Answer: 13

Explanation: Olivia started with 15 stickers. After giving away 2: $15 - 2 = 13$ stickers.

Problem 9: Sebastian has 2 apples. Their friend gives them 3 more apples. How many apples does Sebastian have now?

Answer: 5

Explanation: Sebastian started with 2 apples. Adding 3 more: $2 + 3 = 5$ apples.

Problem 10: Ava has 10 stickers. They give 3 stickers to a friend. How many stickers does Ava have left?

Answer: 7

Explanation: Ava started with 10 stickers. After giving away 3: $10 - 3 = 7$ stickers.

Problem 11: Olivia has 14 stickers. They give 4 stickers to a friend. How many stickers does Olivia have left?

Answer: 10

Explanation: Olivia started with 14 stickers. After giving away 4: $14 - 4 = 10$ stickers.

Problem 12: Harper has 9 stickers. They give 1 stickers to a friend. How many stickers does Harper have left?

Answer: 8

Explanation: Harper started with 9 stickers. After giving away 1: $9 - 1 = 8$ stickers.

Problem 13: Jack has 8 stickers. They give 2 stickers to a friend. How many stickers does Jack have left?

Answer: 6

Explanation: Jack started with 8 stickers. After giving away 2: $8 - 2 = 6$ stickers.

Problem 14: Henry has 15 stickers. They give 3 stickers to a friend. How many stickers does Henry have left?

Answer: 12

Explanation: Henry started with 15 stickers. After giving away 3: $15 - 3 = 12$ stickers.

Problem 15: Sofia has 5 stickers. They give 1 stickers to a friend. How many stickers does Sofia have left?

Answer: 4

Explanation: Sofia started with 5 stickers. After giving away 1: $5 - 1 = 4$ stickers.

Problem 16: Samuel has 10 stickers. They give 5 stickers to a friend. How many stickers does Samuel have left?

Answer: 5

Explanation: Samuel started with 10 stickers. After giving away 5: $10 - 5 = 5$ stickers.

Problem 17: Aiden has 13 stickers. They give 3 stickers to a friend. How many stickers does Aiden have left?

Answer: 10

Explanation: Aiden started with 13 stickers. After giving away 3: $13 - 3 = 10$ stickers.

Problem 18: Olivia has 2 apples. Their friend gives them 5 more apples. How many apples does Olivia have now?

Answer: 7

Explanation: Olivia started with 2 apples. Adding 5 more: $2 + 5 = 7$ apples.

Problem 19: Lucas has 7 apples. Their friend gives them 1 more apples. How many apples does Lucas have now?

Answer: 8

Explanation: Lucas started with 7 apples. Adding 1 more: $7 + 1 = 8$ apples.

Problem 20: Amelia has 6 stickers. They give 3 stickers to a friend. How many stickers does Amelia have left?

Answer: 3

Explanation: Amelia started with 6 stickers. After giving away 3: $6 - 3 = 3$ stickers.

Problem 21: Harper has 6 apples. Their friend gives them 2 more apples. How many apples does Harper have now?

Answer: 8

Explanation: Harper started with 6 apples. Adding 2 more: $6 + 2 = 8$ apples.

Problem 22: Lucas has 8 apples. Their friend gives them 10 more apples. How many apples does Lucas have now?

Answer: 18

Explanation: Lucas started with 8 apples. Adding 10 more: $8 + 10 = 18$ apples.

Problem 23: Ethan has 14 stickers. They give 1 stickers to a friend. How many stickers does Ethan have left?

Answer: 13

Explanation: Ethan started with 14 stickers. After giving away 1: $14 - 1 = 13$ stickers.

Problem 24: Jack has 10 stickers. They give 5 stickers to a friend. How many stickers does Jack have left?

Answer: 5

Explanation: Jack started with 10 stickers. After giving away 5: $10 - 5 = 5$ stickers.

Problem 25: Alexander has 3 apples. Their friend gives them 2 more apples. How many apples does Alexander have now?

Answer: 5

Explanation: Alexander started with 3 apples. Adding 2 more: $3 + 2 = 5$ apples.

Problem 26: Sebastian has 1 apples. Their friend gives them 2 more apples. How many apples does Sebastian have now?

Answer: 3

Explanation: Sebastian started with 1 apples. Adding 2 more: $1 + 2 = 3$ apples.

Problem 27: Mia has 15 stickers. They give 3 stickers to a friend. How many stickers does Mia have left?

Answer: 12

Explanation: Mia started with 15 stickers. After giving away 3: $15 - 3 = 12$ stickers.

Problem 28: Joseph has 1 apples. Their friend gives them 10 more apples. How many apples does Joseph have now?

Answer: 11

Explanation: Joseph started with 1 apples. Adding 10 more: $1 + 10 = 11$ apples.

Problem 29: Ethan has 8 apples. Their friend gives them 7 more apples. How many apples does Ethan have now?

Answer: 15

Explanation: Ethan started with 8 apples. Adding 7 more: $8 + 7 = 15$ apples.

Problem 30: Sofia has 8 apples. Their friend gives them 2 more apples. How many apples does Sofia have now?

Answer: 10

Explanation: Sofia started with 8 apples. Adding 2 more: $8 + 2 = 10$ apples.

Problem 31: Amelia has 4 apples. Their friend gives them 10 more apples. How many apples does Amelia have now?

Answer: 14

Explanation: Amelia started with 4 apples. Adding 10 more: $4 + 10 = 14$ apples.

Problem 32: Noah has 4 apples. Their friend gives them 2 more apples. How many apples does Noah have now?

Answer: 6

Explanation: Noah started with 4 apples. Adding 2 more: $4 + 2 = 6$ apples.

Problem 33: Emily has 8 apples. Their friend gives them 2 more apples. How many apples does Emily have now?

Answer: 10

Explanation: Emily started with 8 apples. Adding 2 more: $8 + 2 = 10$ apples.

Problem 34: Sophia has 7 apples. Their friend gives them 1 more apples. How many apples does Sophia have now?

Answer: 8

Explanation: Sophia started with 7 apples. Adding 1 more: $7 + 1 = 8$ apples.

Problem 35: Emily has 7 apples. Their friend gives them 1 more apples. How many apples does Emily have now?

Answer: 8

Explanation: Emily started with 7 apples. Adding 1 more: $7 + 1 = 8$ apples.

Problem 36: Noah has 5 stickers. They give 1 stickers to a friend. How many stickers does Noah have left?

Answer: 4

Explanation: Noah started with 5 stickers. After giving away 1: $5 - 1 = 4$ stickers.

Problem 37: Evelyn has 3 apples. Their friend gives them 5 more apples. How many apples does Evelyn have now?

Answer: 8

Explanation: Evelyn started with 3 apples. Adding 5 more: $3 + 5 = 8$ apples.

Problem 38: Olivia has 8 apples. Their friend gives them 4 more apples. How many apples does Olivia have now?

Answer: 12

Explanation: Olivia started with 8 apples. Adding 4 more: $8 + 4 = 12$ apples.

Problem 39: Emma has 10 stickers. They give 3 stickers to a friend. How many stickers does Emma have left?

Answer: 7

Explanation: Emma started with 10 stickers. After giving away 3: $10 - 3 = 7$ stickers.

Problem 40: Emily has 15 stickers. They give 1 stickers to a friend. How many stickers does Emily have left?

Answer: 14

Explanation: Emily started with 15 stickers. After giving away 1: $15 - 1 = 14$ stickers.

Problem 41: Harper has 3 apples. Their friend gives them 2 more apples. How many apples does Harper have now?

Answer: 5

Explanation: Harper started with 3 apples. Adding 2 more: $3 + 2 = 5$ apples.

Problem 42: Aiden has 8 apples. Their friend gives them 6 more apples. How many apples does Aiden have now?

Answer: 14

Explanation: Aiden started with 8 apples. Adding 6 more: $8 + 6 = 14$ apples.

Problem 43: Olivia has 2 apples. Their friend gives them 1 more apples. How many apples does Olivia have now?

Answer: 3

Explanation: Olivia started with 2 apples. Adding 1 more: $2 + 1 = 3$ apples.

Problem 44: Jack has 8 stickers. They give 5 stickers to a friend. How many stickers does Jack have left?

Answer: 3

Explanation: Jack started with 8 stickers. After giving away 5: $8 - 5 = 3$ stickers.

Problem 45: Joseph has 9 stickers. They give 1 stickers to a friend. How many stickers does Joseph have left?

Answer: 8

Explanation: Joseph started with 9 stickers. After giving away 1: $9 - 1 = 8$ stickers.

Practice Exercises - Grade 1

Exercise 1: Alexander has 9 stickers. They give 3 stickers to a friend. How many stickers does Alexander have left?

Exercise 2: Olivia has 9 apples. Their friend gives them 3 more apples. How many apples does Olivia have now?

Exercise 3: Sofia has 9 stickers. They give 3 stickers to a friend. How many stickers does Sofia have left?

Exercise 4: Evelyn has 6 stickers. They give 4 stickers to a friend. How many stickers does Evelyn have left?

Exercise 5: Avery has 1 apples. Their friend gives them 7 more apples. How many apples does Avery have now?

Exercise 6: Joseph has 9 stickers. They give 4 stickers to a friend. How many stickers does Joseph have left?

Exercise 7: Jack has 6 apples. Their friend gives them 5 more apples. How many apples does Jack have now?

Exercise 8: Mia has 3 apples. Their friend gives them 6 more apples. How many apples does Mia have now?

Exercise 9: Samuel has 5 stickers. They give 1 stickers to a friend. How many stickers does Samuel have left?

Exercise 10: Emily has 6 stickers. They give 1 stickers to a friend. How many stickers does Emily have left?

Exercise 11: Sebastian has 10 stickers. They give 1 stickers to a friend. How many stickers does Sebastian have left?

Exercise 12: Mason has 13 stickers. They give 2 stickers to a friend. How many stickers does Mason have left?

Exercise 13: Joseph has 5 stickers. They give 3 stickers to a friend. How many stickers does Joseph have left?

Exercise 14: Sebastian has 8 apples. Their friend gives them 2 more apples. How many apples does Sebastian have now?

Exercise 15: Emily has 5 apples. Their friend gives them 9 more apples. How many apples does Emily have now?

Exercise 16: Mia has 9 stickers. They give 1 stickers to a friend. How many stickers does Mia have left?

Exercise 17: Charlotte has 4 apples. Their friend gives them 3 more apples. How many apples does Charlotte have now?

Exercise 18: Liam has 14 stickers. They give 2 stickers to a friend. How many stickers does Liam have left?

Exercise 19: Sebastian has 13 stickers. They give 1 stickers to a friend. How many stickers does Sebastian have left?

Exercise 20: Ava has 4 apples. Their friend gives them 2 more apples. How many apples does Ava have now?

Practice Exercise Answers:

Exercise 1: 6

Exercise 2: 12

Exercise 3: 6

Exercise 4: 2

Exercise 5: 8

Exercise 6: 5

Exercise 7: 11

Exercise 8: 9

Exercise 9: 4

Exercise 10: 5

Exercise 11: 9

Exercise 12: 11

Exercise 13: 2

Exercise 14: 10

Exercise 15: 14

Exercise 16: 8

Exercise 17: 7

Exercise 18: 12

Exercise 19: 12

Exercise 20: 6

Challenge Problems - Grade 1

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Samuel has 8 apples. Their friend gives them 6 more apples. How many apples does Samuel have now?

Answer: 14

Solution: Samuel started with 8 apples. Adding 6 more: $8 + 6 = 14$ apples.

Challenge 2: Ethan has 7 apples. Their friend gives them 2 more apples. How many apples does Ethan have now?

Answer: 9

Solution: Ethan started with 7 apples. Adding 2 more: $7 + 2 = 9$ apples.

Challenge 3: Sophia has 3 apples. Their friend gives them 8 more apples. How many apples does Sophia have now?

Answer: 11

Solution: Sophia started with 3 apples. Adding 8 more: $3 + 8 = 11$ apples.

Challenge 4: Joseph has 12 stickers. They give 1 stickers to a friend. How many stickers does Joseph have left?

Answer: 11

Solution: Joseph started with 12 stickers. After giving away 1: $12 - 1 = 11$ stickers.

Challenge 5: Noah has 7 apples. Their friend gives them 8 more apples. How many apples does Noah have now?

Answer: 15

Solution: Noah started with 7 apples. Adding 8 more: $7 + 8 = 15$ apples.

Challenge 6: Jack has 11 stickers. They give 3 stickers to a friend. How many stickers does Jack have left?

Answer: 8

Solution: Jack started with 11 stickers. After giving away 3: $11 - 3 = 8$ stickers.

Challenge 7: Avery has 2 apples. Their friend gives them 6 more apples. How many apples does Avery have now?

Answer: 8

Solution: Avery started with 2 apples. Adding 6 more: $2 + 6 = 8$ apples.

Challenge 8: Charlotte has 1 apples. Their friend gives them 6 more apples. How many apples does Charlotte have now?

Answer: 7

Solution: Charlotte started with 1 apples. Adding 6 more: $1 + 6 = 7$ apples.

Challenge 9: Avery has 9 apples. Their friend gives them 6 more apples. How many apples does Avery have now?

Answer: 15

Solution: Avery started with 9 apples. Adding 6 more: $9 + 6 = 15$ apples.

Challenge 10: Isabella has 13 stickers. They give 5 stickers to a friend. How many stickers does Isabella have left?

Answer: 8

Solution: Isabella started with 13 stickers. After giving away 5: $13 - 5 = 8$ stickers.

Challenge 11: James has 9 apples. Their friend gives them 4 more apples. How many apples does James have now?

Answer: 13

Solution: James started with 9 apples. Adding 4 more: $9 + 4 = 13$ apples.

Challenge 12: Evelyn has 7 stickers. They give 3 stickers to a friend. How many stickers does Evelyn have left?

Answer: 4

Solution: Evelyn started with 7 stickers. After giving away 3: $7 - 3 = 4$ stickers.

Challenge 13: Abigail has 7 stickers. They give 2 stickers to a friend. How many stickers does Abigail have left?

Answer: 5

Solution: Abigail started with 7 stickers. After giving away 2: $7 - 2 = 5$ stickers.

Challenge 14: Sophia has 10 stickers. They give 3 stickers to a friend. How many stickers does Sophia have left?

Answer: 7

Solution: Sophia started with 10 stickers. After giving away 3: $10 - 3 = 7$ stickers.

Challenge 15: Sofia has 14 stickers. They give 4 stickers to a friend. How many stickers does Sofia have left?

Answer: 10

Solution: Sofia started with 14 stickers. After giving away 4: $14 - 4 = 10$ stickers.

Grade 2 Problems

Problem 1: Amelia has 10 boxes of crayons. Each box has 4 crayons. How many crayons does Amelia have in total?

Answer: 40

Explanation: Multiplying boxes by crayons per box: $10 \times 4 = 40$ crayons.

Problem 2: Alexander collected 48 seashells on Monday and 12 seashells on Tuesday. How many seashells did Alexander collect in total?

Answer: 60

Explanation: Adding the seashells from both days: $48 + 12 = 60$ seashells.

Problem 3: Charlotte collected 37 seashells on Monday and 27 seashells on Tuesday. How many seashells did Charlotte collect in total?

Answer: 64

Explanation: Adding the seashells from both days: $37 + 27 = 64$ seashells.

Problem 4: Mia has 10 boxes of crayons. Each box has 5 crayons. How many crayons does Mia have in total?

Answer: 50

Explanation: Multiplying boxes by crayons per box: $10 \times 5 = 50$ crayons.

Problem 5: Mason has 3 boxes of crayons. Each box has 2 crayons. How many crayons does Mason have in total?

Answer: 6

Explanation: Multiplying boxes by crayons per box: $3 \times 2 = 6$ crayons.

Problem 6: Mason has 9 boxes of crayons. Each box has 3 crayons. How many crayons does Mason have in total?

Answer: 27

Explanation: Multiplying boxes by crayons per box: $9 \times 3 = 27$ crayons.

Problem 7: Lucas has 7 boxes of crayons. Each box has 2 crayons. How many crayons does Lucas have in total?

Answer: 14

Explanation: Multiplying boxes by crayons per box: $7 \times 2 = 14$ crayons.

Problem 8: Emily had 95 candies. They ate 19 candies. How many candies does Emily have left?

Answer: 76

Explanation: Subtracting the candies eaten: $95 - 19 = 76$ candies.

Problem 9: Aiden has 5 boxes of crayons. Each box has 3 crayons. How many crayons does Aiden have in total?

Answer: 15

Explanation: Multiplying boxes by crayons per box: $5 \times 3 = 15$ crayons.

Problem 10: Emily has 9 boxes of crayons. Each box has 5 crayons. How many crayons does Emily have in total?

Answer: 45

Explanation: Multiplying boxes by crayons per box: $9 \times 5 = 45$ crayons.

Problem 11: Elizabeth collected 35 seashells on Monday and 19 seashells on Tuesday. How many seashells did Elizabeth collect in total?

Answer: 54

Explanation: Adding the seashells from both days: $35 + 19 = 54$ seashells.

Problem 12: Sebastian had 58 candies. They ate 23 candies. How many candies does Sebastian have left?

Answer: 35

Explanation: Subtracting the candies eaten: $58 - 23 = 35$ candies.

Problem 13: Abigail had 69 candies. They ate 21 candies. How many candies does Abigail have left?

Answer: 48

Explanation: Subtracting the candies eaten: $69 - 21 = 48$ candies.

Problem 14: Sofia has 10 boxes of crayons. Each box has 5 crayons. How many crayons does Sofia have in total?

Answer: 50

Explanation: Multiplying boxes by crayons per box: $10 \times 5 = 50$ crayons.

Problem 15: Olivia had 90 candies. They ate 16 candies. How many candies does Olivia have left?

Answer: 74

Explanation: Subtracting the candies eaten: $90 - 16 = 74$ candies.

Problem 16: Evelyn collected 25 seashells on Monday and 15 seashells on Tuesday. How many seashells did Evelyn collect in total?

Answer: 40

Explanation: Adding the seashells from both days: $25 + 15 = 40$ seashells.

Problem 17: Joseph collected 20 seashells on Monday and 42 seashells on Tuesday. How many seashells did Joseph collect in total?

Answer: 62

Explanation: Adding the seashells from both days: $20 + 42 = 62$ seashells.

Problem 18: Joseph collected 11 seashells on Monday and 32 seashells on Tuesday. How many seashells did Joseph collect in total?

Answer: 43

Explanation: Adding the seashells from both days: $11 + 32 = 43$ seashells.

Problem 19: Jack has 6 boxes of crayons. Each box has 4 crayons. How many crayons does Jack have in total?

Answer: 24

Explanation: Multiplying boxes by crayons per box: $6 \times 4 = 24$ crayons.

Problem 20: Avery collected 50 seashells on Monday and 31 seashells on Tuesday. How many seashells did Avery collect in total?

Answer: 81

Explanation: Adding the seashells from both days: $50 + 31 = 81$ seashells.

Problem 21: Charlotte had 90 candies. They ate 19 candies. How many candies does Charlotte have left?

Answer: 71

Explanation: Subtracting the candies eaten: $90 - 19 = 71$ candies.

Problem 22: Emily collected 41 seashells on Monday and 23 seashells on Tuesday. How many seashells did Emily collect in total?

Answer: 64

Explanation: Adding the seashells from both days: $41 + 23 = 64$ seashells.

Problem 23: Alexander has 9 boxes of crayons. Each box has 3 crayons. How many crayons does Alexander have in total?

Answer: 27

Explanation: Multiplying boxes by crayons per box: $9 \times 3 = 27$ crayons.

Problem 24: Emma has 9 boxes of crayons. Each box has 2 crayons. How many crayons does Emma have in total?

Answer: 18

Explanation: Multiplying boxes by crayons per box: $9 \times 2 = 18$ crayons.

Problem 25: Benjamin has 9 boxes of crayons. Each box has 4 crayons. How many crayons does Benjamin have in total?

Answer: 36

Explanation: Multiplying boxes by crayons per box: $9 \times 4 = 36$ crayons.

Problem 26: Harper had 99 candies. They ate 21 candies. How many candies does Harper have left?

Answer: 78

Explanation: Subtracting the candies eaten: $99 - 21 = 78$ candies.

Problem 27: Evelyn has 4 boxes of crayons. Each box has 3 crayons. How many crayons does Evelyn have in total?

Answer: 12

Explanation: Multiplying boxes by crayons per box: $4 \times 3 = 12$ crayons.

Problem 28: Emma had 73 candies. They ate 27 candies. How many candies does Emma have left?

Answer: 46

Explanation: Subtracting the candies eaten: $73 - 27 = 46$ candies.

Problem 29: Olivia has 7 boxes of crayons. Each box has 3 crayons. How many crayons does Olivia have in total?

Answer: 21

Explanation: Multiplying boxes by crayons per box: $7 \times 3 = 21$ crayons.

Problem 30: Sofia had 76 candies. They ate 23 candies. How many candies does Sofia have left?

Answer: 53

Explanation: Subtracting the candies eaten: $76 - 23 = 53$ candies.

Problem 31: Emma has 6 boxes of crayons. Each box has 3 crayons. How many crayons does Emma have in total?

Answer: 18

Explanation: Multiplying boxes by crayons per box: $6 \times 3 = 18$ crayons.

Problem 32: Emily has 6 boxes of crayons. Each box has 3 crayons. How many crayons does Emily have in total?

Answer: 18

Explanation: Multiplying boxes by crayons per box: $6 \times 3 = 18$ crayons.

Problem 33: Sophia collected 36 seashells on Monday and 39 seashells on Tuesday. How many seashells did Sophia collect in total?

Answer: 75

Explanation: Adding the seashells from both days: $36 + 39 = 75$ seashells.

Problem 34: Sophia had 68 candies. They ate 22 candies. How many candies does Sophia have left?

Answer: 46

Explanation: Subtracting the candies eaten: $68 - 22 = 46$ candies.

Problem 35: Liam has 5 boxes of crayons. Each box has 5 crayons. How many crayons does Liam have in total?

Answer: 25

Explanation: Multiplying boxes by crayons per box: $5 \times 5 = 25$ crayons.

Problem 36: Henry collected 17 seashells on Monday and 23 seashells on Tuesday. How many seashells did Henry collect in total?

Answer: 40

Explanation: Adding the seashells from both days: $17 + 23 = 40$ seashells.

Problem 37: Isabella has 9 boxes of crayons. Each box has 5 crayons. How many crayons does Isabella have in total?

Answer: 45

Explanation: Multiplying boxes by crayons per box: $9 \times 5 = 45$ crayons.

Problem 38: Jack had 68 candies. They ate 20 candies. How many candies does Jack have left?

Answer: 48

Explanation: Subtracting the candies eaten: $68 - 20 = 48$ candies.

Problem 39: Henry collected 35 seashells on Monday and 38 seashells on Tuesday. How many seashells did Henry collect in total?

Answer: 73

Explanation: Adding the seashells from both days: $35 + 38 = 73$ seashells.

Problem 40: Olivia collected 24 seashells on Monday and 34 seashells on Tuesday. How many seashells did Olivia collect in total?

Answer: 58

Explanation: Adding the seashells from both days: $24 + 34 = 58$ seashells.

Problem 41: Samuel has 9 boxes of crayons. Each box has 2 crayons. How many crayons does Samuel have in total?

Answer: 18

Explanation: Multiplying boxes by crayons per box: $9 \times 2 = 18$ crayons.

Problem 42: Aiden collected 46 seashells on Monday and 23 seashells on Tuesday. How many seashells did Aiden collect in total?

Answer: 69

Explanation: Adding the seashells from both days: $46 + 23 = 69$ seashells.

Problem 43: Mason collected 17 seashells on Monday and 25 seashells on Tuesday. How many seashells did Mason collect in total?

Answer: 42

Explanation: Adding the seashells from both days: $17 + 25 = 42$ seashells.

Problem 44: Isabella had 91 candies. They ate 18 candies. How many candies does Isabella have left?

Answer: 73

Explanation: Subtracting the candies eaten: $91 - 18 = 73$ candies.

Problem 45: Ethan has 6 boxes of crayons. Each box has 5 crayons. How many crayons does Ethan have in total?

Answer: 30

Explanation: Multiplying boxes by crayons per box: $6 \times 5 = 30$ crayons.

Practice Exercises - Grade 2

Exercise 1: Olivia has 8 boxes of crayons. Each box has 3 crayons. How many crayons does Olivia have in total?

Exercise 2: Charlotte had 74 candies. They ate 23 candies. How many candies does Charlotte have left?

Exercise 3: Liam had 35 candies. They ate 17 candies. How many candies does Liam have left?

Exercise 4: Emma has 8 boxes of crayons. Each box has 2 crayons. How many crayons does Emma have in total?

Exercise 5: Charlotte has 6 boxes of crayons. Each box has 2 crayons. How many crayons does Charlotte have in total?

Exercise 6: Harper had 64 candies. They ate 17 candies. How many candies does Harper have left?

Exercise 7: Mason has 4 boxes of crayons. Each box has 4 crayons. How many crayons does Mason have in total?

Exercise 8: Charlotte has 2 boxes of crayons. Each box has 5 crayons. How many crayons does Charlotte have in total?

Exercise 9: Sofia had 57 candies. They ate 21 candies. How many candies does Sofia have left?

Exercise 10: Elizabeth collected 10 seashells on Monday and 48 seashells on Tuesday. How many seashells did Elizabeth collect in total?

Exercise 11: Harper has 6 boxes of crayons. Each box has 4 crayons. How many crayons does Harper have in total?

Exercise 12: Alexander has 8 boxes of crayons. Each box has 3 crayons. How many crayons does Alexander have in total?

Exercise 13: Charlotte had 69 candies. They ate 28 candies. How many candies does Charlotte have left?

Exercise 14: Emily collected 28 seashells on Monday and 38 seashells on Tuesday. How many seashells did Emily collect in total?

Exercise 15: Abigail has 8 boxes of crayons. Each box has 3 crayons. How many crayons does Abigail have in total?

Exercise 16: Ava collected 30 seashells on Monday and 39 seashells on Tuesday. How many seashells did Ava collect in total?

Exercise 17: Ethan collected 23 seashells on Monday and 23 seashells on Tuesday. How many seashells did Ethan collect in total?

Exercise 18: Emily has 7 boxes of crayons. Each box has 2 crayons. How many crayons does Emily have in total?

Exercise 19: Elizabeth has 5 boxes of crayons. Each box has 2 crayons. How many crayons does Elizabeth have in total?

Exercise 20: David had 52 candies. They ate 27 candies. How many candies does David have left?

Practice Exercise Answers:

Exercise 1: 24

Exercise 2: 51

Exercise 3: 18

Exercise 4: 16

Exercise 5: 12

Exercise 6: 47

Exercise 7: 16

Exercise 8: 10

Exercise 9: 36

Exercise 10: 58

Exercise 11: 24

Exercise 12: 24

Exercise 13: 41

Exercise 14: 66

Exercise 15: 24

Exercise 16: 69

Exercise 17: 46

Exercise 18: 14

Exercise 19: 10

Exercise 20: 25

Challenge Problems - Grade 2

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Mason collected 46 seashells on Monday and 15 seashells on Tuesday. How many seashells did Mason collect in total?

Answer: 61

Solution: Adding the seashells from both days: $46 + 15 = 61$ seashells.

Challenge 2: Noah collected 16 seashells on Monday and 34 seashells on Tuesday. How many seashells did Noah collect in total?

Answer: 50

Solution: Adding the seashells from both days: $16 + 34 = 50$ seashells.

Challenge 3: Olivia had 73 candies. They ate 28 candies. How many candies does Olivia have left?

Answer: 45

Solution: Subtracting the candies eaten: $73 - 28 = 45$ candies.

Challenge 4: Evelyn collected 26 seashells on Monday and 27 seashells on Tuesday. How many seashells did Evelyn collect in total?

Answer: 53

Solution: Adding the seashells from both days: $26 + 27 = 53$ seashells.

Challenge 5: Samuel had 41 candies. They ate 26 candies. How many candies does Samuel have left?

Answer: 15

Solution: Subtracting the candies eaten: $41 - 26 = 15$ candies.

Challenge 6: Sebastian has 10 boxes of crayons. Each box has 5 crayons. How many crayons does Sebastian have in total?

Answer: 50

Solution: Multiplying boxes by crayons per box: $10 \times 5 = 50$ crayons.

Challenge 7: Olivia had 57 candies. They ate 14 candies. How many candies does Olivia have left?

Answer: 43

Solution: Subtracting the candies eaten: $57 - 14 = 43$ candies.

Challenge 8: Ethan had 63 candies. They ate 26 candies. How many candies does Ethan have left?

Answer: 37

Solution: Subtracting the candies eaten: $63 - 26 = 37$ candies.

Challenge 9: Ava had 34 candies. They ate 26 candies. How many candies does Ava have left?

Answer: 8

Solution: Subtracting the candies eaten: $34 - 26 = 8$ candies.

Challenge 10: Avery had 59 candies. They ate 18 candies. How many candies does Avery have left?

Answer: 41

Solution: Subtracting the candies eaten: $59 - 18 = 41$ candies.

Challenge 11: Samuel collected 12 seashells on Monday and 15 seashells on Tuesday. How many seashells did Samuel collect in total?

Answer: 27

Solution: Adding the seashells from both days: $12 + 15 = 27$ seashells.

Challenge 12: Elizabeth had 49 candies. They ate 19 candies. How many candies does Elizabeth have left?

Answer: 30

Solution: Subtracting the candies eaten: $49 - 19 = 30$ candies.

Challenge 13: Emily had 91 candies. They ate 27 candies. How many candies does Emily have left?

Answer: 64

Solution: Subtracting the candies eaten: $91 - 27 = 64$ candies.

Challenge 14: Aiden had 44 candies. They ate 24 candies. How many candies does Aiden have left?

Answer: 20

Solution: Subtracting the candies eaten: $44 - 24 = 20$ candies.

Challenge 15: Henry had 69 candies. They ate 26 candies. How many candies does Henry have left?

Answer: 43

Solution: Subtracting the candies eaten: $69 - 26 = 43$ candies.

Grade 3 Problems

Problem 1: Mia buys 48 packs of trading cards. Each pack has 12 cards. How many cards does Mia have?

Answer: 576

Explanation: Multiplying packs by cards per pack: $48 \times 12 = 576$ cards.

Problem 2: A library has 312 fiction books and 135 non-fiction books. How many books does the library have in total?

Answer: 447

Explanation: Adding fiction and non-fiction books: $312 + 135 = 447$ books.

Problem 3: A library has 470 fiction books and 469 non-fiction books. How many books does the library have in total?

Answer: 939

Explanation: Adding fiction and non-fiction books: $470 + 469 = 939$ books.

Problem 4: Aiden has 165 cookies to share equally among 11 friends. How many cookies does each friend get?

Answer: 15

Explanation: Dividing cookies equally: $165 \div 11 = 15$ cookies per friend.

Problem 5: Sebastian has 77 cookies to share equally among 7 friends. How many cookies does each friend get?

Answer: 11

Explanation: Dividing cookies equally: $77 \div 7 = 11$ cookies per friend.

Problem 6: A library has 264 fiction books and 419 non-fiction books. How many books does the library have in total?

Answer: 683

Explanation: Adding fiction and non-fiction books: $264 + 419 = 683$ books.

Problem 7: A library has 458 fiction books and 228 non-fiction books. How many books does the library have in total?

Answer: 686

Explanation: Adding fiction and non-fiction books: $458 + 228 = 686$ books.

Problem 8: Emma has 90 cookies to share equally among 9 friends. How many cookies does each friend get?

Answer: 10

Explanation: Dividing cookies equally: $90 \div 9 = 10$ cookies per friend.

Problem 9: Mia buys 40 packs of trading cards. Each pack has 3 cards. How many cards does Mia have?

Answer: 120

Explanation: Multiplying packs by cards per pack: $40 \times 3 = 120$ cards.

Problem 10: Abigail buys 29 packs of trading cards. Each pack has 12 cards. How many cards does Abigail have?

Answer: 348

Explanation: Multiplying packs by cards per pack: $29 \times 12 = 348$ cards.

Problem 11: Jack has 143 cookies to share equally among 11 friends. How many cookies does each friend get?

Answer: 13

Explanation: Dividing cookies equally: $143 \div 11 = 13$ cookies per friend.

Problem 12: Mason buys 47 packs of trading cards. Each pack has 8 cards. How many cards does Mason have?

Answer: 376

Explanation: Multiplying packs by cards per pack: $47 \times 8 = 376$ cards.

Problem 13: A school has 887 students. 229 students went on a field trip. How many students stayed at school?

Answer: 658

Explanation: Subtracting students on trip: $887 - 229 = 658$ students.

Problem 14: Elizabeth has 80 cookies to share equally among 5 friends. How many cookies does each friend get?

Answer: 16

Explanation: Dividing cookies equally: $80 \div 5 = 16$ cookies per friend.

Problem 15: Henry buys 48 packs of trading cards. Each pack has 3 cards. How many cards does Henry have?

Answer: 144

Explanation: Multiplying packs by cards per pack: $48 \times 3 = 144$ cards.

Problem 16: Abigail has 12 cookies to share equally among 2 friends. How many cookies does each friend get?

Answer: 6

Explanation: Dividing cookies equally: $12 \div 2 = 6$ cookies per friend.

Problem 17: A school has 739 students. 187 students went on a field trip. How many students stayed at school?

Answer: 552

Explanation: Subtracting students on trip: $739 - 187 = 552$ students.

Problem 18: A school has 514 students. 297 students went on a field trip. How many students stayed at school?

Answer: 217

Explanation: Subtracting students on trip: $514 - 297 = 217$ students.

Problem 19: James has 90 cookies to share equally among 6 friends. How many cookies does each friend get?

Answer: 15

Explanation: Dividing cookies equally: $90 \div 6 = 15$ cookies per friend.

Problem 20: A library has 357 fiction books and 486 non-fiction books. How many books does the library have in total?

Answer: 843

Explanation: Adding fiction and non-fiction books: $357 + 486 = 843$ books.

Problem 21: Emily buys 23 packs of trading cards. Each pack has 5 cards. How many cards does Emily have?

Answer: 115

Explanation: Multiplying packs by cards per pack: $23 \times 5 = 115$ cards.

Problem 22: A school has 917 students. 297 students went on a field trip. How many students stayed at school?

Answer: 620

Explanation: Subtracting students on trip: $917 - 297 = 620$ students.

Problem 23: Mason buys 32 packs of trading cards. Each pack has 6 cards. How many cards does Mason have?

Answer: 192

Explanation: Multiplying packs by cards per pack: $32 \times 6 = 192$ cards.

Problem 24: A library has 455 fiction books and 449 non-fiction books. How many books does the library have in total?

Answer: 904

Explanation: Adding fiction and non-fiction books: $455 + 449 = 904$ books.

Problem 25: A school has 405 students. 192 students went on a field trip. How many students stayed at school?

Answer: 213

Explanation: Subtracting students on trip: $405 - 192 = 213$ students.

Problem 26: David has 162 cookies to share equally among 9 friends. How many cookies does each friend get?

Answer: 18

Explanation: Dividing cookies equally: $162 \div 9 = 18$ cookies per friend.

Problem 27: A school has 614 students. 241 students went on a field trip. How many students stayed at school?

Answer: 373

Explanation: Subtracting students on trip: $614 - 241 = 373$ students.

Problem 28: A school has 884 students. 167 students went on a field trip. How many students stayed at school?

Answer: 717

Explanation: Subtracting students on trip: $884 - 167 = 717$ students.

Problem 29: A library has 262 fiction books and 151 non-fiction books. How many books does the library have in total?

Answer: 413

Explanation: Adding fiction and non-fiction books: $262 + 151 = 413$ books.

Problem 30: A library has 156 fiction books and 379 non-fiction books. How many books does the library have in total?

Answer: 535

Explanation: Adding fiction and non-fiction books: $156 + 379 = 535$ books.

Problem 31: Mia buys 10 packs of trading cards. Each pack has 10 cards. How many cards does Mia have?

Answer: 100

Explanation: Multiplying packs by cards per pack: $10 \times 10 = 100$ cards.

Problem 32: Emily buys 20 packs of trading cards. Each pack has 4 cards. How many cards does Emily have?

Answer: 80

Explanation: Multiplying packs by cards per pack: $20 \times 4 = 80$ cards.

Problem 33: Sofia has 35 cookies to share equally among 5 friends. How many cookies does each friend get?

Answer: 7

Explanation: Dividing cookies equally: $35 \div 5 = 7$ cookies per friend.

Problem 34: Noah buys 35 packs of trading cards. Each pack has 12 cards. How many cards does Noah have?

Answer: 420

Explanation: Multiplying packs by cards per pack: $35 \times 12 = 420$ cards.

Problem 35: A library has 411 fiction books and 391 non-fiction books. How many books does the library have in total?

Answer: 802

Explanation: Adding fiction and non-fiction books: $411 + 391 = 802$ books.

Problem 36: A school has 898 students. 215 students went on a field trip. How many students stayed at school?

Answer: 683

Explanation: Subtracting students on trip: $898 - 215 = 683$ students.

Problem 37: A school has 322 students. 256 students went on a field trip. How many students stayed at school?

Answer: 66

Explanation: Subtracting students on trip: $322 - 256 = 66$ students.

Problem 38: Mia has 72 cookies to share equally among 8 friends. How many cookies does each friend get?

Answer: 9

Explanation: Dividing cookies equally: $72 \div 8 = 9$ cookies per friend.

Problem 39: David buys 32 packs of trading cards. Each pack has 7 cards. How many cards does David have?

Answer: 224

Explanation: Multiplying packs by cards per pack: $32 \times 7 = 224$ cards.

Problem 40: A school has 919 students. 184 students went on a field trip. How many students stayed at school?

Answer: 735

Explanation: Subtracting students on trip: $919 - 184 = 735$ students.

Problem 41: A library has 466 fiction books and 256 non-fiction books. How many books does the library have in total?

Answer: 722

Explanation: Adding fiction and non-fiction books: $466 + 256 = 722$ books.

Problem 42: Sebastian has 60 cookies to share equally among 5 friends. How many cookies does each friend get?

Answer: 12

Explanation: Dividing cookies equally: $60 \div 5 = 12$ cookies per friend.

Problem 43: Samuel buys 22 packs of trading cards. Each pack has 9 cards. How many cards does Samuel have?

Answer: 198

Explanation: Multiplying packs by cards per pack: $22 \times 9 = 198$ cards.

Problem 44: A school has 911 students. 202 students went on a field trip. How many students stayed at school?

Answer: 709

Explanation: Subtracting students on trip: $911 - 202 = 709$ students.

Problem 45: Emma has 36 cookies to share equally among 2 friends. How many cookies does each friend get?

Answer: 18

Explanation: Dividing cookies equally: $36 \div 2 = 18$ cookies per friend.

Practice Exercises - Grade 3

Exercise 1: A school has 330 students. 106 students went on a field trip. How many students stayed at school?

Exercise 2: Sophia has 40 cookies to share equally among 8 friends. How many cookies does each friend get?

Exercise 3: Sebastian has 216 cookies to share equally among 12 friends. How many cookies does each friend get?

Exercise 4: A library has 393 fiction books and 155 non-fiction books. How many books does the library have in total?

Exercise 5: A school has 710 students. 181 students went on a field trip. How many students stayed at school?

Exercise 6: A library has 334 fiction books and 371 non-fiction books. How many books does the library have in total?

Exercise 7: Olivia has 170 cookies to share equally among 10 friends. How many cookies does each friend get?

Exercise 8: A school has 807 students. 104 students went on a field trip. How many students stayed at school?

Exercise 9: Harper has 76 cookies to share equally among 4 friends. How many cookies does each friend get?

Exercise 10: David has 117 cookies to share equally among 9 friends. How many cookies does each friend get?

Exercise 11: Sophia buys 44 packs of trading cards. Each pack has 8 cards. How many cards does Sophia have?

Exercise 12: Sebastian has 120 cookies to share equally among 6 friends. How many cookies does each friend get?

Exercise 13: Emma has 162 cookies to share equally among 9 friends. How many cookies does each friend get?

Exercise 14: Noah buys 48 packs of trading cards. Each pack has 9 cards. How many cards does Noah have?

Exercise 15: Sebastian has 81 cookies to share equally among 9 friends. How many cookies does each friend get?

Exercise 16: A school has 994 students. 277 students went on a field trip. How many students stayed at school?

Exercise 17: A library has 352 fiction books and 271 non-fiction books. How many books does the library have in total?

Exercise 18: A school has 388 students. 249 students went on a field trip. How many students stayed at school?

Exercise 19: Elizabeth has 99 cookies to share equally among 9 friends. How many cookies does each friend get?

Exercise 20: A school has 334 students. 218 students went on a field trip. How many students stayed at school?

Practice Exercise Answers:

Exercise 1: 224

Exercise 2: 5

Exercise 3: 18

Exercise 4: 548

Exercise 5: 529

Exercise 6: 705

Exercise 7: 17

Exercise 8: 703

Exercise 9: 19

Exercise 10: 13

Exercise 11: 352

Exercise 12: 20

Exercise 13: 18

Exercise 14: 432

Exercise 15: 9

Exercise 16: 717

Exercise 17: 623

Exercise 18: 139

Exercise 19: 11

Exercise 20: 116

Challenge Problems - Grade 3

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Jack has 128 cookies to share equally among 8 friends. How many cookies does each friend get?

Answer: 16

Solution: Dividing cookies equally: $128 \div 8 = 16$ cookies per friend.

Challenge 2: Isabella buys 16 packs of trading cards. Each pack has 7 cards. How many cards does Isabella have?

Answer: 112

Solution: Multiplying packs by cards per pack: $16 \times 7 = 112$ cards.

Challenge 3: A school has 375 students. 247 students went on a field trip. How many students stayed at school?

Answer: 128

Solution: Subtracting students on trip: $375 - 247 = 128$ students.

Challenge 4: A library has 419 fiction books and 223 non-fiction books. How many books does the library have in total?

Answer: 642

Solution: Adding fiction and non-fiction books: $419 + 223 = 642$ books.

Challenge 5: A school has 879 students. 147 students went on a field trip. How many students stayed at school?

Answer: 732

Solution: Subtracting students on trip: $879 - 147 = 732$ students.

Challenge 6: A library has 212 fiction books and 143 non-fiction books. How many books does the library have in total?

Answer: 355

Solution: Adding fiction and non-fiction books: $212 + 143 = 355$ books.

Challenge 7: Isabella has 99 cookies to share equally among 9 friends. How many cookies does each friend get?

Answer: 11

Solution: Dividing cookies equally: $99 \div 9 = 11$ cookies per friend.

Challenge 8: A school has 804 students. 297 students went on a field trip. How many students stayed at school?

Answer: 507

Solution: Subtracting students on trip: $804 - 297 = 507$ students.

Challenge 9: A library has 195 fiction books and 280 non-fiction books. How many books does the library have in total?

Answer: 475

Solution: Adding fiction and non-fiction books: $195 + 280 = 475$ books.

Challenge 10: A library has 163 fiction books and 225 non-fiction books. How many books does the library have in total?

Answer: 388

Solution: Adding fiction and non-fiction books: $163 + 225 = 388$ books.

Challenge 11: A school has 435 students. 132 students went on a field trip. How many students stayed at school?

Answer: 303

Solution: Subtracting students on trip: $435 - 132 = 303$ students.

Challenge 12: A school has 365 students. 164 students went on a field trip. How many students stayed at school?

Answer: 201

Solution: Subtracting students on trip: $365 - 164 = 201$ students.

Challenge 13: A library has 329 fiction books and 179 non-fiction books. How many books does the library have in total?

Answer: 508

Solution: Adding fiction and non-fiction books: $329 + 179 = 508$ books.

Challenge 14: A school has 898 students. 212 students went on a field trip. How many students stayed at school?

Answer: 686

Solution: Subtracting students on trip: $898 - 212 = 686$ students.

Challenge 15: A school has 475 students. 236 students went on a field trip. How many students stayed at school?

Answer: 239

Solution: Subtracting students on trip: $475 - 236 = 239$ students.

Grade 4 Problems

Problem 1: A factory produces 84 toys each day. How many toys are produced in 48 days?

Answer: 4032

Explanation: Multiplying daily production by days: $84 \times 48 = 4032$ toys.

Problem 2: A store sold 585 items in January, 1754 items in February, and 1488 items in March. How many items were sold in total?

Answer: 3827

Explanation: Adding all three months: $585 + 1754 + 1488 = 3827$ items.

Problem 3: A factory produces 46 toys each day. How many toys are produced in 49 days?

Answer: 2254

Explanation: Multiplying daily production by days: $46 \times 49 = 2254$ toys.

Problem 4: A factory produces 37 toys each day. How many toys are produced in 30 days?

Answer: 1110

Explanation: Multiplying daily production by days: $37 \times 30 = 1110$ toys.

Problem 5: A city has a population of 9317 people. 2310 people moved away. What is the new population?

Answer: 7007

Explanation: Subtracting people who moved: $9317 - 2310 = 7007$ people.

Problem 6: A factory produces 80 toys each day. How many toys are produced in 16 days?

Answer: 1280

Explanation: Multiplying daily production by days: $80 \times 16 = 1280$ toys.

Problem 7: A farmer has 1080 eggs to pack into cartons. Each carton holds 24 eggs. How many cartons can be filled?

Answer: 45

Explanation: Dividing eggs by carton size: $1080 \div 24 = 45$ cartons.

Problem 8: A store sold 1363 items in January, 1043 items in February, and 1646 items in March. How many items were sold in total?

Answer: 4052

Explanation: Adding all three months: $1363 + 1043 + 1646 = 4052$ items.

Problem 9: A store sold 1532 items in January, 1656 items in February, and 632 items in March. How many items were sold in total?

Answer: 3820

Explanation: Adding all three months: $1532 + 1656 + 632 = 3820$ items.

Problem 10: A factory produces 69 toys each day. How many toys are produced in 27 days?

Answer: 1863

Explanation: Multiplying daily production by days: $69 \times 27 = 1863$ toys.

Problem 11: A farmer has 666 eggs to pack into cartons. Each carton holds 18 eggs. How many cartons can be filled?

Answer: 37

Explanation: Dividing eggs by carton size: $666 \div 18 = 37$ cartons.

Problem 12: A factory produces 81 toys each day. How many toys are produced in 40 days?

Answer: 3240

Explanation: Multiplying daily production by days: $81 \times 40 = 3240$ toys.

Problem 13: A store sold 557 items in January, 551 items in February, and 523 items in March. How many items were sold in total?

Answer: 1631

Explanation: Adding all three months: $557 + 551 + 523 = 1631$ items.

Problem 14: A store sold 600 items in January, 1736 items in February, and 1576 items in March. How many items were sold in total?

Answer: 3912

Explanation: Adding all three months: $600 + 1736 + 1576 = 3912$ items.

Problem 15: A factory produces 76 toys each day. How many toys are produced in 38 days?

Answer: 2888

Explanation: Multiplying daily production by days: $76 \times 38 = 2888$ toys.

Problem 16: A city has a population of 8249 people. 1591 people moved away. What is the new population?

Answer: 6658

Explanation: Subtracting people who moved: $8249 - 1591 = 6658$ people.

Problem 17: A city has a population of 5410 people. 4029 people moved away. What is the new population?

Answer: 1381

Explanation: Subtracting people who moved: $5410 - 4029 = 1381$ people.

Problem 18: A farmer has 546 eggs to pack into cartons. Each carton holds 14 eggs. How many cartons can be filled?

Answer: 39

Explanation: Dividing eggs by carton size: $546 \div 14 = 39$ cartons.

Problem 19: A factory produces 38 toys each day. How many toys are produced in 34 days?

Answer: 1292

Explanation: Multiplying daily production by days: $38 \times 34 = 1292$ toys.

Problem 20: A store sold 1548 items in January, 1928 items in February, and 1553 items in March. How many items were sold in total?

Answer: 5029

Explanation: Adding all three months: $1548 + 1928 + 1553 = 5029$ items.

Problem 21: A farmer has 448 eggs to pack into cartons. Each carton holds 14 eggs. How many cartons can be filled?

Answer: 32

Explanation: Dividing eggs by carton size: $448 \div 14 = 32$ cartons.

Problem 22: A store sold 898 items in January, 1916 items in February, and 1758 items in March. How many items were sold in total?

Answer: 4572

Explanation: Adding all three months: $898 + 1916 + 1758 = 4572$ items.

Problem 23: A farmer has 544 eggs to pack into cartons. Each carton holds 16 eggs. How many cartons can be filled?

Answer: 34

Explanation: Dividing eggs by carton size: $544 \div 16 = 34$ cartons.

Problem 24: A store sold 731 items in January, 1776 items in February, and 1305 items in March. How many items were sold in total?

Answer: 3812

Explanation: Adding all three months: $731 + 1776 + 1305 = 3812$ items.

Problem 25: A store sold 833 items in January, 1003 items in February, and 1724 items in March. How many items were sold in total?

Answer: 3560

Explanation: Adding all three months: $833 + 1003 + 1724 = 3560$ items.

Problem 26: A factory produces 59 toys each day. How many toys are produced in 26 days?

Answer: 1534

Explanation: Multiplying daily production by days: $59 \times 26 = 1534$ toys.

Problem 27: A store sold 759 items in January, 1547 items in February, and 1860 items in March. How many items were sold in total?

Answer: 4166

Explanation: Adding all three months: $759 + 1547 + 1860 = 4166$ items.

Problem 28: A factory produces 80 toys each day. How many toys are produced in 18 days?

Answer: 1440

Explanation: Multiplying daily production by days: $80 \times 18 = 1440$ toys.

Problem 29: A factory produces 64 toys each day. How many toys are produced in 13 days?

Answer: 832

Explanation: Multiplying daily production by days: $64 \times 13 = 832$ toys.

Problem 30: A store sold 1111 items in January, 875 items in February, and 1137 items in March. How many items were sold in total?

Answer: 3123

Explanation: Adding all three months: $1111 + 875 + 1137 = 3123$ items.

Problem 31: A store sold 1794 items in January, 1588 items in February, and 1146 items in March. How many items were sold in total?

Answer: 4528

Explanation: Adding all three months: $1794 + 1588 + 1146 = 4528$ items.

Problem 32: A farmer has 705 eggs to pack into cartons. Each carton holds 15 eggs. How many cartons can be filled?

Answer: 47

Explanation: Dividing eggs by carton size: $705 \div 15 = 47$ cartons.

Problem 33: A store sold 512 items in January, 1810 items in February, and 1427 items in March. How many items were sold in total?

Answer: 3749

Explanation: Adding all three months: $512 + 1810 + 1427 = 3749$ items.

Problem 34: A farmer has 490 eggs to pack into cartons. Each carton holds 14 eggs. How many cartons can be filled?

Answer: 35

Explanation: Dividing eggs by carton size: $490 \div 14 = 35$ cartons.

Problem 35: A store sold 1355 items in January, 1071 items in February, and 1626 items in March. How many items were sold in total?

Answer: 4052

Explanation: Adding all three months: $1355 + 1071 + 1626 = 4052$ items.

Problem 36: A farmer has 720 eggs to pack into cartons. Each carton holds 24 eggs. How many cartons can be filled?

Answer: 30

Explanation: Dividing eggs by carton size: $720 \div 24 = 30$ cartons.

Problem 37: A store sold 1016 items in January, 1075 items in February, and 785 items in March. How many items were sold in total?

Answer: 2876

Explanation: Adding all three months: $1016 + 1075 + 785 = 2876$ items.

Problem 38: A store sold 1640 items in January, 1857 items in February, and 1758 items in March. How many items were sold in total?

Answer: 5255

Explanation: Adding all three months: $1640 + 1857 + 1758 = 5255$ items.

Problem 39: A farmer has 720 eggs to pack into cartons. Each carton holds 15 eggs. How many cartons can be filled?

Answer: 48

Explanation: Dividing eggs by carton size: $720 \div 15 = 48$ cartons.

Problem 40: A factory produces 68 toys each day. How many toys are produced in 36 days?

Answer: 2448

Explanation: Multiplying daily production by days: $68 \times 36 = 2448$ toys.

Problem 41: A city has a population of 5781 people. 1074 people moved away. What is the new population?

Answer: 4707

Explanation: Subtracting people who moved: $5781 - 1074 = 4707$ people.

Problem 42: A farmer has 820 eggs to pack into cartons. Each carton holds 20 eggs. How many cartons can be filled?

Answer: 41

Explanation: Dividing eggs by carton size: $820 \div 20 = 41$ cartons.

Problem 43: A factory produces 53 toys each day. How many toys are produced in 35 days?

Answer: 1855

Explanation: Multiplying daily production by days: $53 \times 35 = 1855$ toys.

Problem 44: A city has a population of 6208 people. 1440 people moved away. What is the new population?

Answer: 4768

Explanation: Subtracting people who moved: $6208 - 1440 = 4768$ people.

Problem 45: A store sold 1630 items in January, 625 items in February, and 1178 items in March. How many items were sold in total?

Answer: 3433

Explanation: Adding all three months: $1630 + 625 + 1178 = 3433$ items.

Practice Exercises - Grade 4

Exercise 1: A city has a population of 5595 people. 2688 people moved away. What is the new population?

Exercise 2: A farmer has 540 eggs to pack into cartons. Each carton holds 18 eggs. How many cartons can be filled?

Exercise 3: A city has a population of 9149 people. 3774 people moved away. What is the new population?

Exercise 4: A city has a population of 8118 people. 1118 people moved away. What is the new population?

Exercise 5: A factory produces 86 toys each day. How many toys are produced in 16 days?

Exercise 6: A city has a population of 9704 people. 2842 people moved away. What is the new population?

Exercise 7: A store sold 1531 items in January, 1292 items in February, and 1396 items in March. How many items were sold in total?

Exercise 8: A store sold 1881 items in January, 642 items in February, and 552 items in March. How many items were sold in total?

Exercise 9: A factory produces 77 toys each day. How many toys are produced in 50 days?

Exercise 10: A city has a population of 8274 people. 4710 people moved away. What is the new population?

Exercise 11: A factory produces 58 toys each day. How many toys are produced in 21 days?

Exercise 12: A store sold 1624 items in January, 1501 items in February, and 1557 items in March. How many items were sold in total?

Exercise 13: A city has a population of 7701 people. 1953 people moved away. What is the new population?

Exercise 14: A farmer has 704 eggs to pack into cartons. Each carton holds 22 eggs. How many cartons can be filled?

Exercise 15: A store sold 1566 items in January, 559 items in February, and 794 items in March. How many items were sold in total?

Exercise 16: A city has a population of 8259 people. 4222 people moved away. What is the new population?

Exercise 17: A factory produces 54 toys each day. How many toys are produced in 40 days?

Exercise 18: A factory produces 81 toys each day. How many toys are produced in 26 days?

Exercise 19: A farmer has 931 eggs to pack into cartons. Each carton holds 19 eggs. How many cartons can be filled?

Exercise 20: A store sold 528 items in January, 679 items in February, and 811 items in March. How many items were sold in total?

Practice Exercise Answers:

Exercise 1: 2907

Exercise 2: 30

Exercise 3: 5375

Exercise 4: 7000

Exercise 5: 1376

Exercise 6: 6862

Exercise 7: 4219

Exercise 8: 3075

Exercise 9: 3850

Exercise 10: 3564

Exercise 11: 1218

Exercise 12: 4682

Exercise 13: 5748

Exercise 14: 32

Exercise 15: 2919

Exercise 16: 4037

Exercise 17: 2160

Exercise 18: 2106

Exercise 19: 49

Exercise 20: 2018

Challenge Problems - Grade 4

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: A factory produces 28 toys each day. How many toys are produced in 25 days?

Answer: 700

Solution: Multiplying daily production by days: $28 \times 25 = 700$ toys.

Challenge 2: A farmer has 462 eggs to pack into cartons. Each carton holds 22 eggs. How many cartons can be filled?

Answer: 21

Solution: Dividing eggs by carton size: $462 \div 22 = 21$ cartons.

Challenge 3: A city has a population of 8147 people. 2770 people moved away. What is the new population?

Answer: 5377

Solution: Subtracting people who moved: $8147 - 2770 = 5377$ people.

Challenge 4: A store sold 545 items in January, 1931 items in February, and 1130 items in March. How many items were sold in total?

Answer: 3606

Solution: Adding all three months: $545 + 1931 + 1130 = 3606$ items.

Challenge 5: A factory produces 84 toys each day. How many toys are produced in 16 days?

Answer: 1344

Solution: Multiplying daily production by days: $84 \times 16 = 1344$ toys.

Challenge 6: A city has a population of 7314 people. 4698 people moved away. What is the new population?

Answer: 2616

Solution: Subtracting people who moved: $7314 - 4698 = 2616$ people.

Challenge 7: A farmer has 880 eggs to pack into cartons. Each carton holds 22 eggs. How many cartons can be filled?

Answer: 40

Solution: Dividing eggs by carton size: $880 \div 22 = 40$ cartons.

Challenge 8: A factory produces 51 toys each day. How many toys are produced in 20 days?

Answer: 1020

Solution: Multiplying daily production by days: $51 \times 20 = 1020$ toys.

Challenge 9: A city has a population of 7434 people. 4667 people moved away. What is the new population?

Answer: 2767

Solution: Subtracting people who moved: $7434 - 4667 = 2767$ people.

Challenge 10: A farmer has 384 eggs to pack into cartons. Each carton holds 16 eggs. How many cartons can be filled?

Answer: 24

Solution: Dividing eggs by carton size: $384 \div 16 = 24$ cartons.

Challenge 11: A factory produces 99 toys each day. How many toys are produced in 43 days?

Answer: 4257

Solution: Multiplying daily production by days: $99 \times 43 = 4257$ toys.

Challenge 12: A factory produces 99 toys each day. How many toys are produced in 13 days?

Answer: 1287

Solution: Multiplying daily production by days: $99 \times 13 = 1287$ toys.

Challenge 13: A factory produces 49 toys each day. How many toys are produced in 48 days?

Answer: 2352

Solution: Multiplying daily production by days: $49 \times 48 = 2352$ toys.

Challenge 14: A city has a population of 6805 people. 1251 people moved away. What is the new population?

Answer: 5554

Solution: Subtracting people who moved: $6805 - 1251 = 5554$ people.

Challenge 15: A city has a population of 7097 people. 2900 people moved away. What is the new population?

Answer: 4197

Solution: Subtracting people who moved: $7097 - 2900 = 4197$ people.

Grade 5 Problems

Problem 1: Harper buys a sandwich for \$45.97 and a drink for \$49.42. What is the total cost?

Answer: 95.39

Explanation: Adding the prices: $\$45.97 + \$49.42 = \$95.39$.

Problem 2: Sofia buys 9 books at \$19 each. They have a coupon for \$10 off. How much does Sofia pay in total?

Answer: 161

Explanation: First multiply: $9 \times \$19 = \171 . Then subtract discount: $\$171 - \$10 = \$161$.

Problem 3: Two cities have populations of 44,743 and 25,137 people. What is their combined population?

Answer: 69880

Explanation: Adding both populations: $44,743 + 25,137 = 69,880$ people.

Problem 4: Two cities have populations of 27,707 and 17,973 people. What is their combined population?

Answer: 45680

Explanation: Adding both populations: $27,707 + 17,973 = 45,680$ people.

Problem 5: Charlotte buys a sandwich for \$24.85 and a drink for \$11.04. What is the total cost?

Answer: 35.89

Explanation: Adding the prices: $\$24.85 + \$11.04 = \$35.89$.

Problem 6: David buys 7 books at \$14 each. They have a coupon for \$3 off. How much does David pay in total?

Answer: 95

Explanation: First multiply: $7 \times \$14 = \98 . Then subtract discount: $\$98 - \$3 = \$95$.

Problem 7: Emma buys a sandwich for \$26.59 and a drink for \$36.29. What is the total cost?

Answer: 62.88

Explanation: Adding the prices: $\$26.59 + \$36.29 = \$62.88$.

Problem 8: Charlotte buys a sandwich for \$40.12 and a drink for \$35.12. What is the total cost?

Answer: 75.24

Explanation: Adding the prices: $\$40.12 + \$35.12 = \$75.24$.

Problem 9: David buys a sandwich for \$10.46 and a drink for \$12.75. What is the total cost?

Answer: 23.21

Explanation: Adding the prices: $\$10.46 + \$12.75 = \$23.21$.

Problem 10: Two cities have populations of 21,025 and 21,179 people. What is their combined population?

Answer: 42204

Explanation: Adding both populations: $21,025 + 21,179 = 42,204$ people.

Problem 11: Two cities have populations of 45,720 and 19,352 people. What is their combined population?

Answer: 65072

Explanation: Adding both populations: $45,720 + 19,352 = 65,072$ people.

Problem 12: Ethan buys a sandwich for \$42.22 and a drink for \$29.77. What is the total cost?

Answer: 71.99

Explanation: Adding the prices: $\$42.22 + \$29.77 = \$71.99$.

Problem 13: Elizabeth buys a sandwich for \$18.27 and a drink for \$29.68. What is the total cost?

Answer: 47.95

Explanation: Adding the prices: $\$18.27 + \$29.68 = \$47.95$.

Problem 14: Charlotte buys a sandwich for \$25.58 and a drink for \$24.69. What is the total cost?

Answer: 50.27

Explanation: Adding the prices: $\$25.58 + \$24.69 = \$50.27$.

Problem 15: Sebastian buys a sandwich for \$10.51 and a drink for \$32.07. What is the total cost?

Answer: 42.58

Explanation: Adding the prices: $\$10.51 + \$32.07 = \$42.58$.

Problem 16: Mason buys 10 books at \$11 each. They have a coupon for \$9 off. How much does Mason pay in total?

Answer: 101

Explanation: First multiply: $10 \times \$11 = \110 . Then subtract discount: $\$110 - \$9 = \$101$.

Problem 17: Elizabeth buys a sandwich for \$13.68 and a drink for \$38.16. What is the total cost?

Answer: 51.84

Explanation: Adding the prices: $\$13.68 + \$38.16 = \$51.84$.

Problem 18: Aiden buys 7 books at \$6 each. They have a coupon for \$10 off. How much does Aiden pay in total?

Answer: 32

Explanation: First multiply: $7 \times \$6 = \42 . Then subtract discount: $\$42 - \$10 = \$32$.

Problem 19: Two cities have populations of 24,508 and 36,248 people. What is their combined population?

Answer: 60756

Explanation: Adding both populations: $24,508 + 36,248 = 60,756$ people.

Problem 20: Two cities have populations of 13,944 and 29,035 people. What is their combined population?

Answer: 42979

Explanation: Adding both populations: $13,944 + 29,035 = 42,979$ people.

Problem 21: Aiden buys a sandwich for \$46.21 and a drink for \$14.36. What is the total cost?

Answer: 60.57

Explanation: Adding the prices: $\$46.21 + \$14.36 = \$60.57$.

Problem 22: Sophia buys a sandwich for \$20.43 and a drink for \$33.59. What is the total cost?

Answer: 54.02

Explanation: Adding the prices: $\$20.43 + \$33.59 = \$54.02$.

Problem 23: Lucas buys 3 books at \$18 each. They have a coupon for \$8 off. How much does Lucas pay in total?

Answer: 46

Explanation: First multiply: $3 \times \$18 = \54 . Then subtract discount: $\$54 - \$8 = \$46$.

Problem 24: Two cities have populations of 28,380 and 23,166 people. What is their combined population?

Answer: 51546

Explanation: Adding both populations: $28,380 + 23,166 = 51,546$ people.

Problem 25: Two cities have populations of 49,709 and 34,920 people. What is their combined population?

Answer: 84629

Explanation: Adding both populations: $49,709 + 34,920 = 84,629$ people.

Problem 26: Evelyn buys 4 books at \$12 each. They have a coupon for \$7 off. How much does Evelyn pay in total?

Answer: 41

Explanation: First multiply: $4 \times \$12 = \48 . Then subtract discount: $\$48 - \$7 = \$41$.

Problem 27: James buys a sandwich for \$17.21 and a drink for \$26.24. What is the total cost?

Answer: 43.45

Explanation: Adding the prices: $\$17.21 + \$26.24 = \$43.45$.

Problem 28: Samuel buys a sandwich for \$39.92 and a drink for \$38.52. What is the total cost?

Answer: 78.44

Explanation: Adding the prices: $\$39.92 + \$38.52 = \$78.44$.

Problem 29: Alexander buys 8 books at \$20 each. They have a coupon for \$5 off. How much does Alexander pay in total?

Answer: 155

Explanation: First multiply: $8 \times \$20 = \160 . Then subtract discount: $\$160 - \$5 = \$155$.

Problem 30: Two cities have populations of 34,775 and 19,447 people. What is their combined population?

Answer: 54222

Explanation: Adding both populations: $34,775 + 19,447 = 54,222$ people.

Problem 31: Two cities have populations of 12,271 and 40,675 people. What is their combined population?

Answer: 52946

Explanation: Adding both populations: $12,271 + 40,675 = 52,946$ people.

Problem 32: Sebastian buys a sandwich for \$33.05 and a drink for \$47.03. What is the total cost?

Answer: 80.08

Explanation: Adding the prices: $\$33.05 + \$47.03 = \$80.08$.

Problem 33: Mia buys a sandwich for \$19.32 and a drink for \$20.11. What is the total cost?

Answer: 39.43

Explanation: Adding the prices: $\$19.32 + \$20.11 = \$39.43$.

Problem 34: Joseph buys a sandwich for \$21.5 and a drink for \$33.85. What is the total cost?

Answer: 55.35

Explanation: Adding the prices: $\$21.5 + \$33.85 = \$55.35$.

Problem 35: Samuel buys 6 books at \$6 each. They have a coupon for \$10 off. How much does Samuel pay in total?

Answer: 26

Explanation: First multiply: $6 \times \$6 = \36 . Then subtract discount: $\$36 - \$10 = \$26$.

Problem 36: Olivia buys 4 books at \$8 each. They have a coupon for \$5 off. How much does Olivia pay in total?

Answer: 27

Explanation: First multiply: $4 \times \$8 = \32 . Then subtract discount: $\$32 - \$5 = \$27$.

Problem 37: Sebastian buys 8 books at \$17 each. They have a coupon for \$2 off. How much does Sebastian pay in total?

Answer: 134

Explanation: First multiply: $8 \times \$17 = \136 . Then subtract discount: $\$136 - \$2 = \$134$.

Problem 38: Two cities have populations of 21,563 and 15,527 people. What is their combined population?

Answer: 37090

Explanation: Adding both populations: $21,563 + 15,527 = 37,090$ people.

Problem 39: Mia buys a sandwich for \$36.11 and a drink for \$39.64. What is the total cost?

Answer: 75.75

Explanation: Adding the prices: $\$36.11 + \$39.64 = \$75.75$.

Problem 40: Sophia buys a sandwich for \$37.09 and a drink for \$36.87. What is the total cost?

Answer: 73.96

Explanation: Adding the prices: $\$37.09 + \$36.87 = \$73.96$.

Problem 41: Emma buys 10 books at \$15 each. They have a coupon for \$8 off. How much does Emma pay in total?

Answer: 142

Explanation: First multiply: $10 \times \$15 = \150 . Then subtract discount: $\$150 - \$8 = \$142$.

Problem 42: Ava buys 5 books at \$14 each. They have a coupon for \$10 off. How much does Ava pay in total?

Answer: 60

Explanation: First multiply: $5 \times \$14 = \70 . Then subtract discount: $\$70 - \$10 = \$60$.

Problem 43: Two cities have populations of 30,880 and 26,028 people. What is their combined population?

Answer: 56908

Explanation: Adding both populations: $30,880 + 26,028 = 56,908$ people.

Problem 44: Liam buys a sandwich for \$15.55 and a drink for \$31.69. What is the total cost?

Answer: 47.24

Explanation: Adding the prices: $\$15.55 + \$31.69 = \$47.24$.

Problem 45: Elizabeth buys 8 books at \$6 each. They have a coupon for \$4 off. How much does Elizabeth pay in total?

Answer: 44

Explanation: First multiply: $8 \times \$6 = \48 . Then subtract discount: $\$48 - \$4 = \$44$.

Practice Exercises - Grade 5

Exercise 1: Ava buys a sandwich for \$45.26 and a drink for \$21.16. What is the total cost?

Exercise 2: Two cities have populations of 43,851 and 16,468 people. What is their combined population?

Exercise 3: Isabella buys a sandwich for \$28.84 and a drink for \$46.13. What is the total cost?

Exercise 4: Avery buys 10 books at \$20 each. They have a coupon for \$4 off. How much does Avery pay in total?

Exercise 5: Mia buys a sandwich for \$37.79 and a drink for \$15.18. What is the total cost?

Exercise 6: Avery buys a sandwich for \$27.65 and a drink for \$48.95. What is the total cost?

Exercise 7: James buys 10 books at \$9 each. They have a coupon for \$8 off. How much does James pay in total?

Exercise 8: Alexander buys a sandwich for \$48.73 and a drink for \$16.18. What is the total cost?

Exercise 9: Elizabeth buys a sandwich for \$40.96 and a drink for \$48.81. What is the total cost?

Exercise 10: Two cities have populations of 13,121 and 41,624 people. What is their combined population?

Exercise 11: Abigail buys a sandwich for \$34.97 and a drink for \$28.34. What is the total cost?

Exercise 12: David buys 5 books at \$14 each. They have a coupon for \$6 off. How much does David pay in total?

Exercise 13: Abigail buys 9 books at \$9 each. They have a coupon for \$6 off. How much does Abigail pay in total?

Exercise 14: Abigail buys a sandwich for \$34.28 and a drink for \$29.95. What is the total cost?

Exercise 15: Henry buys 9 books at \$8 each. They have a coupon for \$7 off. How much does Henry pay in total?

Exercise 16: Two cities have populations of 29,722 and 15,491 people. What is their combined population?

Exercise 17: Two cities have populations of 19,157 and 16,850 people. What is their combined population?

Exercise 18: Alexander buys a sandwich for \$26.71 and a drink for \$16.93. What is the total cost?

Exercise 19: Two cities have populations of 38,641 and 24,568 people. What is their combined population?

Exercise 20: Two cities have populations of 29,917 and 27,218 people. What is their combined population?

Practice Exercise Answers:

Exercise 1: 66.42

Exercise 2: 60319

Exercise 3: 74.97

Exercise 4: 196

Exercise 5: 52.97

Exercise 6: 76.6

Exercise 7: 82

Exercise 8: 64.91

Exercise 9: 89.77

Exercise 10: 54745

Exercise 11: 63.31

Exercise 12: 64

Exercise 13: 75

Exercise 14: 64.23

Exercise 15: 65

Exercise 16: 45213

Exercise 17: 36007

Exercise 18: 43.64

Exercise 19: 63209

Exercise 20: 57135

Challenge Problems - Grade 5

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Sebastian buys 5 books at \$14 each. They have a coupon for \$4 off. How much does Sebastian pay in total?

Answer: 66

Solution: First multiply: $5 \times \$14 = \70 . Then subtract discount: $\$70 - \$4 = \$66$.

Challenge 2: Two cities have populations of 27,329 and 10,591 people. What is their combined population?

Answer: 37920

Solution: Adding both populations: $27,329 + 10,591 = 37,920$ people.

Challenge 3: Two cities have populations of 14,639 and 40,547 people. What is their combined population?

Answer: 55186

Solution: Adding both populations: $14,639 + 40,547 = 55,186$ people.

Challenge 4: Two cities have populations of 10,423 and 10,671 people. What is their combined population?

Answer: 21094

Solution: Adding both populations: $10,423 + 10,671 = 21,094$ people.

Challenge 5: Isabella buys a sandwich for \$21.62 and a drink for \$13.52. What is the total cost?

Answer: 35.14

Solution: Adding the prices: $\$21.62 + \$13.52 = \$35.14$.

Challenge 6: Sebastian buys a sandwich for \$46.25 and a drink for \$22.54. What is the total cost?

Answer: 68.79

Solution: Adding the prices: $\$46.25 + \$22.54 = \$68.79$.

Challenge 7: Harper buys 7 books at \$13 each. They have a coupon for \$10 off. How much does Harper pay in total?

Answer: 81

Solution: First multiply: $7 \times \$13 = \91 . Then subtract discount: $\$91 - \$10 = \$81$.

Challenge 8: Sofia buys a sandwich for \$13.08 and a drink for \$10.86. What is the total cost?

Answer: 23.94

Solution: Adding the prices: $\$13.08 + \$10.86 = \$23.94$.

Challenge 9: Aiden buys a sandwich for \$21.31 and a drink for \$17.62. What is the total cost?

Answer: 38.93

Solution: Adding the prices: $\$21.31 + \$17.62 = \$38.93$.

Challenge 10: Two cities have populations of 27,704 and 44,956 people. What is their combined population?

Answer: 72660

Solution: Adding both populations: $27,704 + 44,956 = 72,660$ people.

Challenge 11: Two cities have populations of 34,637 and 23,159 people. What is their combined population?

Answer: 57796

Solution: Adding both populations: $34,637 + 23,159 = 57,796$ people.

Challenge 12: Charlotte buys a sandwich for \$44.5 and a drink for \$26.81. What is the total cost?

Answer: 71.31

Solution: Adding the prices: $\$44.5 + \$26.81 = \$71.31$.

Challenge 13: Two cities have populations of 13,946 and 33,612 people. What is their combined population?

Answer: 47558

Solution: Adding both populations: $13,946 + 33,612 = 47,558$ people.

Challenge 14: Two cities have populations of 42,914 and 39,659 people. What is their combined population?

Answer: 82573

Solution: Adding both populations: $42,914 + 39,659 = 82,573$ people.

Challenge 15: Two cities have populations of 34,709 and 30,649 people. What is their combined population?

Answer: 65358

Solution: Adding both populations: $34,709 + 30,649 = 65,358$ people.

Grade 6 Problems

Problem 1: A teacher orders 3 pencils for each of 28 students, plus 9 extra pencils. How many pencils in total?

Answer: 93

Explanation: First multiply: $28 \times 3 = 84$. Then add extra: $84 + 9 = 93$ pencils.

Problem 2: The high temperature today is 8°C . The low temperature is -11°C . What is the difference between the high and low temperatures?

Answer: 19

Explanation: Difference: $8 - (-11) = 8 + 11 = 19^{\circ}\text{C}$.

Problem 3: A jacket costs \$98. It's on sale for 10% off. What is the sale price?

Answer: 89

Explanation: 10% of \$98 is \$9. Sale price: $\$98 - \$9 = \$89$.

Problem 4: A jacket costs \$130. It's on sale for 10% off. What is the sale price?

Answer: 117

Explanation: 10% of \$130 is \$13. Sale price: $\$130 - \$13 = \$117$.

Problem 5: A teacher orders 7 pencils for each of 32 students, plus 14 extra pencils. How many pencils in total?

Answer: 238

Explanation: First multiply: $32 \times 7 = 224$. Then add extra: $224 + 14 = 238$ pencils.

Problem 6: A teacher orders 7 pencils for each of 22 students, plus 5 extra pencils. How many pencils in total?

Answer: 159

Explanation: First multiply: $22 \times 7 = 154$. Then add extra: $154 + 5 = 159$ pencils.

Problem 7: The high temperature today is 15°C . The low temperature is -15°C . What is the difference between the high and low temperatures?

Answer: 30

Explanation: Difference: $15 - (-15) = 15 + 15 = 30^{\circ}\text{C}$.

Problem 8: A jacket costs \$120. It's on sale for 20% off. What is the sale price?

Answer: 96

Explanation: 20% of \$120 is \$24. Sale price: $\$120 - \$24 = \$96$.

Problem 9: A teacher orders 7 pencils for each of 22 students, plus 9 extra pencils. How many pencils in total?

Answer: 163

Explanation: First multiply: $22 \times 7 = 154$. Then add extra: $154 + 9 = 163$ pencils.

Problem 10: A jacket costs \$140. It's on sale for 50% off. What is the sale price?

Answer: 70

Explanation: 50% of \$140 is \$70. Sale price: $\$140 - \$70 = \$70$.

Problem 11: A teacher orders 5 pencils for each of 32 students, plus 10 extra pencils. How many pencils in total?

Answer: 170

Explanation: First multiply: $32 \times 5 = 160$. Then add extra: $160 + 10 = 170$ pencils.

Problem 12: The high temperature today is 15°C . The low temperature is -11°C . What is the difference between the high and low temperatures?

Answer: 26

Explanation: Difference: $15 - (-11) = 15 + 11 = 26^{\circ}\text{C}$.

Problem 13: A teacher orders 3 pencils for each of 20 students, plus 6 extra pencils. How many pencils in total?

Answer: 66

Explanation: First multiply: $20 \times 3 = 60$. Then add extra: $60 + 6 = 66$ pencils.

Problem 14: A jacket costs \$109. It's on sale for 10% off. What is the sale price?

Answer: 99

Explanation: 10% of \$109 is \$10. Sale price: $\$109 - \$10 = \$99$.

Problem 15: The high temperature today is 7°C . The low temperature is -5°C . What is the difference between the high and low temperatures?

Answer: 12

Explanation: Difference: $7 - (-5) = 7 + 5 = 12^{\circ}\text{C}$.

Problem 16: A teacher orders 4 pencils for each of 26 students, plus 15 extra pencils. How many pencils in total?

Answer: 119

Explanation: First multiply: $26 \times 4 = 104$. Then add extra: $104 + 15 = 119$ pencils.

Problem 17: The high temperature today is 12°C . The low temperature is -14°C . What is the difference between the high and low temperatures?

Answer: 26

Explanation: Difference: $12 - (-14) = 12 + 14 = 26^{\circ}\text{C}$.

Problem 18: A jacket costs \$90. It's on sale for 25% off. What is the sale price?

Answer: 68

Explanation: 25% of \$90 is \$22. Sale price: $\$90 - \$22 = \$68$.

Problem 19: A teacher orders 5 pencils for each of 31 students, plus 8 extra pencils. How many pencils in total?

Answer: 163

Explanation: First multiply: $31 \times 5 = 155$. Then add extra: $155 + 8 = 163$ pencils.

Problem 20: The high temperature today is 14°C . The low temperature is -11°C . What is the difference between the high and low temperatures?

Answer: 25

Explanation: Difference: $14 - (-11) = 14 + 11 = 25^{\circ}\text{C}$.

Problem 21: The high temperature today is 12°C . The low temperature is -5°C . What is the difference between the high and low temperatures?

Answer: 17

Explanation: Difference: $12 - (-5) = 12 + 5 = 17^{\circ}\text{C}$.

Problem 22: A teacher orders 8 pencils for each of 33 students, plus 12 extra pencils. How many pencils in total?

Answer: 276

Explanation: First multiply: $33 \times 8 = 264$. Then add extra: $264 + 12 = 276$ pencils.

Problem 23: A jacket costs \$122. It's on sale for 10% off. What is the sale price?

Answer: 110

Explanation: 10% of \$122 is \$12. Sale price: $\$122 - \$12 = \$110$.

Problem 24: A teacher orders 6 pencils for each of 24 students, plus 7 extra pencils. How many pencils in total?

Answer: 151

Explanation: First multiply: $24 \times 6 = 144$. Then add extra: $144 + 7 = 151$ pencils.

Problem 25: A jacket costs \$56. It's on sale for 25% off. What is the sale price?

Answer: 42

Explanation: 25% of \$56 is \$14. Sale price: $\$56 - \$14 = \$42$.

Problem 26: A jacket costs \$125. It's on sale for 20% off. What is the sale price?

Answer: 100

Explanation: 20% of \$125 is \$25. Sale price: $\$125 - \$25 = \$100$.

Problem 27: A jacket costs \$83. It's on sale for 50% off. What is the sale price?

Answer: 42

Explanation: 50% of \$83 is \$41. Sale price: $\$83 - \$41 = \$42$.

Problem 28: The high temperature today is 13°C . The low temperature is -6°C . What is the difference between the high and low temperatures?

Answer: 19

Explanation: Difference: $13 - (-6) = 13 + 6 = 19^{\circ}\text{C}$.

Problem 29: A teacher orders 6 pencils for each of 25 students, plus 15 extra pencils. How many pencils in total?

Answer: 165

Explanation: First multiply: $25 \times 6 = 150$. Then add extra: $150 + 15 = 165$ pencils.

Problem 30: A teacher orders 3 pencils for each of 31 students, plus 13 extra pencils. How many pencils in total?

Answer: 106

Explanation: First multiply: $31 \times 3 = 93$. Then add extra: $93 + 13 = 106$ pencils.

Problem 31: A teacher orders 3 pencils for each of 35 students, plus 13 extra pencils. How many pencils in total?

Answer: 118

Explanation: First multiply: $35 \times 3 = 105$. Then add extra: $105 + 13 = 118$ pencils.

Problem 32: The high temperature today is 11°C . The low temperature is -14°C . What is the difference between the high and low temperatures?

Answer: 25

Explanation: Difference: $11 - (-14) = 11 + 14 = 25^{\circ}\text{C}$.

Problem 33: A jacket costs \$118. It's on sale for 10% off. What is the sale price?

Answer: 107

Explanation: 10% of \$118 is \$11. Sale price: $\$118 - \$11 = \$107$.

Problem 34: A jacket costs \$104. It's on sale for 20% off. What is the sale price?

Answer: 84

Explanation: 20% of \$104 is \$20. Sale price: $\$104 - \$20 = \$84$.

Problem 35: A jacket costs \$143. It's on sale for 20% off. What is the sale price?

Answer: 115

Explanation: 20% of \$143 is \$28. Sale price: $\$143 - \$28 = \$115$.

Problem 36: A jacket costs \$187. It's on sale for 25% off. What is the sale price?

Answer: 141

Explanation: 25% of \$187 is \$46. Sale price: $\$187 - \$46 = \$141$.

Problem 37: A jacket costs \$65. It's on sale for 25% off. What is the sale price?

Answer: 49

Explanation: 25% of \$65 is \$16. Sale price: $\$65 - \$16 = \$49$.

Problem 38: A teacher orders 5 pencils for each of 32 students, plus 10 extra pencils. How many pencils in total?

Answer: 170

Explanation: First multiply: $32 \times 5 = 160$. Then add extra: $160 + 10 = 170$ pencils.

Problem 39: A jacket costs \$116. It's on sale for 10% off. What is the sale price?

Answer: 105

Explanation: 10% of \$116 is \$11. Sale price: $\$116 - \$11 = \$105$.

Problem 40: The high temperature today is 12°C . The low temperature is -12°C . What is the difference between the high and low temperatures?

Answer: 24

Explanation: Difference: $12 - (-12) = 12 + 12 = 24^{\circ}\text{C}$.

Problem 41: The high temperature today is 12°C . The low temperature is -9°C . What is the difference between the high and low temperatures?

Answer: 21

Explanation: Difference: $12 - (-9) = 12 + 9 = 21^{\circ}\text{C}$.

Problem 42: A jacket costs \$164. It's on sale for 20% off. What is the sale price?

Answer: 132

Explanation: 20% of \$164 is \$32. Sale price: $\$164 - \$32 = \$132$.

Problem 43: A teacher orders 8 pencils for each of 30 students, plus 12 extra pencils. How many pencils in total?

Answer: 252

Explanation: First multiply: $30 \times 8 = 240$. Then add extra: $240 + 12 = 252$ pencils.

Problem 44: A teacher orders 3 pencils for each of 30 students, plus 11 extra pencils. How many pencils in total?

Answer: 101

Explanation: First multiply: $30 \times 3 = 90$. Then add extra: $90 + 11 = 101$ pencils.

Problem 45: A jacket costs \$149. It's on sale for 50% off. What is the sale price?

Answer: 75

Explanation: 50% of \$149 is \$74. Sale price: $\$149 - \$74 = \$75$.

Practice Exercises - Grade 6

Exercise 1: A teacher orders 8 pencils for each of 33 students, plus 10 extra pencils. How many pencils in total?

Exercise 2: A jacket costs \$182. It's on sale for 20% off. What is the sale price?

Exercise 3: A teacher orders 4 pencils for each of 20 students, plus 12 extra pencils. How many pencils in total?

Exercise 4: A teacher orders 7 pencils for each of 22 students, plus 12 extra pencils. How many pencils in total?

Exercise 5: The high temperature today is 14°C . The low temperature is -8°C . What is the difference between the high and low temperatures?

Exercise 6: A teacher orders 4 pencils for each of 22 students, plus 6 extra pencils. How many pencils in total?

Exercise 7: A jacket costs \$149. It's on sale for 20% off. What is the sale price?

Exercise 8: A jacket costs \$109. It's on sale for 25% off. What is the sale price?

Exercise 9: A jacket costs \$54. It's on sale for 10% off. What is the sale price?

Exercise 10: The high temperature today is 12°C . The low temperature is -8°C . What is the difference between the high and low temperatures?

Exercise 11: A jacket costs \$56. It's on sale for 20% off. What is the sale price?

Exercise 12: A jacket costs \$199. It's on sale for 10% off. What is the sale price?

Exercise 13: A jacket costs \$185. It's on sale for 20% off. What is the sale price?

Exercise 14: A teacher orders 4 pencils for each of 33 students, plus 8 extra pencils. How many pencils in total?

Exercise 15: A jacket costs \$72. It's on sale for 20% off. What is the sale price?

Exercise 16: The high temperature today is 12°C . The low temperature is -8°C . What is the difference between the high and low temperatures?

Exercise 17: A jacket costs \$169. It's on sale for 10% off. What is the sale price?

Exercise 18: A teacher orders 6 pencils for each of 32 students, plus 9 extra pencils. How many pencils in total?

Exercise 19: A jacket costs \$169. It's on sale for 50% off. What is the sale price?

Exercise 20: A teacher orders 3 pencils for each of 35 students, plus 10 extra pencils. How many pencils in total?

Practice Exercise Answers:

Exercise 1: 274

Exercise 2: 146

Exercise 3: 92

Exercise 4: 166

Exercise 5: 22

Exercise 6: 94

Exercise 7: 120

Exercise 8: 82

Exercise 9: 49

Exercise 10: 20

Exercise 11: 45

Exercise 12: 180

Exercise 13: 148

Exercise 14: 140

Exercise 15: 58

Exercise 16: 20

Exercise 17: 153

Exercise 18: 201

Exercise 19: 85

Exercise 20: 115

Challenge Problems - Grade 6

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: The high temperature today is 6°C . The low temperature is -9°C . What is the difference between the high and low temperatures?

Answer: 15

Solution: Difference: $6 - (-9) = 6 + 9 = 15^{\circ}\text{C}$.

Challenge 2: A jacket costs \$157. It's on sale for 10% off. What is the sale price?

Answer: 142

Solution: 10% of \$157 is \$15. Sale price: $\$157 - \$15 = \$142$.

Challenge 3: The high temperature today is 7°C . The low temperature is -11°C . What is the difference between the high and low temperatures?

Answer: 18

Solution: Difference: $7 - (-11) = 7 + 11 = 18^{\circ}\text{C}$.

Challenge 4: A jacket costs \$134. It's on sale for 10% off. What is the sale price?

Answer: 121

Solution: 10% of \$134 is \$13. Sale price: $\$134 - \$13 = \$121$.

Challenge 5: A teacher orders 8 pencils for each of 32 students, plus 8 extra pencils. How many pencils in total?

Answer: 264

Solution: First multiply: $32 \times 8 = 256$. Then add extra: $256 + 8 = 264$ pencils.

Challenge 6: A teacher orders 6 pencils for each of 22 students, plus 6 extra pencils. How many pencils in total?

Answer: 138

Solution: First multiply: $22 \times 6 = 132$. Then add extra: $132 + 6 = 138$ pencils.

Challenge 7: A teacher orders 8 pencils for each of 22 students, plus 7 extra pencils. How many pencils in total?

Answer: 183

Solution: First multiply: $22 \times 8 = 176$. Then add extra: $176 + 7 = 183$ pencils.

Challenge 8: A teacher orders 6 pencils for each of 20 students, plus 14 extra pencils. How many pencils in total?

Answer: 134

Solution: First multiply: $20 \times 6 = 120$. Then add extra: $120 + 14 = 134$ pencils.

Challenge 9: A jacket costs \$55. It's on sale for 50% off. What is the sale price?

Answer: 28

Solution: 50% of \$55 is \$27. Sale price: $\$55 - \$27 = \$28$.

Challenge 10: A jacket costs \$117. It's on sale for 50% off. What is the sale price?

Answer: 59

Solution: 50% of \$117 is \$58. Sale price: $\$117 - \$58 = \$59$.

Challenge 11: A teacher orders 4 pencils for each of 25 students, plus 15 extra pencils. How many pencils in total?

Answer: 115

Solution: First multiply: $25 \times 4 = 100$. Then add extra: $100 + 15 = 115$ pencils.

Challenge 12: A teacher orders 6 pencils for each of 35 students, plus 13 extra pencils. How many pencils in total?

Answer: 223

Solution: First multiply: $35 \times 6 = 210$. Then add extra: $210 + 13 = 223$ pencils.

Challenge 13: A teacher orders 6 pencils for each of 30 students, plus 5 extra pencils. How many pencils in total?

Answer: 185

Solution: First multiply: $30 \times 6 = 180$. Then add extra: $180 + 5 = 185$ pencils.

Challenge 14: The high temperature today is 12°C . The low temperature is -5°C . What is the difference between the high and low temperatures?

Answer: 17

Solution: Difference: $12 - (-5) = 12 + 5 = 17^{\circ}\text{C}$.

Challenge 15: A teacher orders 7 pencils for each of 24 students, plus 7 extra pencils. How many pencils in total?

Answer: 175

Solution: First multiply: $24 \times 7 = 168$. Then add extra: $168 + 7 = 175$ pencils.

Chapter 2: Fractions and Decimals

Fractions are an essential part of mathematics. This chapter introduces students to understanding, comparing, adding, subtracting, multiplying, and dividing fractions through practical story-based problems.

Grade 3 Problems

Problem 1: Liam has 6 cookies. They eat $\frac{2}{3}$ of the cookies. How many cookies did Liam eat?

Answer: 4

Explanation: $\frac{2}{3}$ of 6 = $(2 \times 6) \div 3 = 4$ cookies.

Problem 2: Lucas has 6 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Lucas eat?

Answer: 3

Explanation: $\frac{1}{2}$ of 6 = $(1 \times 6) \div 2 = 3$ cookies.

Problem 3: Joseph has 10 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Joseph eat?

Answer: 5

Explanation: $\frac{1}{2}$ of 10 = $(1 \times 10) \div 2 = 5$ cookies.

Problem 4: Amelia has 6 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Amelia eat?

Answer: 3

Explanation: $\frac{1}{2}$ of 6 = $(1 \times 6) \div 2 = 3$ cookies.

Problem 5: Ava has 10 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Ava eat?

Answer: 5

Explanation: $\frac{1}{2}$ of 10 = $(1 \times 10) \div 2 = 5$ cookies.

Problem 6: Evelyn has 12 cookies. They eat $\frac{2}{3}$ of the cookies. How many cookies did Evelyn eat?

Answer: 8

Explanation: $\frac{2}{3}$ of 12 = $(2 \times 12) \div 3 = 8$ cookies.

Problem 7: Charlotte has 6 cookies. They eat $\frac{2}{3}$ of the cookies. How many cookies did Charlotte eat?

Answer: 4

Explanation: $\frac{2}{3}$ of 6 = $(2 \times 6) \div 3 = 4$ cookies.

Problem 8: Ava has 8 cookies. They eat $\frac{3}{4}$ of the cookies. How many cookies did Ava eat?

Answer: 6

Explanation: $\frac{3}{4}$ of 8 = $(3 \times 8) \div 4 = 6$ cookies.

Problem 9: Lucas has 4 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Lucas eat?

Answer: 2

Explanation: $\frac{1}{2}$ of 4 = $(1 \times 4) \div 2 = 2$ cookies.

Problem 10: Emily has 10 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Emily eat?

Answer: 5

Explanation: $\frac{1}{2}$ of 10 = $(1 \times 10) \div 2 = 5$ cookies.

Problem 11: Olivia has 6 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Olivia eat?

Answer: 3

Explanation: $\frac{1}{2}$ of 6 = $(1 \times 6) \div 2 = 3$ cookies.

Problem 12: Harper has 9 cookies. They eat $\frac{1}{3}$ of the cookies. How many cookies did Harper eat?

Answer: 3

Explanation: $\frac{1}{3}$ of 9 = $(1 \times 9) \div 3 = 3$ cookies.

Problem 13: Elizabeth has 12 cookies. They eat $\frac{3}{4}$ of the cookies. How many cookies did Elizabeth eat?

Answer: 9

Explanation: $\frac{3}{4}$ of 12 = $(3 \times 12) \div 4 = 9$ cookies.

Problem 14: Evelyn has 8 cookies. They eat $\frac{2}{4}$ of the cookies. How many cookies did Evelyn eat?

Answer: 4

Explanation: $2/4$ of $8 = (2 \times 8) \div 4 = 4$ cookies.

Problem 15: Alexander has 15 cookies. They eat $2/3$ of the cookies. How many cookies did Alexander eat?

Answer: 10

Explanation: $2/3$ of $15 = (2 \times 15) \div 3 = 10$ cookies.

Problem 16: Noah has 4 cookies. They eat $1/2$ of the cookies. How many cookies did Noah eat?

Answer: 2

Explanation: $1/2$ of $4 = (1 \times 4) \div 2 = 2$ cookies.

Problem 17: Samuel has 4 cookies. They eat $1/2$ of the cookies. How many cookies did Samuel eat?

Answer: 2

Explanation: $1/2$ of $4 = (1 \times 4) \div 2 = 2$ cookies.

Problem 18: Henry has 16 cookies. They eat $2/4$ of the cookies. How many cookies did Henry eat?

Answer: 8

Explanation: $2/4$ of $16 = (2 \times 16) \div 4 = 8$ cookies.

Problem 19: Sebastian has 6 cookies. They eat $1/3$ of the cookies. How many cookies did Sebastian eat?

Answer: 2

Explanation: $1/3$ of $6 = (1 \times 6) \div 3 = 2$ cookies.

Problem 20: Abigail has 20 cookies. They eat $2/4$ of the cookies. How many cookies did Abigail eat?

Answer: 10

Explanation: $2/4$ of $20 = (2 \times 20) \div 4 = 10$ cookies.

Problem 21: Olivia has 4 cookies. They eat $1/2$ of the cookies. How many cookies did Olivia eat?

Answer: 2

Explanation: $1/2$ of $4 = (1 \times 4) \div 2 = 2$ cookies.

Problem 22: Sophia has 20 cookies. They eat $2/4$ of the cookies. How many cookies did Sophia eat?

Answer: 10

Explanation: $2/4$ of 20 = $(2 \times 20) \div 4 = 10$ cookies.

Problem 23: Aiden has 8 cookies. They eat $1/4$ of the cookies. How many cookies did Aiden eat?

Answer: 2

Explanation: $1/4$ of 8 = $(1 \times 8) \div 4 = 2$ cookies.

Problem 24: Aiden has 8 cookies. They eat $3/4$ of the cookies. How many cookies did Aiden eat?

Answer: 6

Explanation: $3/4$ of 8 = $(3 \times 8) \div 4 = 6$ cookies.

Problem 25: Olivia has 6 cookies. They eat $2/3$ of the cookies. How many cookies did Olivia eat?

Answer: 4

Explanation: $2/3$ of 6 = $(2 \times 6) \div 3 = 4$ cookies.

Problem 26: James has 4 cookies. They eat $1/2$ of the cookies. How many cookies did James eat?

Answer: 2

Explanation: $1/2$ of 4 = $(1 \times 4) \div 2 = 2$ cookies.

Problem 27: Noah has 6 cookies. They eat $1/2$ of the cookies. How many cookies did Noah eat?

Answer: 3

Explanation: $1/2$ of 6 = $(1 \times 6) \div 2 = 3$ cookies.

Problem 28: Emily has 8 cookies. They eat $1/2$ of the cookies. How many cookies did Emily eat?

Answer: 4

Explanation: $1/2$ of 8 = $(1 \times 8) \div 2 = 4$ cookies.

Problem 29: Ava has 20 cookies. They eat $1/4$ of the cookies. How many cookies did Ava eat?

Answer: 5

Explanation: $1/4$ of 20 = $(1 \times 20) \div 4 = 5$ cookies.

Problem 30: Noah has 12 cookies. They eat $1/4$ of the cookies. How many cookies did Noah eat?

Answer: 3

Explanation: $1/4$ of 12 = $(1 \times 12) \div 4 = 3$ cookies.

Problem 31: Amelia has 8 cookies. They eat $1/2$ of the cookies. How many cookies did Amelia eat?

Answer: 4

Explanation: $1/2$ of 8 = $(1 \times 8) \div 2 = 4$ cookies.

Problem 32: Evelyn has 9 cookies. They eat $2/3$ of the cookies. How many cookies did Evelyn eat?

Answer: 6

Explanation: $2/3$ of 9 = $(2 \times 9) \div 3 = 6$ cookies.

Problem 33: Avery has 8 cookies. They eat $1/4$ of the cookies. How many cookies did Avery eat?

Answer: 2

Explanation: $1/4$ of 8 = $(1 \times 8) \div 4 = 2$ cookies.

Problem 34: Isabella has 6 cookies. They eat $1/3$ of the cookies. How many cookies did Isabella eat?

Answer: 2

Explanation: $1/3$ of 6 = $(1 \times 6) \div 3 = 2$ cookies.

Problem 35: Abigail has 6 cookies. They eat $2/3$ of the cookies. How many cookies did Abigail eat?

Answer: 4

Explanation: $2/3$ of 6 = $(2 \times 6) \div 3 = 4$ cookies.

Problem 36: Avery has 20 cookies. They eat $1/4$ of the cookies. How many cookies did Avery eat?

Answer: 5

Explanation: $1/4$ of 20 = $(1 \times 20) \div 4 = 5$ cookies.

Problem 37: Aiden has 12 cookies. They eat $1/4$ of the cookies. How many cookies did Aiden eat?

Answer: 3

Explanation: $1/4$ of 12 = $(1 \times 12) \div 4 = 3$ cookies.

Problem 38: Joseph has 8 cookies. They eat $1/4$ of the cookies. How many cookies did Joseph eat?

Answer: 2

Explanation: $1/4$ of 8 = $(1 \times 8) \div 4 = 2$ cookies.

Problem 39: Avery has 12 cookies. They eat $2/3$ of the cookies. How many cookies did Avery eat?

Answer: 8

Explanation: $2/3$ of 12 = $(2 \times 12) \div 3 = 8$ cookies.

Problem 40: Noah has 8 cookies. They eat $2/4$ of the cookies. How many cookies did Noah eat?

Answer: 4

Explanation: $2/4$ of 8 = $(2 \times 8) \div 4 = 4$ cookies.

Problem 41: Harper has 20 cookies. They eat $3/4$ of the cookies. How many cookies did Harper eat?

Answer: 15

Explanation: $3/4$ of 20 = $(3 \times 20) \div 4 = 15$ cookies.

Problem 42: Emma has 12 cookies. They eat $2/4$ of the cookies. How many cookies did Emma eat?

Answer: 6

Explanation: $2/4$ of 12 = $(2 \times 12) \div 4 = 6$ cookies.

Problem 43: Henry has 12 cookies. They eat $1/4$ of the cookies. How many cookies did Henry eat?

Answer: 3

Explanation: $1/4$ of 12 = $(1 \times 12) \div 4 = 3$ cookies.

Problem 44: Sophia has 9 cookies. They eat $2/3$ of the cookies. How many cookies did Sophia eat?

Answer: 6

Explanation: $2/3$ of 9 = $(2 \times 9) \div 3 = 6$ cookies.

Problem 45: Mia has 12 cookies. They eat $2/3$ of the cookies. How many cookies did Mia eat?

Answer: 8

Explanation: $2/3$ of 12 = $(2 \times 12) \div 3 = 8$ cookies.

Practice Exercises - Grade 3

Exercise 1: Emily has 16 cookies. They eat $\frac{2}{4}$ of the cookies. How many cookies did Emily eat?

Exercise 2: Jack has 8 cookies. They eat $\frac{2}{4}$ of the cookies. How many cookies did Jack eat?

Exercise 3: Samuel has 6 cookies. They eat $\frac{2}{3}$ of the cookies. How many cookies did Samuel eat?

Exercise 4: Abigail has 10 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Abigail eat?

Exercise 5: Sofia has 6 cookies. They eat $\frac{2}{3}$ of the cookies. How many cookies did Sofia eat?

Exercise 6: Olivia has 16 cookies. They eat $\frac{1}{4}$ of the cookies. How many cookies did Olivia eat?

Exercise 7: Lucas has 20 cookies. They eat $\frac{1}{4}$ of the cookies. How many cookies did Lucas eat?

Exercise 8: Ava has 8 cookies. They eat $\frac{1}{4}$ of the cookies. How many cookies did Ava eat?

Exercise 9: Mia has 12 cookies. They eat $\frac{3}{4}$ of the cookies. How many cookies did Mia eat?

Exercise 10: Aiden has 6 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Aiden eat?

Exercise 11: Henry has 6 cookies. They eat $\frac{2}{3}$ of the cookies. How many cookies did Henry eat?

Exercise 12: Abigail has 10 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Abigail eat?

Exercise 13: Charlotte has 10 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Charlotte eat?

Exercise 14: Isabella has 6 cookies. They eat $\frac{1}{3}$ of the cookies. How many cookies did Isabella eat?

Exercise 15: Joseph has 9 cookies. They eat $\frac{2}{3}$ of the cookies. How many cookies did Joseph eat?

Exercise 16: Alexander has 8 cookies. They eat $\frac{2}{4}$ of the cookies. How many cookies did Alexander eat?

Exercise 17: Emma has 10 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Emma eat?

Exercise 18: Mason has 15 cookies. They eat $\frac{1}{3}$ of the cookies. How many cookies did Mason eat?

Exercise 19: Liam has 15 cookies. They eat $\frac{1}{3}$ of the cookies. How many cookies did Liam eat?

Exercise 20: Amelia has 6 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Amelia eat?

Practice Exercise Answers:

Exercise 1: 8

Exercise 2: 4

Exercise 3: 4

Exercise 4: 5

Exercise 5: 4

Exercise 6: 4

Exercise 7: 5

Exercise 8: 2

Exercise 9: 9

Exercise 10: 3

Exercise 11: 4

Exercise 12: 5

Exercise 13: 5

Exercise 14: 2

Exercise 15: 6

Exercise 16: 4

Exercise 17: 5

Exercise 18: 5

Exercise 19: 5

Exercise 20: 3

Challenge Problems - Grade 3

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Alexander has 8 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Alexander eat?

Answer: 4

Solution: $\frac{1}{2}$ of 8 = $(1 \times 8) \div 2 = 4$ cookies.

Challenge 2: James has 6 cookies. They eat $\frac{2}{3}$ of the cookies. How many cookies did James eat?

Answer: 4

Solution: $\frac{2}{3}$ of 6 = $(2 \times 6) \div 3 = 4$ cookies.

Challenge 3: Emily has 16 cookies. They eat $\frac{3}{4}$ of the cookies. How many cookies did Emily eat?

Answer: 12

Solution: $\frac{3}{4}$ of 16 = $(3 \times 16) \div 4 = 12$ cookies.

Challenge 4: Emma has 12 cookies. They eat $\frac{2}{3}$ of the cookies. How many cookies did Emma eat?

Answer: 8

Solution: $\frac{2}{3}$ of 12 = $(2 \times 12) \div 3 = 8$ cookies.

Challenge 5: Charlotte has 10 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Charlotte eat?

Answer: 5

Solution: $\frac{1}{2}$ of 10 = $(1 \times 10) \div 2 = 5$ cookies.

Challenge 6: Alexander has 8 cookies. They eat $\frac{1}{4}$ of the cookies. How many cookies did Alexander eat?

Answer: 2

Solution: $\frac{1}{4}$ of 8 = $(1 \times 8) \div 4 = 2$ cookies.

Challenge 7: Lucas has 15 cookies. They eat $\frac{2}{3}$ of the cookies. How many cookies did Lucas eat?

Answer: 10

Solution: $\frac{2}{3}$ of 15 = $(2 \times 15) \div 3 = 10$ cookies.

Challenge 8: Noah has 8 cookies. They eat $\frac{1}{2}$ of the cookies. How many cookies did Noah eat?

Answer: 4

Solution: $1/2$ of $8 = (1 \times 8) \div 2 = 4$ cookies.

Challenge 9: David has 20 cookies. They eat $2/4$ of the cookies. How many cookies did David eat?

Answer: 10

Solution: $2/4$ of $20 = (2 \times 20) \div 4 = 10$ cookies.

Challenge 10: Lucas has 6 cookies. They eat $2/3$ of the cookies. How many cookies did Lucas eat?

Answer: 4

Solution: $2/3$ of $6 = (2 \times 6) \div 3 = 4$ cookies.

Challenge 11: Mason has 4 cookies. They eat $1/2$ of the cookies. How many cookies did Mason eat?

Answer: 2

Solution: $1/2$ of $4 = (1 \times 4) \div 2 = 2$ cookies.

Challenge 12: Ethan has 6 cookies. They eat $1/2$ of the cookies. How many cookies did Ethan eat?

Answer: 3

Solution: $1/2$ of $6 = (1 \times 6) \div 2 = 3$ cookies.

Challenge 13: Ethan has 15 cookies. They eat $2/3$ of the cookies. How many cookies did Ethan eat?

Answer: 10

Solution: $2/3$ of $15 = (2 \times 15) \div 3 = 10$ cookies.

Challenge 14: Harper has 6 cookies. They eat $1/2$ of the cookies. How many cookies did Harper eat?

Answer: 3

Solution: $1/2$ of $6 = (1 \times 6) \div 2 = 3$ cookies.

Challenge 15: Avery has 20 cookies. They eat $2/4$ of the cookies. How many cookies did Avery eat?

Answer: 10

Solution: $2/4$ of $20 = (2 \times 20) \div 4 = 10$ cookies.

Grade 4 Problems

Problem 1: Liam walks $\frac{5}{8}$ of a mile to school and $\frac{2}{8}$ of a mile back home. How far does Liam walk in total?

Answer: $\frac{7}{8}$

Explanation: Adding fractions with same denominator: $\frac{5}{8} + \frac{2}{8} = \frac{7}{8}$ miles.

Problem 2: Jack walks $\frac{2}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Jack walk in total?

Answer: $\frac{3}{4}$

Explanation: Adding fractions with same denominator: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ miles.

Problem 3: Harper walks $\frac{4}{8}$ of a mile to school and $\frac{3}{8}$ of a mile back home. How far does Harper walk in total?

Answer: $\frac{7}{8}$

Explanation: Adding fractions with same denominator: $\frac{4}{8} + \frac{3}{8} = \frac{7}{8}$ miles.

Problem 4: Olivia walks $\frac{4}{6}$ of a mile to school and $\frac{1}{6}$ of a mile back home. How far does Olivia walk in total?

Answer: $\frac{5}{6}$

Explanation: Adding fractions with same denominator: $\frac{4}{6} + \frac{1}{6} = \frac{5}{6}$ miles.

Problem 5: Noah walks $\frac{2}{6}$ of a mile to school and $\frac{3}{6}$ of a mile back home. How far does Noah walk in total?

Answer: $\frac{5}{6}$

Explanation: Adding fractions with same denominator: $\frac{2}{6} + \frac{3}{6} = \frac{5}{6}$ miles.

Problem 6: Samuel walks $\frac{5}{8}$ of a mile to school and $\frac{1}{8}$ of a mile back home. How far does Samuel walk in total?

Answer: $\frac{6}{8}$

Explanation: Adding fractions with same denominator: $\frac{5}{8} + \frac{1}{8} = \frac{6}{8}$ miles.

Problem 7: Harper walks $\frac{1}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Harper walk in total?

Answer: $\frac{2}{4}$

Explanation: Adding fractions with same denominator: $\frac{1}{4} + \frac{1}{4} = \frac{2}{4}$ miles.

Problem 8: Mason walks $\frac{2}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Mason walk in total?

Answer: $\frac{3}{4}$

Explanation: Adding fractions with same denominator: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ miles.

Problem 9: David walks $\frac{4}{6}$ of a mile to school and $\frac{1}{6}$ of a mile back home. How far does David walk in total?

Answer: $\frac{5}{6}$

Explanation: Adding fractions with same denominator: $\frac{4}{6} + \frac{1}{6} = \frac{5}{6}$ miles.

Problem 10: Mia walks $\frac{2}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Mia walk in total?

Answer: $\frac{3}{4}$

Explanation: Adding fractions with same denominator: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ miles.

Problem 11: Aiden walks $\frac{1}{5}$ of a mile to school and $\frac{2}{5}$ of a mile back home. How far does Aiden walk in total?

Answer: $\frac{3}{5}$

Explanation: Adding fractions with same denominator: $\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$ miles.

Problem 12: Lucas walks $\frac{3}{5}$ of a mile to school and $\frac{1}{5}$ of a mile back home. How far does Lucas walk in total?

Answer: $\frac{4}{5}$

Explanation: Adding fractions with same denominator: $\frac{3}{5} + \frac{1}{5} = \frac{4}{5}$ miles.

Problem 13: Joseph walks $\frac{2}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Joseph walk in total?

Answer: $\frac{3}{4}$

Explanation: Adding fractions with same denominator: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ miles.

Problem 14: Evelyn walks $\frac{2}{5}$ of a mile to school and $\frac{2}{5}$ of a mile back home. How far does Evelyn walk in total?

Answer: $\frac{4}{5}$

Explanation: Adding fractions with same denominator: $\frac{2}{5} + \frac{2}{5} = \frac{4}{5}$ miles.

Problem 15: Avery walks $\frac{2}{6}$ of a mile to school and $\frac{3}{6}$ of a mile back home. How far does Avery walk in total?

Answer: $\frac{5}{6}$

Explanation: Adding fractions with same denominator: $\frac{2}{6} + \frac{3}{6} = \frac{5}{6}$ miles.

Problem 16: James walks $\frac{4}{8}$ of a mile to school and $\frac{1}{8}$ of a mile back home. How far does James walk in total?

Answer: $\frac{5}{8}$

Explanation: Adding fractions with same denominator: $\frac{4}{8} + \frac{1}{8} = \frac{5}{8}$ miles.

Problem 17: Amelia walks $\frac{2}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Amelia walk in total?

Answer: $\frac{3}{4}$

Explanation: Adding fractions with same denominator: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ miles.

Problem 18: Noah walks $\frac{1}{5}$ of a mile to school and $\frac{2}{5}$ of a mile back home. How far does Noah walk in total?

Answer: $\frac{3}{5}$

Explanation: Adding fractions with same denominator: $\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$ miles.

Problem 19: Charlotte walks $\frac{2}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Charlotte walk in total?

Answer: $\frac{3}{4}$

Explanation: Adding fractions with same denominator: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ miles.

Problem 20: Joseph walks $\frac{2}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Joseph walk in total?

Answer: $\frac{3}{4}$

Explanation: Adding fractions with same denominator: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ miles.

Problem 21: James walks $\frac{4}{8}$ of a mile to school and $\frac{3}{8}$ of a mile back home. How far does James walk in total?

Answer: $\frac{7}{8}$

Explanation: Adding fractions with same denominator: $\frac{4}{8} + \frac{3}{8} = \frac{7}{8}$ miles.

Problem 22: Liam walks $\frac{3}{6}$ of a mile to school and $\frac{1}{6}$ of a mile back home. How far does Liam walk in total?

Answer: $\frac{4}{6}$

Explanation: Adding fractions with same denominator: $\frac{3}{6} + \frac{1}{6} = \frac{4}{6}$ miles.

Problem 23: Olivia walks $\frac{1}{5}$ of a mile to school and $\frac{1}{5}$ of a mile back home. How far does Olivia walk in total?

Answer: $\frac{2}{5}$

Explanation: Adding fractions with same denominator: $\frac{1}{5} + \frac{1}{5} = \frac{2}{5}$ miles.

Problem 24: Jack walks $\frac{2}{6}$ of a mile to school and $\frac{1}{6}$ of a mile back home. How far does Jack walk in total?

Answer: $\frac{3}{6}$

Explanation: Adding fractions with same denominator: $\frac{2}{6} + \frac{1}{6} = \frac{3}{6}$ miles.

Problem 25: Aiden walks $\frac{1}{4}$ of a mile to school and $\frac{2}{4}$ of a mile back home. How far does Aiden walk in total?

Answer: $\frac{3}{4}$

Explanation: Adding fractions with same denominator: $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$ miles.

Problem 26: Elizabeth walks $\frac{1}{6}$ of a mile to school and $\frac{2}{6}$ of a mile back home. How far does Elizabeth walk in total?

Answer: $\frac{3}{6}$

Explanation: Adding fractions with same denominator: $\frac{1}{6} + \frac{2}{6} = \frac{3}{6}$ miles.

Problem 27: David walks $\frac{2}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does David walk in total?

Answer: $\frac{3}{4}$

Explanation: Adding fractions with same denominator: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ miles.

Problem 28: Ethan walks $\frac{4}{8}$ of a mile to school and $\frac{3}{8}$ of a mile back home. How far does Ethan walk in total?

Answer: $\frac{7}{8}$

Explanation: Adding fractions with same denominator: $\frac{4}{8} + \frac{3}{8} = \frac{7}{8}$ miles.

Problem 29: Harper walks $\frac{5}{8}$ of a mile to school and $\frac{2}{8}$ of a mile back home. How far does Harper walk in total?

Answer: $\frac{7}{8}$

Explanation: Adding fractions with same denominator: $\frac{5}{8} + \frac{2}{8} = \frac{7}{8}$ miles.

Problem 30: Isabella walks $\frac{4}{8}$ of a mile to school and $\frac{3}{8}$ of a mile back home. How far does Isabella walk in total?

Answer: $\frac{7}{8}$

Explanation: Adding fractions with same denominator: $\frac{4}{8} + \frac{3}{8} = \frac{7}{8}$ miles.

Problem 31: Sophia walks $\frac{2}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Sophia walk in total?

Answer: $\frac{3}{4}$

Explanation: Adding fractions with same denominator: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ miles.

Problem 32: Mia walks $\frac{1}{5}$ of a mile to school and $\frac{2}{5}$ of a mile back home. How far does Mia walk in total?

Answer: $\frac{3}{5}$

Explanation: Adding fractions with same denominator: $\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$ miles.

Problem 33: Joseph walks $\frac{2}{5}$ of a mile to school and $\frac{1}{5}$ of a mile back home. How far does Joseph walk in total?

Answer: $\frac{3}{5}$

Explanation: Adding fractions with same denominator: $\frac{2}{5} + \frac{1}{5} = \frac{3}{5}$ miles.

Problem 34: Emily walks $\frac{1}{6}$ of a mile to school and $\frac{3}{6}$ of a mile back home. How far does Emily walk in total?

Answer: $\frac{4}{6}$

Explanation: Adding fractions with same denominator: $1/6 + 3/6 = 4/6$ miles.

Problem 35: David walks $2/5$ of a mile to school and $1/5$ of a mile back home. How far does David walk in total?

Answer: $3/5$

Explanation: Adding fractions with same denominator: $2/5 + 1/5 = 3/5$ miles.

Problem 36: Jack walks $3/8$ of a mile to school and $1/8$ of a mile back home. How far does Jack walk in total?

Answer: $4/8$

Explanation: Adding fractions with same denominator: $3/8 + 1/8 = 4/8$ miles.

Problem 37: Noah walks $2/5$ of a mile to school and $2/5$ of a mile back home. How far does Noah walk in total?

Answer: $4/5$

Explanation: Adding fractions with same denominator: $2/5 + 2/5 = 4/5$ miles.

Problem 38: David walks $4/8$ of a mile to school and $2/8$ of a mile back home. How far does David walk in total?

Answer: $6/8$

Explanation: Adding fractions with same denominator: $4/8 + 2/8 = 6/8$ miles.

Problem 39: Alexander walks $2/5$ of a mile to school and $1/5$ of a mile back home. How far does Alexander walk in total?

Answer: $3/5$

Explanation: Adding fractions with same denominator: $2/5 + 1/5 = 3/5$ miles.

Problem 40: Noah walks $3/5$ of a mile to school and $1/5$ of a mile back home. How far does Noah walk in total?

Answer: $4/5$

Explanation: Adding fractions with same denominator: $3/5 + 1/5 = 4/5$ miles.

Problem 41: James walks $3/8$ of a mile to school and $1/8$ of a mile back home. How far does James walk in total?

Answer: $\frac{4}{8}$

Explanation: Adding fractions with same denominator: $\frac{3}{8} + \frac{1}{8} = \frac{4}{8}$ miles.

Problem 42: Mia walks $\frac{2}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Mia walk in total?

Answer: $\frac{3}{4}$

Explanation: Adding fractions with same denominator: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ miles.

Problem 43: Aiden walks $\frac{2}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Aiden walk in total?

Answer: $\frac{3}{4}$

Explanation: Adding fractions with same denominator: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ miles.

Problem 44: Mason walks $\frac{2}{5}$ of a mile to school and $\frac{2}{5}$ of a mile back home. How far does Mason walk in total?

Answer: $\frac{4}{5}$

Explanation: Adding fractions with same denominator: $\frac{2}{5} + \frac{2}{5} = \frac{4}{5}$ miles.

Problem 45: Mason walks $\frac{2}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Mason walk in total?

Answer: $\frac{3}{4}$

Explanation: Adding fractions with same denominator: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ miles.

Practice Exercises - Grade 4

Exercise 1: Aiden walks $\frac{2}{6}$ of a mile to school and $\frac{2}{6}$ of a mile back home. How far does Aiden walk in total?

Exercise 2: David walks $\frac{5}{8}$ of a mile to school and $\frac{1}{8}$ of a mile back home. How far does David walk in total?

Exercise 3: Charlotte walks $\frac{1}{5}$ of a mile to school and $\frac{1}{5}$ of a mile back home. How far does Charlotte walk in total?

Exercise 4: Lucas walks $\frac{2}{5}$ of a mile to school and $\frac{2}{5}$ of a mile back home. How far does Lucas walk in total?

Exercise 5: Emily walks $\frac{2}{5}$ of a mile to school and $\frac{2}{5}$ of a mile back home. How far does Emily walk in total?

Exercise 6: David walks $\frac{2}{5}$ of a mile to school and $\frac{2}{5}$ of a mile back home. How far does David walk in total?

Exercise 7: Mia walks $\frac{3}{6}$ of a mile to school and $\frac{1}{6}$ of a mile back home. How far does Mia walk in total?

Exercise 8: Ava walks $\frac{1}{5}$ of a mile to school and $\frac{3}{5}$ of a mile back home. How far does Ava walk in total?

Exercise 9: David walks $\frac{1}{5}$ of a mile to school and $\frac{3}{5}$ of a mile back home. How far does David walk in total?

Exercise 10: Liam walks $\frac{1}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Liam walk in total?

Exercise 11: Evelyn walks $\frac{5}{8}$ of a mile to school and $\frac{2}{8}$ of a mile back home. How far does Evelyn walk in total?

Exercise 12: Alexander walks $\frac{1}{8}$ of a mile to school and $\frac{2}{8}$ of a mile back home. How far does Alexander walk in total?

Exercise 13: Emma walks $\frac{3}{6}$ of a mile to school and $\frac{1}{6}$ of a mile back home. How far does Emma walk in total?

Exercise 14: Sophia walks $\frac{1}{5}$ of a mile to school and $\frac{3}{5}$ of a mile back home. How far does Sophia walk in total?

Exercise 15: Alexander walks $\frac{1}{6}$ of a mile to school and $\frac{1}{6}$ of a mile back home. How far does Alexander walk in total?

Exercise 16: Ethan walks $\frac{2}{8}$ of a mile to school and $\frac{2}{8}$ of a mile back home. How far does Ethan walk in total?

Exercise 17: Aiden walks $\frac{1}{6}$ of a mile to school and $\frac{2}{6}$ of a mile back home. How far does Aiden walk in total?

Exercise 18: Sophia walks $\frac{4}{6}$ of a mile to school and $\frac{1}{6}$ of a mile back home. How far does Sophia walk in total?

Exercise 19: Samuel walks $\frac{3}{8}$ of a mile to school and $\frac{4}{8}$ of a mile back home. How far does Samuel walk in total?

Exercise 20: David walks $\frac{2}{5}$ of a mile to school and $\frac{2}{5}$ of a mile back home. How far does David walk in total?

Practice Exercise Answers:

Exercise 1: $\frac{4}{6}$

Exercise 2: $\frac{6}{8}$

Exercise 3: $\frac{2}{5}$

Exercise 4: $\frac{4}{5}$

Exercise 5: $\frac{4}{5}$

Exercise 6: $\frac{4}{5}$

Exercise 7: $\frac{4}{6}$

Exercise 8: $\frac{4}{5}$

Exercise 9: $\frac{4}{5}$

Exercise 10: $\frac{2}{4}$

Exercise 11: $\frac{7}{8}$

Exercise 12: $\frac{3}{8}$

Exercise 13: $\frac{4}{6}$

Exercise 14: $\frac{4}{5}$

Exercise 15: $\frac{2}{6}$

Exercise 16: $\frac{4}{8}$

Exercise 17: $\frac{3}{6}$

Exercise 18: $\frac{5}{6}$

Exercise 19: $\frac{7}{8}$

Exercise 20: $\frac{4}{5}$

Challenge Problems - Grade 4

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Samuel walks $\frac{1}{5}$ of a mile to school and $\frac{2}{5}$ of a mile back home. How far does Samuel walk in total?

Answer: $\frac{3}{5}$

Solution: Adding fractions with same denominator: $\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$ miles.

Challenge 2: James walks $\frac{6}{8}$ of a mile to school and $\frac{1}{8}$ of a mile back home. How far does James walk in total?

Answer: $\frac{7}{8}$

Solution: Adding fractions with same denominator: $\frac{6}{8} + \frac{1}{8} = \frac{7}{8}$ miles.

Challenge 3: Emily walks $\frac{2}{5}$ of a mile to school and $\frac{1}{5}$ of a mile back home. How far does Emily walk in total?

Answer: $\frac{3}{5}$

Solution: Adding fractions with same denominator: $\frac{2}{5} + \frac{1}{5} = \frac{3}{5}$ miles.

Challenge 4: Emily walks $\frac{1}{8}$ of a mile to school and $\frac{6}{8}$ of a mile back home. How far does Emily walk in total?

Answer: $\frac{7}{8}$

Solution: Adding fractions with same denominator: $\frac{1}{8} + \frac{6}{8} = \frac{7}{8}$ miles.

Challenge 5: Mia walks $\frac{2}{5}$ of a mile to school and $\frac{1}{5}$ of a mile back home. How far does Mia walk in total?

Answer: $\frac{3}{5}$

Solution: Adding fractions with same denominator: $\frac{2}{5} + \frac{1}{5} = \frac{3}{5}$ miles.

Challenge 6: Elizabeth walks $\frac{4}{6}$ of a mile to school and $\frac{1}{6}$ of a mile back home. How far does Elizabeth walk in total?

Answer: $\frac{5}{6}$

Solution: Adding fractions with same denominator: $\frac{4}{6} + \frac{1}{6} = \frac{5}{6}$ miles.

Challenge 7: Aiden walks $\frac{3}{8}$ of a mile to school and $\frac{2}{8}$ of a mile back home. How far does Aiden walk in total?

Answer: $\frac{5}{8}$

Solution: Adding fractions with same denominator: $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$ miles.

Challenge 8: James walks $\frac{1}{5}$ of a mile to school and $\frac{3}{5}$ of a mile back home. How far does James walk in total?

Answer: $\frac{4}{5}$

Solution: Adding fractions with same denominator: $\frac{1}{5} + \frac{3}{5} = \frac{4}{5}$ miles.

Challenge 9: Samuel walks $\frac{1}{5}$ of a mile to school and $\frac{1}{5}$ of a mile back home. How far does Samuel walk in total?

Answer: $\frac{2}{5}$

Solution: Adding fractions with same denominator: $\frac{1}{5} + \frac{1}{5} = \frac{2}{5}$ miles.

Challenge 10: David walks $\frac{4}{8}$ of a mile to school and $\frac{2}{8}$ of a mile back home. How far does David walk in total?

Answer: $\frac{6}{8}$

Solution: Adding fractions with same denominator: $\frac{4}{8} + \frac{2}{8} = \frac{6}{8}$ miles.

Challenge 11: Samuel walks $\frac{3}{5}$ of a mile to school and $\frac{1}{5}$ of a mile back home. How far does Samuel walk in total?

Answer: $\frac{4}{5}$

Solution: Adding fractions with same denominator: $\frac{3}{5} + \frac{1}{5} = \frac{4}{5}$ miles.

Challenge 12: Charlotte walks $\frac{2}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Charlotte walk in total?

Answer: $\frac{3}{4}$

Solution: Adding fractions with same denominator: $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ miles.

Challenge 13: Emily walks $\frac{2}{6}$ of a mile to school and $\frac{2}{6}$ of a mile back home. How far does Emily walk in total?

Answer: $\frac{4}{6}$

Solution: Adding fractions with same denominator: $\frac{2}{6} + \frac{2}{6} = \frac{4}{6}$ miles.

Challenge 14: Liam walks $\frac{3}{8}$ of a mile to school and $\frac{2}{8}$ of a mile back home. How far does Liam walk in total?

Answer: $\frac{5}{8}$

Solution: Adding fractions with same denominator: $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$ miles.

Challenge 15: Emma walks $\frac{1}{4}$ of a mile to school and $\frac{1}{4}$ of a mile back home. How far does Emma walk in total?

Answer: $\frac{2}{4}$

Solution: Adding fractions with same denominator: $\frac{1}{4} + \frac{1}{4} = \frac{2}{4}$ miles.

Grade 5 Problems

Problem 1: Avery used $\frac{1}{6}$ of a cup of flour for cookies and $\frac{2}{3}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{5}{6}$

Explanation: Converting to common denominator 6: $\frac{1}{6} = \frac{1}{6}$ and $\frac{2}{3} = \frac{4}{6}$. Adding: $\frac{1}{6} + \frac{4}{6} = \frac{5}{6}$ cups.

Problem 2: Sofia used $\frac{2}{5}$ of a cup of flour for cookies and $\frac{3}{6}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{27}{30}$

Explanation: Converting to common denominator 30: $\frac{2}{5} = \frac{12}{30}$ and $\frac{3}{6} = \frac{15}{30}$. Adding: $\frac{12}{30} + \frac{15}{30} = \frac{27}{30}$ cups.

Problem 3: Avery used $\frac{1}{5}$ of a cup of flour for cookies and $\frac{1}{4}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{9}{20}$

Explanation: Converting to common denominator 20: $\frac{1}{5} = \frac{4}{20}$ and $\frac{1}{4} = \frac{5}{20}$. Adding: $\frac{4}{20} + \frac{5}{20} = \frac{9}{20}$ cups.

Problem 4: Evelyn used $\frac{2}{6}$ of a cup of flour for cookies and $\frac{4}{7}$ of a cup for a cake. How much flour was used in total?

Answer: $38/42$

Explanation: Converting to common denominator 42: $2/6 = 14/42$ and $4/7 = 24/42$. Adding: $14/42 + 24/42 = 38/42$ cups.

Problem 5: Samuel used $1/5$ of a cup of flour for cookies and $5/6$ of a cup for a cake. How much flour was used in total?

Answer: $31/30$

Explanation: Converting to common denominator 30: $1/5 = 6/30$ and $5/6 = 25/30$. Adding: $6/30 + 25/30 = 31/30$ cups.

Problem 6: Henry used $1/2$ of a cup of flour for cookies and $2/3$ of a cup for a cake. How much flour was used in total?

Answer: $7/6$

Explanation: Converting to common denominator 6: $1/2 = 3/6$ and $2/3 = 4/6$. Adding: $3/6 + 4/6 = 7/6$ cups.

Problem 7: Henry used $1/3$ of a cup of flour for cookies and $1/2$ of a cup for a cake. How much flour was used in total?

Answer: $5/6$

Explanation: Converting to common denominator 6: $1/3 = 2/6$ and $1/2 = 3/6$. Adding: $2/6 + 3/6 = 5/6$ cups.

Problem 8: Avery used $2/4$ of a cup of flour for cookies and $2/3$ of a cup for a cake. How much flour was used in total?

Answer: $14/12$

Explanation: Converting to common denominator 12: $2/4 = 6/12$ and $2/3 = 8/12$. Adding: $6/12 + 8/12 = 14/12$ cups.

Problem 9: David used $3/4$ of a cup of flour for cookies and $2/5$ of a cup for a cake. How much flour was used in total?

Answer: $23/20$

Explanation: Converting to common denominator 20: $3/4 = 15/20$ and $2/5 = 8/20$. Adding: $15/20 + 8/20 = 23/20$ cups.

Problem 10: Aiden used $1/2$ of a cup of flour for cookies and $2/4$ of a cup for a cake. How much flour was used in total?

Answer: $4/4$

Explanation: Converting to common denominator 4: $\frac{1}{2} = \frac{2}{4}$ and $\frac{2}{4} = \frac{2}{4}$. Adding: $\frac{2}{4} + \frac{2}{4} = \frac{4}{4}$ cups.

Problem 11: Benjamin used $\frac{2}{4}$ of a cup of flour for cookies and $\frac{4}{5}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{26}{20}$

Explanation: Converting to common denominator 20: $\frac{2}{4} = \frac{10}{20}$ and $\frac{4}{5} = \frac{16}{20}$. Adding: $\frac{10}{20} + \frac{16}{20} = \frac{26}{20}$ cups.

Problem 12: Henry used $\frac{1}{6}$ of a cup of flour for cookies and $\frac{1}{2}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{4}{6}$

Explanation: Converting to common denominator 6: $\frac{1}{6} = \frac{1}{6}$ and $\frac{1}{2} = \frac{3}{6}$. Adding: $\frac{1}{6} + \frac{3}{6} = \frac{4}{6}$ cups.

Problem 13: Sofia used $\frac{4}{5}$ of a cup of flour for cookies and $\frac{1}{6}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{29}{30}$

Explanation: Converting to common denominator 30: $\frac{4}{5} = \frac{24}{30}$ and $\frac{1}{6} = \frac{5}{30}$. Adding: $\frac{24}{30} + \frac{5}{30} = \frac{29}{30}$ cups.

Problem 14: Mia used $\frac{1}{2}$ of a cup of flour for cookies and $\frac{2}{3}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{7}{6}$

Explanation: Converting to common denominator 6: $\frac{1}{2} = \frac{3}{6}$ and $\frac{2}{3} = \frac{4}{6}$. Adding: $\frac{3}{6} + \frac{4}{6} = \frac{7}{6}$ cups.

Problem 15: Olivia used $\frac{4}{6}$ of a cup of flour for cookies and $\frac{1}{2}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{7}{6}$

Explanation: Converting to common denominator 6: $\frac{4}{6} = \frac{4}{6}$ and $\frac{1}{2} = \frac{3}{6}$. Adding: $\frac{4}{6} + \frac{3}{6} = \frac{7}{6}$ cups.

Problem 16: Charlotte used $\frac{1}{4}$ of a cup of flour for cookies and $\frac{3}{5}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{17}{20}$

Explanation: Converting to common denominator 20: $\frac{1}{4} = \frac{5}{20}$ and $\frac{3}{5} = \frac{12}{20}$. Adding: $\frac{5}{20} + \frac{12}{20} = \frac{17}{20}$ cups.

Problem 17: James used $\frac{1}{2}$ of a cup of flour for cookies and $\frac{1}{3}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{5}{6}$

Explanation: Converting to common denominator 6: $\frac{1}{2} = \frac{3}{6}$ and $\frac{1}{3} = \frac{2}{6}$. Adding: $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$ cups.

Problem 18: Samuel used $\frac{2}{3}$ of a cup of flour for cookies and $\frac{3}{5}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{19}{15}$

Explanation: Converting to common denominator 15: $\frac{2}{3} = \frac{10}{15}$ and $\frac{3}{5} = \frac{9}{15}$. Adding: $\frac{10}{15} + \frac{9}{15} = \frac{19}{15}$ cups.

Problem 19: Isabella used $\frac{3}{4}$ of a cup of flour for cookies and $\frac{1}{6}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{11}{12}$

Explanation: Converting to common denominator 12: $\frac{3}{4} = \frac{9}{12}$ and $\frac{1}{6} = \frac{2}{12}$. Adding: $\frac{9}{12} + \frac{2}{12} = \frac{11}{12}$ cups.

Problem 20: Samuel used $\frac{2}{3}$ of a cup of flour for cookies and $\frac{1}{4}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{11}{12}$

Explanation: Converting to common denominator 12: $\frac{2}{3} = \frac{8}{12}$ and $\frac{1}{4} = \frac{3}{12}$. Adding: $\frac{8}{12} + \frac{3}{12} = \frac{11}{12}$ cups.

Problem 21: Joseph used $\frac{1}{5}$ of a cup of flour for cookies and $\frac{2}{3}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{13}{15}$

Explanation: Converting to common denominator 15: $\frac{1}{5} = \frac{3}{15}$ and $\frac{2}{3} = \frac{10}{15}$. Adding: $\frac{3}{15} + \frac{10}{15} = \frac{13}{15}$ cups.

Problem 22: Aiden used $\frac{5}{6}$ of a cup of flour for cookies and $\frac{6}{7}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{71}{42}$

Explanation: Converting to common denominator 42: $\frac{5}{6} = \frac{35}{42}$ and $\frac{6}{7} = \frac{36}{42}$. Adding: $\frac{35}{42} + \frac{36}{42} = \frac{71}{42}$ cups.

Problem 23: Henry used $\frac{3}{6}$ of a cup of flour for cookies and $\frac{1}{3}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{5}{6}$

Explanation: Converting to common denominator 6: $\frac{3}{6} = \frac{3}{6}$ and $\frac{1}{3} = \frac{2}{6}$. Adding: $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$ cups.

Problem 24: Benjamin used $\frac{3}{4}$ of a cup of flour for cookies and $\frac{1}{2}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{5}{4}$

Explanation: Converting to common denominator 4: $\frac{3}{4} = \frac{3}{4}$ and $\frac{1}{2} = \frac{2}{4}$. Adding: $\frac{3}{4} + \frac{2}{4} = \frac{5}{4}$ cups.

Problem 25: Samuel used $\frac{5}{6}$ of a cup of flour for cookies and $\frac{3}{4}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{19}{12}$

Explanation: Converting to common denominator 12: $\frac{5}{6} = \frac{10}{12}$ and $\frac{3}{4} = \frac{9}{12}$. Adding: $\frac{10}{12} + \frac{9}{12} = \frac{19}{12}$ cups.

Problem 26: Ethan used $\frac{2}{4}$ of a cup of flour for cookies and $\frac{1}{6}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{8}{12}$

Explanation: Converting to common denominator 12: $\frac{2}{4} = \frac{6}{12}$ and $\frac{1}{6} = \frac{2}{12}$. Adding: $\frac{6}{12} + \frac{2}{12} = \frac{8}{12}$ cups.

Problem 27: Ethan used $\frac{3}{6}$ of a cup of flour for cookies and $\frac{3}{5}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{33}{30}$

Explanation: Converting to common denominator 30: $\frac{3}{6} = \frac{15}{30}$ and $\frac{3}{5} = \frac{18}{30}$. Adding: $\frac{15}{30} + \frac{18}{30} = \frac{33}{30}$ cups.

Problem 28: Lucas used $\frac{3}{4}$ of a cup of flour for cookies and $\frac{1}{3}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{13}{12}$

Explanation: Converting to common denominator 12: $\frac{3}{4} = \frac{9}{12}$ and $\frac{1}{3} = \frac{4}{12}$. Adding: $\frac{9}{12} + \frac{4}{12} = \frac{13}{12}$ cups.

Problem 29: Jack used $\frac{1}{2}$ of a cup of flour for cookies and $\frac{2}{3}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{7}{6}$

Explanation: Converting to common denominator 6: $\frac{1}{2} = \frac{3}{6}$ and $\frac{2}{3} = \frac{4}{6}$. Adding: $\frac{3}{6} + \frac{4}{6} = \frac{7}{6}$ cups.

Problem 30: Benjamin used $\frac{1}{3}$ of a cup of flour for cookies and $\frac{2}{4}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{10}{12}$

Explanation: Converting to common denominator 12: $\frac{1}{3} = \frac{4}{12}$ and $\frac{2}{4} = \frac{6}{12}$. Adding: $\frac{4}{12} + \frac{6}{12} = \frac{10}{12}$ cups.

Problem 31: Olivia used $\frac{1}{3}$ of a cup of flour for cookies and $\frac{1}{2}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{5}{6}$

Explanation: Converting to common denominator 6: $\frac{1}{3} = \frac{2}{6}$ and $\frac{1}{2} = \frac{3}{6}$. Adding: $\frac{2}{6} + \frac{3}{6} = \frac{5}{6}$ cups.

Problem 32: Sophia used $\frac{1}{3}$ of a cup of flour for cookies and $\frac{1}{2}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{5}{6}$

Explanation: Converting to common denominator 6: $\frac{1}{3} = \frac{2}{6}$ and $\frac{1}{2} = \frac{3}{6}$. Adding: $\frac{2}{6} + \frac{3}{6} = \frac{5}{6}$ cups.

Problem 33: Evelyn used $\frac{4}{6}$ of a cup of flour for cookies and $\frac{2}{4}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{14}{12}$

Explanation: Converting to common denominator 12: $\frac{4}{6} = \frac{8}{12}$ and $\frac{2}{4} = \frac{6}{12}$. Adding: $\frac{8}{12} + \frac{6}{12} = \frac{14}{12}$ cups.

Problem 34: Henry used $\frac{1}{4}$ of a cup of flour for cookies and $\frac{2}{6}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{7}{12}$

Explanation: Converting to common denominator 12: $\frac{1}{4} = \frac{3}{12}$ and $\frac{2}{6} = \frac{4}{12}$. Adding: $\frac{3}{12} + \frac{4}{12} = \frac{7}{12}$ cups.

Problem 35: Abigail used $\frac{1}{5}$ of a cup of flour for cookies and $\frac{1}{6}$ of a cup for a cake. How much flour was used in total?

Answer: 11/30

Explanation: Converting to common denominator 30: $\frac{1}{5} = \frac{6}{30}$ and $\frac{1}{6} = \frac{5}{30}$. Adding: $\frac{6}{30} + \frac{5}{30} = \frac{11}{30}$ cups.

Problem 36: Sebastian used $\frac{1}{2}$ of a cup of flour for cookies and $\frac{4}{6}$ of a cup for a cake. How much flour was used in total?

Answer: 7/6

Explanation: Converting to common denominator 6: $\frac{1}{2} = \frac{3}{6}$ and $\frac{4}{6} = \frac{4}{6}$. Adding: $\frac{3}{6} + \frac{4}{6} = \frac{7}{6}$ cups.

Problem 37: Harper used $\frac{1}{6}$ of a cup of flour for cookies and $\frac{1}{2}$ of a cup for a cake. How much flour was used in total?

Answer: 4/6

Explanation: Converting to common denominator 6: $\frac{1}{6} = \frac{1}{6}$ and $\frac{1}{2} = \frac{3}{6}$. Adding: $\frac{1}{6} + \frac{3}{6} = \frac{4}{6}$ cups.

Problem 38: Avery used $\frac{1}{2}$ of a cup of flour for cookies and $\frac{1}{3}$ of a cup for a cake. How much flour was used in total?

Answer: 5/6

Explanation: Converting to common denominator 6: $\frac{1}{2} = \frac{3}{6}$ and $\frac{1}{3} = \frac{2}{6}$. Adding: $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$ cups.

Problem 39: Henry used $\frac{2}{5}$ of a cup of flour for cookies and $\frac{2}{6}$ of a cup for a cake. How much flour was used in total?

Answer: 22/30

Explanation: Converting to common denominator 30: $\frac{2}{5} = \frac{12}{30}$ and $\frac{2}{6} = \frac{10}{30}$. Adding: $\frac{12}{30} + \frac{10}{30} = \frac{22}{30}$ cups.

Problem 40: Benjamin used $\frac{1}{3}$ of a cup of flour for cookies and $\frac{1}{4}$ of a cup for a cake. How much flour was used in total?

Answer: 7/12

Explanation: Converting to common denominator 12: $\frac{1}{3} = \frac{4}{12}$ and $\frac{1}{4} = \frac{3}{12}$. Adding: $\frac{4}{12} + \frac{3}{12} = \frac{7}{12}$ cups.

Problem 41: Alexander used $\frac{1}{2}$ of a cup of flour for cookies and $\frac{2}{6}$ of a cup for a cake. How much flour was used in total?

Answer: 5/6

Explanation: Converting to common denominator 6: $\frac{1}{2} = \frac{3}{6}$ and $\frac{2}{6} = \frac{2}{6}$. Adding: $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$ cups.

Problem 42: Sebastian used $\frac{1}{3}$ of a cup of flour for cookies and $\frac{2}{6}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{4}{6}$

Explanation: Converting to common denominator 6: $\frac{1}{3} = \frac{2}{6}$ and $\frac{2}{6} = \frac{2}{6}$. Adding: $\frac{2}{6} + \frac{2}{6} = \frac{4}{6}$ cups.

Problem 43: Benjamin used $\frac{1}{3}$ of a cup of flour for cookies and $\frac{4}{5}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{17}{15}$

Explanation: Converting to common denominator 15: $\frac{1}{3} = \frac{5}{15}$ and $\frac{4}{5} = \frac{12}{15}$. Adding: $\frac{5}{15} + \frac{12}{15} = \frac{17}{15}$ cups.

Problem 44: Mia used $\frac{3}{5}$ of a cup of flour for cookies and $\frac{1}{3}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{14}{15}$

Explanation: Converting to common denominator 15: $\frac{3}{5} = \frac{9}{15}$ and $\frac{1}{3} = \frac{5}{15}$. Adding: $\frac{9}{15} + \frac{5}{15} = \frac{14}{15}$ cups.

Problem 45: Sofia used $\frac{2}{5}$ of a cup of flour for cookies and $\frac{3}{4}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{23}{20}$

Explanation: Converting to common denominator 20: $\frac{2}{5} = \frac{8}{20}$ and $\frac{3}{4} = \frac{15}{20}$. Adding: $\frac{8}{20} + \frac{15}{20} = \frac{23}{20}$ cups.

Practice Exercises - Grade 5

Exercise 1: Harper used $\frac{2}{4}$ of a cup of flour for cookies and $\frac{4}{5}$ of a cup for a cake. How much flour was used in total?

Exercise 2: Emily used $\frac{3}{6}$ of a cup of flour for cookies and $\frac{6}{7}$ of a cup for a cake. How much flour was used in total?

Exercise 3: Ava used $\frac{2}{6}$ of a cup of flour for cookies and $\frac{3}{4}$ of a cup for a cake. How much flour was used in total?

Exercise 4: Joseph used $\frac{1}{2}$ of a cup of flour for cookies and $\frac{1}{4}$ of a cup for a cake. How much flour was used in total?

Exercise 5: Aiden used $\frac{2}{4}$ of a cup of flour for cookies and $\frac{1}{3}$ of a cup for a cake. How much flour was used in total?

Exercise 6: James used $\frac{1}{5}$ of a cup of flour for cookies and $\frac{2}{3}$ of a cup for a cake. How much flour was used in total?

Exercise 7: Joseph used $\frac{1}{5}$ of a cup of flour for cookies and $\frac{1}{2}$ of a cup for a cake. How much flour was used in total?

Exercise 8: Ava used $\frac{2}{4}$ of a cup of flour for cookies and $\frac{1}{3}$ of a cup for a cake. How much flour was used in total?

Exercise 9: Jack used $\frac{2}{4}$ of a cup of flour for cookies and $\frac{2}{3}$ of a cup for a cake. How much flour was used in total?

Exercise 10: Noah used $\frac{1}{5}$ of a cup of flour for cookies and $\frac{1}{3}$ of a cup for a cake. How much flour was used in total?

Exercise 11: Joseph used $\frac{2}{3}$ of a cup of flour for cookies and $\frac{1}{4}$ of a cup for a cake. How much flour was used in total?

Exercise 12: Lucas used $\frac{3}{4}$ of a cup of flour for cookies and $\frac{1}{3}$ of a cup for a cake. How much flour was used in total?

Exercise 13: Ethan used $\frac{4}{5}$ of a cup of flour for cookies and $\frac{5}{6}$ of a cup for a cake. How much flour was used in total?

Exercise 14: Charlotte used $\frac{1}{5}$ of a cup of flour for cookies and $\frac{1}{6}$ of a cup for a cake. How much flour was used in total?

Exercise 15: Avery used $\frac{2}{4}$ of a cup of flour for cookies and $\frac{2}{6}$ of a cup for a cake. How much flour was used in total?

Exercise 16: Aiden used $\frac{1}{2}$ of a cup of flour for cookies and $\frac{5}{6}$ of a cup for a cake. How much flour was used in total?

Exercise 17: Harper used $\frac{2}{4}$ of a cup of flour for cookies and $\frac{2}{3}$ of a cup for a cake. How much flour was used in total?

Exercise 18: Ethan used $\frac{3}{5}$ of a cup of flour for cookies and $\frac{1}{2}$ of a cup for a cake. How much flour was used in total?

Exercise 19: James used $\frac{1}{4}$ of a cup of flour for cookies and $\frac{1}{2}$ of a cup for a cake. How much flour was used in total?

Exercise 20: Avery used $\frac{1}{6}$ of a cup of flour for cookies and $\frac{1}{3}$ of a cup for a cake. How much flour was used in total?

Practice Exercise Answers:

Exercise 1: $\frac{26}{20}$

Exercise 2: $\frac{57}{42}$

Exercise 3: $\frac{13}{12}$

Exercise 4: $\frac{3}{4}$

Exercise 5: $\frac{10}{12}$

Exercise 6: $\frac{13}{15}$

Exercise 7: $\frac{7}{10}$

Exercise 8: $\frac{10}{12}$

Exercise 9: $\frac{14}{12}$

Exercise 10: $\frac{8}{15}$

Exercise 11: $\frac{11}{12}$

Exercise 12: $\frac{13}{12}$

Exercise 13: $\frac{49}{30}$

Exercise 14: $\frac{11}{30}$

Exercise 15: $\frac{10}{12}$

Exercise 16: $\frac{8}{6}$

Exercise 17: $\frac{14}{12}$

Exercise 18: $\frac{11}{10}$

Exercise 19: $\frac{3}{4}$

Exercise 20: $\frac{3}{6}$

Challenge Problems - Grade 5

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Henry used $\frac{1}{6}$ of a cup of flour for cookies and $\frac{6}{7}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{43}{42}$

Solution: Converting to common denominator 42: $\frac{1}{6} = \frac{7}{42}$ and $\frac{6}{7} = \frac{36}{42}$. Adding: $\frac{7}{42} + \frac{36}{42} = \frac{43}{42}$ cups.

Challenge 2: Charlotte used $\frac{2}{4}$ of a cup of flour for cookies and $\frac{1}{2}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{4}{4}$

Solution: Converting to common denominator 4: $\frac{2}{4} = \frac{2}{4}$ and $\frac{1}{2} = \frac{2}{4}$. Adding: $\frac{2}{4} + \frac{2}{4} = \frac{4}{4}$ cups.

Challenge 3: Liam used $\frac{1}{3}$ of a cup of flour for cookies and $\frac{1}{2}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{5}{6}$

Solution: Converting to common denominator 6: $\frac{1}{3} = \frac{2}{6}$ and $\frac{1}{2} = \frac{3}{6}$. Adding: $\frac{2}{6} + \frac{3}{6} = \frac{5}{6}$ cups.

Challenge 4: Benjamin used $\frac{1}{4}$ of a cup of flour for cookies and $\frac{1}{2}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{3}{4}$

Solution: Converting to common denominator 4: $\frac{1}{4} = \frac{1}{4}$ and $\frac{1}{2} = \frac{2}{4}$. Adding: $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$ cups.

Challenge 5: Alexander used $\frac{1}{4}$ of a cup of flour for cookies and $\frac{1}{3}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{7}{12}$

Solution: Converting to common denominator 12: $\frac{1}{4} = \frac{3}{12}$ and $\frac{1}{3} = \frac{4}{12}$. Adding: $\frac{3}{12} + \frac{4}{12} = \frac{7}{12}$ cups.

Challenge 6: Liam used $\frac{2}{6}$ of a cup of flour for cookies and $\frac{5}{7}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{44}{42}$

Solution: Converting to common denominator 42: $\frac{2}{6} = \frac{14}{42}$ and $\frac{5}{7} = \frac{30}{42}$. Adding: $\frac{14}{42} + \frac{30}{42} = \frac{44}{42}$ cups.

Challenge 7: Sofia used $\frac{2}{3}$ of a cup of flour for cookies and $\frac{2}{6}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{6}{6}$

Solution: Converting to common denominator 6: $\frac{2}{3} = \frac{4}{6}$ and $\frac{2}{6} = \frac{2}{6}$. Adding: $\frac{4}{6} + \frac{2}{6} = \frac{6}{6}$ cups.

Challenge 8: Ava used $\frac{1}{3}$ of a cup of flour for cookies and $\frac{1}{4}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{7}{12}$

Solution: Converting to common denominator 12: $\frac{1}{3} = \frac{4}{12}$ and $\frac{1}{4} = \frac{3}{12}$. Adding: $\frac{4}{12} + \frac{3}{12} = \frac{7}{12}$ cups.

Challenge 9: Amelia used $\frac{1}{6}$ of a cup of flour for cookies and $\frac{1}{7}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{13}{42}$

Solution: Converting to common denominator 42: $\frac{1}{6} = \frac{7}{42}$ and $\frac{1}{7} = \frac{6}{42}$. Adding: $\frac{7}{42} + \frac{6}{42} = \frac{13}{42}$ cups.

Challenge 10: Ethan used $\frac{3}{5}$ of a cup of flour for cookies and $\frac{2}{6}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{28}{30}$

Solution: Converting to common denominator 30: $\frac{3}{5} = \frac{18}{30}$ and $\frac{2}{6} = \frac{10}{30}$. Adding: $\frac{18}{30} + \frac{10}{30} = \frac{28}{30}$ cups.

Challenge 11: Jack used $\frac{1}{4}$ of a cup of flour for cookies and $\frac{3}{5}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{17}{20}$

Solution: Converting to common denominator 20: $\frac{1}{4} = \frac{5}{20}$ and $\frac{3}{5} = \frac{12}{20}$. Adding: $\frac{5}{20} + \frac{12}{20} = \frac{17}{20}$ cups.

Challenge 12: Abigail used $\frac{1}{2}$ of a cup of flour for cookies and $\frac{2}{5}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{9}{10}$

Solution: Converting to common denominator 10: $\frac{1}{2} = \frac{5}{10}$ and $\frac{2}{5} = \frac{4}{10}$. Adding: $\frac{5}{10} + \frac{4}{10} = \frac{9}{10}$ cups.

Challenge 13: Mia used $\frac{4}{5}$ of a cup of flour for cookies and $\frac{2}{3}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{22}{15}$

Solution: Converting to common denominator 15: $\frac{4}{5} = \frac{12}{15}$ and $\frac{2}{3} = \frac{10}{15}$. Adding: $\frac{12}{15} + \frac{10}{15} = \frac{22}{15}$ cups.

Challenge 14: Sofia used $\frac{2}{6}$ of a cup of flour for cookies and $\frac{1}{7}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{20}{42}$

Solution: Converting to common denominator 42: $\frac{2}{6} = \frac{14}{42}$ and $\frac{1}{7} = \frac{6}{42}$. Adding: $\frac{14}{42} + \frac{6}{42} = \frac{20}{42}$ cups.

Challenge 15: Amelia used $\frac{3}{4}$ of a cup of flour for cookies and $\frac{1}{2}$ of a cup for a cake. How much flour was used in total?

Answer: $\frac{5}{4}$

Solution: Converting to common denominator 4: $\frac{3}{4} = \frac{3}{4}$ and $\frac{1}{2} = \frac{2}{4}$. Adding: $\frac{3}{4} + \frac{2}{4} = \frac{5}{4}$ cups.

Grade 6 Problems

Problem 1: Ethan has $\frac{4}{2}$ of a pizza. They give $\frac{5}{6}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{20}{12}$

Explanation: Multiplying fractions: $(\frac{4}{2}) \times (\frac{5}{6}) = \frac{20}{12}$ of the pizza.

Problem 2: Evelyn has $\frac{1}{3}$ of a pizza. They give $\frac{5}{7}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{5}{21}$

Explanation: Multiplying fractions: $(\frac{1}{3}) \times (\frac{5}{7}) = \frac{5}{21}$ of the pizza.

Problem 3: Elizabeth has $\frac{2}{3}$ pounds of trail mix to divide into bags of $\frac{3}{2}$ pounds each. How many bags can be made?

Answer: 4/9

Explanation: Dividing fractions: $(2/3) \div (3/2) = (2/3) \times (2/3) = 4/9$ bags.

Problem 4: Samuel has $2/5$ of a pizza. They give $3/6$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 6/30

Explanation: Multiplying fractions: $(2/5) \times (3/6) = 6/30$ of the pizza.

Problem 5: Samuel has $3/3$ of a pizza. They give $2/6$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 6/18

Explanation: Multiplying fractions: $(3/3) \times (2/6) = 6/18$ of the pizza.

Problem 6: Ethan has $3/8$ of a pizza. They give $4/4$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 12/32

Explanation: Multiplying fractions: $(3/8) \times (4/4) = 12/32$ of the pizza.

Problem 7: Abigail has $1/3$ of a pizza. They give $1/3$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 1/9

Explanation: Multiplying fractions: $(1/3) \times (1/3) = 1/9$ of the pizza.

Problem 8: Sophia has $5/6$ pounds of trail mix to divide into bags of $1/6$ pounds each. How many bags can be made?

Answer: 30/6

Explanation: Dividing fractions: $(5/6) \div (1/6) = (5/6) \times (6/1) = 30/6$ bags.

Problem 9: Ethan has $3/7$ of a pizza. They give $4/7$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 12/49

Explanation: Multiplying fractions: $(3/7) \times (4/7) = 12/49$ of the pizza.

Problem 10: Mia has $\frac{3}{6}$ of a pizza. They give $\frac{4}{5}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{12}{30}$

Explanation: Multiplying fractions: $(\frac{3}{6}) \times (\frac{4}{5}) = \frac{12}{30}$ of the pizza.

Problem 11: Mason has $\frac{4}{6}$ of a pizza. They give $\frac{3}{3}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{12}{18}$

Explanation: Multiplying fractions: $(\frac{4}{6}) \times (\frac{3}{3}) = \frac{12}{18}$ of the pizza.

Problem 12: Charlotte has $\frac{4}{5}$ pounds of trail mix to divide into bags of $\frac{4}{5}$ pounds each. How many bags can be made?

Answer: $\frac{20}{20}$

Explanation: Dividing fractions: $(\frac{4}{5}) \div (\frac{4}{5}) = (\frac{4}{5}) \times (\frac{5}{4}) = \frac{20}{20}$ bags.

Problem 13: Amelia has $\frac{1}{6}$ of a pizza. They give $\frac{5}{6}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{5}{36}$

Explanation: Multiplying fractions: $(\frac{1}{6}) \times (\frac{5}{6}) = \frac{5}{36}$ of the pizza.

Problem 14: Ethan has $\frac{4}{8}$ of a pizza. They give $\frac{5}{8}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{20}{64}$

Explanation: Multiplying fractions: $(\frac{4}{8}) \times (\frac{5}{8}) = \frac{20}{64}$ of the pizza.

Problem 15: Ava has $\frac{3}{6}$ pounds of trail mix to divide into bags of $\frac{3}{5}$ pounds each. How many bags can be made?

Answer: $\frac{15}{18}$

Explanation: Dividing fractions: $(\frac{3}{6}) \div (\frac{3}{5}) = (\frac{3}{6}) \times (\frac{5}{3}) = \frac{15}{18}$ bags.

Problem 16: Joseph has $\frac{1}{7}$ of a pizza. They give $\frac{2}{8}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{2}{56}$

Explanation: Multiplying fractions: $(\frac{1}{7}) \times (\frac{2}{8}) = \frac{2}{56}$ of the pizza.

Problem 17: David has $\frac{2}{8}$ pounds of trail mix to divide into bags of $\frac{2}{5}$ pounds each. How many bags can be made?

Answer: 10/16

Explanation: Dividing fractions: $(\frac{2}{8}) \div (\frac{2}{5}) = (\frac{2}{8}) \times (\frac{5}{2}) = 10/16$ bags.

Problem 18: Jack has $\frac{6}{6}$ pounds of trail mix to divide into bags of $\frac{2}{3}$ pounds each. How many bags can be made?

Answer: 18/12

Explanation: Dividing fractions: $(\frac{6}{6}) \div (\frac{2}{3}) = (\frac{6}{6}) \times (\frac{3}{2}) = 18/12$ bags.

Problem 19: Henry has $\frac{5}{6}$ of a pizza. They give $\frac{5}{6}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 25/36

Explanation: Multiplying fractions: $(\frac{5}{6}) \times (\frac{5}{6}) = 25/36$ of the pizza.

Problem 20: David has $\frac{6}{4}$ pounds of trail mix to divide into bags of $\frac{4}{4}$ pounds each. How many bags can be made?

Answer: 24/16

Explanation: Dividing fractions: $(\frac{6}{4}) \div (\frac{4}{4}) = (\frac{6}{4}) \times (\frac{4}{4}) = 24/16$ bags.

Problem 21: Olivia has $\frac{4}{2}$ of a pizza. They give $\frac{5}{3}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 20/6

Explanation: Multiplying fractions: $(\frac{4}{2}) \times (\frac{5}{3}) = 20/6$ of the pizza.

Problem 22: Mia has $\frac{3}{7}$ pounds of trail mix to divide into bags of $\frac{1}{5}$ pounds each. How many bags can be made?

Answer: 15/7

Explanation: Dividing fractions: $(\frac{3}{7}) \div (\frac{1}{5}) = (\frac{3}{7}) \times (\frac{5}{1}) = 15/7$ bags.

Problem 23: Sebastian has $\frac{1}{6}$ of a pizza. They give $\frac{4}{5}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 4/30

Explanation: Multiplying fractions: $(1/6) \times (4/5) = 4/30$ of the pizza.

Problem 24: Aiden has $4/7$ of a pizza. They give $4/6$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $16/42$

Explanation: Multiplying fractions: $(4/7) \times (4/6) = 16/42$ of the pizza.

Problem 25: Sophia has $3/4$ pounds of trail mix to divide into bags of $3/2$ pounds each. How many bags can be made?

Answer: $6/12$

Explanation: Dividing fractions: $(3/4) \div (3/2) = (3/4) \times (2/3) = 6/12$ bags.

Problem 26: Henry has $2/5$ pounds of trail mix to divide into bags of $3/5$ pounds each. How many bags can be made?

Answer: $10/15$

Explanation: Dividing fractions: $(2/5) \div (3/5) = (2/5) \times (5/3) = 10/15$ bags.

Problem 27: Joseph has $5/2$ of a pizza. They give $5/6$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $25/12$

Explanation: Multiplying fractions: $(5/2) \times (5/6) = 25/12$ of the pizza.

Problem 28: Isabella has $6/7$ pounds of trail mix to divide into bags of $4/6$ pounds each. How many bags can be made?

Answer: $36/28$

Explanation: Dividing fractions: $(6/7) \div (4/6) = (6/7) \times (6/4) = 36/28$ bags.

Problem 29: Liam has $2/4$ pounds of trail mix to divide into bags of $1/5$ pounds each. How many bags can be made?

Answer: $10/4$

Explanation: Dividing fractions: $(2/4) \div (1/5) = (2/4) \times (5/1) = 10/4$ bags.

Problem 30: Sofia has $5/4$ of a pizza. They give $1/6$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 5/24

Explanation: Multiplying fractions: $(5/4) \times (1/6) = 5/24$ of the pizza.

Problem 31: Olivia has $3/8$ of a pizza. They give $5/3$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 15/24

Explanation: Multiplying fractions: $(3/8) \times (5/3) = 15/24$ of the pizza.

Problem 32: Elizabeth has $6/4$ pounds of trail mix to divide into bags of $2/5$ pounds each. How many bags can be made?

Answer: 30/8

Explanation: Dividing fractions: $(6/4) \div (2/5) = (6/4) \times (5/2) = 30/8$ bags.

Problem 33: Sophia has $3/3$ of a pizza. They give $5/4$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 15/12

Explanation: Multiplying fractions: $(3/3) \times (5/4) = 15/12$ of the pizza.

Problem 34: Liam has $5/7$ pounds of trail mix to divide into bags of $2/4$ pounds each. How many bags can be made?

Answer: 20/14

Explanation: Dividing fractions: $(5/7) \div (2/4) = (5/7) \times (4/2) = 20/14$ bags.

Problem 35: Aiden has $5/2$ of a pizza. They give $2/2$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 10/4

Explanation: Multiplying fractions: $(5/2) \times (2/2) = 10/4$ of the pizza.

Problem 36: Joseph has $3/4$ pounds of trail mix to divide into bags of $1/2$ pounds each. How many bags can be made?

Answer: 6/4

Explanation: Dividing fractions: $(3/4) \div (1/2) = (3/4) \times (2/1) = 6/4$ bags.

Problem 37: Samuel has $\frac{4}{7}$ of a pizza. They give $\frac{2}{7}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{8}{49}$

Explanation: Multiplying fractions: $(\frac{4}{7}) \times (\frac{2}{7}) = \frac{8}{49}$ of the pizza.

Problem 38: Elizabeth has $\frac{5}{6}$ of a pizza. They give $\frac{4}{4}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{20}{24}$

Explanation: Multiplying fractions: $(\frac{5}{6}) \times (\frac{4}{4}) = \frac{20}{24}$ of the pizza.

Problem 39: Ava has $\frac{3}{4}$ pounds of trail mix to divide into bags of $\frac{2}{6}$ pounds each. How many bags can be made?

Answer: $\frac{18}{8}$

Explanation: Dividing fractions: $(\frac{3}{4}) \div (\frac{2}{6}) = (\frac{3}{4}) \times (\frac{6}{2}) = \frac{18}{8}$ bags.

Problem 40: Mason has $\frac{2}{7}$ pounds of trail mix to divide into bags of $\frac{1}{5}$ pounds each. How many bags can be made?

Answer: $\frac{10}{7}$

Explanation: Dividing fractions: $(\frac{2}{7}) \div (\frac{1}{5}) = (\frac{2}{7}) \times (\frac{5}{1}) = \frac{10}{7}$ bags.

Problem 41: Mason has $\frac{3}{7}$ of a pizza. They give $\frac{2}{2}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{6}{14}$

Explanation: Multiplying fractions: $(\frac{3}{7}) \times (\frac{2}{2}) = \frac{6}{14}$ of the pizza.

Problem 42: Olivia has $\frac{2}{7}$ pounds of trail mix to divide into bags of $\frac{3}{6}$ pounds each. How many bags can be made?

Answer: $\frac{12}{21}$

Explanation: Dividing fractions: $(\frac{2}{7}) \div (\frac{3}{6}) = (\frac{2}{7}) \times (\frac{6}{3}) = \frac{12}{21}$ bags.

Problem 43: Ava has $\frac{4}{4}$ of a pizza. They give $\frac{3}{3}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{12}{12}$

Explanation: Multiplying fractions: $(\frac{4}{4}) \times (\frac{3}{3}) = \frac{12}{12}$ of the pizza.

Problem 44: Olivia has $\frac{5}{6}$ pounds of trail mix to divide into bags of $\frac{2}{2}$ pounds each. How many bags can be made?

Answer: 10/12

Explanation: Dividing fractions: $(\frac{5}{6}) \div (\frac{2}{2}) = (\frac{5}{6}) \times (\frac{2}{2}) = 10/12$ bags.

Problem 45: Olivia has $\frac{2}{6}$ of a pizza. They give $\frac{3}{8}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 6/48

Explanation: Multiplying fractions: $(\frac{2}{6}) \times (\frac{3}{8}) = 6/48$ of the pizza.

Practice Exercises - Grade 6

Exercise 1: Sebastian has $\frac{5}{7}$ pounds of trail mix to divide into bags of $\frac{2}{3}$ pounds each. How many bags can be made?

Exercise 2: Isabella has $\frac{5}{3}$ pounds of trail mix to divide into bags of $\frac{2}{6}$ pounds each. How many bags can be made?

Exercise 3: Mason has $\frac{5}{2}$ of a pizza. They give $\frac{5}{6}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Exercise 4: Abigail has $\frac{5}{7}$ of a pizza. They give $\frac{3}{7}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Exercise 5: Mason has $\frac{2}{8}$ of a pizza. They give $\frac{2}{2}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Exercise 6: Sebastian has $\frac{5}{6}$ pounds of trail mix to divide into bags of $\frac{4}{5}$ pounds each. How many bags can be made?

Exercise 7: Samuel has $\frac{3}{3}$ pounds of trail mix to divide into bags of $\frac{3}{3}$ pounds each. How many bags can be made?

Exercise 8: Mia has $\frac{2}{3}$ pounds of trail mix to divide into bags of $\frac{4}{4}$ pounds each. How many bags can be made?

Exercise 9: Joseph has $\frac{5}{3}$ of a pizza. They give $\frac{4}{6}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Exercise 10: Isabella has $\frac{3}{2}$ of a pizza. They give $\frac{4}{2}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Exercise 11: Harper has $\frac{5}{8}$ of a pizza. They give $\frac{3}{3}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Exercise 12: Joseph has $\frac{2}{7}$ of a pizza. They give $\frac{5}{8}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Exercise 13: Henry has $\frac{4}{8}$ pounds of trail mix to divide into bags of $\frac{3}{5}$ pounds each. How many bags can be made?

Exercise 14: Harper has $\frac{4}{8}$ pounds of trail mix to divide into bags of $\frac{4}{3}$ pounds each. How many bags can be made?

Exercise 15: Alexander has $\frac{5}{6}$ of a pizza. They give $\frac{2}{3}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Exercise 16: Amelia has $\frac{5}{2}$ of a pizza. They give $\frac{2}{2}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Exercise 17: Emma has $\frac{3}{3}$ of a pizza. They give $\frac{1}{5}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Exercise 18: Ava has $\frac{4}{5}$ of a pizza. They give $\frac{3}{6}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Exercise 19: Emily has $\frac{4}{3}$ pounds of trail mix to divide into bags of $\frac{3}{2}$ pounds each. How many bags can be made?

Exercise 20: Amelia has $\frac{3}{7}$ of a pizza. They give $\frac{5}{2}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Practice Exercise Answers:

Exercise 1: $\frac{15}{14}$

Exercise 2: $\frac{30}{6}$

Exercise 3: 25/12

Exercise 4: 15/49

Exercise 5: 4/16

Exercise 6: 25/24

Exercise 7: 9/9

Exercise 8: 8/12

Exercise 9: 20/18

Exercise 10: 12/4

Exercise 11: 15/24

Exercise 12: 10/56

Exercise 13: 20/24

Exercise 14: 12/32

Exercise 15: 10/18

Exercise 16: 10/4

Exercise 17: 3/15

Exercise 18: 12/30

Exercise 19: 8/9

Exercise 20: 15/14

Challenge Problems - Grade 6

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Sophia has $\frac{2}{7}$ of a pizza. They give $\frac{2}{7}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 4/49

Solution: Multiplying fractions: $(\frac{2}{7}) \times (\frac{2}{7}) = \frac{4}{49}$ of the pizza.

Challenge 2: Joseph has $\frac{4}{5}$ pounds of trail mix to divide into bags of $\frac{1}{6}$ pounds each. How many bags can be made?

Answer: 24/5

Solution: Dividing fractions: $(4/5) \div (1/6) = (4/5) \times (6/1) = 24/5$ bags.

Challenge 3: Mia has $1/2$ of a pizza. They give $5/7$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 5/14

Solution: Multiplying fractions: $(1/2) \times (5/7) = 5/14$ of the pizza.

Challenge 4: Aiden has $1/6$ of a pizza. They give $3/4$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 3/24

Solution: Multiplying fractions: $(1/6) \times (3/4) = 3/24$ of the pizza.

Challenge 5: Ava has $3/4$ pounds of trail mix to divide into bags of $1/6$ pounds each. How many bags can be made?

Answer: 18/4

Solution: Dividing fractions: $(3/4) \div (1/6) = (3/4) \times (6/1) = 18/4$ bags.

Challenge 6: Mia has $5/4$ of a pizza. They give $4/8$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 20/32

Solution: Multiplying fractions: $(5/4) \times (4/8) = 20/32$ of the pizza.

Challenge 7: Aiden has $3/5$ of a pizza. They give $1/3$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: 3/15

Solution: Multiplying fractions: $(3/5) \times (1/3) = 3/15$ of the pizza.

Challenge 8: Aiden has $6/4$ pounds of trail mix to divide into bags of $4/5$ pounds each. How many bags can be made?

Answer: 30/16

Solution: Dividing fractions: $(6/4) \div (4/5) = (6/4) \times (5/4) = 30/16$ bags.

Challenge 9: Amelia has $\frac{3}{7}$ of a pizza. They give $\frac{4}{7}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{12}{49}$

Solution: Multiplying fractions: $(\frac{3}{7}) \times (\frac{4}{7}) = \frac{12}{49}$ of the pizza.

Challenge 10: Jack has $\frac{2}{6}$ pounds of trail mix to divide into bags of $\frac{4}{4}$ pounds each. How many bags can be made?

Answer: $\frac{8}{24}$

Solution: Dividing fractions: $(\frac{2}{6}) \div (\frac{4}{4}) = (\frac{2}{6}) \times (\frac{4}{4}) = \frac{8}{24}$ bags.

Challenge 11: Olivia has $\frac{1}{2}$ of a pizza. They give $\frac{4}{3}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{4}{6}$

Solution: Multiplying fractions: $(\frac{1}{2}) \times (\frac{4}{3}) = \frac{4}{6}$ of the pizza.

Challenge 12: Samuel has $\frac{2}{7}$ of a pizza. They give $\frac{5}{2}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{10}{14}$

Solution: Multiplying fractions: $(\frac{2}{7}) \times (\frac{5}{2}) = \frac{10}{14}$ of the pizza.

Challenge 13: Alexander has $\frac{2}{7}$ pounds of trail mix to divide into bags of $\frac{1}{3}$ pounds each. How many bags can be made?

Answer: $\frac{6}{7}$

Solution: Dividing fractions: $(\frac{2}{7}) \div (\frac{1}{3}) = (\frac{2}{7}) \times (\frac{3}{1}) = \frac{6}{7}$ bags.

Challenge 14: Olivia has $\frac{3}{8}$ of a pizza. They give $\frac{4}{6}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{12}{48}$

Solution: Multiplying fractions: $(\frac{3}{8}) \times (\frac{4}{6}) = \frac{12}{48}$ of the pizza.

Challenge 15: Ethan has $\frac{4}{7}$ of a pizza. They give $\frac{5}{3}$ of that amount to a friend. What fraction of the whole pizza did the friend get?

Answer: $\frac{20}{21}$

Solution: Multiplying fractions: $(\frac{4}{7}) \times (\frac{5}{3}) = \frac{20}{21}$ of the pizza.

Chapter 3: Geometry Basics

Geometry helps us understand shapes, sizes, and spatial relationships. Students will explore perimeter, area, volume, and properties of various geometric figures through hands-on story problems.

Grade 1 Problems

Problem 1: Henry draws 13 circles. How many corners do all the circles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Explanation: Each circle has 0 corners. $13 \times 0 = 0$ corners.

Problem 2: James draws 4 circles. How many corners do all the circles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Explanation: Each circle has 0 corners. $4 \times 0 = 0$ corners.

Problem 3: Sebastian draws 9 squares. How many corners do all the squares have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 36

Explanation: Each square has 4 corners. $9 \times 4 = 36$ corners.

Problem 4: Lucas draws 13 rectangles. How many corners do all the rectangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 52

Explanation: Each rectangle has 4 corners. $13 \times 4 = 52$ corners.

Problem 5: Avery draws 6 rectangles. How many corners do all the rectangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 24

Explanation: Each rectangle has 4 corners. $6 \times 4 = 24$ corners.

Problem 6: Sophia draws 13 rectangles. How many corners do all the rectangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 52

Explanation: Each rectangle has 4 corners. $13 \times 4 = 52$ corners.

Problem 7: Alexander draws 8 circles. How many corners do all the circles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Explanation: Each circle has 0 corners. $8 \times 0 = 0$ corners.

Problem 8: Olivia draws 4 triangles. How many corners do all the triangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 12

Explanation: Each triangle has 3 corners. $4 \times 3 = 12$ corners.

Problem 9: Lucas draws 12 triangles. How many corners do all the triangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 36

Explanation: Each triangle has 3 corners. $12 \times 3 = 36$ corners.

Problem 10: Isabella draws 13 triangles. How many corners do all the triangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 39

Explanation: Each triangle has 3 corners. $13 \times 3 = 39$ corners.

Problem 11: Lucas draws 5 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 20

Explanation: Each square has 4 corners. $5 \times 4 = 20$ corners.

Problem 12: Charlotte draws 6 circles. How many corners do all the circles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Explanation: Each circle has 0 corners. $6 \times 0 = 0$ corners.

Problem 13: Abigail draws 9 circles. How many corners do all the circles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Explanation: Each circle has 0 corners. $9 \times 0 = 0$ corners.

Problem 14: Aiden draws 5 triangles. How many corners do all the triangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 15

Explanation: Each triangle has 3 corners. $5 \times 3 = 15$ corners.

Problem 15: Jack draws 12 rectangles. How many corners do all the rectangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 48

Explanation: Each rectangle has 4 corners. $12 \times 4 = 48$ corners.

Problem 16: Olivia draws 10 rectangles. How many corners do all the rectangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 40

Explanation: Each rectangle has 4 corners. $10 \times 4 = 40$ corners.

Problem 17: Sofia draws 14 squares. How many corners do all the squares have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 56

Explanation: Each square has 4 corners. $14 \times 4 = 56$ corners.

Problem 18: Avery draws 13 rectangles. How many corners do all the rectangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 52

Explanation: Each rectangle has 4 corners. $13 \times 4 = 52$ corners.

Problem 19: Sophia draws 8 squares. How many corners do all the squares have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 32

Explanation: Each square has 4 corners. $8 \times 4 = 32$ corners.

Problem 20: Harper draws 13 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 52

Explanation: Each square has 4 corners. $13 \times 4 = 52$ corners.

Problem 21: James draws 12 triangles. How many corners do all the triangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 36

Explanation: Each triangle has 3 corners. $12 \times 3 = 36$ corners.

Problem 22: Samuel draws 4 rectangles. How many corners do all the rectangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 16

Explanation: Each rectangle has 4 corners. $4 \times 4 = 16$ corners.

Problem 23: Ethan draws 5 triangles. How many corners do all the triangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 15

Explanation: Each triangle has 3 corners. $5 \times 3 = 15$ corners.

Problem 24: Harper draws 8 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 32

Explanation: Each square has 4 corners. $8 \times 4 = 32$ corners.

Problem 25: Liam draws 14 rectangles. How many corners do all the rectangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 56

Explanation: Each rectangle has 4 corners. $14 \times 4 = 56$ corners.

Problem 26: Emma draws 13 triangles. How many corners do all the triangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 39

Explanation: Each triangle has 3 corners. $13 \times 3 = 39$ corners.

Problem 27: Ava draws 11 squares. How many corners do all the squares have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 44

Explanation: Each square has 4 corners. $11 \times 4 = 44$ corners.

Problem 28: Abigail draws 15 circles. How many corners do all the circles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Explanation: Each circle has 0 corners. $15 \times 0 = 0$ corners.

Problem 29: Jack draws 5 circles. How many corners do all the circles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Explanation: Each circle has 0 corners. $5 \times 0 = 0$ corners.

Problem 30: Emily draws 12 squares. How many corners do all the squares have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 48

Explanation: Each square has 4 corners. $12 \times 4 = 48$ corners.

Problem 31: Sebastian draws 12 squares. How many corners do all the squares have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 48

Explanation: Each square has 4 corners. $12 \times 4 = 48$ corners.

Problem 32: Henry draws 14 squares. How many corners do all the squares have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 56

Explanation: Each square has 4 corners. $14 \times 4 = 56$ corners.

Problem 33: Emily draws 4 circles. How many corners do all the circles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Explanation: Each circle has 0 corners. $4 \times 0 = 0$ corners.

Problem 34: Alexander draws 14 rectangles. How many corners do all the rectangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 56

Explanation: Each rectangle has 4 corners. $14 \times 4 = 56$ corners.

Problem 35: Evelyn draws 7 circles. How many corners do all the circles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Explanation: Each circle has 0 corners. $7 \times 0 = 0$ corners.

Problem 36: Lucas draws 15 triangles. How many corners do all the triangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 45

Explanation: Each triangle has 3 corners. $15 \times 3 = 45$ corners.

Problem 37: James draws 10 rectangles. How many corners do all the rectangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 40

Explanation: Each rectangle has 4 corners. $10 \times 4 = 40$ corners.

Problem 38: Harper draws 10 rectangles. How many corners do all the rectangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 40

Explanation: Each rectangle has 4 corners. $10 \times 4 = 40$ corners.

Problem 39: Emily draws 7 squares. How many corners do all the squares have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 28

Explanation: Each square has 4 corners. $7 \times 4 = 28$ corners.

Problem 40: Mia draws 12 circles. How many corners do all the circles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Explanation: Each circle has 0 corners. $12 \times 0 = 0$ corners.

Problem 41: Lucas draws 8 circles. How many corners do all the circles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Explanation: Each circle has 0 corners. $8 \times 0 = 0$ corners.

Problem 42: Mason draws 11 rectangles. How many corners do all the rectangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 44

Explanation: Each rectangle has 4 corners. $11 \times 4 = 44$ corners.

Problem 43: James draws 6 squares. How many corners do all the squares have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 24

Explanation: Each square has 4 corners. $6 \times 4 = 24$ corners.

Problem 44: Ethan draws 8 circles. How many corners do all the circles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Explanation: Each circle has 0 corners. $8 \times 0 = 0$ corners.

Problem 45: Mason draws 13 squares. How many corners do all the squares have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 52

Explanation: Each square has 4 corners. $13 \times 4 = 52$ corners.

Practice Exercises - Grade 1

Exercise 1: Abigail draws 14 triangles. How many corners do all the triangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 2: Aiden draws 8 triangles. How many corners do all the triangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 3: Joseph draws 13 rectangles. How many corners do all the rectangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 4: Charlotte draws 14 circles. How many corners do all the circles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 5: Sofia draws 8 squares. How many corners do all the squares have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 6: Alexander draws 3 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 7: Emma draws 11 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 8: Charlotte draws 8 rectangles. How many corners do all the rectangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 9: Alexander draws 8 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 10: Mia draws 6 squares. How many corners do all the squares have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 11: Avery draws 15 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 12: James draws 9 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 13: Mason draws 5 rectangles. How many corners do all the rectangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 14: Abigail draws 6 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 15: Noah draws 9 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 16: Amelia draws 11 rectangles. How many corners do all the rectangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 17: Mason draws 11 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 18: Noah draws 10 circles. How many corners do all the circles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 19: Charlotte draws 13 circles. How many corners do all the circles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Exercise 20: Avery draws 14 triangles. How many corners do all the triangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Practice Exercise Answers:

Exercise 1: 42

Exercise 2: 24

Exercise 3: 52

Exercise 4: 0

Exercise 5: 32

Exercise 6: 12

Exercise 7: 44

Exercise 8: 32

Exercise 9: 32

Exercise 10: 24

Exercise 11: 60

Exercise 12: 36

Exercise 13: 20

Exercise 14: 24

Exercise 15: 36

Exercise 16: 44

Exercise 17: 44

Exercise 18: 0

Exercise 19: 0

Exercise 20: 42

Challenge Problems - Grade 1

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: James draws 12 circles. How many corners do all the circles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Solution: Each circle has 0 corners. $12 \times 0 = 0$ corners.

Challenge 2: Charlotte draws 3 triangles. How many corners do all the triangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 9

Solution: Each triangle has 3 corners. $3 \times 3 = 9$ corners.

Challenge 3: Lucas draws 5 triangles. How many corners do all the triangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 15

Solution: Each triangle has 3 corners. $5 \times 3 = 15$ corners.

Challenge 4: Emma draws 8 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 32

Solution: Each square has 4 corners. $8 \times 4 = 32$ corners.

Challenge 5: Sofia draws 7 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 28

Solution: Each square has 4 corners. $7 \times 4 = 28$ corners.

Challenge 6: Ava draws 13 triangles. How many corners do all the triangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 39

Solution: Each triangle has 3 corners. $13 \times 3 = 39$ corners.

Challenge 7: Mason draws 13 circles. How many corners do all the circles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Solution: Each circle has 0 corners. $13 \times 0 = 0$ corners.

Challenge 8: Charlotte draws 15 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 60

Solution: Each square has 4 corners. $15 \times 4 = 60$ corners.

Challenge 9: Liam draws 14 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 56

Solution: Each square has 4 corners. $14 \times 4 = 56$ corners.

Challenge 10: Avery draws 12 squares. How many corners do all the squares have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 48

Solution: Each square has 4 corners. $12 \times 4 = 48$ corners.

Challenge 11: Charlotte draws 11 triangles. How many corners do all the triangles have in total?
(Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 33

Solution: Each triangle has 3 corners. $11 \times 3 = 33$ corners.

Challenge 12: Liam draws 5 circles. How many corners do all the circles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 0

Solution: Each circle has 0 corners. $5 \times 0 = 0$ corners.

Challenge 13: Harper draws 7 rectangles. How many corners do all the rectangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 28

Solution: Each rectangle has 4 corners. $7 \times 4 = 28$ corners.

Challenge 14: Mia draws 5 squares. How many corners do all the squares have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 20

Solution: Each square has 4 corners. $5 \times 4 = 20$ corners.

Challenge 15: Emma draws 10 rectangles. How many corners do all the rectangles have in total? (Remember: circles have 0 corners, triangles have 3, squares and rectangles have 4)

Answer: 40

Solution: Each rectangle has 4 corners. $10 \times 4 = 40$ corners.

Grade 2 Problems

Problem 1: Joseph has a rectangular garden that is 8 meters long and 4 meters wide. What is the perimeter of the garden?

Answer: 24

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (8 + 4) = 2 \times 12 = 24$ meters.

Problem 2: Elizabeth has a rectangular garden that is 7 meters long and 4 meters wide. What is the perimeter of the garden?

Answer: 22

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (7 + 4) = 2 \times 11 = 22$ meters.

Problem 3: Elizabeth has a rectangular garden that is 5 meters long and 4 meters wide. What is the perimeter of the garden?

Answer: 18

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (5 + 4) = 2 \times 9 = 18$ meters.

Problem 4: Ava has a rectangular garden that is 9 meters long and 5 meters wide. What is the perimeter of the garden?

Answer: 28

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (9 + 5) = 2 \times 14 = 28$ meters.

Problem 5: Alexander has a rectangular garden that is 4 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 20

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (4 + 6) = 2 \times 10 = 20$ meters.

Problem 6: James has a rectangular garden that is 8 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 28

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (8 + 6) = 2 \times 14 = 28$ meters.

Problem 7: Noah has a rectangular garden that is 4 meters long and 7 meters wide. What is the perimeter of the garden?

Answer: 22

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (4 + 7) = 2 \times 11 = 22$ meters.

Problem 8: James has a rectangular garden that is 4 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 20

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (4 + 6) = 2 \times 10 = 20$ meters.

Problem 9: Benjamin has a rectangular garden that is 6 meters long and 3 meters wide. What is the perimeter of the garden?

Answer: 18

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (6 + 3) = 2 \times 9 = 18$ meters.

Problem 10: Emily has a rectangular garden that is 9 meters long and 8 meters wide. What is the perimeter of the garden?

Answer: 34

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (9 + 8) = 2 \times 17 = 34$ meters.

Problem 11: Sophia has a rectangular garden that is 5 meters long and 8 meters wide. What is the perimeter of the garden?

Answer: 26

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (5 + 8) = 2 \times 13 = 26$ meters.

Problem 12: Mason has a rectangular garden that is 10 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 32

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (10 + 6) = 2 \times 16 = 32$ meters.

Problem 13: Ethan has a rectangular garden that is 10 meters long and 8 meters wide. What is the perimeter of the garden?

Answer: 36

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (10 + 8) = 2 \times 18 = 36$ meters.

Problem 14: Alexander has a rectangular garden that is 5 meters long and 5 meters wide. What is the perimeter of the garden?

Answer: 20

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (5 + 5) = 2 \times 10 = 20$ meters.

Problem 15: Ethan has a rectangular garden that is 5 meters long and 3 meters wide. What is the perimeter of the garden?

Answer: 16

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (5 + 3) = 2 \times 8 = 16$ meters.

Problem 16: Ethan has a rectangular garden that is 5 meters long and 3 meters wide. What is the perimeter of the garden?

Answer: 16

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (5 + 3) = 2 \times 8 = 16$ meters.

Problem 17: David has a rectangular garden that is 6 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 24

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (6 + 6) = 2 \times 12 = 24$ meters.

Problem 18: Olivia has a rectangular garden that is 10 meters long and 8 meters wide. What is the perimeter of the garden?

Answer: 36

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (10 + 8) = 2 \times 18 = 36$ meters.

Problem 19: David has a rectangular garden that is 6 meters long and 5 meters wide. What is the perimeter of the garden?

Answer: 22

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (6 + 5) = 2 \times 11 = 22$ meters.

Problem 20: Charlotte has a rectangular garden that is 4 meters long and 5 meters wide. What is the perimeter of the garden?

Answer: 18

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (4 + 5) = 2 \times 9 = 18$ meters.

Problem 21: Ava has a rectangular garden that is 3 meters long and 5 meters wide. What is the perimeter of the garden?

Answer: 16

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (3 + 5) = 2 \times 8 = 16$ meters.

Problem 22: Elizabeth has a rectangular garden that is 6 meters long and 8 meters wide. What is the perimeter of the garden?

Answer: 28

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (6 + 8) = 2 \times 14 = 28$ meters.

Problem 23: Mason has a rectangular garden that is 7 meters long and 5 meters wide. What is the perimeter of the garden?

Answer: 24

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (7 + 5) = 2 \times 12 = 24$ meters.

Problem 24: Samuel has a rectangular garden that is 5 meters long and 2 meters wide. What is the perimeter of the garden?

Answer: 14

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (5 + 2) = 2 \times 7 = 14$ meters.

Problem 25: Harper has a rectangular garden that is 5 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 22

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (5 + 6) = 2 \times 11 = 22$ meters.

Problem 26: Jack has a rectangular garden that is 8 meters long and 7 meters wide. What is the perimeter of the garden?

Answer: 30

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (8 + 7) = 2 \times 15 = 30$ meters.

Problem 27: Olivia has a rectangular garden that is 6 meters long and 7 meters wide. What is the perimeter of the garden?

Answer: 26

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (6 + 7) = 2 \times 13 = 26$ meters.

Problem 28: Emily has a rectangular garden that is 10 meters long and 8 meters wide. What is the perimeter of the garden?

Answer: 36

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (10 + 8) = 2 \times 18 = 36$ meters.

Problem 29: Olivia has a rectangular garden that is 6 meters long and 8 meters wide. What is the perimeter of the garden?

Answer: 28

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (6 + 8) = 2 \times 14 = 28$ meters.

Problem 30: Charlotte has a rectangular garden that is 7 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 26

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (7 + 6) = 2 \times 13 = 26$ meters.

Problem 31: Charlotte has a rectangular garden that is 9 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 30

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (9 + 6) = 2 \times 15 = 30$ meters.

Problem 32: Sebastian has a rectangular garden that is 8 meters long and 3 meters wide. What is the perimeter of the garden?

Answer: 22

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (8 + 3) = 2 \times 11 = 22$ meters.

Problem 33: Noah has a rectangular garden that is 8 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 28

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (8 + 6) = 2 \times 14 = 28$ meters.

Problem 34: Sebastian has a rectangular garden that is 3 meters long and 3 meters wide. What is the perimeter of the garden?

Answer: 12

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (3 + 3) = 2 \times 6 = 12$ meters.

Problem 35: Jack has a rectangular garden that is 9 meters long and 5 meters wide. What is the perimeter of the garden?

Answer: 28

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (9 + 5) = 2 \times 14 = 28$ meters.

Problem 36: Emma has a rectangular garden that is 9 meters long and 5 meters wide. What is the perimeter of the garden?

Answer: 28

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (9 + 5) = 2 \times 14 = 28$ meters.

Problem 37: Noah has a rectangular garden that is 8 meters long and 4 meters wide. What is the perimeter of the garden?

Answer: 24

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (8 + 4) = 2 \times 12 = 24$ meters.

Problem 38: Charlotte has a rectangular garden that is 4 meters long and 4 meters wide. What is the perimeter of the garden?

Answer: 16

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (4 + 4) = 2 \times 8 = 16$ meters.

Problem 39: Emma has a rectangular garden that is 10 meters long and 8 meters wide. What is the perimeter of the garden?

Answer: 36

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (10 + 8) = 2 \times 18 = 36$ meters.

Problem 40: Emma has a rectangular garden that is 10 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 32

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (10 + 6) = 2 \times 16 = 32$ meters.

Problem 41: Amelia has a rectangular garden that is 8 meters long and 2 meters wide. What is the perimeter of the garden?

Answer: 20

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (8 + 2) = 2 \times 10 = 20$ meters.

Problem 42: Alexander has a rectangular garden that is 3 meters long and 7 meters wide. What is the perimeter of the garden?

Answer: 20

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (3 + 7) = 2 \times 10 = 20$ meters.

Problem 43: Noah has a rectangular garden that is 6 meters long and 2 meters wide. What is the perimeter of the garden?

Answer: 16

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (6 + 2) = 2 \times 8 = 16$ meters.

Problem 44: Charlotte has a rectangular garden that is 10 meters long and 4 meters wide. What is the perimeter of the garden?

Answer: 28

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (10 + 4) = 2 \times 14 = 28$ meters.

Problem 45: Amelia has a rectangular garden that is 5 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 22

Explanation: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (5 + 6) = 2 \times 11 = 22$ meters.

Practice Exercises - Grade 2

Exercise 1: David has a rectangular garden that is 4 meters long and 6 meters wide. What is the perimeter of the garden?

Exercise 2: Isabella has a rectangular garden that is 6 meters long and 4 meters wide. What is the perimeter of the garden?

Exercise 3: Jack has a rectangular garden that is 4 meters long and 4 meters wide. What is the perimeter of the garden?

Exercise 4: Joseph has a rectangular garden that is 10 meters long and 5 meters wide. What is the perimeter of the garden?

Exercise 5: Mason has a rectangular garden that is 5 meters long and 6 meters wide. What is the perimeter of the garden?

Exercise 6: Emily has a rectangular garden that is 6 meters long and 5 meters wide. What is the perimeter of the garden?

Exercise 7: Amelia has a rectangular garden that is 10 meters long and 8 meters wide. What is the perimeter of the garden?

Exercise 8: Evelyn has a rectangular garden that is 7 meters long and 8 meters wide. What is the perimeter of the garden?

Exercise 9: Alexander has a rectangular garden that is 10 meters long and 3 meters wide. What is the perimeter of the garden?

Exercise 10: Ethan has a rectangular garden that is 3 meters long and 4 meters wide. What is the perimeter of the garden?

Exercise 11: Henry has a rectangular garden that is 10 meters long and 7 meters wide. What is the perimeter of the garden?

Exercise 12: Mia has a rectangular garden that is 3 meters long and 3 meters wide. What is the perimeter of the garden?

Exercise 13: Noah has a rectangular garden that is 6 meters long and 2 meters wide. What is the perimeter of the garden?

Exercise 14: Samuel has a rectangular garden that is 10 meters long and 5 meters wide. What is the perimeter of the garden?

Exercise 15: Samuel has a rectangular garden that is 8 meters long and 8 meters wide. What is the perimeter of the garden?

Exercise 16: Emma has a rectangular garden that is 9 meters long and 4 meters wide. What is the perimeter of the garden?

Exercise 17: Ava has a rectangular garden that is 8 meters long and 3 meters wide. What is the perimeter of the garden?

Exercise 18: Liam has a rectangular garden that is 6 meters long and 4 meters wide. What is the perimeter of the garden?

Exercise 19: Amelia has a rectangular garden that is 10 meters long and 5 meters wide. What is the perimeter of the garden?

Exercise 20: Benjamin has a rectangular garden that is 8 meters long and 4 meters wide. What is the perimeter of the garden?

Practice Exercise Answers:

Exercise 1: 20

Exercise 2: 20

Exercise 3: 16

Exercise 4: 30

Exercise 5: 22

Exercise 6: 22

Exercise 7: 36

Exercise 8: 30

Exercise 9: 26

Exercise 10: 14

Exercise 11: 34

Exercise 12: 12

Exercise 13: 16

Exercise 14: 30

Exercise 15: 32

Exercise 16: 26

Exercise 17: 22

Exercise 18: 20

Exercise 19: 30

Exercise 20: 24

Challenge Problems - Grade 2

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Liam has a rectangular garden that is 3 meters long and 5 meters wide. What is the perimeter of the garden?

Answer: 16

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (3 + 5) = 2 \times 8 = 16$ meters.

Challenge 2: Elizabeth has a rectangular garden that is 6 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 24

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (6 + 6) = 2 \times 12 = 24$ meters.

Challenge 3: Mason has a rectangular garden that is 7 meters long and 7 meters wide. What is the perimeter of the garden?

Answer: 28

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (7 + 7) = 2 \times 14 = 28$ meters.

Challenge 4: Isabella has a rectangular garden that is 8 meters long and 3 meters wide. What is the perimeter of the garden?

Answer: 22

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (8 + 3) = 2 \times 11 = 22$ meters.

Challenge 5: Sophia has a rectangular garden that is 3 meters long and 2 meters wide. What is the perimeter of the garden?

Answer: 10

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (3 + 2) = 2 \times 5 = 10$ meters.

Challenge 6: Isabella has a rectangular garden that is 6 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 24

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (6 + 6) = 2 \times 12 = 24$ meters.

Challenge 7: Olivia has a rectangular garden that is 8 meters long and 6 meters wide. What is the perimeter of the garden?

Answer: 28

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (8 + 6) = 2 \times 14 = 28$ meters.

Challenge 8: David has a rectangular garden that is 8 meters long and 8 meters wide. What is the perimeter of the garden?

Answer: 32

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (8 + 8) = 2 \times 16 = 32$ meters.

Challenge 9: Elizabeth has a rectangular garden that is 8 meters long and 7 meters wide. What is the perimeter of the garden?

Answer: 30

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (8 + 7) = 2 \times 15 = 30$ meters.

Challenge 10: Aiden has a rectangular garden that is 4 meters long and 4 meters wide. What is the perimeter of the garden?

Answer: 16

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (4 + 4) = 2 \times 8 = 16$ meters.

Challenge 11: Liam has a rectangular garden that is 7 meters long and 4 meters wide. What is the perimeter of the garden?

Answer: 22

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (7 + 4) = 2 \times 11 = 22$ meters.

Challenge 12: Liam has a rectangular garden that is 6 meters long and 2 meters wide. What is the perimeter of the garden?

Answer: 16

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (6 + 2) = 2 \times 8 = 16$ meters.

Challenge 13: Liam has a rectangular garden that is 7 meters long and 4 meters wide. What is the perimeter of the garden?

Answer: 22

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (7 + 4) = 2 \times 11 = 22$ meters.

Challenge 14: David has a rectangular garden that is 6 meters long and 5 meters wide. What is the perimeter of the garden?

Answer: 22

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (6 + 5) = 2 \times 11 = 22$ meters.

Challenge 15: Amelia has a rectangular garden that is 10 meters long and 3 meters wide. What is the perimeter of the garden?

Answer: 26

Solution: Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (10 + 3) = 2 \times 13 = 26$ meters.

Grade 3 Problems

Problem 1: Sophia is tiling a rectangular floor that is 6 feet long and 6 feet wide. What is the area of the floor?

Answer: 36

Explanation: Area = length $\times$ width = $6 \times 6 = 36$ square feet.

Problem 2: Benjamin is tiling a rectangular floor that is 12 feet long and 3 feet wide. What is the area of the floor?

Answer: 36

Explanation: Area = length $\times$ width = $12 \times 3 = 36$ square feet.

Problem 3: Evelyn is tiling a rectangular floor that is 6 feet long and 3 feet wide. What is the area of the floor?

Answer: 18

Explanation: Area = length $\times$ width = $6 \times 3 = 18$ square feet.

Problem 4: Benjamin is tiling a rectangular floor that is 10 feet long and 4 feet wide. What is the area of the floor?

Answer: 40

Explanation: Area = length $\times$ width = $10 \times 4 = 40$ square feet.

Problem 5: Mia is tiling a rectangular floor that is 12 feet long and 8 feet wide. What is the area of the floor?

Answer: 96

Explanation: Area = length $\times$ width = $12 \times 8 = 96$ square feet.

Problem 6: Sebastian is tiling a rectangular floor that is 6 feet long and 9 feet wide. What is the area of the floor?

Answer: 54

Explanation: Area = length $\times$ width = $6 \times 9 = 54$ square feet.

Problem 7: Avery is tiling a rectangular floor that is 7 feet long and 6 feet wide. What is the area of the floor?

Answer: 42

Explanation: Area = length $\times$ width = $7 \times 6 = 42$ square feet.

Problem 8: Liam is tiling a rectangular floor that is 7 feet long and 6 feet wide. What is the area of the floor?

Answer: 42

Explanation: Area = length $\times$ width = $7 \times 6 = 42$ square feet.

Problem 9: Isabella is tiling a rectangular floor that is 11 feet long and 3 feet wide. What is the area of the floor?

Answer: 33

Explanation: Area = length $\times$ width = $11 \times 3 = 33$ square feet.

Problem 10: Charlotte is tiling a rectangular floor that is 5 feet long and 7 feet wide. What is the area of the floor?

Answer: 35

Explanation: Area = length $\times$ width = $5 \times 7 = 35$ square feet.

Problem 11: Lucas is tiling a rectangular floor that is 5 feet long and 5 feet wide. What is the area of the floor?

Answer: 25

Explanation: Area = length $\times$ width = $5 \times 5 = 25$ square feet.

Problem 12: Evelyn is tiling a rectangular floor that is 5 feet long and 7 feet wide. What is the area of the floor?

Answer: 35

Explanation: Area = length $\times$ width = $5 \times 7 = 35$ square feet.

Problem 13: Ethan is tiling a rectangular floor that is 6 feet long and 8 feet wide. What is the area of the floor?

Answer: 48

Explanation: Area = length $\times$ width = $6 \times 8 = 48$ square feet.

Problem 14: Isabella is tiling a rectangular floor that is 10 feet long and 4 feet wide. What is the area of the floor?

Answer: 40

Explanation: Area = length $\times$ width = $10 \times 4 = 40$ square feet.

Problem 15: Sofia is tiling a rectangular floor that is 11 feet long and 10 feet wide. What is the area of the floor?

Answer: 110

Explanation: Area = length $\times$ width = $11 \times 10 = 110$ square feet.

Problem 16: Charlotte is tiling a rectangular floor that is 4 feet long and 9 feet wide. What is the area of the floor?

Answer: 36

Explanation: Area = length $\times$ width = $4 \times 9 = 36$ square feet.

Problem 17: Noah is tiling a rectangular floor that is 11 feet long and 4 feet wide. What is the area of the floor?

Answer: 44

Explanation: Area = length $\times$ width = $11 \times 4 = 44$ square feet.

Problem 18: Joseph is tiling a rectangular floor that is 7 feet long and 3 feet wide. What is the area of the floor?

Answer: 21

Explanation: Area = length $\times$ width = $7 \times 3 = 21$ square feet.

Problem 19: Emma is tiling a rectangular floor that is 12 feet long and 7 feet wide. What is the area of the floor?

Answer: 84

Explanation: Area = length $\times$ width = $12 \times 7 = 84$ square feet.

Problem 20: Lucas is tiling a rectangular floor that is 9 feet long and 5 feet wide. What is the area of the floor?

Answer: 45

Explanation: Area = length $\times$ width = $9 \times 5 = 45$ square feet.

Problem 21: Sebastian is tiling a rectangular floor that is 7 feet long and 7 feet wide. What is the area of the floor?

Answer: 49

Explanation: Area = length $\times$ width = $7 \times 7 = 49$ square feet.

Problem 22: Mia is tiling a rectangular floor that is 5 feet long and 5 feet wide. What is the area of the floor?

Answer: 25

Explanation: Area = length $\times$ width = $5 \times 5 = 25$ square feet.

Problem 23: Samuel is tiling a rectangular floor that is 12 feet long and 4 feet wide. What is the area of the floor?

Answer: 48

Explanation: Area = length $\times$ width = $12 \times 4 = 48$ square feet.

Problem 24: Sofia is tiling a rectangular floor that is 5 feet long and 10 feet wide. What is the area of the floor?

Answer: 50

Explanation: Area = length $\times$ width = $5 \times 10 = 50$ square feet.

Problem 25: Amelia is tiling a rectangular floor that is 9 feet long and 8 feet wide. What is the area of the floor?

Answer: 72

Explanation: Area = length $\times$ width = $9 \times 8 = 72$ square feet.

Problem 26: Henry is tiling a rectangular floor that is 4 feet long and 10 feet wide. What is the area of the floor?

Answer: 40

Explanation: Area = length $\times$ width = $4 \times 10 = 40$ square feet.

Problem 27: Olivia is tiling a rectangular floor that is 11 feet long and 3 feet wide. What is the area of the floor?

Answer: 33

Explanation: Area = length $\times$ width = $11 \times 3 = 33$ square feet.

Problem 28: Elizabeth is tiling a rectangular floor that is 11 feet long and 4 feet wide. What is the area of the floor?

Answer: 44

Explanation: Area = length $\times$ width = $11 \times 4 = 44$ square feet.

Problem 29: James is tiling a rectangular floor that is 11 feet long and 10 feet wide. What is the area of the floor?

Answer: 110

Explanation: Area = length $\times$ width = $11 \times 10 = 110$ square feet.

Problem 30: James is tiling a rectangular floor that is 4 feet long and 7 feet wide. What is the area of the floor?

Answer: 28

Explanation: Area = length $\times$ width = $4 \times 7 = 28$ square feet.

Problem 31: Abigail is tiling a rectangular floor that is 11 feet long and 3 feet wide. What is the area of the floor?

Answer: 33

Explanation: Area = length $\times$ width = $11 \times 3 = 33$ square feet.

Problem 32: Amelia is tiling a rectangular floor that is 6 feet long and 10 feet wide. What is the area of the floor?

Answer: 60

Explanation: Area = length $\times$ width = $6 \times 10 = 60$ square feet.

Problem 33: Alexander is tiling a rectangular floor that is 12 feet long and 3 feet wide. What is the area of the floor?

Answer: 36

Explanation: Area = length $\times$ width = $12 \times 3 = 36$ square feet.

Problem 34: Elizabeth is tiling a rectangular floor that is 6 feet long and 8 feet wide. What is the area of the floor?

Answer: 48

Explanation: Area = length $\times$ width = $6 \times 8 = 48$ square feet.

Problem 35: James is tiling a rectangular floor that is 4 feet long and 10 feet wide. What is the area of the floor?

Answer: 40

Explanation: Area = length $\times$ width = $4 \times 10 = 40$ square feet.

Problem 36: Isabella is tiling a rectangular floor that is 8 feet long and 10 feet wide. What is the area of the floor?

Answer: 80

Explanation: Area = length $\times$ width = $8 \times 10 = 80$ square feet.

Problem 37: Ava is tiling a rectangular floor that is 6 feet long and 8 feet wide. What is the area of the floor?

Answer: 48

Explanation: Area = length $\times$ width = $6 \times 8 = 48$ square feet.

Problem 38: Abigail is tiling a rectangular floor that is 4 feet long and 7 feet wide. What is the area of the floor?

Answer: 28

Explanation: Area = length $\times$ width = $4 \times 7 = 28$ square feet.

Problem 39: Sophia is tiling a rectangular floor that is 9 feet long and 7 feet wide. What is the area of the floor?

Answer: 63

Explanation: Area = length $\times$ width = $9 \times 7 = 63$ square feet.

Problem 40: Liam is tiling a rectangular floor that is 9 feet long and 4 feet wide. What is the area of the floor?

Answer: 36

Explanation: Area = length $\times$ width = $9 \times 4 = 36$ square feet.

Problem 41: Liam is tiling a rectangular floor that is 12 feet long and 9 feet wide. What is the area of the floor?

Answer: 108

Explanation: Area = length $\times$ width = $12 \times 9 = 108$ square feet.

Problem 42: Evelyn is tiling a rectangular floor that is 9 feet long and 7 feet wide. What is the area of the floor?

Answer: 63

Explanation: Area = length $\times$ width = $9 \times 7 = 63$ square feet.

Problem 43: James is tiling a rectangular floor that is 11 feet long and 8 feet wide. What is the area of the floor?

Answer: 88

Explanation: Area = length $\times$ width = $11 \times 8 = 88$ square feet.

Problem 44: Ethan is tiling a rectangular floor that is 12 feet long and 6 feet wide. What is the area of the floor?

Answer: 72

Explanation: Area = length $\times$ width = $12 \times 6 = 72$ square feet.

Problem 45: Joseph is tiling a rectangular floor that is 6 feet long and 3 feet wide. What is the area of the floor?

Answer: 18

Explanation: Area = length $\times$ width = $6 \times 3 = 18$ square feet.

Practice Exercises - Grade 3

Exercise 1: Olivia is tiling a rectangular floor that is 11 feet long and 3 feet wide. What is the area of the floor?

Exercise 2: Amelia is tiling a rectangular floor that is 10 feet long and 9 feet wide. What is the area of the floor?

Exercise 3: Aiden is tiling a rectangular floor that is 6 feet long and 3 feet wide. What is the area of the floor?

Exercise 4: Lucas is tiling a rectangular floor that is 10 feet long and 4 feet wide. What is the area of the floor?

Exercise 5: Mason is tiling a rectangular floor that is 5 feet long and 4 feet wide. What is the area of the floor?

Exercise 6: Sebastian is tiling a rectangular floor that is 10 feet long and 10 feet wide. What is the area of the floor?

Exercise 7: Henry is tiling a rectangular floor that is 12 feet long and 9 feet wide. What is the area of the floor?

Exercise 8: Mason is tiling a rectangular floor that is 10 feet long and 3 feet wide. What is the area of the floor?

Exercise 9: Charlotte is tiling a rectangular floor that is 11 feet long and 4 feet wide. What is the area of the floor?

Exercise 10: Mia is tiling a rectangular floor that is 11 feet long and 5 feet wide. What is the area of the floor?

Exercise 11: Sebastian is tiling a rectangular floor that is 6 feet long and 10 feet wide. What is the area of the floor?

Exercise 12: Sofia is tiling a rectangular floor that is 8 feet long and 9 feet wide. What is the area of the floor?

Exercise 13: Lucas is tiling a rectangular floor that is 8 feet long and 9 feet wide. What is the area of the floor?

Exercise 14: Amelia is tiling a rectangular floor that is 12 feet long and 5 feet wide. What is the area of the floor?

Exercise 15: Sophia is tiling a rectangular floor that is 11 feet long and 3 feet wide. What is the area of the floor?

Exercise 16: Olivia is tiling a rectangular floor that is 6 feet long and 5 feet wide. What is the area of the floor?

Exercise 17: Charlotte is tiling a rectangular floor that is 7 feet long and 5 feet wide. What is the area of the floor?

Exercise 18: Mason is tiling a rectangular floor that is 5 feet long and 3 feet wide. What is the area of the floor?

Exercise 19: Jack is tiling a rectangular floor that is 8 feet long and 6 feet wide. What is the area of the floor?

Exercise 20: Mason is tiling a rectangular floor that is 10 feet long and 3 feet wide. What is the area of the floor?

Practice Exercise Answers:

Exercise 1: 33

Exercise 2: 90

Exercise 3: 18

Exercise 4: 40

Exercise 5: 20

Exercise 6: 100

Exercise 7: 108

Exercise 8: 30

Exercise 9: 44

Exercise 10: 55

Exercise 11: 60

Exercise 12: 72

Exercise 13: 72

Exercise 14: 60

Exercise 15: 33

Exercise 16: 30

Exercise 17: 35

Exercise 18: 15

Exercise 19: 48

Exercise 20: 30

Challenge Problems - Grade 3

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Benjamin is tiling a rectangular floor that is 11 feet long and 5 feet wide. What is the area of the floor?

Answer: 55

Solution: Area = length $\times$ width = $11 \times 5 = 55$ square feet.

Challenge 2: Henry is tiling a rectangular floor that is 9 feet long and 5 feet wide. What is the area of the floor?

Answer: 45

Solution: Area = length $\times$ width = $9 \times 5 = 45$ square feet.

Challenge 3: Emma is tiling a rectangular floor that is 11 feet long and 4 feet wide. What is the area of the floor?

Answer: 44

Solution: Area = length $\times$ width = $11 \times 4 = 44$ square feet.

Challenge 4: Joseph is tiling a rectangular floor that is 9 feet long and 5 feet wide. What is the area of the floor?

Answer: 45

Solution: Area = length $\times$ width = $9 \times 5 = 45$ square feet.

Challenge 5: Benjamin is tiling a rectangular floor that is 8 feet long and 8 feet wide. What is the area of the floor?

Answer: 64

Solution: Area = length $\times$ width = $8 \times 8 = 64$ square feet.

Challenge 6: Isabella is tiling a rectangular floor that is 4 feet long and 5 feet wide. What is the area of the floor?

Answer: 20

Solution: Area = length $\times$ width = $4 \times 5 = 20$ square feet.

Challenge 7: Isabella is tiling a rectangular floor that is 12 feet long and 9 feet wide. What is the area of the floor?

Answer: 108

Solution: Area = length $\times$ width = $12 \times 9 = 108$ square feet.

Challenge 8: Noah is tiling a rectangular floor that is 4 feet long and 4 feet wide. What is the area of the floor?

Answer: 16

Solution: Area = length $\times$ width = $4 \times 4 = 16$ square feet.

Challenge 9: Lucas is tiling a rectangular floor that is 5 feet long and 4 feet wide. What is the area of the floor?

Answer: 20

Solution: Area = length $\times$ width = $5 \times 4 = 20$ square feet.

Challenge 10: Noah is tiling a rectangular floor that is 11 feet long and 8 feet wide. What is the area of the floor?

Answer: 88

Solution: Area = length $\times$ width = $11 \times 8 = 88$ square feet.

Challenge 11: Jack is tiling a rectangular floor that is 10 feet long and 6 feet wide. What is the area of the floor?

Answer: 60

Solution: Area = length $\times$ width = $10 \times 6 = 60$ square feet.

Challenge 12: Alexander is tiling a rectangular floor that is 10 feet long and 9 feet wide. What is the area of the floor?

Answer: 90

Solution: Area = length $\times$ width = $10 \times 9 = 90$ square feet.

Challenge 13: Ethan is tiling a rectangular floor that is 12 feet long and 10 feet wide. What is the area of the floor?

Answer: 120

Solution: Area = length $\times$ width = $12 \times 10 = 120$ square feet.

Challenge 14: Sebastian is tiling a rectangular floor that is 8 feet long and 8 feet wide. What is the area of the floor?

Answer: 64

Solution: Area = length $\times$ width = $8 \times 8 = 64$ square feet.

Challenge 15: Evelyn is tiling a rectangular floor that is 11 feet long and 8 feet wide. What is the area of the floor?

Answer: 88

Solution: Area = length $\times$ width = $11 \times 8 = 88$ square feet.

Grade 4 Problems

Problem 1: Lucas has a square playground with sides of 13 yards. What is the area and perimeter of the playground?

Answer: Area: 169 sq yards, Perimeter: 52 yards

Explanation: For a square with side 13: Area = $13 \times 13 = 169$ square yards. Perimeter = $4 \times 13 = 52$ yards.

Problem 2: Henry has a square playground with sides of 12 yards. What is the area and perimeter of the playground?

Answer: Area: 144 sq yards, Perimeter: 48 yards

Explanation: For a square with side 12: Area = $12 \times 12 = 144$ square yards. Perimeter = $4 \times 12 = 48$ yards.

Problem 3: Mia has a square playground with sides of 15 yards. What is the area and perimeter of the playground?

Answer: Area: 225 sq yards, Perimeter: 60 yards

Explanation: For a square with side 15: Area = $15 \times 15 = 225$ square yards. Perimeter = $4 \times 15 = 60$ yards.

Problem 4: Amelia has a square playground with sides of 11 yards. What is the area and perimeter of the playground?

Answer: Area: 121 sq yards, Perimeter: 44 yards

Explanation: For a square with side 11: Area = $11 \times 11 = 121$ square yards. Perimeter = $4 \times 11 = 44$ yards.

Problem 5: Amelia has a square playground with sides of 13 yards. What is the area and perimeter of the playground?

Answer: Area: 169 sq yards, Perimeter: 52 yards

Explanation: For a square with side 13: Area = $13 \times 13 = 169$ square yards. Perimeter = $4 \times 13 = 52$ yards.

Problem 6: Charlotte has a square playground with sides of 5 yards. What is the area and perimeter of the playground?

Answer: Area: 25 sq yards, Perimeter: 20 yards

Explanation: For a square with side 5: Area = $5 \times 5 = 25$ square yards. Perimeter = $4 \times 5 = 20$ yards.

Problem 7: Harper has a square playground with sides of 12 yards. What is the area and perimeter of the playground?

Answer: Area: 144 sq yards, Perimeter: 48 yards

Explanation: For a square with side 12: Area = $12 \times 12 = 144$ square yards. Perimeter = $4 \times 12 = 48$ yards.

Problem 8: Amelia has a square playground with sides of 13 yards. What is the area and perimeter of the playground?

Answer: Area: 169 sq yards, Perimeter: 52 yards

Explanation: For a square with side 13: Area = $13 \times 13 = 169$ square yards. Perimeter = $4 \times 13 = 52$ yards.

Problem 9: Benjamin has a square playground with sides of 9 yards. What is the area and perimeter of the playground?

Answer: Area: 81 sq yards, Perimeter: 36 yards

Explanation: For a square with side 9: Area = $9 \times 9 = 81$ square yards. Perimeter = $4 \times 9 = 36$ yards.

Problem 10: Abigail has a square playground with sides of 9 yards. What is the area and perimeter of the playground?

Answer: Area: 81 sq yards, Perimeter: 36 yards

Explanation: For a square with side 9: Area = $9 \times 9 = 81$ square yards. Perimeter = $4 \times 9 = 36$ yards.

Problem 11: Sebastian has a square playground with sides of 13 yards. What is the area and perimeter of the playground?

Answer: Area: 169 sq yards, Perimeter: 52 yards

Explanation: For a square with side 13: Area = $13 \times 13 = 169$ square yards. Perimeter = $4 \times 13 = 52$ yards.

Problem 12: Emma has a square playground with sides of 10 yards. What is the area and perimeter of the playground?

Answer: Area: 100 sq yards, Perimeter: 40 yards

Explanation: For a square with side 10: Area = $10 \times 10 = 100$ square yards. Perimeter = $4 \times 10 = 40$ yards.

Problem 13: Abigail has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Answer: Area: 196 sq yards, Perimeter: 56 yards

Explanation: For a square with side 14: Area = $14 \times 14 = 196$ square yards. Perimeter = $4 \times 14 = 56$ yards.

Problem 14: Mia has a square playground with sides of 11 yards. What is the area and perimeter of the playground?

Answer: Area: 121 sq yards, Perimeter: 44 yards

Explanation: For a square with side 11: Area = $11 \times 11 = 121$ square yards. Perimeter = $4 \times 11 = 44$ yards.

Problem 15: Mason has a square playground with sides of 13 yards. What is the area and perimeter of the playground?

Answer: Area: 169 sq yards, Perimeter: 52 yards

Explanation: For a square with side 13: Area = $13 \times 13 = 169$ square yards. Perimeter = $4 \times 13 = 52$ yards.

Problem 16: Mia has a square playground with sides of 15 yards. What is the area and perimeter of the playground?

Answer: Area: 225 sq yards, Perimeter: 60 yards

Explanation: For a square with side 15: Area = $15 \times 15 = 225$ square yards. Perimeter = $4 \times 15 = 60$ yards.

Problem 17: Elizabeth has a square playground with sides of 5 yards. What is the area and perimeter of the playground?

Answer: Area: 25 sq yards, Perimeter: 20 yards

Explanation: For a square with side 5: Area = $5 \times 5 = 25$ square yards. Perimeter = $4 \times 5 = 20$ yards.

Problem 18: Avery has a square playground with sides of 15 yards. What is the area and perimeter of the playground?

Answer: Area: 225 sq yards, Perimeter: 60 yards

Explanation: For a square with side 15: Area = $15 \times 15 = 225$ square yards. Perimeter = $4 \times 15 = 60$ yards.

Problem 19: Samuel has a square playground with sides of 12 yards. What is the area and perimeter of the playground?

Answer: Area: 144 sq yards, Perimeter: 48 yards

Explanation: For a square with side 12: Area = $12 \times 12 = 144$ square yards. Perimeter = $4 \times 12 = 48$ yards.

Problem 20: Emma has a square playground with sides of 6 yards. What is the area and perimeter of the playground?

Answer: Area: 36 sq yards, Perimeter: 24 yards

Explanation: For a square with side 6: Area = $6 \times 6 = 36$ square yards. Perimeter = $4 \times 6 = 24$ yards.

Problem 21: Charlotte has a square playground with sides of 6 yards. What is the area and perimeter of the playground?

Answer: Area: 36 sq yards, Perimeter: 24 yards

Explanation: For a square with side 6: Area = $6 \times 6 = 36$ square yards. Perimeter = $4 \times 6 = 24$ yards.

Problem 22: Avery has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Answer: Area: 196 sq yards, Perimeter: 56 yards

Explanation: For a square with side 14: Area = $14 \times 14 = 196$ square yards. Perimeter = $4 \times 14 = 56$ yards.

Problem 23: Avery has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Answer: Area: 196 sq yards, Perimeter: 56 yards

Explanation: For a square with side 14: Area = $14 \times 14 = 196$ square yards. Perimeter = $4 \times 14 = 56$ yards.

Problem 24: Alexander has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Answer: Area: 196 sq yards, Perimeter: 56 yards

Explanation: For a square with side 14: Area = $14 \times 14 = 196$ square yards. Perimeter = $4 \times 14 = 56$ yards.

Problem 25: Charlotte has a square playground with sides of 8 yards. What is the area and perimeter of the playground?

Answer: Area: 64 sq yards, Perimeter: 32 yards

Explanation: For a square with side 8: Area = $8 \times 8 = 64$ square yards. Perimeter = $4 \times 8 = 32$ yards.

Problem 26: Mia has a square playground with sides of 15 yards. What is the area and perimeter of the playground?

Answer: Area: 225 sq yards, Perimeter: 60 yards

Explanation: For a square with side 15: Area = $15 \times 15 = 225$ square yards. Perimeter = $4 \times 15 = 60$ yards.

Problem 27: Lucas has a square playground with sides of 15 yards. What is the area and perimeter of the playground?

Answer: Area: 225 sq yards, Perimeter: 60 yards

Explanation: For a square with side 15: Area = $15 \times 15 = 225$ square yards. Perimeter = $4 \times 15 = 60$ yards.

Problem 28: Emma has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Answer: Area: 196 sq yards, Perimeter: 56 yards

Explanation: For a square with side 14: Area = $14 \times 14 = 196$ square yards. Perimeter = $4 \times 14 = 56$ yards.

Problem 29: Benjamin has a square playground with sides of 6 yards. What is the area and perimeter of the playground?

Answer: Area: 36 sq yards, Perimeter: 24 yards

Explanation: For a square with side 6: Area = $6 \times 6 = 36$ square yards. Perimeter = $4 \times 6 = 24$ yards.

Problem 30: Elizabeth has a square playground with sides of 8 yards. What is the area and perimeter of the playground?

Answer: Area: 64 sq yards, Perimeter: 32 yards

Explanation: For a square with side 8: Area = $8 \times 8 = 64$ square yards. Perimeter = $4 \times 8 = 32$ yards.

Problem 31: Emma has a square playground with sides of 12 yards. What is the area and perimeter of the playground?

Answer: Area: 144 sq yards, Perimeter: 48 yards

Explanation: For a square with side 12: Area = $12 \times 12 = 144$ square yards. Perimeter = $4 \times 12 = 48$ yards.

Problem 32: James has a square playground with sides of 10 yards. What is the area and perimeter of the playground?

Answer: Area: 100 sq yards, Perimeter: 40 yards

Explanation: For a square with side 10: Area = $10 \times 10 = 100$ square yards. Perimeter = $4 \times 10 = 40$ yards.

Problem 33: Olivia has a square playground with sides of 10 yards. What is the area and perimeter of the playground?

Answer: Area: 100 sq yards, Perimeter: 40 yards

Explanation: For a square with side 10: Area = $10 \times 10 = 100$ square yards. Perimeter = $4 \times 10 = 40$ yards.

Problem 34: Ethan has a square playground with sides of 8 yards. What is the area and perimeter of the playground?

Answer: Area: 64 sq yards, Perimeter: 32 yards

Explanation: For a square with side 8: Area = $8 \times 8 = 64$ square yards. Perimeter = $4 \times 8 = 32$ yards.

Problem 35: Alexander has a square playground with sides of 15 yards. What is the area and perimeter of the playground?

Answer: Area: 225 sq yards, Perimeter: 60 yards

Explanation: For a square with side 15: Area = $15 \times 15 = 225$ square yards. Perimeter = $4 \times 15 = 60$ yards.

Problem 36: Samuel has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Answer: Area: 196 sq yards, Perimeter: 56 yards

Explanation: For a square with side 14: Area = $14 \times 14 = 196$ square yards. Perimeter = $4 \times 14 = 56$ yards.

Problem 37: Ava has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Answer: Area: 196 sq yards, Perimeter: 56 yards

Explanation: For a square with side 14: Area = $14 \times 14 = 196$ square yards. Perimeter = $4 \times 14 = 56$ yards.

Problem 38: Samuel has a square playground with sides of 6 yards. What is the area and perimeter of the playground?

Answer: Area: 36 sq yards, Perimeter: 24 yards

Explanation: For a square with side 6: Area = $6 \times 6 = 36$ square yards. Perimeter = $4 \times 6 = 24$ yards.

Problem 39: James has a square playground with sides of 10 yards. What is the area and perimeter of the playground?

Answer: Area: 100 sq yards, Perimeter: 40 yards

Explanation: For a square with side 10: Area = $10 \times 10 = 100$ square yards. Perimeter = $4 \times 10 = 40$ yards.

Problem 40: Samuel has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Answer: Area: 196 sq yards, Perimeter: 56 yards

Explanation: For a square with side 14: Area = $14 \times 14 = 196$ square yards. Perimeter = $4 \times 14 = 56$ yards.

Problem 41: Emily has a square playground with sides of 10 yards. What is the area and perimeter of the playground?

Answer: Area: 100 sq yards, Perimeter: 40 yards

Explanation: For a square with side 10: Area = $10 \times 10 = 100$ square yards. Perimeter = $4 \times 10 = 40$ yards.

Problem 42: Benjamin has a square playground with sides of 5 yards. What is the area and perimeter of the playground?

Answer: Area: 25 sq yards, Perimeter: 20 yards

Explanation: For a square with side 5: Area = $5 \times 5 = 25$ square yards. Perimeter = $4 \times 5 = 20$ yards.

Problem 43: Charlotte has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Answer: Area: 196 sq yards, Perimeter: 56 yards

Explanation: For a square with side 14: Area = $14 \times 14 = 196$ square yards. Perimeter = $4 \times 14 = 56$ yards.

Problem 44: Emma has a square playground with sides of 6 yards. What is the area and perimeter of the playground?

Answer: Area: 36 sq yards, Perimeter: 24 yards

Explanation: For a square with side 6: Area = $6 \times 6 = 36$ square yards. Perimeter = $4 \times 6 = 24$ yards.

Problem 45: Sebastian has a square playground with sides of 9 yards. What is the area and perimeter of the playground?

Answer: Area: 81 sq yards, Perimeter: 36 yards

Explanation: For a square with side 9: Area = $9 \times 9 = 81$ square yards. Perimeter = $4 \times 9 = 36$ yards.

Practice Exercises - Grade 4

Exercise 1: Ethan has a square playground with sides of 8 yards. What is the area and perimeter of the playground?

Exercise 2: Joseph has a square playground with sides of 12 yards. What is the area and perimeter of the playground?

Exercise 3: Evelyn has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Exercise 4: Olivia has a square playground with sides of 11 yards. What is the area and perimeter of the playground?

Exercise 5: Benjamin has a square playground with sides of 12 yards. What is the area and perimeter of the playground?

Exercise 6: Joseph has a square playground with sides of 6 yards. What is the area and perimeter of the playground?

Exercise 7: Mason has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Exercise 8: Samuel has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Exercise 9: Mason has a square playground with sides of 15 yards. What is the area and perimeter of the playground?

Exercise 10: Alexander has a square playground with sides of 6 yards. What is the area and perimeter of the playground?

Exercise 11: Samuel has a square playground with sides of 7 yards. What is the area and perimeter of the playground?

Exercise 12: Sebastian has a square playground with sides of 12 yards. What is the area and perimeter of the playground?

Exercise 13: Mia has a square playground with sides of 11 yards. What is the area and perimeter of the playground?

Exercise 14: James has a square playground with sides of 10 yards. What is the area and perimeter of the playground?

Exercise 15: Liam has a square playground with sides of 7 yards. What is the area and perimeter of the playground?

Exercise 16: Mia has a square playground with sides of 7 yards. What is the area and perimeter of the playground?

Exercise 17: Harper has a square playground with sides of 5 yards. What is the area and perimeter of the playground?

Exercise 18: Liam has a square playground with sides of 6 yards. What is the area and perimeter of the playground?

Exercise 19: Aiden has a square playground with sides of 9 yards. What is the area and perimeter of the playground?

Exercise 20: Samuel has a square playground with sides of 12 yards. What is the area and perimeter of the playground?

Practice Exercise Answers:

Exercise 1: Area: 64 sq yards, Perimeter: 32 yards

Exercise 2: Area: 144 sq yards, Perimeter: 48 yards

Exercise 3: Area: 196 sq yards, Perimeter: 56 yards

Exercise 4: Area: 121 sq yards, Perimeter: 44 yards

Exercise 5: Area: 144 sq yards, Perimeter: 48 yards

Exercise 6: Area: 36 sq yards, Perimeter: 24 yards

Exercise 7: Area: 196 sq yards, Perimeter: 56 yards

Exercise 8: Area: 196 sq yards, Perimeter: 56 yards

Exercise 9: Area: 225 sq yards, Perimeter: 60 yards

Exercise 10: Area: 36 sq yards, Perimeter: 24 yards

Exercise 11: Area: 49 sq yards, Perimeter: 28 yards

Exercise 12: Area: 144 sq yards, Perimeter: 48 yards

Exercise 13: Area: 121 sq yards, Perimeter: 44 yards

Exercise 14: Area: 100 sq yards, Perimeter: 40 yards

Exercise 15: Area: 49 sq yards, Perimeter: 28 yards

Exercise 16: Area: 49 sq yards, Perimeter: 28 yards

Exercise 17: Area: 25 sq yards, Perimeter: 20 yards

Exercise 18: Area: 36 sq yards, Perimeter: 24 yards

Exercise 19: Area: 81 sq yards, Perimeter: 36 yards

Exercise 20: Area: 144 sq yards, Perimeter: 48 yards

Challenge Problems - Grade 4

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Avery has a square playground with sides of 12 yards. What is the area and perimeter of the playground?

Answer: Area: 144 sq yards, Perimeter: 48 yards

Solution: For a square with side 12: Area = $12 \times 12 = 144$ square yards. Perimeter = $4 \times 12 = 48$ yards.

Challenge 2: Elizabeth has a square playground with sides of 10 yards. What is the area and perimeter of the playground?

Answer: Area: 100 sq yards, Perimeter: 40 yards

Solution: For a square with side 10: Area = $10 \times 10 = 100$ square yards. Perimeter = $4 \times 10 = 40$ yards.

Challenge 3: Sophia has a square playground with sides of 9 yards. What is the area and perimeter of the playground?

Answer: Area: 81 sq yards, Perimeter: 36 yards

Solution: For a square with side 9: Area = $9 \times 9 = 81$ square yards. Perimeter = $4 \times 9 = 36$ yards.

Challenge 4: Harper has a square playground with sides of 8 yards. What is the area and perimeter of the playground?

Answer: Area: 64 sq yards, Perimeter: 32 yards

Solution: For a square with side 8: Area = $8 \times 8 = 64$ square yards. Perimeter = $4 \times 8 = 32$ yards.

Challenge 5: David has a square playground with sides of 15 yards. What is the area and perimeter of the playground?

Answer: Area: 225 sq yards, Perimeter: 60 yards

Solution: For a square with side 15: Area = $15 \times 15 = 225$ square yards. Perimeter = $4 \times 15 = 60$ yards.

Challenge 6: James has a square playground with sides of 15 yards. What is the area and perimeter of the playground?

Answer: Area: 225 sq yards, Perimeter: 60 yards

Solution: For a square with side 15: Area = $15 \times 15 = 225$ square yards. Perimeter = $4 \times 15 = 60$ yards.

Challenge 7: Sofia has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Answer: Area: 196 sq yards, Perimeter: 56 yards

Solution: For a square with side 14: Area = $14 \times 14 = 196$ square yards. Perimeter = $4 \times 14 = 56$ yards.

Challenge 8: Sophia has a square playground with sides of 7 yards. What is the area and perimeter of the playground?

Answer: Area: 49 sq yards, Perimeter: 28 yards

Solution: For a square with side 7: Area = $7 \times 7 = 49$ square yards. Perimeter = $4 \times 7 = 28$ yards.

Challenge 9: James has a square playground with sides of 9 yards. What is the area and perimeter of the playground?

Answer: Area: 81 sq yards, Perimeter: 36 yards

Solution: For a square with side 9: Area = $9 \times 9 = 81$ square yards. Perimeter = $4 \times 9 = 36$ yards.

Challenge 10: Benjamin has a square playground with sides of 7 yards. What is the area and perimeter of the playground?

Answer: Area: 49 sq yards, Perimeter: 28 yards

Solution: For a square with side 7: Area = $7 \times 7 = 49$ square yards. Perimeter = $4 \times 7 = 28$ yards.

Challenge 11: Sofia has a square playground with sides of 8 yards. What is the area and perimeter of the playground?

Answer: Area: 64 sq yards, Perimeter: 32 yards

Solution: For a square with side 8: Area = $8 \times 8 = 64$ square yards. Perimeter = $4 \times 8 = 32$ yards.

Challenge 12: Avery has a square playground with sides of 14 yards. What is the area and perimeter of the playground?

Answer: Area: 196 sq yards, Perimeter: 56 yards

Solution: For a square with side 14: Area = $14 \times 14 = 196$ square yards. Perimeter = $4 \times 14 = 56$ yards.

Challenge 13: Abigail has a square playground with sides of 7 yards. What is the area and perimeter of the playground?

Answer: Area: 49 sq yards, Perimeter: 28 yards

Solution: For a square with side 7: Area = $7 \times 7 = 49$ square yards. Perimeter = $4 \times 7 = 28$ yards.

Challenge 14: Harper has a square playground with sides of 13 yards. What is the area and perimeter of the playground?

Answer: Area: 169 sq yards, Perimeter: 52 yards

Solution: For a square with side 13: Area = $13 \times 13 = 169$ square yards. Perimeter = $4 \times 13 = 52$ yards.

Challenge 15: Sebastian has a square playground with sides of 7 yards. What is the area and perimeter of the playground?

Answer: Area: 49 sq yards, Perimeter: 28 yards

Solution: For a square with side 7: Area = $7 \times 7 = 49$ square yards. Perimeter = $4 \times 7 = 28$ yards.

Grade 5 Problems

Problem 1: Aiden has a box that is 10 inches long, 4 inches wide, and 10 inches tall. What is the volume of the box?

Answer: 400

Explanation: Volume = length $\times$ width $\times$ height = $10 \times 4 \times 10 = 400$ cubic inches.

Problem 2: Jack has a box that is 8 inches long, 4 inches wide, and 10 inches tall. What is the volume of the box?

Answer: 320

Explanation: Volume = length $\times$ width $\times$ height = $8 \times 4 \times 10 = 320$ cubic inches.

Problem 3: Evelyn has a box that is 3 inches long, 7 inches wide, and 7 inches tall. What is the volume of the box?

Answer: 147

Explanation: Volume = length $\times$ width $\times$ height = $3 \times 7 \times 7 = 147$ cubic inches.

Problem 4: Avery has a box that is 3 inches long, 4 inches wide, and 5 inches tall. What is the volume of the box?

Answer: 60

Explanation: Volume = length $\times$ width $\times$ height = $3 \times 4 \times 5 = 60$ cubic inches.

Problem 5: Alexander has a box that is 3 inches long, 8 inches wide, and 5 inches tall. What is the volume of the box?

Answer: 120

Explanation: Volume = length $\times$ width $\times$ height = $3 \times 8 \times 5 = 120$ cubic inches.

Problem 6: Liam has a box that is 5 inches long, 4 inches wide, and 8 inches tall. What is the volume of the box?

Answer: 160

Explanation: Volume = length $\times$ width $\times$ height = $5 \times 4 \times 8 = 160$ cubic inches.

Problem 7: Charlotte has a box that is 5 inches long, 4 inches wide, and 10 inches tall. What is the volume of the box?

Answer: 200

Explanation: Volume = length $\times$ width $\times$ height = $5 \times 4 \times 10 = 200$ cubic inches.

Problem 8: Abigail has a box that is 6 inches long, 10 inches wide, and 8 inches tall. What is the volume of the box?

Answer: 480

Explanation: Volume = length $\times$ width $\times$ height = $6 \times 10 \times 8 = 480$ cubic inches.

Problem 9: Liam has a box that is 8 inches long, 6 inches wide, and 6 inches tall. What is the volume of the box?

Answer: 288

Explanation: Volume = length $\times$ width $\times$ height = $8 \times 6 \times 6 = 288$ cubic inches.

Problem 10: David has a box that is 5 inches long, 4 inches wide, and 4 inches tall. What is the volume of the box?

Answer: 80

Explanation: Volume = length $\times$ width $\times$ height = $5 \times 4 \times 4 = 80$ cubic inches.

Problem 11: Benjamin has a box that is 4 inches long, 7 inches wide, and 6 inches tall. What is the volume of the box?

Answer: 168

Explanation: Volume = length $\times$ width $\times$ height = $4 \times 7 \times 6 = 168$ cubic inches.

Problem 12: Olivia has a box that is 5 inches long, 9 inches wide, and 6 inches tall. What is the volume of the box?

Answer: 270

Explanation: Volume = length $\times$ width $\times$ height = $5 \times 9 \times 6 = 270$ cubic inches.

Problem 13: Mason has a box that is 5 inches long, 5 inches wide, and 4 inches tall. What is the volume of the box?

Answer: 100

Explanation: Volume = length $\times$ width $\times$ height = $5 \times 5 \times 4 = 100$ cubic inches.

Problem 14: Elizabeth has a box that is 5 inches long, 7 inches wide, and 8 inches tall. What is the volume of the box?

Answer: 280

Explanation: Volume = length $\times$ width $\times$ height = $5 \times 7 \times 8 = 280$ cubic inches.

Problem 15: Sofia has a box that is 3 inches long, 9 inches wide, and 4 inches tall. What is the volume of the box?

Answer: 108

Explanation: Volume = length $\times$ width $\times$ height = $3 \times 9 \times 4 = 108$ cubic inches.

Problem 16: Jack has a box that is 6 inches long, 9 inches wide, and 4 inches tall. What is the volume of the box?

Answer: 216

Explanation: Volume = length $\times$ width $\times$ height = $6 \times 9 \times 4 = 216$ cubic inches.

Problem 17: Emma has a box that is 4 inches long, 9 inches wide, and 10 inches tall. What is the volume of the box?

Answer: 360

Explanation: Volume = length $\times$ width $\times$ height = $4 \times 9 \times 10 = 360$ cubic inches.

Problem 18: James has a box that is 5 inches long, 8 inches wide, and 10 inches tall. What is the volume of the box?

Answer: 400

Explanation: Volume = length $\times$ width $\times$ height = $5 \times 8 \times 10 = 400$ cubic inches.

Problem 19: Aiden has a box that is 10 inches long, 3 inches wide, and 4 inches tall. What is the volume of the box?

Answer: 120

Explanation: Volume = length $\times$ width $\times$ height = $10 \times 3 \times 4 = 120$ cubic inches.

Problem 20: Mia has a box that is 5 inches long, 4 inches wide, and 6 inches tall. What is the volume of the box?

Answer: 120

Explanation: Volume = length $\times$ width $\times$ height = $5 \times 4 \times 6 = 120$ cubic inches.

Problem 21: Amelia has a box that is 8 inches long, 3 inches wide, and 7 inches tall. What is the volume of the box?

Answer: 168

Explanation: Volume = length $\times$ width $\times$ height = $8 \times 3 \times 7 = 168$ cubic inches.

Problem 22: Sofia has a box that is 6 inches long, 4 inches wide, and 7 inches tall. What is the volume of the box?

Answer: 168

Explanation: Volume = length $\times$ width $\times$ height = $6 \times 4 \times 7 = 168$ cubic inches.

Problem 23: Emily has a box that is 4 inches long, 9 inches wide, and 9 inches tall. What is the volume of the box?

Answer: 324

Explanation: Volume = length $\times$ width $\times$ height = $4 \times 9 \times 9 = 324$ cubic inches.

Problem 24: Liam has a box that is 7 inches long, 9 inches wide, and 3 inches tall. What is the volume of the box?

Answer: 189

Explanation: Volume = length $\times$ width $\times$ height = $7 \times 9 \times 3 = 189$ cubic inches.

Problem 25: Henry has a box that is 3 inches long, 6 inches wide, and 4 inches tall. What is the volume of the box?

Answer: 72

Explanation: Volume = length $\times$ width $\times$ height = $3 \times 6 \times 4 = 72$ cubic inches.

Problem 26: James has a box that is 10 inches long, 6 inches wide, and 6 inches tall. What is the volume of the box?

Answer: 360

Explanation: Volume = length $\times$ width $\times$ height = $10 \times 6 \times 6 = 360$ cubic inches.

Problem 27: Abigail has a box that is 8 inches long, 9 inches wide, and 4 inches tall. What is the volume of the box?

Answer: 288

Explanation: Volume = length $\times$ width $\times$ height = $8 \times 9 \times 4 = 288$ cubic inches.

Problem 28: David has a box that is 5 inches long, 8 inches wide, and 10 inches tall. What is the volume of the box?

Answer: 400

Explanation: Volume = length $\times$ width $\times$ height = $5 \times 8 \times 10 = 400$ cubic inches.

Problem 29: David has a box that is 4 inches long, 4 inches wide, and 10 inches tall. What is the volume of the box?

Answer: 160

Explanation: Volume = length $\times$ width $\times$ height = $4 \times 4 \times 10 = 160$ cubic inches.

Problem 30: Mason has a box that is 5 inches long, 8 inches wide, and 9 inches tall. What is the volume of the box?

Answer: 360

Explanation: Volume = length $\times$ width $\times$ height = $5 \times 8 \times 9 = 360$ cubic inches.

Problem 31: Joseph has a box that is 3 inches long, 8 inches wide, and 3 inches tall. What is the volume of the box?

Answer: 72

Explanation: Volume = length $\times$ width $\times$ height = $3 \times 8 \times 3 = 72$ cubic inches.

Problem 32: Emily has a box that is 4 inches long, 7 inches wide, and 5 inches tall. What is the volume of the box?

Answer: 140

Explanation: Volume = length $\times$ width $\times$ height = $4 \times 7 \times 5 = 140$ cubic inches.

Problem 33: Mason has a box that is 10 inches long, 8 inches wide, and 7 inches tall. What is the volume of the box?

Answer: 560

Explanation: Volume = length $\times$ width $\times$ height = $10 \times 8 \times 7 = 560$ cubic inches.

Problem 34: Olivia has a box that is 7 inches long, 6 inches wide, and 6 inches tall. What is the volume of the box?

Answer: 252

Explanation: Volume = length $\times$ width $\times$ height = $7 \times 6 \times 6 = 252$ cubic inches.

Problem 35: Sebastian has a box that is 9 inches long, 6 inches wide, and 9 inches tall. What is the volume of the box?

Answer: 486

Explanation: Volume = length $\times$ width $\times$ height = $9 \times 6 \times 9 = 486$ cubic inches.

Problem 36: Olivia has a box that is 8 inches long, 8 inches wide, and 8 inches tall. What is the volume of the box?

Answer: 512

Explanation: Volume = length $\times$ width $\times$ height = $8 \times 8 \times 8 = 512$ cubic inches.

Problem 37: Mia has a box that is 8 inches long, 7 inches wide, and 4 inches tall. What is the volume of the box?

Answer: 224

Explanation: Volume = length $\times$ width $\times$ height = $8 \times 7 \times 4 = 224$ cubic inches.

Problem 38: Abigail has a box that is 5 inches long, 10 inches wide, and 5 inches tall. What is the volume of the box?

Answer: 250

Explanation: Volume = length $\times$ width $\times$ height = $5 \times 10 \times 5 = 250$ cubic inches.

Problem 39: Olivia has a box that is 4 inches long, 9 inches wide, and 3 inches tall. What is the volume of the box?

Answer: 108

Explanation: Volume = length $\times$ width $\times$ height = $4 \times 9 \times 3 = 108$ cubic inches.

Problem 40: David has a box that is 4 inches long, 8 inches wide, and 10 inches tall. What is the volume of the box?

Answer: 320

Explanation: Volume = length $\times$ width $\times$ height = $4 \times 8 \times 10 = 320$ cubic inches.

Problem 41: Mason has a box that is 4 inches long, 5 inches wide, and 7 inches tall. What is the volume of the box?

Answer: 140

Explanation: Volume = length $\times$ width $\times$ height = $4 \times 5 \times 7 = 140$ cubic inches.

Problem 42: Emily has a box that is 4 inches long, 6 inches wide, and 7 inches tall. What is the volume of the box?

Answer: 168

Explanation: Volume = length $\times$ width $\times$ height = $4 \times 6 \times 7 = 168$ cubic inches.

Problem 43: Emily has a box that is 4 inches long, 6 inches wide, and 5 inches tall. What is the volume of the box?

Answer: 120

Explanation: Volume = length $\times$ width $\times$ height = $4 \times 6 \times 5 = 120$ cubic inches.

Problem 44: Emily has a box that is 9 inches long, 8 inches wide, and 6 inches tall. What is the volume of the box?

Answer: 432

Explanation: Volume = length $\times$ width $\times$ height = $9 \times 8 \times 6 = 432$ cubic inches.

Problem 45: Emily has a box that is 9 inches long, 4 inches wide, and 8 inches tall. What is the volume of the box?

Answer: 288

Explanation: Volume = length $\times$ width $\times$ height = $9 \times 4 \times 8 = 288$ cubic inches.

Practice Exercises - Grade 5

Exercise 1: Samuel has a box that is 9 inches long, 10 inches wide, and 9 inches tall. What is the volume of the box?

Exercise 2: Samuel has a box that is 7 inches long, 7 inches wide, and 3 inches tall. What is the volume of the box?

Exercise 3: James has a box that is 10 inches long, 4 inches wide, and 4 inches tall. What is the volume of the box?

Exercise 4: Alexander has a box that is 5 inches long, 6 inches wide, and 8 inches tall. What is the volume of the box?

Exercise 5: Emma has a box that is 9 inches long, 7 inches wide, and 4 inches tall. What is the volume of the box?

Exercise 6: Harper has a box that is 7 inches long, 7 inches wide, and 6 inches tall. What is the volume of the box?

Exercise 7: Aiden has a box that is 7 inches long, 8 inches wide, and 9 inches tall. What is the volume of the box?

Exercise 8: Harper has a box that is 6 inches long, 5 inches wide, and 3 inches tall. What is the volume of the box?

Exercise 9: Charlotte has a box that is 6 inches long, 8 inches wide, and 7 inches tall. What is the volume of the box?

Exercise 10: Evelyn has a box that is 9 inches long, 10 inches wide, and 5 inches tall. What is the volume of the box?

Exercise 11: Jack has a box that is 4 inches long, 5 inches wide, and 9 inches tall. What is the volume of the box?

Exercise 12: Alexander has a box that is 9 inches long, 5 inches wide, and 9 inches tall. What is the volume of the box?

Exercise 13: Mason has a box that is 9 inches long, 7 inches wide, and 7 inches tall. What is the volume of the box?

Exercise 14: Olivia has a box that is 6 inches long, 4 inches wide, and 5 inches tall. What is the volume of the box?

Exercise 15: Mason has a box that is 5 inches long, 9 inches wide, and 5 inches tall. What is the volume of the box?

Exercise 16: Amelia has a box that is 7 inches long, 6 inches wide, and 8 inches tall. What is the volume of the box?

Exercise 17: Charlotte has a box that is 8 inches long, 7 inches wide, and 7 inches tall. What is the volume of the box?

Exercise 18: Emma has a box that is 4 inches long, 8 inches wide, and 6 inches tall. What is the volume of the box?

Exercise 19: James has a box that is 8 inches long, 3 inches wide, and 5 inches tall. What is the volume of the box?

Exercise 20: James has a box that is 7 inches long, 6 inches wide, and 3 inches tall. What is the volume of the box?

Practice Exercise Answers:

Exercise 1: 810

Exercise 2: 147

Exercise 3: 160

Exercise 4: 240

Exercise 5: 252

Exercise 6: 294

Exercise 7: 504

Exercise 8: 90

Exercise 9: 336

Exercise 10: 450

Exercise 11: 180

Exercise 12: 405

Exercise 13: 441

Exercise 14: 120

Exercise 15: 225

Exercise 16: 336

Exercise 17: 392

Exercise 18: 192

Exercise 19: 120

Exercise 20: 126

Challenge Problems - Grade 5

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Benjamin has a box that is 3 inches long, 10 inches wide, and 7 inches tall. What is the volume of the box?

Answer: 210

Solution: Volume = length $\times$ width $\times$ height = $3 \times 10 \times 7 = 210$ cubic inches.

Challenge 2: Avery has a box that is 8 inches long, 5 inches wide, and 4 inches tall. What is the volume of the box?

Answer: 160

Solution: Volume = length $\times$ width $\times$ height = $8 \times 5 \times 4 = 160$ cubic inches.

Challenge 3: Lucas has a box that is 8 inches long, 6 inches wide, and 9 inches tall. What is the volume of the box?

Answer: 432

Solution: Volume = length $\times$ width $\times$ height = $8 \times 6 \times 9 = 432$ cubic inches.

Challenge 4: Jack has a box that is 9 inches long, 5 inches wide, and 8 inches tall. What is the volume of the box?

Answer: 360

Solution: Volume = length $\times$ width $\times$ height = $9 \times 5 \times 8 = 360$ cubic inches.

Challenge 5: Emily has a box that is 4 inches long, 3 inches wide, and 5 inches tall. What is the volume of the box?

Answer: 60

Solution: Volume = length $\times$ width $\times$ height = $4 \times 3 \times 5 = 60$ cubic inches.

Challenge 6: Olivia has a box that is 6 inches long, 3 inches wide, and 10 inches tall. What is the volume of the box?

Answer: 180

Solution: Volume = length $\times$ width $\times$ height = $6 \times 3 \times 10 = 180$ cubic inches.

Challenge 7: Sophia has a box that is 10 inches long, 3 inches wide, and 5 inches tall. What is the volume of the box?

Answer: 150

Solution: Volume = length $\times$ width $\times$ height = $10 \times 3 \times 5 = 150$ cubic inches.

Challenge 8: Mason has a box that is 5 inches long, 9 inches wide, and 7 inches tall. What is the volume of the box?

Answer: 315

Solution: Volume = length $\times$ width $\times$ height = $5 \times 9 \times 7 = 315$ cubic inches.

Challenge 9: Joseph has a box that is 9 inches long, 8 inches wide, and 6 inches tall. What is the volume of the box?

Answer: 432

Solution: Volume = length $\times$ width $\times$ height = $9 \times 8 \times 6 = 432$ cubic inches.

Challenge 10: Ethan has a box that is 5 inches long, 8 inches wide, and 8 inches tall. What is the volume of the box?

Answer: 320

Solution: Volume = length $\times$ width $\times$ height = $5 \times 8 \times 8 = 320$ cubic inches.

Challenge 11: Benjamin has a box that is 6 inches long, 4 inches wide, and 4 inches tall. What is the volume of the box?

Answer: 96

Solution: Volume = length $\times$ width $\times$ height = $6 \times 4 \times 4 = 96$ cubic inches.

Challenge 12: Sofia has a box that is 3 inches long, 6 inches wide, and 6 inches tall. What is the volume of the box?

Answer: 108

Solution: Volume = length $\times$ width $\times$ height = $3 \times 6 \times 6 = 108$ cubic inches.

Challenge 13: David has a box that is 4 inches long, 8 inches wide, and 6 inches tall. What is the volume of the box?

Answer: 192

Solution: Volume = length $\times$ width $\times$ height = $4 \times 8 \times 6 = 192$ cubic inches.

Challenge 14: Joseph has a box that is 9 inches long, 7 inches wide, and 5 inches tall. What is the volume of the box?

Answer: 315

Solution: Volume = length $\times$ width $\times$ height = $9 \times 7 \times 5 = 315$ cubic inches.

Challenge 15: Sebastian has a box that is 10 inches long, 5 inches wide, and 8 inches tall. What is the volume of the box?

Answer: 400

Solution: Volume = length $\times$ width $\times$ height = $10 \times 5 \times 8 = 400$ cubic inches.

Grade 6 Problems

Problem 1: Alexander has a circular pond with a radius of 5 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 78.5

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 5^2 = 3.14 \times 25 = 78.5$ square meters.

Problem 2: Abigail has a triangular garden with a base of 11 meters and height of 9 meters. What is the area?

Answer: 49

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (11 \times 9) \div 2 = 49$ square meters.

Problem 3: Mason has a triangular garden with a base of 8 meters and height of 10 meters. What is the area?

Answer: 40

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (8 \times 10) \div 2 = 40$ square meters.

Problem 4: Harper has a circular pond with a radius of 6 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 113.04

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 6^2 = 3.14 \times 36 = 113.04$ square meters.

Problem 5: Jack has a triangular garden with a base of 6 meters and height of 9 meters. What is the area?

Answer: 27

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (6 \times 9) \div 2 = 27$ square meters.

Problem 6: Mason has a triangular garden with a base of 8 meters and height of 4 meters. What is the area?

Answer: 16

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (8 \times 4) \div 2 = 16$ square meters.

Problem 7: James has a triangular garden with a base of 15 meters and height of 6 meters. What is the area?

Answer: 45

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (15 \times 6) \div 2 = 45$ square meters.

Problem 8: Liam has a circular pond with a radius of 9 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 254.34

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 9^2 = 3.14 \times 81 = 254.34$ square meters.

Problem 9: Noah has a circular pond with a radius of 5 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 78.5

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 5^2 = 3.14 \times 25 = 78.5$ square meters.

Problem 10: James has a triangular garden with a base of 6 meters and height of 9 meters. What is the area?

Answer: 27

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (6 \times 9) \div 2 = 27$ square meters.

Problem 11: Lucas has a triangular garden with a base of 10 meters and height of 6 meters. What is the area?

Answer: 30

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (10 \times 6) \div 2 = 30$ square meters.

Problem 12: Joseph has a circular pond with a radius of 7 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 153.86

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 7^2 = 3.14 \times 49 = 153.86$ square meters.

Problem 13: Mia has a triangular garden with a base of 7 meters and height of 12 meters. What is the area?

Answer: 42

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (7 \times 12) \div 2 = 42$ square meters.

Problem 14: Emma has a circular pond with a radius of 8 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 200.96

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 8^2 = 3.14 \times 64 = 200.96$ square meters.

Problem 15: Emma has a circular pond with a radius of 7 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 153.86

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 7^2 = 3.14 \times 49 = 153.86$ square meters.

Problem 16: Abigail has a circular pond with a radius of 7 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 153.86

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 7^2 = 3.14 \times 49 = 153.86$ square meters.

Problem 17: Sofia has a triangular garden with a base of 12 meters and height of 7 meters. What is the area?

Answer: 42

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (12 \times 7) \div 2 = 42$ square meters.

Problem 18: Olivia has a triangular garden with a base of 12 meters and height of 7 meters. What is the area?

Answer: 42

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (12 \times 7) \div 2 = 42$ square meters.

Problem 19: Abigail has a circular pond with a radius of 3 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 28.26

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 3^2 = 3.14 \times 9 = 28.26$ square meters.

Problem 20: Benjamin has a triangular garden with a base of 12 meters and height of 7 meters. What is the area?

Answer: 42

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (12 \times 7) \div 2 = 42$ square meters.

Problem 21: Alexander has a circular pond with a radius of 6 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 113.04

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 6^2 = 3.14 \times 36 = 113.04$ square meters.

Problem 22: Alexander has a triangular garden with a base of 10 meters and height of 10 meters. What is the area?

Answer: 50

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (10 \times 10) \div 2 = 50$ square meters.

Problem 23: Liam has a circular pond with a radius of 8 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 200.96

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 8^2 = 3.14 \times 64 = 200.96$ square meters.

Problem 24: Ethan has a triangular garden with a base of 11 meters and height of 11 meters. What is the area?

Answer: 60

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (11 \times 11) \div 2 = 60$ square meters.

Problem 25: Harper has a triangular garden with a base of 14 meters and height of 6 meters. What is the area?

Answer: 42

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (14 \times 6) \div 2 = 42$ square meters.

Problem 26: Henry has a circular pond with a radius of 3 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 28.26

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 3^2 = 3.14 \times 9 = 28.26$ square meters.

Problem 27: Emma has a triangular garden with a base of 6 meters and height of 8 meters. What is the area?

Answer: 24

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (6 \times 8) \div 2 = 24$ square meters.

Problem 28: Jack has a triangular garden with a base of 7 meters and height of 10 meters. What is the area?

Answer: 35

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (7 \times 10) \div 2 = 35$ square meters.

Problem 29: Liam has a circular pond with a radius of 10 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 314.0

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 10^2 = 3.14 \times 100 = 314.0$ square meters.

Problem 30: Alexander has a triangular garden with a base of 11 meters and height of 10 meters. What is the area?

Answer: 55

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (11 \times 10) \div 2 = 55$ square meters.

Problem 31: David has a circular pond with a radius of 7 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 153.86

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 7^2 = 3.14 \times 49 = 153.86$ square meters.

Problem 32: Henry has a triangular garden with a base of 7 meters and height of 12 meters. What is the area?

Answer: 42

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (7 \times 12) \div 2 = 42$ square meters.

Problem 33: Avery has a triangular garden with a base of 10 meters and height of 5 meters. What is the area?

Answer: 25

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (10 \times 5) \div 2 = 25$ square meters.

Problem 34: David has a triangular garden with a base of 14 meters and height of 5 meters. What is the area?

Answer: 35

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (14 \times 5) \div 2 = 35$ square meters.

Problem 35: Aiden has a triangular garden with a base of 12 meters and height of 5 meters. What is the area?

Answer: 30

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (12 \times 5) \div 2 = 30$ square meters.

Problem 36: Joseph has a circular pond with a radius of 6 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 113.04

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 6^2 = 3.14 \times 36 = 113.04$ square meters.

Problem 37: Abigail has a triangular garden with a base of 8 meters and height of 5 meters. What is the area?

Answer: 20

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (8 \times 5) \div 2 = 20$ square meters.

Problem 38: Joseph has a triangular garden with a base of 15 meters and height of 11 meters. What is the area?

Answer: 82

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (15 \times 11) \div 2 = 82$ square meters.

Problem 39: Isabella has a triangular garden with a base of 14 meters and height of 5 meters. What is the area?

Answer: 35

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (14 \times 5) \div 2 = 35$ square meters.

Problem 40: Aiden has a triangular garden with a base of 11 meters and height of 10 meters. What is the area?

Answer: 55

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (11 \times 10) \div 2 = 55$ square meters.

Problem 41: David has a triangular garden with a base of 6 meters and height of 5 meters. What is the area?

Answer: 15

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (6 \times 5) \div 2 = 15$ square meters.

Problem 42: Olivia has a circular pond with a radius of 10 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 314.0

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 10^2 = 3.14 \times 100 = 314.0$ square meters.

Problem 43: Henry has a triangular garden with a base of 13 meters and height of 11 meters. What is the area?

Answer: 71

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (13 \times 11) \div 2 = 71$ square meters.

Problem 44: Joseph has a circular pond with a radius of 6 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 113.04

Explanation: Area of circle = $\pi \times r^2 = 3.14 \times 6^2 = 3.14 \times 36 = 113.04$ square meters.

Problem 45: David has a triangular garden with a base of 7 meters and height of 6 meters. What is the area?

Answer: 21

Explanation: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (7 \times 6) \div 2 = 21$ square meters.

Practice Exercises - Grade 6

Exercise 1: Liam has a circular pond with a radius of 9 meters. What is the area? (Use $\pi \approx 3.14$)

Exercise 2: Lucas has a triangular garden with a base of 6 meters and height of 11 meters. What is the area?

Exercise 3: Evelyn has a circular pond with a radius of 9 meters. What is the area? (Use $\pi \approx 3.14$)

Exercise 4: Joseph has a circular pond with a radius of 3 meters. What is the area? (Use $\pi \approx 3.14$)

Exercise 5: Olivia has a circular pond with a radius of 8 meters. What is the area? (Use $\pi \approx 3.14$)

Exercise 6: Aiden has a triangular garden with a base of 7 meters and height of 10 meters. What is the area?

Exercise 7: Elizabeth has a circular pond with a radius of 10 meters. What is the area? (Use $\pi \approx 3.14$)

Exercise 8: Emma has a triangular garden with a base of 15 meters and height of 11 meters. What is the area?

Exercise 9: Emma has a triangular garden with a base of 11 meters and height of 12 meters. What is the area?

Exercise 10: Mason has a circular pond with a radius of 4 meters. What is the area? (Use $\pi \approx 3.14$)

Exercise 11: Olivia has a circular pond with a radius of 4 meters. What is the area? (Use $\pi \approx 3.14$)

Exercise 12: Isabella has a circular pond with a radius of 9 meters. What is the area? (Use $\pi \approx 3.14$)

Exercise 13: Alexander has a circular pond with a radius of 3 meters. What is the area? (Use $\pi \approx 3.14$)

Exercise 14: Elizabeth has a circular pond with a radius of 8 meters. What is the area? (Use $\pi \approx 3.14$)

Exercise 15: Emily has a circular pond with a radius of 10 meters. What is the area? (Use $\pi \approx 3.14$)

Exercise 16: Isabella has a circular pond with a radius of 6 meters. What is the area? (Use $\pi \approx 3.14$)

Exercise 17: Avery has a circular pond with a radius of 8 meters. What is the area? (Use $\pi \approx 3.14$)

Exercise 18: Avery has a triangular garden with a base of 10 meters and height of 9 meters. What is the area?

Exercise 19: Olivia has a triangular garden with a base of 7 meters and height of 6 meters. What is the area?

Exercise 20: David has a triangular garden with a base of 11 meters and height of 4 meters. What is the area?

Practice Exercise Answers:

Exercise 1: 254.34

Exercise 2: 33

Exercise 3: 254.34

Exercise 4: 28.26

Exercise 5: 200.96

Exercise 6: 35

Exercise 7: 314.0

Exercise 8: 82

Exercise 9: 66

Exercise 10: 50.24

Exercise 11: 50.24

Exercise 12: 254.34

Exercise 13: 28.26

Exercise 14: 200.96

Exercise 15: 314.0

Exercise 16: 113.04

Exercise 17: 200.96

Exercise 18: 45

Exercise 19: 21

Exercise 20: 22

Challenge Problems - Grade 6

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Jack has a circular pond with a radius of 3 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 28.26

Solution: Area of circle = $\pi \times r^2 = 3.14 \times 3^2 = 3.14 \times 9 = 28.26$ square meters.

Challenge 2: Joseph has a triangular garden with a base of 11 meters and height of 5 meters. What is the area?

Answer: 27

Solution: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (11 \times 5) \div 2 = 27$ square meters.

Challenge 3: Liam has a circular pond with a radius of 4 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 50.24

Solution: Area of circle = $\pi \times r^2 = 3.14 \times 4^2 = 3.14 \times 16 = 50.24$ square meters.

Challenge 4: Sophia has a triangular garden with a base of 10 meters and height of 4 meters. What is the area?

Answer: 20

Solution: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (10 \times 4) \div 2 = 20$ square meters.

Challenge 5: Henry has a circular pond with a radius of 7 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 153.86

Solution: Area of circle = $\pi \times r^2 = 3.14 \times 7^2 = 3.14 \times 49 = 153.86$ square meters.

Challenge 6: Samuel has a circular pond with a radius of 3 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 28.26

Solution: Area of circle = $\pi \times r^2 = 3.14 \times 3^2 = 3.14 \times 9 = 28.26$ square meters.

Challenge 7: Liam has a circular pond with a radius of 7 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 153.86

Solution: Area of circle = $\pi \times r^2 = 3.14 \times 7^2 = 3.14 \times 49 = 153.86$ square meters.

Challenge 8: Samuel has a triangular garden with a base of 10 meters and height of 7 meters. What is the area?

Answer: 35

Solution: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (10 \times 7) \div 2 = 35$ square meters.

Challenge 9: Sebastian has a circular pond with a radius of 6 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 113.04

Solution: Area of circle = $\pi \times r^2 = 3.14 \times 6^2 = 3.14 \times 36 = 113.04$ square meters.

Challenge 10: Henry has a triangular garden with a base of 11 meters and height of 4 meters. What is the area?

Answer: 22

Solution: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (11 \times 4) \div 2 = 22$ square meters.

Challenge 11: David has a circular pond with a radius of 4 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 50.24

Solution: Area of circle = $\pi \times r^2 = 3.14 \times 4^2 = 3.14 \times 16 = 50.24$ square meters.

Challenge 12: James has a circular pond with a radius of 3 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 28.26

Solution: Area of circle = $\pi \times r^2 = 3.14 \times 3^2 = 3.14 \times 9 = 28.26$ square meters.

Challenge 13: Emma has a triangular garden with a base of 6 meters and height of 12 meters. What is the area?

Answer: 36

Solution: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (6 \times 12) \div 2 = 36$ square meters.

Challenge 14: Emma has a triangular garden with a base of 14 meters and height of 7 meters. What is the area?

Answer: 49

Solution: Area of triangle = $(\text{base} \times \text{height}) \div 2 = (14 \times 7) \div 2 = 49$ square meters.

Challenge 15: James has a circular pond with a radius of 9 meters. What is the area? (Use $\pi \approx 3.14$)

Answer: 254.34

Solution: Area of circle = $\pi \times r^2 = 3.14 \times 9^2 = 3.14 \times 81 = 254.34$ square meters.

Chapter 4: Ratios and Proportions

Ratios and proportions are powerful tools for comparing quantities. This chapter develops students' ability to work with ratios, rates, and proportional relationships in practical contexts.

Grade 4 Problems

Problem 1: In David's class, the ratio of boys to girls is 4:2. If there are 12 students, how many are boys?

Answer: 8

Explanation: The ratio 4:2 means 4 boys for every 2 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 2: In Emily's class, the ratio of boys to girls is 5:4. If there are 18 students, how many are boys?

Answer: 10

Explanation: The ratio 5:4 means 5 boys for every 4 girls. Total ratio parts: 9. Each part = $18 \div 9 = 2$ students. Boys = $5 \times 2 = 10$.

Problem 3: In Harper's class, the ratio of boys to girls is 3:5. If there are 16 students, how many are boys?

Answer: 6

Explanation: The ratio 3:5 means 3 boys for every 5 girls. Total ratio parts: 8. Each part = $16 \div 8 = 2$ students. Boys = $3 \times 2 = 6$.

Problem 4: In Alexander's class, the ratio of boys to girls is 3:3. If there are 12 students, how many are boys?

Answer: 6

Explanation: The ratio 3:3 means 3 boys for every 3 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students. Boys = $3 \times 2 = 6$.

Problem 5: In David's class, the ratio of boys to girls is 2:2. If there are 8 students, how many are boys?

Answer: 4

Explanation: The ratio 2:2 means 2 boys for every 2 girls. Total ratio parts: 4. Each part = $8 \div 4 = 2$ students. Boys = $2 \times 2 = 4$.

Problem 6: In Liam's class, the ratio of boys to girls is 2:3. If there are 10 students, how many are boys?

Answer: 4

Explanation: The ratio 2:3 means 2 boys for every 3 girls. Total ratio parts: 5. Each part = $10 \div 5 = 2$ students. Boys = $2 \times 2 = 4$.

Problem 7: In Noah's class, the ratio of boys to girls is 4:2. If there are 12 students, how many are boys?

Answer: 8

Explanation: The ratio 4:2 means 4 boys for every 2 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 8: In Evelyn's class, the ratio of boys to girls is 4:3. If there are 14 students, how many are boys?

Answer: 8

Explanation: The ratio 4:3 means 4 boys for every 3 girls. Total ratio parts: 7. Each part = $14 \div 7 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 9: In Sebastian's class, the ratio of boys to girls is 3:2. If there are 10 students, how many are boys?

Answer: 6

Explanation: The ratio 3:2 means 3 boys for every 2 girls. Total ratio parts: 5. Each part = $10 \div 5 = 2$ students. Boys = $3 \times 2 = 6$.

Problem 10: In Samuel's class, the ratio of boys to girls is 4:3. If there are 14 students, how many are boys?

Answer: 8

Explanation: The ratio 4:3 means 4 boys for every 3 girls. Total ratio parts: 7. Each part = $14 \div 7 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 11: In Sebastian's class, the ratio of boys to girls is 5:4. If there are 18 students, how many are boys?

Answer: 10

Explanation: The ratio 5:4 means 5 boys for every 4 girls. Total ratio parts: 9. Each part = $18 \div 9 = 2$ students. Boys = $5 \times 2 = 10$.

Problem 12: In Isabella's class, the ratio of boys to girls is 2:3. If there are 10 students, how many are boys?

Answer: 4

Explanation: The ratio 2:3 means 2 boys for every 3 girls. Total ratio parts: 5. Each part = $10 \div 5 = 2$ students. Boys = $2 \times 2 = 4$.

Problem 13: In Emma's class, the ratio of boys to girls is 2:3. If there are 10 students, how many are boys?

Answer: 4

Explanation: The ratio 2:3 means 2 boys for every 3 girls. Total ratio parts: 5. Each part = $10 \div 5 = 2$ students. Boys = $2 \times 2 = 4$.

Problem 14: In Evelyn's class, the ratio of boys to girls is 4:2. If there are 12 students, how many are boys?

Answer: 8

Explanation: The ratio 4:2 means 4 boys for every 2 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 15: In Olivia's class, the ratio of boys to girls is 5:3. If there are 16 students, how many are boys?

Answer: 10

Explanation: The ratio 5:3 means 5 boys for every 3 girls. Total ratio parts: 8. Each part = $16 \div 8 = 2$ students. Boys = $5 \times 2 = 10$.

Problem 16: In Avery's class, the ratio of boys to girls is 2:3. If there are 10 students, how many are boys?

Answer: 4

Explanation: The ratio 2:3 means 2 boys for every 3 girls. Total ratio parts: 5. Each part = $10 \div 5 = 2$ students. Boys = $2 \times 2 = 4$.

Problem 17: In Henry's class, the ratio of boys to girls is 2:4. If there are 12 students, how many are boys?

Answer: 4

Explanation: The ratio 2:4 means 2 boys for every 4 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students. Boys = $2 \times 2 = 4$.

Problem 18: In Aiden's class, the ratio of boys to girls is 4:2. If there are 12 students, how many are boys?

Answer: 8

Explanation: The ratio 4:2 means 4 boys for every 2 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 19: In Abigail's class, the ratio of boys to girls is 2:5. If there are 14 students, how many are boys?

Answer: 4

Explanation: The ratio 2:5 means 2 boys for every 5 girls. Total ratio parts: 7. Each part = $14 \div 7 = 2$ students. Boys = $2 \times 2 = 4$.

Problem 20: In Noah's class, the ratio of boys to girls is 5:4. If there are 18 students, how many are boys?

Answer: 10

Explanation: The ratio 5:4 means 5 boys for every 4 girls. Total ratio parts: 9. Each part = $18 \div 9 = 2$ students. Boys = $5 \times 2 = 10$.

Problem 21: In Evelyn's class, the ratio of boys to girls is 2:3. If there are 10 students, how many are boys?

Answer: 4

Explanation: The ratio 2:3 means 2 boys for every 3 girls. Total ratio parts: 5. Each part = $10 \div 5 = 2$ students. Boys = $2 \times 2 = 4$.

Problem 22: In Joseph's class, the ratio of boys to girls is 5:4. If there are 18 students, how many are boys?

Answer: 10

Explanation: The ratio 5:4 means 5 boys for every 4 girls. Total ratio parts: 9. Each part = $18 \div 9 = 2$ students. Boys = $5 \times 2 = 10$.

Problem 23: In Amelia's class, the ratio of boys to girls is 3:5. If there are 16 students, how many are boys?

Answer: 6

Explanation: The ratio 3:5 means 3 boys for every 5 girls. Total ratio parts: 8. Each part = $16 \div 8 = 2$ students. Boys = $3 \times 2 = 6$.

Problem 24: In Isabella's class, the ratio of boys to girls is 3:4. If there are 14 students, how many are boys?

Answer: 6

Explanation: The ratio 3:4 means 3 boys for every 4 girls. Total ratio parts: 7. Each part = $14 \div 7 = 2$ students. Boys = $3 \times 2 = 6$.

Problem 25: In Sophia's class, the ratio of boys to girls is 2:4. If there are 12 students, how many are boys?

Answer: 4

Explanation: The ratio 2:4 means 2 boys for every 4 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students. Boys = $2 \times 2 = 4$.

Problem 26: In Liam's class, the ratio of boys to girls is 5:5. If there are 20 students, how many are boys?

Answer: 10

Explanation: The ratio 5:5 means 5 boys for every 5 girls. Total ratio parts: 10. Each part = $20 \div 10 = 2$ students. Boys = $5 \times 2 = 10$.

Problem 27: In Lucas's class, the ratio of boys to girls is 4:2. If there are 12 students, how many are boys?

Answer: 8

Explanation: The ratio 4:2 means 4 boys for every 2 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 28: In Sebastian's class, the ratio of boys to girls is 4:2. If there are 12 students, how many are boys?

Answer: 8

Explanation: The ratio 4:2 means 4 boys for every 2 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 29: In Jack's class, the ratio of boys to girls is 5:4. If there are 18 students, how many are boys?

Answer: 10

Explanation: The ratio 5:4 means 5 boys for every 4 girls. Total ratio parts: 9. Each part = $18 \div 9 = 2$ students. Boys = $5 \times 2 = 10$.

Problem 30: In David's class, the ratio of boys to girls is 4:3. If there are 14 students, how many are boys?

Answer: 8

Explanation: The ratio 4:3 means 4 boys for every 3 girls. Total ratio parts: 7. Each part = $14 \div 7 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 31: In David's class, the ratio of boys to girls is 4:2. If there are 12 students, how many are boys?

Answer: 8

Explanation: The ratio 4:2 means 4 boys for every 2 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 32: In James's class, the ratio of boys to girls is 2:4. If there are 12 students, how many are boys?

Answer: 4

Explanation: The ratio 2:4 means 2 boys for every 4 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students. Boys = $2 \times 2 = 4$.

Problem 33: In Sofia's class, the ratio of boys to girls is 2:2. If there are 8 students, how many are boys?

Answer: 4

Explanation: The ratio 2:2 means 2 boys for every 2 girls. Total ratio parts: 4. Each part = $8 \div 4 = 2$ students. Boys = $2 \times 2 = 4$.

Problem 34: In Ava's class, the ratio of boys to girls is 4:4. If there are 16 students, how many are boys?

Answer: 8

Explanation: The ratio 4:4 means 4 boys for every 4 girls. Total ratio parts: 8. Each part = $16 \div 8 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 35: In Elizabeth's class, the ratio of boys to girls is 4:3. If there are 14 students, how many are boys?

Answer: 8

Explanation: The ratio 4:3 means 4 boys for every 3 girls. Total ratio parts: 7. Each part = $14 \div 7 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 36: In Sebastian's class, the ratio of boys to girls is 4:4. If there are 16 students, how many are boys?

Answer: 8

Explanation: The ratio 4:4 means 4 boys for every 4 girls. Total ratio parts: 8. Each part = $16 \div 8 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 37: In Emily's class, the ratio of boys to girls is 5:2. If there are 14 students, how many are boys?

Answer: 10

Explanation: The ratio 5:2 means 5 boys for every 2 girls. Total ratio parts: 7. Each part = $14 \div 7 = 2$ students. Boys = $5 \times 2 = 10$.

Problem 38: In Samuel's class, the ratio of boys to girls is 3:2. If there are 10 students, how many are boys?

Answer: 6

Explanation: The ratio 3:2 means 3 boys for every 2 girls. Total ratio parts: 5. Each part = $10 \div 5 = 2$ students. Boys = $3 \times 2 = 6$.

Problem 39: In Jack's class, the ratio of boys to girls is 4:3. If there are 14 students, how many are boys?

Answer: 8

Explanation: The ratio 4:3 means 4 boys for every 3 girls. Total ratio parts: 7. Each part = $14 \div 7 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 40: In Emma's class, the ratio of boys to girls is 4:2. If there are 12 students, how many are boys?

Answer: 8

Explanation: The ratio 4:2 means 4 boys for every 2 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students. Boys = $4 \times 2 = 8$.

Problem 41: In Henry's class, the ratio of boys to girls is 3:2. If there are 10 students, how many are boys?

Answer: 6

Explanation: The ratio 3:2 means 3 boys for every 2 girls. Total ratio parts: 5. Each part = $10 \div 5 = 2$ students. Boys = $3 \times 2 = 6$.

Problem 42: In Samuel's class, the ratio of boys to girls is 5:3. If there are 16 students, how many are boys?

Answer: 10

Explanation: The ratio 5:3 means 5 boys for every 3 girls. Total ratio parts: 8. Each part = $16 \div 8 = 2$ students. Boys = $5 \times 2 = 10$.

Problem 43: In Sebastian's class, the ratio of boys to girls is 2:3. If there are 10 students, how many are boys?

Answer: 4

Explanation: The ratio 2:3 means 2 boys for every 3 girls. Total ratio parts: 5. Each part = $10 \div 5 = 2$ students. Boys = $2 \times 2 = 4$.

Problem 44: In Amelia's class, the ratio of boys to girls is 2:3. If there are 10 students, how many are boys?

Answer: 4

Explanation: The ratio 2:3 means 2 boys for every 3 girls. Total ratio parts: 5. Each part = $10 \div 5 = 2$ students. Boys = $2 \times 2 = 4$.

Problem 45: In Jack's class, the ratio of boys to girls is 5:3. If there are 16 students, how many are boys?

Answer: 10

Explanation: The ratio 5:3 means 5 boys for every 3 girls. Total ratio parts: 8. Each part = $16 \div 8 = 2$ students. Boys = $5 \times 2 = 10$.

Practice Exercises - Grade 4

Exercise 1: In David's class, the ratio of boys to girls is 3:2. If there are 10 students, how many are boys?

Exercise 2: In Noah's class, the ratio of boys to girls is 2:5. If there are 14 students, how many are boys?

Exercise 3: In Amelia's class, the ratio of boys to girls is 2:5. If there are 14 students, how many are boys?

Exercise 4: In Elizabeth's class, the ratio of boys to girls is 2:4. If there are 12 students, how many are boys?

Exercise 5: In Evelyn's class, the ratio of boys to girls is 3:5. If there are 16 students, how many are boys?

Exercise 6: In David's class, the ratio of boys to girls is 2:4. If there are 12 students, how many are boys?

Exercise 7: In Amelia's class, the ratio of boys to girls is 5:5. If there are 20 students, how many are boys?

Exercise 8: In Emma's class, the ratio of boys to girls is 2:3. If there are 10 students, how many are boys?

Exercise 9: In Liam's class, the ratio of boys to girls is 5:5. If there are 20 students, how many are boys?

Exercise 10: In Henry's class, the ratio of boys to girls is 4:4. If there are 16 students, how many are boys?

Exercise 11: In Sebastian's class, the ratio of boys to girls is 3:3. If there are 12 students, how many are boys?

Exercise 12: In Evelyn's class, the ratio of boys to girls is 2:2. If there are 8 students, how many are boys?

Exercise 13: In Emily's class, the ratio of boys to girls is 5:4. If there are 18 students, how many are boys?

Exercise 14: In Ava's class, the ratio of boys to girls is 2:2. If there are 8 students, how many are boys?

Exercise 15: In Ethan's class, the ratio of boys to girls is 4:4. If there are 16 students, how many are boys?

Exercise 16: In Sophia's class, the ratio of boys to girls is 4:4. If there are 16 students, how many are boys?

Exercise 17: In Noah's class, the ratio of boys to girls is 3:4. If there are 14 students, how many are boys?

Exercise 18: In Liam's class, the ratio of boys to girls is 5:2. If there are 14 students, how many are boys?

Exercise 19: In Evelyn's class, the ratio of boys to girls is 2:5. If there are 14 students, how many are boys?

Exercise 20: In Sebastian's class, the ratio of boys to girls is 3:3. If there are 12 students, how many are boys?

Practice Exercise Answers:

Exercise 1: 6

Exercise 2: 4

Exercise 3: 4

Exercise 4: 4

Exercise 5: 6

Exercise 6: 4

Exercise 7: 10

Exercise 8: 4

Exercise 9: 10

Exercise 10: 8

Exercise 11: 6

Exercise 12: 4

Exercise 13: 10

Exercise 14: 4

Exercise 15: 8

Exercise 16: 8

Exercise 17: 6

Exercise 18: 10

Exercise 19: 4

Exercise 20: 6

Challenge Problems - Grade 4

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: In Amelia's class, the ratio of boys to girls is 2:4. If there are 12 students, how many are boys?

Answer: 4

Solution: The ratio 2:4 means 2 boys for every 4 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students.
Boys = $2 \times 2 = 4$.

Challenge 2: In Ethan's class, the ratio of boys to girls is 3:4. If there are 14 students, how many are boys?

Answer: 6

Solution: The ratio 3:4 means 3 boys for every 4 girls. Total ratio parts: 7. Each part = $14 \div 7 = 2$ students.
Boys = $3 \times 2 = 6$.

Challenge 3: In Emma's class, the ratio of boys to girls is 4:3. If there are 14 students, how many are boys?

Answer: 8

Solution: The ratio 4:3 means 4 boys for every 3 girls. Total ratio parts: 7. Each part = $14 \div 7 = 2$ students.
Boys = $4 \times 2 = 8$.

Challenge 4: In Ava's class, the ratio of boys to girls is 3:3. If there are 12 students, how many are boys?

Answer: 6

Solution: The ratio 3:3 means 3 boys for every 3 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students.
Boys = $3 \times 2 = 6$.

Challenge 5: In Amelia's class, the ratio of boys to girls is 3:5. If there are 16 students, how many are boys?

Answer: 6

Solution: The ratio 3:5 means 3 boys for every 5 girls. Total ratio parts: 8. Each part = $16 \div 8 = 2$ students.
Boys = $3 \times 2 = 6$.

Challenge 6: In Joseph's class, the ratio of boys to girls is 3:5. If there are 16 students, how many are boys?

Answer: 6

Solution: The ratio 3:5 means 3 boys for every 5 girls. Total ratio parts: 8. Each part = $16 \div 8 = 2$ students.
Boys = $3 \times 2 = 6$.

Challenge 7: In Alexander's class, the ratio of boys to girls is 5:3. If there are 16 students, how many are boys?

Answer: 10

Solution: The ratio 5:3 means 5 boys for every 3 girls. Total ratio parts: 8. Each part = $16 \div 8 = 2$ students.
Boys = $5 \times 2 = 10$.

Challenge 8: In Avery's class, the ratio of boys to girls is 4:5. If there are 18 students, how many are boys?

Answer: 8

Solution: The ratio 4:5 means 4 boys for every 5 girls. Total ratio parts: 9. Each part = $18 \div 9 = 2$ students.
Boys = $4 \times 2 = 8$.

Challenge 9: In Aiden's class, the ratio of boys to girls is 5:5. If there are 20 students, how many are boys?

Answer: 10

Solution: The ratio 5:5 means 5 boys for every 5 girls. Total ratio parts: 10. Each part = $20 \div 10 = 2$ students.
Boys = $5 \times 2 = 10$.

Challenge 10: In Liam's class, the ratio of boys to girls is 5:5. If there are 20 students, how many are boys?

Answer: 10

Solution: The ratio 5:5 means 5 boys for every 5 girls. Total ratio parts: 10. Each part = $20 \div 10 = 2$ students.
Boys = $5 \times 2 = 10$.

Challenge 11: In Aiden's class, the ratio of boys to girls is 5:4. If there are 18 students, how many are boys?

Answer: 10

Solution: The ratio 5:4 means 5 boys for every 4 girls. Total ratio parts: 9. Each part = $18 \div 9 = 2$ students.
Boys = $5 \times 2 = 10$.

Challenge 12: In Mason's class, the ratio of boys to girls is 2:3. If there are 10 students, how many are boys?

Answer: 4

Solution: The ratio 2:3 means 2 boys for every 3 girls. Total ratio parts: 5. Each part = $10 \div 5 = 2$ students.
Boys = $2 \times 2 = 4$.

Challenge 13: In Ethan's class, the ratio of boys to girls is 2:3. If there are 10 students, how many are boys?

Answer: 4

Solution: The ratio 2:3 means 2 boys for every 3 girls. Total ratio parts: 5. Each part = $10 \div 5 = 2$ students.
Boys = $2 \times 2 = 4$.

Challenge 14: In Mason's class, the ratio of boys to girls is 4:5. If there are 18 students, how many are boys?

Answer: 8

Solution: The ratio 4:5 means 4 boys for every 5 girls. Total ratio parts: 9. Each part = $18 \div 9 = 2$ students.
Boys = $4 \times 2 = 8$.

Challenge 15: In Isabella's class, the ratio of boys to girls is 2:4. If there are 12 students, how many are boys?

Answer: 4

Solution: The ratio 2:4 means 2 boys for every 4 girls. Total ratio parts: 6. Each part = $12 \div 6 = 2$ students.
Boys = $2 \times 2 = 4$.

Grade 5 Problems

Problem 1: Noah drives 142 miles in 3 hours. What is the average speed in miles per hour?

Answer: 47

Explanation: Speed = distance $\div$ time = $142 \div 3 = 47$ miles per hour.

Problem 2: Emma drives 69 miles in 5 hours. What is the average speed in miles per hour?

Answer: 13

Explanation: Speed = distance $\div$ time = $69 \div 5 = 13$ miles per hour.

Problem 3: Harper drives 101 miles in 5 hours. What is the average speed in miles per hour?

Answer: 20

Explanation: Speed = distance $\div$ time = $101 \div 5 = 20$ miles per hour.

Problem 4: Sophia drives 157 miles in 5 hours. What is the average speed in miles per hour?

Answer: 31

Explanation: Speed = distance $\div$ time = $157 \div 5 = 31$ miles per hour.

Problem 5: Olivia drives 87 miles in 5 hours. What is the average speed in miles per hour?

Answer: 17

Explanation: Speed = distance $\div$ time = $87 \div 5 = 17$ miles per hour.

Problem 6: David drives 131 miles in 5 hours. What is the average speed in miles per hour?

Answer: 26

Explanation: Speed = distance $\div$ time = $131 \div 5 = 26$ miles per hour.

Problem 7: Isabella drives 64 miles in 4 hours. What is the average speed in miles per hour?

Answer: 16

Explanation: Speed = distance $\div$ time = $64 \div 4 = 16$ miles per hour.

Problem 8: Elizabeth drives 120 miles in 4 hours. What is the average speed in miles per hour?

Answer: 30

Explanation: Speed = distance $\div$ time = $120 \div 4 = 30$ miles per hour.

Problem 9: Alexander drives 175 miles in 2 hours. What is the average speed in miles per hour?

Answer: 87

Explanation: Speed = distance $\div$ time = $175 \div 2 = 87$ miles per hour.

Problem 10: Ava drives 179 miles in 2 hours. What is the average speed in miles per hour?

Answer: 89

Explanation: Speed = distance $\div$ time = $179 \div 2 = 89$ miles per hour.

Problem 11: Avery drives 71 miles in 3 hours. What is the average speed in miles per hour?

Answer: 23

Explanation: Speed = distance $\div$ time = $71 \div 3 = 23$ miles per hour.

Problem 12: Liam drives 169 miles in 4 hours. What is the average speed in miles per hour?

Answer: 42

Explanation: Speed = distance $\div$ time = $169 \div 4 = 42$ miles per hour.

Problem 13: David drives 88 miles in 4 hours. What is the average speed in miles per hour?

Answer: 22

Explanation: Speed = distance $\div$ time = $88 \div 4 = 22$ miles per hour.

Problem 14: Evelyn drives 120 miles in 3 hours. What is the average speed in miles per hour?

Answer: 40

Explanation: Speed = distance $\div$ time = $120 \div 3 = 40$ miles per hour.

Problem 15: Ava drives 140 miles in 3 hours. What is the average speed in miles per hour?

Answer: 46

Explanation: Speed = distance $\div$ time = $140 \div 3 = 46$ miles per hour.

Problem 16: Evelyn drives 180 miles in 3 hours. What is the average speed in miles per hour?

Answer: 60

Explanation: Speed = distance $\div$ time = $180 \div 3 = 60$ miles per hour.

Problem 17: Ethan drives 167 miles in 3 hours. What is the average speed in miles per hour?

Answer: 55

Explanation: Speed = distance $\div$ time = $167 \div 3 = 55$ miles per hour.

Problem 18: Aiden drives 75 miles in 4 hours. What is the average speed in miles per hour?

Answer: 18

Explanation: Speed = distance $\div$ time = $75 \div 4 = 18$ miles per hour.

Problem 19: Joseph drives 110 miles in 2 hours. What is the average speed in miles per hour?

Answer: 55

Explanation: Speed = distance $\div$ time = $110 \div 2 = 55$ miles per hour.

Problem 20: Lucas drives 87 miles in 3 hours. What is the average speed in miles per hour?

Answer: 29

Explanation: Speed = distance $\div$ time = $87 \div 3 = 29$ miles per hour.

Problem 21: Mia drives 62 miles in 2 hours. What is the average speed in miles per hour?

Answer: 31

Explanation: Speed = distance $\div$ time = $62 \div 2 = 31$ miles per hour.

Problem 22: Abigail drives 174 miles in 3 hours. What is the average speed in miles per hour?

Answer: 58

Explanation: Speed = distance $\div$ time = $174 \div 3 = 58$ miles per hour.

Problem 23: Charlotte drives 77 miles in 3 hours. What is the average speed in miles per hour?

Answer: 25

Explanation: Speed = distance $\div$ time = $77 \div 3 = 25$ miles per hour.

Problem 24: Joseph drives 141 miles in 3 hours. What is the average speed in miles per hour?

Answer: 47

Explanation: Speed = distance $\div$ time = $141 \div 3 = 47$ miles per hour.

Problem 25: Henry drives 114 miles in 3 hours. What is the average speed in miles per hour?

Answer: 38

Explanation: Speed = distance $\div$ time = $114 \div 3 = 38$ miles per hour.

Problem 26: Sophia drives 138 miles in 4 hours. What is the average speed in miles per hour?

Answer: 34

Explanation: Speed = distance $\div$ time = $138 \div 4 = 34$ miles per hour.

Problem 27: James drives 104 miles in 2 hours. What is the average speed in miles per hour?

Answer: 52

Explanation: Speed = distance $\div$ time = $104 \div 2 = 52$ miles per hour.

Problem 28: Samuel drives 178 miles in 4 hours. What is the average speed in miles per hour?

Answer: 44

Explanation: Speed = distance $\div$ time = $178 \div 4 = 44$ miles per hour.

Problem 29: Noah drives 117 miles in 5 hours. What is the average speed in miles per hour?

Answer: 23

Explanation: Speed = distance $\div$ time = $117 \div 5 = 23$ miles per hour.

Problem 30: Amelia drives 143 miles in 5 hours. What is the average speed in miles per hour?

Answer: 28

Explanation: Speed = distance $\div$ time = $143 \div 5 = 28$ miles per hour.

Problem 31: James drives 69 miles in 2 hours. What is the average speed in miles per hour?

Answer: 34

Explanation: Speed = distance $\div$ time = $69 \div 2 = 34$ miles per hour.

Problem 32: Elizabeth drives 66 miles in 5 hours. What is the average speed in miles per hour?

Answer: 13

Explanation: Speed = distance $\div$ time = $66 \div 5 = 13$ miles per hour.

Problem 33: Benjamin drives 164 miles in 2 hours. What is the average speed in miles per hour?

Answer: 82

Explanation: Speed = distance $\div$ time = $164 \div 2 = 82$ miles per hour.

Problem 34: Olivia drives 88 miles in 3 hours. What is the average speed in miles per hour?

Answer: 29

Explanation: Speed = distance $\div$ time = $88 \div 3 = 29$ miles per hour.

Problem 35: Sebastian drives 91 miles in 5 hours. What is the average speed in miles per hour?

Answer: 18

Explanation: Speed = distance $\div$ time = $91 \div 5 = 18$ miles per hour.

Problem 36: James drives 180 miles in 3 hours. What is the average speed in miles per hour?

Answer: 60

Explanation: Speed = distance $\div$ time = $180 \div 3 = 60$ miles per hour.

Problem 37: Henry drives 155 miles in 4 hours. What is the average speed in miles per hour?

Answer: 38

Explanation: Speed = distance $\div$ time = $155 \div 4 = 38$ miles per hour.

Problem 38: Mia drives 69 miles in 4 hours. What is the average speed in miles per hour?

Answer: 17

Explanation: Speed = distance $\div$ time = $69 \div 4 = 17$ miles per hour.

Problem 39: Lucas drives 80 miles in 3 hours. What is the average speed in miles per hour?

Answer: 26

Explanation: Speed = distance $\div$ time = $80 \div 3 = 26$ miles per hour.

Problem 40: Benjamin drives 72 miles in 5 hours. What is the average speed in miles per hour?

Answer: 14

Explanation: Speed = distance $\div$ time = $72 \div 5 = 14$ miles per hour.

Problem 41: Henry drives 110 miles in 5 hours. What is the average speed in miles per hour?

Answer: 22

Explanation: Speed = distance $\div$ time = $110 \div 5 = 22$ miles per hour.

Problem 42: David drives 136 miles in 4 hours. What is the average speed in miles per hour?

Answer: 34

Explanation: Speed = distance $\div$ time = $136 \div 4 = 34$ miles per hour.

Problem 43: Sophia drives 159 miles in 4 hours. What is the average speed in miles per hour?

Answer: 39

Explanation: Speed = distance $\div$ time = $159 \div 4 = 39$ miles per hour.

Problem 44: Elizabeth drives 70 miles in 3 hours. What is the average speed in miles per hour?

Answer: 23

Explanation: Speed = distance $\div$ time = $70 \div 3 = 23$ miles per hour.

Problem 45: Ethan drives 72 miles in 5 hours. What is the average speed in miles per hour?

Answer: 14

Explanation: Speed = distance $\div$ time = $72 \div 5 = 14$ miles per hour.

Practice Exercises - Grade 5

Exercise 1: James drives 163 miles in 3 hours. What is the average speed in miles per hour?

Exercise 2: Noah drives 130 miles in 3 hours. What is the average speed in miles per hour?

Exercise 3: Charlotte drives 69 miles in 2 hours. What is the average speed in miles per hour?

Exercise 4: Avery drives 99 miles in 2 hours. What is the average speed in miles per hour?

Exercise 5: Aiden drives 77 miles in 3 hours. What is the average speed in miles per hour?

Exercise 6: Elizabeth drives 87 miles in 4 hours. What is the average speed in miles per hour?

Exercise 7: Olivia drives 148 miles in 2 hours. What is the average speed in miles per hour?

Exercise 8: Isabella drives 115 miles in 4 hours. What is the average speed in miles per hour?

Exercise 9: Mia drives 129 miles in 4 hours. What is the average speed in miles per hour?

Exercise 10: Avery drives 169 miles in 4 hours. What is the average speed in miles per hour?

Exercise 11: Aiden drives 98 miles in 2 hours. What is the average speed in miles per hour?

Exercise 12: Benjamin drives 65 miles in 2 hours. What is the average speed in miles per hour?

Exercise 13: Henry drives 102 miles in 5 hours. What is the average speed in miles per hour?

Exercise 14: Lucas drives 156 miles in 2 hours. What is the average speed in miles per hour?

Exercise 15: Avery drives 68 miles in 4 hours. What is the average speed in miles per hour?

Exercise 16: Emma drives 74 miles in 5 hours. What is the average speed in miles per hour?

Exercise 17: Abigail drives 116 miles in 3 hours. What is the average speed in miles per hour?

Exercise 18: Lucas drives 80 miles in 3 hours. What is the average speed in miles per hour?

Exercise 19: Jack drives 84 miles in 3 hours. What is the average speed in miles per hour?

Exercise 20: Alexander drives 138 miles in 2 hours. What is the average speed in miles per hour?

Practice Exercise Answers:

Exercise 1: 54

Exercise 2: 43

Exercise 3: 34

Exercise 4: 49

Exercise 5: 25

Exercise 6: 21

Exercise 7: 74

Exercise 8: 28

Exercise 9: 32

Exercise 10: 42

Exercise 11: 49

Exercise 12: 32

Exercise 13: 20

Exercise 14: 78

Exercise 15: 17

Exercise 16: 14

Exercise 17: 38

Exercise 18: 26

Exercise 19: 28

Exercise 20: 69

Challenge Problems - Grade 5

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Avery drives 148 miles in 3 hours. What is the average speed in miles per hour?

Answer: 49

Solution: Speed = distance $\div$ time = $148 \div 3 = 49$ miles per hour.

Challenge 2: Olivia drives 129 miles in 5 hours. What is the average speed in miles per hour?

Answer: 25

Solution: Speed = distance $\div$ time = $129 \div 5 = 25$ miles per hour.

Challenge 3: Jack drives 151 miles in 2 hours. What is the average speed in miles per hour?

Answer: 75

Solution: Speed = distance $\div$ time = $151 \div 2 = 75$ miles per hour.

Challenge 4: Joseph drives 128 miles in 4 hours. What is the average speed in miles per hour?

Answer: 32

Solution: Speed = distance $\div$ time = $128 \div 4 = 32$ miles per hour.

Challenge 5: Olivia drives 67 miles in 3 hours. What is the average speed in miles per hour?

Answer: 22

Solution: Speed = distance $\div$ time = $67 \div 3 = 22$ miles per hour.

Challenge 6: Henry drives 84 miles in 5 hours. What is the average speed in miles per hour?

Answer: 16

Solution: Speed = distance $\div$ time = $84 \div 5 = 16$ miles per hour.

Challenge 7: Ethan drives 100 miles in 5 hours. What is the average speed in miles per hour?

Answer: 20

Solution: Speed = distance $\div$ time = $100 \div 5 = 20$ miles per hour.

Challenge 8: Avery drives 75 miles in 2 hours. What is the average speed in miles per hour?

Answer: 37

Solution: Speed = distance $\div$ time = $75 \div 2 = 37$ miles per hour.

Challenge 9: Henry drives 111 miles in 2 hours. What is the average speed in miles per hour?

Answer: 55

Solution: Speed = distance $\div$ time = $111 \div 2 = 55$ miles per hour.

Challenge 10: Sofia drives 71 miles in 5 hours. What is the average speed in miles per hour?

Answer: 14

Solution: Speed = distance $\div$ time = $71 \div 5 = 14$ miles per hour.

Challenge 11: Noah drives 110 miles in 3 hours. What is the average speed in miles per hour?

Answer: 36

Solution: Speed = distance $\div$ time = $110 \div 3 = 36$ miles per hour.

Challenge 12: Joseph drives 167 miles in 5 hours. What is the average speed in miles per hour?

Answer: 33

Solution: Speed = distance $\div$ time = $167 \div 5 = 33$ miles per hour.

Challenge 13: Benjamin drives 107 miles in 3 hours. What is the average speed in miles per hour?

Answer: 35

Solution: Speed = distance $\div$ time = $107 \div 3 = 35$ miles per hour.

Challenge 14: Jack drives 105 miles in 3 hours. What is the average speed in miles per hour?

Answer: 35

Solution: Speed = distance $\div$ time = $105 \div 3 = 35$ miles per hour.

Challenge 15: Abigail drives 87 miles in 4 hours. What is the average speed in miles per hour?

Answer: 21

Solution: Speed = distance $\div$ time = $87 \div 4 = 21$ miles per hour.

Grade 6 Problems

Problem 1: If 2 apples cost \$6, how much do 6 apples cost?

Answer: 18

Explanation: Set up proportion: $2/6 = 6/x$. Cross multiply: $x = (6 \times 6) \div 2 = \18 .

Problem 2: If 4 apples cost \$5, how much do 9 apples cost?

Answer: 11

Explanation: Set up proportion: $4/5 = 9/x$. Cross multiply: $x = (5 \times 9) \div 4 = \11 .

Problem 3: If 3 apples cost \$8, how much do 11 apples cost?

Answer: 29

Explanation: Set up proportion: $3/8 = 11/x$. Cross multiply: $x = (8 \times 11) \div 3 = \29 .

Problem 4: If 5 apples cost \$8, how much do 6 apples cost?

Answer: 9

Explanation: Set up proportion: $5/8 = 6/x$. Cross multiply: $x = (8 \times 6) \div 5 = \9 .

Problem 5: If 2 apples cost \$4, how much do 7 apples cost?

Answer: 14

Explanation: Set up proportion: $2/4 = 7/x$. Cross multiply: $x = (4 \times 7) \div 2 = \14 .

Problem 6: If 2 apples cost \$7, how much do 10 apples cost?

Answer: 35

Explanation: Set up proportion: $2/7 = 10/x$. Cross multiply: $x = (7 \times 10) \div 2 = \35 .

Problem 7: If 5 apples cost \$6, how much do 12 apples cost?

Answer: 14

Explanation: Set up proportion: $5/6 = 12/x$. Cross multiply: $x = (6 \times 12) \div 5 = \14 .

Problem 8: If 5 apples cost \$3, how much do 6 apples cost?

Answer: 3

Explanation: Set up proportion: $5/3 = 6/x$. Cross multiply: $x = (3 \times 6) \div 5 = \3 .

Problem 9: If 4 apples cost \$5, how much do 12 apples cost?

Answer: 15

Explanation: Set up proportion: $4/5 = 12/x$. Cross multiply: $x = (5 \times 12) \div 4 = \15 .

Problem 10: If 3 apples cost \$8, how much do 6 apples cost?

Answer: 16

Explanation: Set up proportion: $3/8 = 6/x$. Cross multiply: $x = (8 \times 6) \div 3 = \16 .

Problem 11: If 5 apples cost \$5, how much do 12 apples cost?

Answer: 12

Explanation: Set up proportion: $5/5 = 12/x$. Cross multiply: $x = (5 \times 12) \div 5 = \12 .

Problem 12: If 2 apples cost \$3, how much do 7 apples cost?

Answer: 10

Explanation: Set up proportion: $2/3 = 7/x$. Cross multiply: $x = (3 \times 7) \div 2 = \10 .

Problem 13: If 3 apples cost \$7, how much do 10 apples cost?

Answer: 23

Explanation: Set up proportion: $3/7 = 10/x$. Cross multiply: $x = (7 \times 10) \div 3 = \23 .

Problem 14: If 5 apples cost \$4, how much do 6 apples cost?

Answer: 4

Explanation: Set up proportion: $5/4 = 6/x$. Cross multiply: $x = (4 \times 6) \div 5 = \4 .

Problem 15: If 3 apples cost \$8, how much do 8 apples cost?

Answer: 21

Explanation: Set up proportion: $3/8 = 8/x$. Cross multiply: $x = (8 \times 8) \div 3 = \21 .

Problem 16: If 3 apples cost \$8, how much do 10 apples cost?

Answer: 26

Explanation: Set up proportion: $3/8 = 10/x$. Cross multiply: $x = (8 \times 10) \div 3 = \26 .

Problem 17: If 3 apples cost \$3, how much do 8 apples cost?

Answer: 8

Explanation: Set up proportion: $3/3 = 8/x$. Cross multiply: $x = (3 \times 8) \div 3 = \8 .

Problem 18: If 5 apples cost \$3, how much do 6 apples cost?

Answer: 3

Explanation: Set up proportion: $5/3 = 6/x$. Cross multiply: $x = (3 \times 6) \div 5 = \3 .

Problem 19: If 5 apples cost \$8, how much do 9 apples cost?

Answer: 14

Explanation: Set up proportion: $5/8 = 9/x$. Cross multiply: $x = (8 \times 9) \div 5 = \14 .

Problem 20: If 2 apples cost \$6, how much do 6 apples cost?

Answer: 18

Explanation: Set up proportion: $2/6 = 6/x$. Cross multiply: $x = (6 \times 6) \div 2 = \18 .

Problem 21: If 5 apples cost \$7, how much do 9 apples cost?

Answer: 12

Explanation: Set up proportion: $5/7 = 9/x$. Cross multiply: $x = (7 \times 9) \div 5 = \12 .

Problem 22: If 4 apples cost \$6, how much do 6 apples cost?

Answer: 9

Explanation: Set up proportion: $4/6 = 6/x$. Cross multiply: $x = (6 \times 6) \div 4 = \9 .

Problem 23: If 2 apples cost \$7, how much do 7 apples cost?

Answer: 24

Explanation: Set up proportion: $2/7 = 7/x$. Cross multiply: $x = (7 \times 7) \div 2 = \24 .

Problem 24: If 4 apples cost \$5, how much do 9 apples cost?

Answer: 11

Explanation: Set up proportion: $4/5 = 9/x$. Cross multiply: $x = (5 \times 9) \div 4 = \11 .

Problem 25: If 3 apples cost \$6, how much do 9 apples cost?

Answer: 18

Explanation: Set up proportion: $3/6 = 9/x$. Cross multiply: $x = (6 \times 9) \div 3 = \18 .

Problem 26: If 3 apples cost \$3, how much do 11 apples cost?

Answer: 11

Explanation: Set up proportion: $3/3 = 11/x$. Cross multiply: $x = (3 \times 11) \div 3 = \11 .

Problem 27: If 4 apples cost \$3, how much do 7 apples cost?

Answer: 5

Explanation: Set up proportion: $4/3 = 7/x$. Cross multiply: $x = (3 \times 7) \div 4 = \5 .

Problem 28: If 3 apples cost \$8, how much do 7 apples cost?

Answer: 18

Explanation: Set up proportion: $3/8 = 7/x$. Cross multiply: $x = (8 \times 7) \div 3 = \18 .

Problem 29: If 5 apples cost \$4, how much do 9 apples cost?

Answer: 7

Explanation: Set up proportion: $5/4 = 9/x$. Cross multiply: $x = (4 \times 9) \div 5 = \7 .

Problem 30: If 4 apples cost \$7, how much do 6 apples cost?

Answer: 10

Explanation: Set up proportion: $4/7 = 6/x$. Cross multiply: $x = (7 \times 6) \div 4 = \10 .

Problem 31: If 2 apples cost \$3, how much do 12 apples cost?

Answer: 18

Explanation: Set up proportion: $2/3 = 12/x$. Cross multiply: $x = (3 \times 12) \div 2 = \18 .

Problem 32: If 4 apples cost \$7, how much do 9 apples cost?

Answer: 15

Explanation: Set up proportion: $4/7 = 9/x$. Cross multiply: $x = (7 \times 9) \div 4 = \15 .

Problem 33: If 4 apples cost \$7, how much do 6 apples cost?

Answer: 10

Explanation: Set up proportion: $4/7 = 6/x$. Cross multiply: $x = (7 \times 6) \div 4 = \10 .

Problem 34: If 5 apples cost \$7, how much do 12 apples cost?

Answer: 16

Explanation: Set up proportion: $5/7 = 12/x$. Cross multiply: $x = (7 \times 12) \div 5 = \16 .

Problem 35: If 4 apples cost \$8, how much do 10 apples cost?

Answer: 20

Explanation: Set up proportion: $4/8 = 10/x$. Cross multiply: $x = (8 \times 10) \div 4 = \20 .

Problem 36: If 5 apples cost \$7, how much do 10 apples cost?

Answer: 14

Explanation: Set up proportion: $5/7 = 10/x$. Cross multiply: $x = (7 \times 10) \div 5 = \14 .

Problem 37: If 5 apples cost \$8, how much do 10 apples cost?

Answer: 16

Explanation: Set up proportion: $5/8 = 10/x$. Cross multiply: $x = (8 \times 10) \div 5 = \16 .

Problem 38: If 2 apples cost \$4, how much do 7 apples cost?

Answer: 14

Explanation: Set up proportion: $2/4 = 7/x$. Cross multiply: $x = (4 \times 7) \div 2 = \14 .

Problem 39: If 2 apples cost \$5, how much do 9 apples cost?

Answer: 22

Explanation: Set up proportion: $2/5 = 9/x$. Cross multiply: $x = (5 \times 9) \div 2 = \22 .

Problem 40: If 4 apples cost \$4, how much do 10 apples cost?

Answer: 10

Explanation: Set up proportion: $4/4 = 10/x$. Cross multiply: $x = (4 \times 10) \div 4 = \10 .

Problem 41: If 5 apples cost \$6, how much do 6 apples cost?

Answer: 7

Explanation: Set up proportion: $5/6 = 6/x$. Cross multiply: $x = (6 \times 6) \div 5 = \7 .

Problem 42: If 2 apples cost \$6, how much do 12 apples cost?

Answer: 36

Explanation: Set up proportion: $2/6 = 12/x$. Cross multiply: $x = (6 \times 12) \div 2 = \36 .

Problem 43: If 3 apples cost \$8, how much do 9 apples cost?

Answer: 24

Explanation: Set up proportion: $3/8 = 9/x$. Cross multiply: $x = (8 \times 9) \div 3 = \24 .

Problem 44: If 5 apples cost \$7, how much do 7 apples cost?

Answer: 9

Explanation: Set up proportion: $5/7 = 7/x$. Cross multiply: $x = (7 \times 7) \div 5 = \9 .

Problem 45: If 5 apples cost \$7, how much do 6 apples cost?

Answer: 8

Explanation: Set up proportion: $5/7 = 6/x$. Cross multiply: $x = (7 \times 6) \div 5 = \8 .

Practice Exercises - Grade 6

Exercise 1: If 2 apples cost \$5, how much do 12 apples cost?

Exercise 2: If 5 apples cost \$6, how much do 11 apples cost?

Exercise 3: If 3 apples cost \$4, how much do 7 apples cost?

Exercise 4: If 2 apples cost \$6, how much do 6 apples cost?

Exercise 5: If 4 apples cost \$6, how much do 12 apples cost?

Exercise 6: If 3 apples cost \$6, how much do 11 apples cost?

Exercise 7: If 4 apples cost \$3, how much do 9 apples cost?

Exercise 8: If 4 apples cost \$7, how much do 7 apples cost?

Exercise 9: If 2 apples cost \$8, how much do 9 apples cost?

Exercise 10: If 5 apples cost \$4, how much do 8 apples cost?

Exercise 11: If 2 apples cost \$8, how much do 12 apples cost?

Exercise 12: If 2 apples cost \$4, how much do 6 apples cost?

Exercise 13: If 5 apples cost \$6, how much do 7 apples cost?

Exercise 14: If 2 apples cost \$4, how much do 6 apples cost?

Exercise 15: If 5 apples cost \$3, how much do 12 apples cost?

Exercise 16: If 4 apples cost \$8, how much do 11 apples cost?

Exercise 17: If 2 apples cost \$3, how much do 6 apples cost?

Exercise 18: If 5 apples cost \$7, how much do 12 apples cost?

Exercise 19: If 5 apples cost \$8, how much do 6 apples cost?

Exercise 20: If 5 apples cost \$6, how much do 6 apples cost?

Practice Exercise Answers:

Exercise 1: 30

Exercise 2: 13

Exercise 3: 9

Exercise 4: 18

Exercise 5: 18

Exercise 6: 22

Exercise 7: 6

Exercise 8: 12

Exercise 9: 36

Exercise 10: 6

Exercise 11: 48

Exercise 12: 12

Exercise 13: 8

Exercise 14: 12

Exercise 15: 7

Exercise 16: 22

Exercise 17: 9

Exercise 18: 16

Exercise 19: 9

Exercise 20: 7

Challenge Problems - Grade 6

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: If 3 apples cost \$4, how much do 8 apples cost?

Answer: 10

Solution: Set up proportion: $\frac{3}{4} = \frac{8}{x}$. Cross multiply: $x = (4 \times 8) \div 3 = \10 .

Challenge 2: If 5 apples cost \$6, how much do 10 apples cost?

Answer: 12

Solution: Set up proportion: $\frac{5}{6} = \frac{10}{x}$. Cross multiply: $x = (6 \times 10) \div 5 = \12 .

Challenge 3: If 5 apples cost \$7, how much do 6 apples cost?

Answer: 8

Solution: Set up proportion: $\frac{5}{7} = \frac{6}{x}$. Cross multiply: $x = (7 \times 6) \div 5 = \8 .

Challenge 4: If 4 apples cost \$3, how much do 12 apples cost?

Answer: 9

Solution: Set up proportion: $\frac{4}{3} = \frac{12}{x}$. Cross multiply: $x = (3 \times 12) \div 4 = \9 .

Challenge 5: If 3 apples cost \$8, how much do 11 apples cost?

Answer: 29

Solution: Set up proportion: $\frac{3}{8} = \frac{11}{x}$. Cross multiply: $x = (8 \times 11) \div 3 = \29 .

Challenge 6: If 5 apples cost \$4, how much do 11 apples cost?

Answer: 8

Solution: Set up proportion: $\frac{5}{4} = \frac{11}{x}$. Cross multiply: $x = (4 \times 11) \div 5 = \8 .

Challenge 7: If 5 apples cost \$6, how much do 9 apples cost?

Answer: 10

Solution: Set up proportion: $\frac{5}{6} = \frac{9}{x}$. Cross multiply: $x = (6 \times 9) \div 5 = \10 .

Challenge 8: If 3 apples cost \$8, how much do 6 apples cost?

Answer: 16

Solution: Set up proportion: $\frac{3}{8} = \frac{6}{x}$. Cross multiply: $x = (8 \times 6) \div 3 = \16 .

Challenge 9: If 4 apples cost \$4, how much do 6 apples cost?

Answer: 6

Solution: Set up proportion: $4/4 = 6/x$. Cross multiply: $x = (4 \times 6) \div 4 = \6 .

Challenge 10: If 4 apples cost \$6, how much do 9 apples cost?

Answer: 13

Solution: Set up proportion: $4/6 = 9/x$. Cross multiply: $x = (6 \times 9) \div 4 = \13 .

Challenge 11: If 5 apples cost \$8, how much do 11 apples cost?

Answer: 17

Solution: Set up proportion: $5/8 = 11/x$. Cross multiply: $x = (8 \times 11) \div 5 = \17 .

Challenge 12: If 3 apples cost \$7, how much do 10 apples cost?

Answer: 23

Solution: Set up proportion: $3/7 = 10/x$. Cross multiply: $x = (7 \times 10) \div 3 = \23 .

Challenge 13: If 5 apples cost \$8, how much do 6 apples cost?

Answer: 9

Solution: Set up proportion: $5/8 = 6/x$. Cross multiply: $x = (8 \times 6) \div 5 = \9 .

Challenge 14: If 2 apples cost \$4, how much do 7 apples cost?

Answer: 14

Solution: Set up proportion: $2/4 = 7/x$. Cross multiply: $x = (4 \times 7) \div 2 = \14 .

Challenge 15: If 2 apples cost \$7, how much do 12 apples cost?

Answer: 42

Solution: Set up proportion: $2/7 = 12/x$. Cross multiply: $x = (7 \times 12) \div 2 = \42 .

Chapter 5: Word Problems

Word problems require students to apply multiple mathematical concepts and develop problem-solving strategies. This chapter presents diverse scenarios that integrate various mathematical operations and concepts.

Grade 1 Problems

Problem 1: Evelyn has 11 marbles. They find 2 more marbles, then give 2 marbles to a friend. How many marbles does Evelyn have now?

Answer: 11

Explanation: Start with 11, add 2: $11 + 2 = 13$. Then subtract 2: $13 - 2 = 11$ marbles.

Problem 2: Sebastian has 11 marbles. They find 7 more marbles, then give 3 marbles to a friend. How many marbles does Sebastian have now?

Answer: 15

Explanation: Start with 11, add 7: $11 + 7 = 18$. Then subtract 3: $18 - 3 = 15$ marbles.

Problem 3: Henry has 11 marbles. They find 7 more marbles, then give 3 marbles to a friend. How many marbles does Henry have now?

Answer: 15

Explanation: Start with 11, add 7: $11 + 7 = 18$. Then subtract 3: $18 - 3 = 15$ marbles.

Problem 4: Mia has 11 marbles. They find 5 more marbles, then give 5 marbles to a friend. How many marbles does Mia have now?

Answer: 11

Explanation: Start with 11, add 5: $11 + 5 = 16$. Then subtract 5: $16 - 5 = 11$ marbles.

Problem 5: Harper has 6 marbles. They find 3 more marbles, then give 5 marbles to a friend. How many marbles does Harper have now?

Answer: 4

Explanation: Start with 6, add 3: $6 + 3 = 9$. Then subtract 5: $9 - 5 = 4$ marbles.

Problem 6: David has 8 marbles. They find 4 more marbles, then give 1 marbles to a friend. How many marbles does David have now?

Answer: 11

Explanation: Start with 8, add 4: $8 + 4 = 12$. Then subtract 1: $12 - 1 = 11$ marbles.

Problem 7: Joseph has 10 marbles. They find 3 more marbles, then give 3 marbles to a friend. How many marbles does Joseph have now?

Answer: 10

Explanation: Start with 10, add 3: $10 + 3 = 13$. Then subtract 3: $13 - 3 = 10$ marbles.

Problem 8: Alexander has 6 marbles. They find 5 more marbles, then give 1 marbles to a friend. How many marbles does Alexander have now?

Answer: 10

Explanation: Start with 6, add 5: $6 + 5 = 11$. Then subtract 1: $11 - 1 = 10$ marbles.

Problem 9: Sebastian has 6 marbles. They find 3 more marbles, then give 2 marbles to a friend. How many marbles does Sebastian have now?

Answer: 7

Explanation: Start with 6, add 3: $6 + 3 = 9$. Then subtract 2: $9 - 2 = 7$ marbles.

Problem 10: Lucas has 14 marbles. They find 7 more marbles, then give 3 marbles to a friend. How many marbles does Lucas have now?

Answer: 18

Explanation: Start with 14, add 7: $14 + 7 = 21$. Then subtract 3: $21 - 3 = 18$ marbles.

Problem 11: Harper has 13 marbles. They find 3 more marbles, then give 1 marbles to a friend. How many marbles does Harper have now?

Answer: 15

Explanation: Start with 13, add 3: $13 + 3 = 16$. Then subtract 1: $16 - 1 = 15$ marbles.

Problem 12: Jack has 13 marbles. They find 5 more marbles, then give 2 marbles to a friend. How many marbles does Jack have now?

Answer: 16

Explanation: Start with 13, add 5: $13 + 5 = 18$. Then subtract 2: $18 - 2 = 16$ marbles.

Problem 13: Abigail has 10 marbles. They find 8 more marbles, then give 3 marbles to a friend. How many marbles does Abigail have now?

Answer: 15

Explanation: Start with 10, add 8: $10 + 8 = 18$. Then subtract 3: $18 - 3 = 15$ marbles.

Problem 14: Liam has 8 marbles. They find 2 more marbles, then give 2 marbles to a friend. How many marbles does Liam have now?

Answer: 8

Explanation: Start with 8, add 2: $8 + 2 = 10$. Then subtract 2: $10 - 2 = 8$ marbles.

Problem 15: Aiden has 10 marbles. They find 7 more marbles, then give 3 marbles to a friend. How many marbles does Aiden have now?

Answer: 14

Explanation: Start with 10, add 7: $10 + 7 = 17$. Then subtract 3: $17 - 3 = 14$ marbles.

Problem 16: Isabella has 12 marbles. They find 4 more marbles, then give 4 marbles to a friend. How many marbles does Isabella have now?

Answer: 12

Explanation: Start with 12, add 4: $12 + 4 = 16$. Then subtract 4: $16 - 4 = 12$ marbles.

Problem 17: Emma has 8 marbles. They find 7 more marbles, then give 5 marbles to a friend. How many marbles does Emma have now?

Answer: 10

Explanation: Start with 8, add 7: $8 + 7 = 15$. Then subtract 5: $15 - 5 = 10$ marbles.

Problem 18: Mason has 14 marbles. They find 6 more marbles, then give 2 marbles to a friend. How many marbles does Mason have now?

Answer: 18

Explanation: Start with 14, add 6: $14 + 6 = 20$. Then subtract 2: $20 - 2 = 18$ marbles.

Problem 19: Alexander has 5 marbles. They find 7 more marbles, then give 3 marbles to a friend. How many marbles does Alexander have now?

Answer: 9

Explanation: Start with 5, add 7: $5 + 7 = 12$. Then subtract 3: $12 - 3 = 9$ marbles.

Problem 20: Jack has 12 marbles. They find 4 more marbles, then give 5 marbles to a friend. How many marbles does Jack have now?

Answer: 11

Explanation: Start with 12, add 4: $12 + 4 = 16$. Then subtract 5: $16 - 5 = 11$ marbles.

Problem 21: Lucas has 13 marbles. They find 2 more marbles, then give 2 marbles to a friend. How many marbles does Lucas have now?

Answer: 13

Explanation: Start with 13, add 2: $13 + 2 = 15$. Then subtract 2: $15 - 2 = 13$ marbles.

Problem 22: Elizabeth has 12 marbles. They find 4 more marbles, then give 5 marbles to a friend. How many marbles does Elizabeth have now?

Answer: 11

Explanation: Start with 12, add 4: $12 + 4 = 16$. Then subtract 5: $16 - 5 = 11$ marbles.

Problem 23: Sophia has 14 marbles. They find 7 more marbles, then give 1 marbles to a friend. How many marbles does Sophia have now?

Answer: 20

Explanation: Start with 14, add 7: $14 + 7 = 21$. Then subtract 1: $21 - 1 = 20$ marbles.

Problem 24: Lucas has 5 marbles. They find 7 more marbles, then give 1 marbles to a friend. How many marbles does Lucas have now?

Answer: 11

Explanation: Start with 5, add 7: $5 + 7 = 12$. Then subtract 1: $12 - 1 = 11$ marbles.

Problem 25: Avery has 7 marbles. They find 7 more marbles, then give 4 marbles to a friend. How many marbles does Avery have now?

Answer: 10

Explanation: Start with 7, add 7: $7 + 7 = 14$. Then subtract 4: $14 - 4 = 10$ marbles.

Problem 26: James has 6 marbles. They find 4 more marbles, then give 3 marbles to a friend. How many marbles does James have now?

Answer: 7

Explanation: Start with 6, add 4: $6 + 4 = 10$. Then subtract 3: $10 - 3 = 7$ marbles.

Problem 27: Emily has 10 marbles. They find 5 more marbles, then give 3 marbles to a friend. How many marbles does Emily have now?

Answer: 12

Explanation: Start with 10, add 5: $10 + 5 = 15$. Then subtract 3: $15 - 3 = 12$ marbles.

Problem 28: Evelyn has 8 marbles. They find 5 more marbles, then give 4 marbles to a friend. How many marbles does Evelyn have now?

Answer: 9

Explanation: Start with 8, add 5: $8 + 5 = 13$. Then subtract 4: $13 - 4 = 9$ marbles.

Problem 29: David has 11 marbles. They find 6 more marbles, then give 1 marbles to a friend. How many marbles does David have now?

Answer: 16

Explanation: Start with 11, add 6: $11 + 6 = 17$. Then subtract 1: $17 - 1 = 16$ marbles.

Problem 30: Benjamin has 10 marbles. They find 7 more marbles, then give 5 marbles to a friend. How many marbles does Benjamin have now?

Answer: 12

Explanation: Start with 10, add 7: $10 + 7 = 17$. Then subtract 5: $17 - 5 = 12$ marbles.

Problem 31: James has 12 marbles. They find 3 more marbles, then give 4 marbles to a friend. How many marbles does James have now?

Answer: 11

Explanation: Start with 12, add 3: $12 + 3 = 15$. Then subtract 4: $15 - 4 = 11$ marbles.

Problem 32: Jack has 11 marbles. They find 6 more marbles, then give 2 marbles to a friend. How many marbles does Jack have now?

Answer: 15

Explanation: Start with 11, add 6: $11 + 6 = 17$. Then subtract 2: $17 - 2 = 15$ marbles.

Problem 33: David has 5 marbles. They find 5 more marbles, then give 3 marbles to a friend. How many marbles does David have now?

Answer: 7

Explanation: Start with 5, add 5: $5 + 5 = 10$. Then subtract 3: $10 - 3 = 7$ marbles.

Problem 34: Evelyn has 13 marbles. They find 4 more marbles, then give 2 marbles to a friend. How many marbles does Evelyn have now?

Answer: 15

Explanation: Start with 13, add 4: $13 + 4 = 17$. Then subtract 2: $17 - 2 = 15$ marbles.

Problem 35: Emily has 11 marbles. They find 7 more marbles, then give 2 marbles to a friend. How many marbles does Emily have now?

Answer: 16

Explanation: Start with 11, add 7: $11 + 7 = 18$. Then subtract 2: $18 - 2 = 16$ marbles.

Problem 36: Emma has 7 marbles. They find 6 more marbles, then give 3 marbles to a friend. How many marbles does Emma have now?

Answer: 10

Explanation: Start with 7, add 6: $7 + 6 = 13$. Then subtract 3: $13 - 3 = 10$ marbles.

Problem 37: Harper has 15 marbles. They find 5 more marbles, then give 3 marbles to a friend. How many marbles does Harper have now?

Answer: 17

Explanation: Start with 15, add 5: $15 + 5 = 20$. Then subtract 3: $20 - 3 = 17$ marbles.

Problem 38: Samuel has 8 marbles. They find 5 more marbles, then give 3 marbles to a friend. How many marbles does Samuel have now?

Answer: 10

Explanation: Start with 8, add 5: $8 + 5 = 13$. Then subtract 3: $13 - 3 = 10$ marbles.

Problem 39: Emily has 11 marbles. They find 7 more marbles, then give 5 marbles to a friend. How many marbles does Emily have now?

Answer: 13

Explanation: Start with 11, add 7: $11 + 7 = 18$. Then subtract 5: $18 - 5 = 13$ marbles.

Problem 40: Mason has 10 marbles. They find 2 more marbles, then give 1 marbles to a friend. How many marbles does Mason have now?

Answer: 11

Explanation: Start with 10, add 2: $10 + 2 = 12$. Then subtract 1: $12 - 1 = 11$ marbles.

Problem 41: Henry has 8 marbles. They find 4 more marbles, then give 3 marbles to a friend. How many marbles does Henry have now?

Answer: 9

Explanation: Start with 8, add 4: $8 + 4 = 12$. Then subtract 3: $12 - 3 = 9$ marbles.

Problem 42: James has 8 marbles. They find 7 more marbles, then give 4 marbles to a friend. How many marbles does James have now?

Answer: 11

Explanation: Start with 8, add 7: $8 + 7 = 15$. Then subtract 4: $15 - 4 = 11$ marbles.

Problem 43: Avery has 14 marbles. They find 6 more marbles, then give 4 marbles to a friend. How many marbles does Avery have now?

Answer: 16

Explanation: Start with 14, add 6: $14 + 6 = 20$. Then subtract 4: $20 - 4 = 16$ marbles.

Problem 44: Emily has 12 marbles. They find 5 more marbles, then give 4 marbles to a friend. How many marbles does Emily have now?

Answer: 13

Explanation: Start with 12, add 5: $12 + 5 = 17$. Then subtract 4: $17 - 4 = 13$ marbles.

Problem 45: Ethan has 5 marbles. They find 7 more marbles, then give 2 marbles to a friend. How many marbles does Ethan have now?

Answer: 10

Explanation: Start with 5, add 7: $5 + 7 = 12$. Then subtract 2: $12 - 2 = 10$ marbles.

Practice Exercises - Grade 1

Exercise 1: Sophia has 14 marbles. They find 4 more marbles, then give 5 marbles to a friend. How many marbles does Sophia have now?

Exercise 2: Sophia has 10 marbles. They find 3 more marbles, then give 2 marbles to a friend. How many marbles does Sophia have now?

Exercise 3: Avery has 13 marbles. They find 7 more marbles, then give 3 marbles to a friend. How many marbles does Avery have now?

Exercise 4: Samuel has 8 marbles. They find 5 more marbles, then give 4 marbles to a friend. How many marbles does Samuel have now?

Exercise 5: Samuel has 12 marbles. They find 6 more marbles, then give 2 marbles to a friend. How many marbles does Samuel have now?

Exercise 6: Aiden has 10 marbles. They find 8 more marbles, then give 1 marbles to a friend. How many marbles does Aiden have now?

Exercise 7: Aiden has 13 marbles. They find 5 more marbles, then give 5 marbles to a friend. How many marbles does Aiden have now?

Exercise 8: Sebastian has 14 marbles. They find 4 more marbles, then give 4 marbles to a friend. How many marbles does Sebastian have now?

Exercise 9: Alexander has 9 marbles. They find 7 more marbles, then give 2 marbles to a friend. How many marbles does Alexander have now?

Exercise 10: Olivia has 15 marbles. They find 4 more marbles, then give 2 marbles to a friend. How many marbles does Olivia have now?

Exercise 11: David has 11 marbles. They find 6 more marbles, then give 4 marbles to a friend. How many marbles does David have now?

Exercise 12: Jack has 14 marbles. They find 6 more marbles, then give 4 marbles to a friend. How many marbles does Jack have now?

Exercise 13: David has 10 marbles. They find 3 more marbles, then give 5 marbles to a friend. How many marbles does David have now?

Exercise 14: Isabella has 8 marbles. They find 3 more marbles, then give 2 marbles to a friend. How many marbles does Isabella have now?

Exercise 15: Sofia has 8 marbles. They find 7 more marbles, then give 1 marbles to a friend. How many marbles does Sofia have now?

Exercise 16: Evelyn has 13 marbles. They find 2 more marbles, then give 5 marbles to a friend. How many marbles does Evelyn have now?

Exercise 17: Aiden has 7 marbles. They find 4 more marbles, then give 5 marbles to a friend. How many marbles does Aiden have now?

Exercise 18: Harper has 8 marbles. They find 2 more marbles, then give 4 marbles to a friend. How many marbles does Harper have now?

Exercise 19: Alexander has 12 marbles. They find 8 more marbles, then give 4 marbles to a friend. How many marbles does Alexander have now?

Exercise 20: Olivia has 11 marbles. They find 8 more marbles, then give 1 marbles to a friend. How many marbles does Olivia have now?

Practice Exercise Answers:

Exercise 1: 13

Exercise 2: 11

Exercise 3: 17

Exercise 4: 9

Exercise 5: 16

Exercise 6: 17

Exercise 7: 13

Exercise 8: 14

Exercise 9: 14

Exercise 10: 17

Exercise 11: 13

Exercise 12: 16

Exercise 13: 8

Exercise 14: 9

Exercise 15: 14

Exercise 16: 10

Exercise 17: 6

Exercise 18: 6

Exercise 19: 16

Exercise 20: 18

Challenge Problems - Grade 1

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: David has 9 marbles. They find 5 more marbles, then give 4 marbles to a friend. How many marbles does David have now?

Answer: 10

Solution: Start with 9, add 5: $9 + 5 = 14$. Then subtract 4: $14 - 4 = 10$ marbles.

Challenge 2: Avery has 15 marbles. They find 4 more marbles, then give 4 marbles to a friend. How many marbles does Avery have now?

Answer: 15

Solution: Start with 15, add 4: $15 + 4 = 19$. Then subtract 4: $19 - 4 = 15$ marbles.

Challenge 3: Harper has 14 marbles. They find 3 more marbles, then give 5 marbles to a friend. How many marbles does Harper have now?

Answer: 12

Solution: Start with 14, add 3: $14 + 3 = 17$. Then subtract 5: $17 - 5 = 12$ marbles.

Challenge 4: Abigail has 7 marbles. They find 3 more marbles, then give 3 marbles to a friend. How many marbles does Abigail have now?

Answer: 7

Solution: Start with 7, add 3: $7 + 3 = 10$. Then subtract 3: $10 - 3 = 7$ marbles.

Challenge 5: Noah has 6 marbles. They find 7 more marbles, then give 3 marbles to a friend. How many marbles does Noah have now?

Answer: 10

Solution: Start with 6, add 7: $6 + 7 = 13$. Then subtract 3: $13 - 3 = 10$ marbles.

Challenge 6: Avery has 8 marbles. They find 6 more marbles, then give 1 marbles to a friend. How many marbles does Avery have now?

Answer: 13

Solution: Start with 8, add 6: $8 + 6 = 14$. Then subtract 1: $14 - 1 = 13$ marbles.

Challenge 7: Sophia has 13 marbles. They find 3 more marbles, then give 3 marbles to a friend. How many marbles does Sophia have now?

Answer: 13

Solution: Start with 13, add 3: $13 + 3 = 16$. Then subtract 3: $16 - 3 = 13$ marbles.

Challenge 8: Charlotte has 8 marbles. They find 5 more marbles, then give 1 marbles to a friend. How many marbles does Charlotte have now?

Answer: 12

Solution: Start with 8, add 5: $8 + 5 = 13$. Then subtract 1: $13 - 1 = 12$ marbles.

Challenge 9: Olivia has 6 marbles. They find 7 more marbles, then give 2 marbles to a friend. How many marbles does Olivia have now?

Answer: 11

Solution: Start with 6, add 7: $6 + 7 = 13$. Then subtract 2: $13 - 2 = 11$ marbles.

Challenge 10: Mason has 9 marbles. They find 5 more marbles, then give 4 marbles to a friend. How many marbles does Mason have now?

Answer: 10

Solution: Start with 9, add 5: $9 + 5 = 14$. Then subtract 4: $14 - 4 = 10$ marbles.

Challenge 11: Lucas has 11 marbles. They find 3 more marbles, then give 5 marbles to a friend. How many marbles does Lucas have now?

Answer: 9

Solution: Start with 11, add 3: $11 + 3 = 14$. Then subtract 5: $14 - 5 = 9$ marbles.

Challenge 12: Emma has 14 marbles. They find 6 more marbles, then give 2 marbles to a friend. How many marbles does Emma have now?

Answer: 18

Solution: Start with 14, add 6: $14 + 6 = 20$. Then subtract 2: $20 - 2 = 18$ marbles.

Challenge 13: Charlotte has 13 marbles. They find 5 more marbles, then give 3 marbles to a friend. How many marbles does Charlotte have now?

Answer: 15

Solution: Start with 13, add 5: $13 + 5 = 18$. Then subtract 3: $18 - 3 = 15$ marbles.

Challenge 14: Sebastian has 7 marbles. They find 4 more marbles, then give 5 marbles to a friend. How many marbles does Sebastian have now?

Answer: 6

Solution: Start with 7, add 4: $7 + 4 = 11$. Then subtract 5: $11 - 5 = 6$ marbles.

Challenge 15: Charlotte has 9 marbles. They find 4 more marbles, then give 4 marbles to a friend. How many marbles does Charlotte have now?

Answer: 9

Solution: Start with 9, add 4: $9 + 4 = 13$. Then subtract 4: $13 - 4 = 9$ marbles.

Grade 2 Problems

Problem 1: Olivia buys a toy for \$15 and a book for \$2. They pay with \$33. How much change does Olivia get?

Answer: 16

Explanation: Total spent: $\$15 + \$2 = \$17$. Change: $\$33 - \$17 = \$16$.

Problem 2: Henry buys a toy for \$11 and a book for \$8. They pay with \$44. How much change does Henry get?

Answer: 25

Explanation: Total spent: $\$11 + \$8 = \$19$. Change: $\$44 - \$19 = \$25$.

Problem 3: Avery buys a toy for \$9 and a book for \$5. They pay with \$34. How much change does Avery get?

Answer: 20

Explanation: Total spent: $\$9 + \$5 = \$14$. Change: $\$34 - \$14 = \$20$.

Problem 4: Liam buys a toy for \$3 and a book for \$6. They pay with \$35. How much change does Liam get?

Answer: 26

Explanation: Total spent: $\$3 + \$6 = \$9$. Change: $\$35 - \$9 = \$26$.

Problem 5: David buys a toy for \$3 and a book for \$12. They pay with \$20. How much change does David get?

Answer: 5

Explanation: Total spent: $\$3 + \$12 = \$15$. Change: $\$20 - \$15 = \$5$.

Problem 6: Sebastian buys a toy for \$12 and a book for \$9. They pay with \$50. How much change does Sebastian get?

Answer: 29

Explanation: Total spent: $\$12 + \$9 = \$21$. Change: $\$50 - \$21 = \$29$.

Problem 7: David buys a toy for \$3 and a book for \$11. They pay with \$42. How much change does David get?

Answer: 28

Explanation: Total spent: $\$3 + \$11 = \$14$. Change: $\$42 - \$14 = \$28$.

Problem 8: Elizabeth buys a toy for \$8 and a book for \$10. They pay with \$20. How much change does Elizabeth get?

Answer: 2

Explanation: Total spent: $\$8 + \$10 = \$18$. Change: $\$20 - \$18 = \$2$.

Problem 9: Henry buys a toy for \$7 and a book for \$5. They pay with \$20. How much change does Henry get?

Answer: 8

Explanation: Total spent: $\$7 + \$5 = \$12$. Change: $\$20 - \$12 = \$8$.

Problem 10: Sofia buys a toy for \$15 and a book for \$5. They pay with \$23. How much change does Sofia get?

Answer: 3

Explanation: Total spent: $\$15 + \$5 = \$20$. Change: $\$23 - \$20 = \$3$.

Problem 11: Benjamin buys a toy for \$9 and a book for \$6. They pay with \$46. How much change does Benjamin get?

Answer: 31

Explanation: Total spent: $\$9 + \$6 = \$15$. Change: $\$46 - \$15 = \$31$.

Problem 12: Aiden buys a toy for \$8 and a book for \$10. They pay with \$43. How much change does Aiden get?

Answer: 25

Explanation: Total spent: $\$8 + \$10 = \$18$. Change: $\$43 - \$18 = \$25$.

Problem 13: Ethan buys a toy for \$12 and a book for \$12. They pay with \$27. How much change does Ethan get?

Answer: 3

Explanation: Total spent: $\$12 + \$12 = \$24$. Change: $\$27 - \$24 = \$3$.

Problem 14: Ethan buys a toy for \$12 and a book for \$11. They pay with \$25. How much change does Ethan get?

Answer: 2

Explanation: Total spent: $\$12 + \$11 = \$23$. Change: $\$25 - \$23 = \$2$.

Problem 15: Olivia buys a toy for \$14 and a book for \$7. They pay with \$50. How much change does Olivia get?

Answer: 29

Explanation: Total spent: $\$14 + \$7 = \$21$. Change: $\$50 - \$21 = \$29$.

Problem 16: Alexander buys a toy for \$15 and a book for \$11. They pay with \$45. How much change does Alexander get?

Answer: 19

Explanation: Total spent: $\$15 + \$11 = \$26$. Change: $\$45 - \$26 = \$19$.

Problem 17: Benjamin buys a toy for \$9 and a book for \$4. They pay with \$37. How much change does Benjamin get?

Answer: 24

Explanation: Total spent: $\$9 + \$4 = \$13$. Change: $\$37 - \$13 = \$24$.

Problem 18: Lucas buys a toy for \$9 and a book for \$8. They pay with \$24. How much change does Lucas get?

Answer: 7

Explanation: Total spent: $\$9 + \$8 = \$17$. Change: $\$24 - \$17 = \$7$.

Problem 19: Mason buys a toy for \$7 and a book for \$4. They pay with \$38. How much change does Mason get?

Answer: 27

Explanation: Total spent: $\$7 + \$4 = \$11$. Change: $\$38 - \$11 = \$27$.

Problem 20: Jack buys a toy for \$9 and a book for \$7. They pay with \$25. How much change does Jack get?

Answer: 9

Explanation: Total spent: $\$9 + \$7 = \$16$. Change: $\$25 - \$16 = \$9$.

Problem 21: Amelia buys a toy for \$3 and a book for \$2. They pay with \$33. How much change does Amelia get?

Answer: 28

Explanation: Total spent: $\$3 + \$2 = \$5$. Change: $\$33 - \$5 = \$28$.

Problem 22: Isabella buys a toy for \$7 and a book for \$9. They pay with \$39. How much change does Isabella get?

Answer: 23

Explanation: Total spent: $\$7 + \$9 = \$16$. Change: $\$39 - \$16 = \$23$.

Problem 23: Henry buys a toy for \$15 and a book for \$2. They pay with \$41. How much change does Henry get?

Answer: 24

Explanation: Total spent: $\$15 + \$2 = \$17$. Change: $\$41 - \$17 = \$24$.

Problem 24: Sophia buys a toy for \$15 and a book for \$5. They pay with \$47. How much change does Sophia get?

Answer: 27

Explanation: Total spent: $\$15 + \$5 = \$20$. Change: $\$47 - \$20 = \$27$.

Problem 25: Samuel buys a toy for \$5 and a book for \$12. They pay with \$42. How much change does Samuel get?

Answer: 25

Explanation: Total spent: $\$5 + \$12 = \$17$. Change: $\$42 - \$17 = \$25$.

Problem 26: Joseph buys a toy for \$13 and a book for \$7. They pay with \$49. How much change does Joseph get?

Answer: 29

Explanation: Total spent: $\$13 + \$7 = \$20$. Change: $\$49 - \$20 = \$29$.

Problem 27: Elizabeth buys a toy for \$6 and a book for \$11. They pay with \$31. How much change does Elizabeth get?

Answer: 14

Explanation: Total spent: $\$6 + \$11 = \$17$. Change: $\$31 - \$17 = \$14$.

Problem 28: Olivia buys a toy for \$3 and a book for \$10. They pay with \$30. How much change does Olivia get?

Answer: 17

Explanation: Total spent: $\$3 + \$10 = \$13$. Change: $\$30 - \$13 = \$17$.

Problem 29: Amelia buys a toy for \$10 and a book for \$11. They pay with \$39. How much change does Amelia get?

Answer: 18

Explanation: Total spent: $\$10 + \$11 = \$21$. Change: $\$39 - \$21 = \$18$.

Problem 30: Charlotte buys a toy for \$13 and a book for \$4. They pay with \$30. How much change does Charlotte get?

Answer: 13

Explanation: Total spent: $\$13 + \$4 = \$17$. Change: $\$30 - \$17 = \$13$.

Problem 31: Emily buys a toy for \$15 and a book for \$9. They pay with \$33. How much change does Emily get?

Answer: 9

Explanation: Total spent: $\$15 + \$9 = \$24$. Change: $\$33 - \$24 = \$9$.

Problem 32: Sophia buys a toy for \$4 and a book for \$11. They pay with \$47. How much change does Sophia get?

Answer: 32

Explanation: Total spent: $\$4 + \$11 = \$15$. Change: $\$47 - \$15 = \$32$.

Problem 33: Harper buys a toy for \$10 and a book for \$8. They pay with \$50. How much change does Harper get?

Answer: 32

Explanation: Total spent: $\$10 + \$8 = \$18$. Change: $\$50 - \$18 = \$32$.

Problem 34: Jack buys a toy for \$11 and a book for \$4. They pay with \$28. How much change does Jack get?

Answer: 13

Explanation: Total spent: $\$11 + \$4 = \$15$. Change: $\$28 - \$15 = \$13$.

Problem 35: Aiden buys a toy for \$3 and a book for \$3. They pay with \$30. How much change does Aiden get?

Answer: 24

Explanation: Total spent: $\$3 + \$3 = \$6$. Change: $\$30 - \$6 = \$24$.

Problem 36: Joseph buys a toy for \$6 and a book for \$9. They pay with \$46. How much change does Joseph get?

Answer: 31

Explanation: Total spent: $\$6 + \$9 = \$15$. Change: $\$46 - \$15 = \$31$.

Problem 37: Sebastian buys a toy for \$5 and a book for \$7. They pay with \$36. How much change does Sebastian get?

Answer: 24

Explanation: Total spent: $\$5 + \$7 = \$12$. Change: $\$36 - \$12 = \$24$.

Problem 38: Noah buys a toy for \$8 and a book for \$5. They pay with \$31. How much change does Noah get?

Answer: 18

Explanation: Total spent: $\$8 + \$5 = \$13$. Change: $\$31 - \$13 = \$18$.

Problem 39: Ethan buys a toy for \$12 and a book for \$10. They pay with \$46. How much change does Ethan get?

Answer: 24

Explanation: Total spent: $\$12 + \$10 = \$22$. Change: $\$46 - \$22 = \$24$.

Problem 40: Samuel buys a toy for \$5 and a book for \$11. They pay with \$43. How much change does Samuel get?

Answer: 27

Explanation: Total spent: $\$5 + \$11 = \$16$. Change: $\$43 - \$16 = \$27$.

Problem 41: Harper buys a toy for \$3 and a book for \$7. They pay with \$40. How much change does Harper get?

Answer: 30

Explanation: Total spent: $\$3 + \$7 = \$10$. Change: $\$40 - \$10 = \$30$.

Problem 42: Evelyn buys a toy for \$5 and a book for \$11. They pay with \$28. How much change does Evelyn get?

Answer: 12

Explanation: Total spent: $\$5 + \$11 = \$16$. Change: $\$28 - \$16 = \$12$.

Problem 43: Joseph buys a toy for \$4 and a book for \$7. They pay with \$43. How much change does Joseph get?

Answer: 32

Explanation: Total spent: $\$4 + \$7 = \$11$. Change: $\$43 - \$11 = \$32$.

Problem 44: Sebastian buys a toy for \$11 and a book for \$7. They pay with \$33. How much change does Sebastian get?

Answer: 15

Explanation: Total spent: $\$11 + \$7 = \$18$. Change: $\$33 - \$18 = \$15$.

Problem 45: Jack buys a toy for \$14 and a book for \$2. They pay with \$47. How much change does Jack get?

Answer: 31

Explanation: Total spent: $\$14 + \$2 = \$16$. Change: $\$47 - \$16 = \$31$.

Practice Exercises - Grade 2

Exercise 1: Lucas buys a toy for \$5 and a book for \$12. They pay with \$33. How much change does Lucas get?

Exercise 2: Avery buys a toy for \$3 and a book for \$5. They pay with \$39. How much change does Avery get?

Exercise 3: Mia buys a toy for \$6 and a book for \$7. They pay with \$31. How much change does Mia get?

Exercise 4: Mia buys a toy for \$14 and a book for \$4. They pay with \$20. How much change does Mia get?

Exercise 5: Jack buys a toy for \$8 and a book for \$7. They pay with \$27. How much change does Jack get?

Exercise 6: Olivia buys a toy for \$8 and a book for \$3. They pay with \$43. How much change does Olivia get?

Exercise 7: Noah buys a toy for \$14 and a book for \$10. They pay with \$25. How much change does Noah get?

Exercise 8: Noah buys a toy for \$3 and a book for \$12. They pay with \$25. How much change does Noah get?

Exercise 9: James buys a toy for \$5 and a book for \$2. They pay with \$41. How much change does James get?

Exercise 10: Elizabeth buys a toy for \$7 and a book for \$11. They pay with \$45. How much change does Elizabeth get?

Exercise 11: Isabella buys a toy for \$15 and a book for \$4. They pay with \$33. How much change does Isabella get?

Exercise 12: Harper buys a toy for \$8 and a book for \$3. They pay with \$41. How much change does Harper get?

Exercise 13: Ethan buys a toy for \$11 and a book for \$12. They pay with \$24. How much change does Ethan get?

Exercise 14: Sebastian buys a toy for \$11 and a book for \$5. They pay with \$33. How much change does Sebastian get?

Exercise 15: Henry buys a toy for \$12 and a book for \$10. They pay with \$29. How much change does Henry get?

Exercise 16: Henry buys a toy for \$6 and a book for \$6. They pay with \$36. How much change does Henry get?

Exercise 17: Evelyn buys a toy for \$10 and a book for \$10. They pay with \$28. How much change does Evelyn get?

Exercise 18: Alexander buys a toy for \$15 and a book for \$8. They pay with \$39. How much change does Alexander get?

Exercise 19: Mia buys a toy for \$12 and a book for \$5. They pay with \$23. How much change does Mia get?

Exercise 20: Liam buys a toy for \$12 and a book for \$11. They pay with \$49. How much change does Liam get?

Practice Exercise Answers:

Exercise 1: 16

Exercise 2: 31

Exercise 3: 18

Exercise 4: 2

Exercise 5: 12

Exercise 6: 32

Exercise 7: 1

Exercise 8: 10

Exercise 9: 34

Exercise 10: 27

Exercise 11: 14

Exercise 12: 30

Exercise 13: 1

Exercise 14: 17

Exercise 15: 7

Exercise 16: 24

Exercise 17: 8

Exercise 18: 16

Exercise 19: 6

Exercise 20: 26

Challenge Problems - Grade 2

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Ava buys a toy for \$9 and a book for \$10. They pay with \$40. How much change does Ava get?

Answer: 21

Solution: Total spent: $\$9 + \$10 = \$19$. Change: $\$40 - \$19 = \$21$.

Challenge 2: Harper buys a toy for \$12 and a book for \$5. They pay with \$26. How much change does Harper get?

Answer: 9

Solution: Total spent: $\$12 + \$5 = \$17$. Change: $\$26 - \$17 = \$9$.

Challenge 3: Noah buys a toy for \$9 and a book for \$2. They pay with \$44. How much change does Noah get?

Answer: 33

Solution: Total spent: $\$9 + \$2 = \$11$. Change: $\$44 - \$11 = \$33$.

Challenge 4: Isabella buys a toy for \$10 and a book for \$11. They pay with \$49. How much change does Isabella get?

Answer: 28

Solution: Total spent: $\$10 + \$11 = \$21$. Change: $\$49 - \$21 = \$28$.

Challenge 5: Sofia buys a toy for \$15 and a book for \$9. They pay with \$22. How much change does Sofia get?

Answer: -2

Solution: Total spent: $\$15 + \$9 = \$24$. Change: $\$22 - \$24 = \$-2$.

Challenge 6: James buys a toy for \$4 and a book for \$7. They pay with \$25. How much change does James get?

Answer: 14

Solution: Total spent: $\$4 + \$7 = \$11$. Change: $\$25 - \$11 = \$14$.

Challenge 7: Elizabeth buys a toy for \$13 and a book for \$7. They pay with \$36. How much change does Elizabeth get?

Answer: 16

Solution: Total spent: $\$13 + \$7 = \$20$. Change: $\$36 - \$20 = \$16$.

Challenge 8: Ava buys a toy for \$4 and a book for \$9. They pay with \$25. How much change does Ava get?

Answer: 12

Solution: Total spent: $\$4 + \$9 = \$13$. Change: $\$25 - \$13 = \$12$.

Challenge 9: Liam buys a toy for \$12 and a book for \$12. They pay with \$27. How much change does Liam get?

Answer: 3

Solution: Total spent: $\$12 + \$12 = \$24$. Change: $\$27 - \$24 = \$3$.

Challenge 10: Harper buys a toy for \$3 and a book for \$10. They pay with \$22. How much change does Harper get?

Answer: 9

Solution: Total spent: $\$3 + \$10 = \$13$. Change: $\$22 - \$13 = \$9$.

Challenge 11: Aiden buys a toy for \$13 and a book for \$6. They pay with \$30. How much change does Aiden get?

Answer: 11

Solution: Total spent: $\$13 + \$6 = \$19$. Change: $\$30 - \$19 = \$11$.

Challenge 12: Evelyn buys a toy for \$11 and a book for \$10. They pay with \$29. How much change does Evelyn get?

Answer: 8

Solution: Total spent: $\$11 + \$10 = \$21$. Change: $\$29 - \$21 = \$8$.

Challenge 13: Joseph buys a toy for \$3 and a book for \$8. They pay with \$36. How much change does Joseph get?

Answer: 25

Solution: Total spent: $\$3 + \$8 = \$11$. Change: $\$36 - \$11 = \$25$.

Challenge 14: Harper buys a toy for \$9 and a book for \$4. They pay with \$33. How much change does Harper get?

Answer: 20

Solution: Total spent: $\$9 + \$4 = \$13$. Change: $\$33 - \$13 = \$20$.

Challenge 15: Emily buys a toy for \$10 and a book for \$3. They pay with \$27. How much change does Emily get?

Answer: 14

Solution: Total spent: $\$10 + \$3 = \$13$. Change: $\$27 - \$13 = \$14$.

Grade 3 Problems

Problem 1: Jack and 4 friends buy 6 snacks at \$9 each. If they split the cost equally, how much does each person pay?

Answer: 10

Explanation: Total cost: $6 \times \$9 = \54 . Each person pays: $\$54 \div 5 = \10 .

Problem 2: Isabella and 2 friends buy 5 snacks at \$8 each. If they split the cost equally, how much does each person pay?

Answer: 13

Explanation: Total cost: $5 \times \$8 = \40 . Each person pays: $\$40 \div 3 = \13 .

Problem 3: Harper and 3 friends buy 5 snacks at \$11 each. If they split the cost equally, how much does each person pay?

Answer: 13

Explanation: Total cost: $5 \times \$11 = \55 . Each person pays: $\$55 \div 4 = \13 .

Problem 4: Ethan and 3 friends buy 4 snacks at \$10 each. If they split the cost equally, how much does each person pay?

Answer: 10

Explanation: Total cost: $4 \times \$10 = \40 . Each person pays: $\$40 \div 4 = \10 .

Problem 5: Samuel and 1 friends buy 4 snacks at \$9 each. If they split the cost equally, how much does each person pay?

Answer: 18

Explanation: Total cost: $4 \times \$9 = \36 . Each person pays: $\$36 \div 2 = \18 .

Problem 6: Emily and 2 friends buy 8 snacks at \$8 each. If they split the cost equally, how much does each person pay?

Answer: 21

Explanation: Total cost: $8 \times \$8 = \64 . Each person pays: $\$64 \div 3 = \21 .

Problem 7: Noah and 2 friends buy 9 snacks at \$8 each. If they split the cost equally, how much does each person pay?

Answer: 24

Explanation: Total cost: $9 \times \$8 = \72 . Each person pays: $\$72 \div 3 = \24 .

Problem 8: Samuel and 4 friends buy 4 snacks at \$8 each. If they split the cost equally, how much does each person pay?

Answer: 6

Explanation: Total cost: $4 \times \$8 = \32 . Each person pays: $\$32 \div 5 = \6 .

Problem 9: Ava and 1 friends buy 4 snacks at \$8 each. If they split the cost equally, how much does each person pay?

Answer: 16

Explanation: Total cost: $4 \times \$8 = \32 . Each person pays: $\$32 \div 2 = \16 .

Problem 10: Mason and 3 friends buy 8 snacks at \$3 each. If they split the cost equally, how much does each person pay?

Answer: 6

Explanation: Total cost: $8 \times \$3 = \24 . Each person pays: $\$24 \div 4 = \6 .

Problem 11: Benjamin and 1 friends buy 6 snacks at \$11 each. If they split the cost equally, how much does each person pay?

Answer: 33

Explanation: Total cost: $6 \times \$11 = \66 . Each person pays: $\$66 \div 2 = \33 .

Problem 12: Noah and 2 friends buy 9 snacks at \$6 each. If they split the cost equally, how much does each person pay?

Answer: 18

Explanation: Total cost: $9 \times \$6 = \54 . Each person pays: $\$54 \div 3 = \18 .

Problem 13: Isabella and 1 friends buy 6 snacks at \$12 each. If they split the cost equally, how much does each person pay?

Answer: 36

Explanation: Total cost: $6 \times \$12 = \72 . Each person pays: $\$72 \div 2 = \36 .

Problem 14: Sebastian and 1 friends buy 7 snacks at \$9 each. If they split the cost equally, how much does each person pay?

Answer: 31

Explanation: Total cost: $7 \times \$9 = \63 . Each person pays: $\$63 \div 2 = \31 .

Problem 15: Evelyn and 2 friends buy 4 snacks at \$6 each. If they split the cost equally, how much does each person pay?

Answer: 8

Explanation: Total cost: $4 \times \$6 = \24 . Each person pays: $\$24 \div 3 = \8 .

Problem 16: Henry and 2 friends buy 10 snacks at \$10 each. If they split the cost equally, how much does each person pay?

Answer: 33

Explanation: Total cost: $10 \times \$10 = \100 . Each person pays: $\$100 \div 3 = \33 .

Problem 17: Ava and 4 friends buy 10 snacks at \$8 each. If they split the cost equally, how much does each person pay?

Answer: 16

Explanation: Total cost: $10 \times \$8 = \80 . Each person pays: $\$80 \div 5 = \16 .

Problem 18: James and 2 friends buy 5 snacks at \$3 each. If they split the cost equally, how much does each person pay?

Answer: 5

Explanation: Total cost: $5 \times \$3 = \15 . Each person pays: $\$15 \div 3 = \5 .

Problem 19: Amelia and 2 friends buy 7 snacks at \$7 each. If they split the cost equally, how much does each person pay?

Answer: 16

Explanation: Total cost: $7 \times \$7 = \49 . Each person pays: $\$49 \div 3 = \16 .

Problem 20: Abigail and 4 friends buy 6 snacks at \$4 each. If they split the cost equally, how much does each person pay?

Answer: 4

Explanation: Total cost: $6 \times \$4 = \24 . Each person pays: $\$24 \div 5 = \4 .

Problem 21: Samuel and 2 friends buy 6 snacks at \$4 each. If they split the cost equally, how much does each person pay?

Answer: 8

Explanation: Total cost: $6 \times \$4 = \24 . Each person pays: $\$24 \div 3 = \8 .

Problem 22: Liam and 4 friends buy 5 snacks at \$12 each. If they split the cost equally, how much does each person pay?

Answer: 12

Explanation: Total cost: $5 \times \$12 = \60 . Each person pays: $\$60 \div 5 = \12 .

Problem 23: Isabella and 1 friends buy 5 snacks at \$5 each. If they split the cost equally, how much does each person pay?

Answer: 12

Explanation: Total cost: $5 \times \$5 = \25 . Each person pays: $\$25 \div 2 = \12 .

Problem 24: Lucas and 3 friends buy 5 snacks at \$12 each. If they split the cost equally, how much does each person pay?

Answer: 15

Explanation: Total cost: $5 \times \$12 = \60 . Each person pays: $\$60 \div 4 = \15 .

Problem 25: Joseph and 2 friends buy 8 snacks at \$10 each. If they split the cost equally, how much does each person pay?

Answer: 26

Explanation: Total cost: $8 \times \$10 = \80 . Each person pays: $\$80 \div 3 = \26 .

Problem 26: Liam and 4 friends buy 8 snacks at \$5 each. If they split the cost equally, how much does each person pay?

Answer: 8

Explanation: Total cost: $8 \times \$5 = \40 . Each person pays: $\$40 \div 5 = \8 .

Problem 27: Alexander and 3 friends buy 8 snacks at \$4 each. If they split the cost equally, how much does each person pay?

Answer: 8

Explanation: Total cost: $8 \times \$4 = \32 . Each person pays: $\$32 \div 4 = \8 .

Problem 28: Mia and 4 friends buy 7 snacks at \$11 each. If they split the cost equally, how much does each person pay?

Answer: 15

Explanation: Total cost: $7 \times \$11 = \77 . Each person pays: $\$77 \div 5 = \15 .

Problem 29: Abigail and 2 friends buy 9 snacks at \$3 each. If they split the cost equally, how much does each person pay?

Answer: 9

Explanation: Total cost: $9 \times \$3 = \27 . Each person pays: $\$27 \div 3 = \9 .

Problem 30: Mia and 3 friends buy 10 snacks at \$3 each. If they split the cost equally, how much does each person pay?

Answer: 7

Explanation: Total cost: $10 \times \$3 = \30 . Each person pays: $\$30 \div 4 = \7 .

Problem 31: Joseph and 4 friends buy 10 snacks at \$9 each. If they split the cost equally, how much does each person pay?

Answer: 18

Explanation: Total cost: $10 \times \$9 = \90 . Each person pays: $\$90 \div 5 = \18 .

Problem 32: Charlotte and 1 friends buy 6 snacks at \$12 each. If they split the cost equally, how much does each person pay?

Answer: 36

Explanation: Total cost: $6 \times \$12 = \72 . Each person pays: $\$72 \div 2 = \36 .

Problem 33: Henry and 4 friends buy 5 snacks at \$9 each. If they split the cost equally, how much does each person pay?

Answer: 9

Explanation: Total cost: $5 \times \$9 = \45 . Each person pays: $\$45 \div 5 = \9 .

Problem 34: Benjamin and 2 friends buy 6 snacks at \$6 each. If they split the cost equally, how much does each person pay?

Answer: 12

Explanation: Total cost: $6 \times \$6 = \36 . Each person pays: $\$36 \div 3 = \12 .

Problem 35: Amelia and 3 friends buy 9 snacks at \$10 each. If they split the cost equally, how much does each person pay?

Answer: 22

Explanation: Total cost: $9 \times \$10 = \90 . Each person pays: $\$90 \div 4 = \22 .

Problem 36: Benjamin and 2 friends buy 7 snacks at \$4 each. If they split the cost equally, how much does each person pay?

Answer: 9

Explanation: Total cost: $7 \times \$4 = \28 . Each person pays: $\$28 \div 3 = \9 .

Problem 37: Olivia and 2 friends buy 7 snacks at \$3 each. If they split the cost equally, how much does each person pay?

Answer: 7

Explanation: Total cost: $7 \times \$3 = \21 . Each person pays: $\$21 \div 3 = \7 .

Problem 38: Elizabeth and 3 friends buy 10 snacks at \$6 each. If they split the cost equally, how much does each person pay?

Answer: 15

Explanation: Total cost: $10 \times \$6 = \60 . Each person pays: $\$60 \div 4 = \15 .

Problem 39: Mia and 1 friends buy 6 snacks at \$10 each. If they split the cost equally, how much does each person pay?

Answer: 30

Explanation: Total cost: $6 \times \$10 = \60 . Each person pays: $\$60 \div 2 = \30 .

Problem 40: Lucas and 4 friends buy 7 snacks at \$5 each. If they split the cost equally, how much does each person pay?

Answer: 7

Explanation: Total cost: $7 \times \$5 = \35 . Each person pays: $\$35 \div 5 = \7 .

Problem 41: Jack and 3 friends buy 4 snacks at \$4 each. If they split the cost equally, how much does each person pay?

Answer: 4

Explanation: Total cost: $4 \times \$4 = \16 . Each person pays: $\$16 \div 4 = \4 .

Problem 42: Aiden and 4 friends buy 9 snacks at \$7 each. If they split the cost equally, how much does each person pay?

Answer: 12

Explanation: Total cost: $9 \times \$7 = \63 . Each person pays: $\$63 \div 5 = \12 .

Problem 43: Isabella and 4 friends buy 9 snacks at \$5 each. If they split the cost equally, how much does each person pay?

Answer: 9

Explanation: Total cost: $9 \times \$5 = \45 . Each person pays: $\$45 \div 5 = \9 .

Problem 44: David and 2 friends buy 4 snacks at \$11 each. If they split the cost equally, how much does each person pay?

Answer: 14

Explanation: Total cost: $4 \times \$11 = \44 . Each person pays: $\$44 \div 3 = \14 .

Problem 45: Sebastian and 2 friends buy 10 snacks at \$6 each. If they split the cost equally, how much does each person pay?

Answer: 20

Explanation: Total cost: $10 \times \$6 = \60 . Each person pays: $\$60 \div 3 = \20 .

Practice Exercises - Grade 3

Exercise 1: Sebastian and 2 friends buy 4 snacks at \$8 each. If they split the cost equally, how much does each person pay?

Exercise 2: Sophia and 4 friends buy 10 snacks at \$3 each. If they split the cost equally, how much does each person pay?

Exercise 3: Emma and 1 friends buy 6 snacks at \$11 each. If they split the cost equally, how much does each person pay?

Exercise 4: Lucas and 1 friends buy 4 snacks at \$10 each. If they split the cost equally, how much does each person pay?

Exercise 5: Ethan and 4 friends buy 5 snacks at \$10 each. If they split the cost equally, how much does each person pay?

Exercise 6: Sofia and 1 friends buy 9 snacks at \$11 each. If they split the cost equally, how much does each person pay?

Exercise 7: James and 2 friends buy 10 snacks at \$8 each. If they split the cost equally, how much does each person pay?

Exercise 8: Alexander and 3 friends buy 10 snacks at \$11 each. If they split the cost equally, how much does each person pay?

Exercise 9: Emily and 3 friends buy 9 snacks at \$3 each. If they split the cost equally, how much does each person pay?

Exercise 10: Emma and 2 friends buy 6 snacks at \$9 each. If they split the cost equally, how much does each person pay?

Exercise 11: Lucas and 1 friends buy 4 snacks at \$9 each. If they split the cost equally, how much does each person pay?

Exercise 12: Henry and 2 friends buy 9 snacks at \$6 each. If they split the cost equally, how much does each person pay?

Exercise 13: Emily and 2 friends buy 7 snacks at \$6 each. If they split the cost equally, how much does each person pay?

Exercise 14: Ethan and 1 friends buy 5 snacks at \$9 each. If they split the cost equally, how much does each person pay?

Exercise 15: Joseph and 3 friends buy 8 snacks at \$11 each. If they split the cost equally, how much does each person pay?

Exercise 16: Mia and 3 friends buy 10 snacks at \$7 each. If they split the cost equally, how much does each person pay?

Exercise 17: Emma and 1 friends buy 7 snacks at \$7 each. If they split the cost equally, how much does each person pay?

Exercise 18: Elizabeth and 2 friends buy 10 snacks at \$8 each. If they split the cost equally, how much does each person pay?

Exercise 19: Benjamin and 2 friends buy 4 snacks at \$12 each. If they split the cost equally, how much does each person pay?

Exercise 20: Benjamin and 2 friends buy 7 snacks at \$12 each. If they split the cost equally, how much does each person pay?

Practice Exercise Answers:

Exercise 1: 10

Exercise 2: 6

Exercise 3: 33

Exercise 4: 20

Exercise 5: 10

Exercise 6: 49

Exercise 7: 26

Exercise 8: 27

Exercise 9: 6

Exercise 10: 18

Exercise 11: 18

Exercise 12: 18

Exercise 13: 14

Exercise 14: 22

Exercise 15: 22

Exercise 16: 17

Exercise 17: 24

Exercise 18: 26

Exercise 19: 16

Exercise 20: 28

Challenge Problems - Grade 3

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Jack and 4 friends buy 4 snacks at \$9 each. If they split the cost equally, how much does each person pay?

Answer: 7

Solution: Total cost: $4 \times \$9 = \36 . Each person pays: $\$36 \div 5 = \7 .

Challenge 2: Liam and 2 friends buy 7 snacks at \$10 each. If they split the cost equally, how much does each person pay?

Answer: 23

Solution: Total cost: $7 \times \$10 = \70 . Each person pays: $\$70 \div 3 = \23 .

Challenge 3: Harper and 2 friends buy 6 snacks at \$4 each. If they split the cost equally, how much does each person pay?

Answer: 8

Solution: Total cost: $6 \times \$4 = \24 . Each person pays: $\$24 \div 3 = \8 .

Challenge 4: Harper and 1 friends buy 7 snacks at \$4 each. If they split the cost equally, how much does each person pay?

Answer: 14

Solution: Total cost: $7 \times \$4 = \28 . Each person pays: $\$28 \div 2 = \14 .

Challenge 5: Elizabeth and 3 friends buy 4 snacks at \$8 each. If they split the cost equally, how much does each person pay?

Answer: 8

Solution: Total cost: $4 \times \$8 = \32 . Each person pays: $\$32 \div 4 = \8 .

Challenge 6: Elizabeth and 2 friends buy 8 snacks at \$11 each. If they split the cost equally, how much does each person pay?

Answer: 29

Solution: Total cost: $8 \times \$11 = \88 . Each person pays: $\$88 \div 3 = \29 .

Challenge 7: David and 4 friends buy 7 snacks at \$5 each. If they split the cost equally, how much does each person pay?

Answer: 7

Solution: Total cost: $7 \times \$5 = \35 . Each person pays: $\$35 \div 5 = \7 .

Challenge 8: Henry and 1 friends buy 6 snacks at \$9 each. If they split the cost equally, how much does each person pay?

Answer: 27

Solution: Total cost: $6 \times \$9 = \54 . Each person pays: $\$54 \div 2 = \27 .

Challenge 9: Harper and 1 friends buy 10 snacks at \$6 each. If they split the cost equally, how much does each person pay?

Answer: 30

Solution: Total cost: $10 \times \$6 = \60 . Each person pays: $\$60 \div 2 = \30 .

Challenge 10: Mason and 3 friends buy 9 snacks at \$11 each. If they split the cost equally, how much does each person pay?

Answer: 24

Solution: Total cost: $9 \times \$11 = \99 . Each person pays: $\$99 \div 4 = \24 .

Challenge 11: David and 1 friends buy 7 snacks at \$10 each. If they split the cost equally, how much does each person pay?

Answer: 35

Solution: Total cost: $7 \times \$10 = \70 . Each person pays: $\$70 \div 2 = \35 .

Challenge 12: Jack and 2 friends buy 7 snacks at \$9 each. If they split the cost equally, how much does each person pay?

Answer: 21

Solution: Total cost: $7 \times \$9 = \63 . Each person pays: $\$63 \div 3 = \21 .

Challenge 13: Ava and 1 friends buy 10 snacks at \$7 each. If they split the cost equally, how much does each person pay?

Answer: 35

Solution: Total cost: $10 \times \$7 = \70 . Each person pays: $\$70 \div 2 = \35 .

Challenge 14: Charlotte and 4 friends buy 7 snacks at \$8 each. If they split the cost equally, how much does each person pay?

Answer: 11

Solution: Total cost: $7 \times \$8 = \56 . Each person pays: $\$56 \div 5 = \11 .

Challenge 15: David and 1 friends buy 5 snacks at \$3 each. If they split the cost equally, how much does each person pay?

Answer: 7

Solution: Total cost: $5 \times \$3 = \15 . Each person pays: $\$15 \div 2 = \7 .

Grade 4 Problems

Problem 1: Emma starts homework at 11:00 AM. They spend 82 minutes on math and 66 minutes on reading. What time do they finish?

Answer: 13:28 AM

Explanation: Total time: $82 + 66 = 148$ minutes = 2 hours 28 minutes. Finish time: $11:00 + 2:28 = 13:28$ AM.

Problem 2: Jack starts homework at 9:00 AM. They spend 85 minutes on math and 41 minutes on reading. What time do they finish?

Answer: 11:06 AM

Explanation: Total time: $85 + 41 = 126$ minutes = 2 hours 6 minutes. Finish time: $9:00 + 2:06 = 11:06$ AM.

Problem 3: Mia starts homework at 10:00 AM. They spend 57 minutes on math and 35 minutes on reading. What time do they finish?

Answer: 11:32 AM

Explanation: Total time: $57 + 35 = 92$ minutes = 1 hours 32 minutes. Finish time: $10:00 + 1:32 = 11:32$ AM.

Problem 4: Benjamin starts homework at 8:00 AM. They spend 49 minutes on math and 46 minutes on reading. What time do they finish?

Answer: 9:35 AM

Explanation: Total time: $49 + 46 = 95$ minutes = 1 hours 35 minutes. Finish time: $8:00 + 1:35 = 9:35$ AM.

Problem 5: Joseph starts homework at 9:00 AM. They spend 94 minutes on math and 86 minutes on reading. What time do they finish?

Answer: 12:00 AM

Explanation: Total time: $94 + 86 = 180$ minutes = 3 hours 0 minutes. Finish time: $9:00 + 3:00 = 12:00$ AM.

Problem 6: Isabella starts homework at 9:00 AM. They spend 75 minutes on math and 53 minutes on reading. What time do they finish?

Answer: 11:08 AM

Explanation: Total time: $75 + 53 = 128$ minutes = 2 hours 8 minutes. Finish time: $9:00 + 2:08 = 11:08$ AM.

Problem 7: Avery starts homework at 10:00 AM. They spend 65 minutes on math and 84 minutes on reading. What time do they finish?

Answer: 12:29 AM

Explanation: Total time: $65 + 84 = 149$ minutes = 2 hours 29 minutes. Finish time: $10:00 + 2:29 = 12:29$ AM.

Problem 8: Lucas starts homework at 10:00 AM. They spend 45 minutes on math and 68 minutes on reading. What time do they finish?

Answer: 11:53 AM

Explanation: Total time: $45 + 68 = 113$ minutes = 1 hours 53 minutes. Finish time: $10:00 + 1:53 = 11:53$ AM.

Problem 9: Charlotte starts homework at 11:00 AM. They spend 95 minutes on math and 58 minutes on reading. What time do they finish?

Answer: 13:33 AM

Explanation: Total time: $95 + 58 = 153$ minutes = 2 hours 33 minutes. Finish time: $11:00 + 2:33 = 13:33$ AM.

Problem 10: Alexander starts homework at 11:00 AM. They spend 72 minutes on math and 63 minutes on reading. What time do they finish?

Answer: 13:15 AM

Explanation: Total time: $72 + 63 = 135$ minutes = 2 hours 15 minutes. Finish time: $11:00 + 2:15 = 13:15$ AM.

Problem 11: Alexander starts homework at 8:00 AM. They spend 99 minutes on math and 39 minutes on reading. What time do they finish?

Answer: 10:18 AM

Explanation: Total time: $99 + 39 = 138$ minutes = 2 hours 18 minutes. Finish time: $8:00 + 2:18 = 10:18$ AM.

Problem 12: Benjamin starts homework at 11:00 AM. They spend 46 minutes on math and 43 minutes on reading. What time do they finish?

Answer: 12:29 AM

Explanation: Total time: $46 + 43 = 89$ minutes = 1 hours 29 minutes. Finish time: $11:00 + 1:29 = 12:29$ AM.

Problem 13: Mason starts homework at 11:00 AM. They spend 106 minutes on math and 55 minutes on reading. What time do they finish?

Answer: 13:41 AM

Explanation: Total time: $106 + 55 = 161$ minutes = 2 hours 41 minutes. Finish time: $11:00 + 2:41 = 13:41$ AM.

Problem 14: Jack starts homework at 9:00 AM. They spend 105 minutes on math and 58 minutes on reading. What time do they finish?

Answer: 11:43 AM

Explanation: Total time: $105 + 58 = 163$ minutes = 2 hours 43 minutes. Finish time: $9:00 + 2:43 = 11:43$ AM.

Problem 15: Avery starts homework at 10:00 AM. They spend 63 minutes on math and 43 minutes on reading. What time do they finish?

Answer: 11:46 AM

Explanation: Total time: $63 + 43 = 106$ minutes = 1 hours 46 minutes. Finish time: $10:00 + 1:46 = 11:46$ AM.

Problem 16: Mason starts homework at 9:00 AM. They spend 73 minutes on math and 90 minutes on reading. What time do they finish?

Answer: 11:43 AM

Explanation: Total time: $73 + 90 = 163$ minutes = 2 hours 43 minutes. Finish time: $9:00 + 2:43 = 11:43$ AM.

Problem 17: Sophia starts homework at 9:00 AM. They spend 66 minutes on math and 74 minutes on reading. What time do they finish?

Answer: 11:20 AM

Explanation: Total time: $66 + 74 = 140$ minutes = 2 hours 20 minutes. Finish time: $9:00 + 2:20 = 11:20$ AM.

Problem 18: Benjamin starts homework at 9:00 AM. They spend 80 minutes on math and 66 minutes on reading. What time do they finish?

Answer: 11:26 AM

Explanation: Total time: $80 + 66 = 146$ minutes = 2 hours 26 minutes. Finish time: $9:00 + 2:26 = 11:26$ AM.

Problem 19: Sebastian starts homework at 10:00 AM. They spend 105 minutes on math and 64 minutes on reading. What time do they finish?

Answer: 12:49 AM

Explanation: Total time: $105 + 64 = 169$ minutes = 2 hours 49 minutes. Finish time: $10:00 + 2:49 = 12:49$ AM.

Problem 20: Liam starts homework at 10:00 AM. They spend 98 minutes on math and 47 minutes on reading. What time do they finish?

Answer: 12:25 AM

Explanation: Total time: $98 + 47 = 145$ minutes = 2 hours 25 minutes. Finish time: $10:00 + 2:25 = 12:25$ AM.

Problem 21: Isabella starts homework at 11:00 AM. They spend 86 minutes on math and 77 minutes on reading. What time do they finish?

Answer: 13:43 AM

Explanation: Total time: $86 + 77 = 163$ minutes = 2 hours 43 minutes. Finish time: $11:00 + 2:43 = 13:43$ AM.

Problem 22: Avery starts homework at 11:00 AM. They spend 92 minutes on math and 38 minutes on reading. What time do they finish?

Answer: 13:10 AM

Explanation: Total time: $92 + 38 = 130$ minutes = 2 hours 10 minutes. Finish time: $11:00 + 2:10 = 13:10$ AM.

Problem 23: Joseph starts homework at 9:00 AM. They spend 56 minutes on math and 85 minutes on reading. What time do they finish?

Answer: 11:21 AM

Explanation: Total time: $56 + 85 = 141$ minutes = 2 hours 21 minutes. Finish time: $9:00 + 2:21 = 11:21$ AM.

Problem 24: Ethan starts homework at 10:00 AM. They spend 104 minutes on math and 43 minutes on reading. What time do they finish?

Answer: 12:27 AM

Explanation: Total time: $104 + 43 = 147$ minutes = 2 hours 27 minutes. Finish time: $10:00 + 2:27 = 12:27$ AM.

Problem 25: Charlotte starts homework at 11:00 AM. They spend 64 minutes on math and 77 minutes on reading. What time do they finish?

Answer: 13:21 AM

Explanation: Total time: $64 + 77 = 141$ minutes = 2 hours 21 minutes. Finish time: $11:00 + 2:21 = 13:21$ AM.

Problem 26: Isabella starts homework at 11:00 AM. They spend 100 minutes on math and 65 minutes on reading. What time do they finish?

Answer: 13:45 AM

Explanation: Total time: $100 + 65 = 165$ minutes = 2 hours 45 minutes. Finish time: $11:00 + 2:45 = 13:45$ AM.

Problem 27: David starts homework at 9:00 AM. They spend 111 minutes on math and 48 minutes on reading. What time do they finish?

Answer: 11:39 AM

Explanation: Total time: $111 + 48 = 159$ minutes = 2 hours 39 minutes. Finish time: $9:00 + 2:39 = 11:39$ AM.

Problem 28: Emily starts homework at 9:00 AM. They spend 94 minutes on math and 66 minutes on reading. What time do they finish?

Answer: 11:40 AM

Explanation: Total time: $94 + 66 = 160$ minutes = 2 hours 40 minutes. Finish time: $9:00 + 2:40 = 11:40$ AM.

Problem 29: Sofia starts homework at 9:00 AM. They spend 78 minutes on math and 70 minutes on reading. What time do they finish?

Answer: 11:28 AM

Explanation: Total time: $78 + 70 = 148$ minutes = 2 hours 28 minutes. Finish time: $9:00 + 2:28 = 11:28$ AM.

Problem 30: Ava starts homework at 10:00 AM. They spend 100 minutes on math and 57 minutes on reading. What time do they finish?

Answer: 12:37 AM

Explanation: Total time: $100 + 57 = 157$ minutes = 2 hours 37 minutes. Finish time: $10:00 + 2:37 = 12:37$ AM.

Problem 31: Avery starts homework at 10:00 AM. They spend 55 minutes on math and 45 minutes on reading. What time do they finish?

Answer: 11:40 AM

Explanation: Total time: $55 + 45 = 100$ minutes = 1 hours 40 minutes. Finish time: $10:00 + 1:40 = 11:40$ AM.

Problem 32: David starts homework at 10:00 AM. They spend 101 minutes on math and 64 minutes on reading. What time do they finish?

Answer: 12:45 AM

Explanation: Total time: $101 + 64 = 165$ minutes = 2 hours 45 minutes. Finish time: $10:00 + 2:45 = 12:45$ AM.

Problem 33: Sofia starts homework at 10:00 AM. They spend 89 minutes on math and 75 minutes on reading. What time do they finish?

Answer: 12:44 AM

Explanation: Total time: $89 + 75 = 164$ minutes = 2 hours 44 minutes. Finish time: $10:00 + 2:44 = 12:44$ AM.

Problem 34: Samuel starts homework at 11:00 AM. They spend 51 minutes on math and 48 minutes on reading. What time do they finish?

Answer: 12:39 AM

Explanation: Total time: $51 + 48 = 99$ minutes = 1 hours 39 minutes. Finish time: $11:00 + 1:39 = 12:39$ AM.

Problem 35: Avery starts homework at 11:00 AM. They spend 55 minutes on math and 75 minutes on reading. What time do they finish?

Answer: 13:10 AM

Explanation: Total time: $55 + 75 = 130$ minutes = 2 hours 10 minutes. Finish time: $11:00 + 2:10 = 13:10$ AM.

Problem 36: Aiden starts homework at 10:00 AM. They spend 110 minutes on math and 31 minutes on reading. What time do they finish?

Answer: 12:21 AM

Explanation: Total time: $110 + 31 = 141$ minutes = 2 hours 21 minutes. Finish time: $10:00 + 2:21 = 12:21$ AM.

Problem 37: Olivia starts homework at 11:00 AM. They spend 72 minutes on math and 79 minutes on reading. What time do they finish?

Answer: 13:31 AM

Explanation: Total time: $72 + 79 = 151$ minutes = 2 hours 31 minutes. Finish time: $11:00 + 2:31 = 13:31$ AM.

Problem 38: Elizabeth starts homework at 10:00 AM. They spend 84 minutes on math and 65 minutes on reading. What time do they finish?

Answer: 12:29 AM

Explanation: Total time: $84 + 65 = 149$ minutes = 2 hours 29 minutes. Finish time: $10:00 + 2:29 = 12:29$ AM.

Problem 39: James starts homework at 8:00 AM. They spend 91 minutes on math and 37 minutes on reading. What time do they finish?

Answer: 10:08 AM

Explanation: Total time: $91 + 37 = 128$ minutes = 2 hours 8 minutes. Finish time: $8:00 + 2:08 = 10:08$ AM.

Problem 40: Joseph starts homework at 9:00 AM. They spend 120 minutes on math and 89 minutes on reading. What time do they finish?

Answer: 12:29 AM

Explanation: Total time: $120 + 89 = 209$ minutes = 3 hours 29 minutes. Finish time: $9:00 + 3:29 = 12:29$ AM.

Problem 41: David starts homework at 10:00 AM. They spend 104 minutes on math and 84 minutes on reading. What time do they finish?

Answer: 13:08 AM

Explanation: Total time: $104 + 84 = 188$ minutes = 3 hours 8 minutes. Finish time: $10:00 + 3:08 = 13:08$ AM.

Problem 42: Amelia starts homework at 8:00 AM. They spend 97 minutes on math and 56 minutes on reading. What time do they finish?

Answer: 10:33 AM

Explanation: Total time: $97 + 56 = 153$ minutes = 2 hours 33 minutes. Finish time: $8:00 + 2:33 = 10:33$ AM.

Problem 43: Sofia starts homework at 8:00 AM. They spend 97 minutes on math and 79 minutes on reading. What time do they finish?

Answer: 10:56 AM

Explanation: Total time: $97 + 79 = 176$ minutes = 2 hours 56 minutes. Finish time: $8:00 + 2:56 = 10:56$ AM.

Problem 44: Emma starts homework at 8:00 AM. They spend 98 minutes on math and 43 minutes on reading. What time do they finish?

Answer: 10:21 AM

Explanation: Total time: $98 + 43 = 141$ minutes = 2 hours 21 minutes. Finish time: $8:00 + 2:21 = 10:21$ AM.

Problem 45: Ava starts homework at 9:00 AM. They spend 82 minutes on math and 88 minutes on reading. What time do they finish?

Answer: 11:50 AM

Explanation: Total time: $82 + 88 = 170$ minutes = 2 hours 50 minutes. Finish time: $9:00 + 2:50 = 11:50$ AM.

Practice Exercises - Grade 4

Exercise 1: Elizabeth starts homework at 8:00 AM. They spend 98 minutes on math and 52 minutes on reading. What time do they finish?

Exercise 2: Noah starts homework at 9:00 AM. They spend 110 minutes on math and 82 minutes on reading. What time do they finish?

Exercise 3: Evelyn starts homework at 10:00 AM. They spend 98 minutes on math and 40 minutes on reading. What time do they finish?

Exercise 4: Lucas starts homework at 8:00 AM. They spend 74 minutes on math and 43 minutes on reading. What time do they finish?

Exercise 5: Avery starts homework at 8:00 AM. They spend 92 minutes on math and 62 minutes on reading. What time do they finish?

Exercise 6: Abigail starts homework at 9:00 AM. They spend 112 minutes on math and 71 minutes on reading. What time do they finish?

Exercise 7: Amelia starts homework at 9:00 AM. They spend 86 minutes on math and 41 minutes on reading. What time do they finish?

Exercise 8: Sophia starts homework at 9:00 AM. They spend 111 minutes on math and 34 minutes on reading. What time do they finish?

Exercise 9: Jack starts homework at 10:00 AM. They spend 118 minutes on math and 80 minutes on reading. What time do they finish?

Exercise 10: Lucas starts homework at 9:00 AM. They spend 86 minutes on math and 56 minutes on reading. What time do they finish?

Exercise 11: Olivia starts homework at 11:00 AM. They spend 99 minutes on math and 76 minutes on reading. What time do they finish?

Exercise 12: Lucas starts homework at 9:00 AM. They spend 61 minutes on math and 44 minutes on reading. What time do they finish?

Exercise 13: Avery starts homework at 10:00 AM. They spend 120 minutes on math and 41 minutes on reading. What time do they finish?

Exercise 14: Samuel starts homework at 10:00 AM. They spend 63 minutes on math and 84 minutes on reading. What time do they finish?

Exercise 15: Abigail starts homework at 9:00 AM. They spend 107 minutes on math and 37 minutes on reading. What time do they finish?

Exercise 16: Sofia starts homework at 11:00 AM. They spend 71 minutes on math and 63 minutes on reading. What time do they finish?

Exercise 17: Samuel starts homework at 11:00 AM. They spend 52 minutes on math and 83 minutes on reading. What time do they finish?

Exercise 18: Sofia starts homework at 9:00 AM. They spend 50 minutes on math and 42 minutes on reading. What time do they finish?

Exercise 19: Alexander starts homework at 8:00 AM. They spend 75 minutes on math and 31 minutes on reading. What time do they finish?

Exercise 20: Joseph starts homework at 11:00 AM. They spend 81 minutes on math and 83 minutes on reading. What time do they finish?

Practice Exercise Answers:

Exercise 1: 10:30 AM

Exercise 2: 12:12 AM

Exercise 3: 12:18 AM

Exercise 4: 9:57 AM

Exercise 5: 10:34 AM

Exercise 6: 12:03 AM

Exercise 7: 11:07 AM

Exercise 8: 11:25 AM

Exercise 9: 13:18 AM

Exercise 10: 11:22 AM

Exercise 11: 13:55 AM

Exercise 12: 10:45 AM

Exercise 13: 12:41 AM

Exercise 14: 12:27 AM

Exercise 15: 11:24 AM

Exercise 16: 13:14 AM

Exercise 17: 13:15 AM

Exercise 18: 10:32 AM

Exercise 19: 9:46 AM

Exercise 20: 13:44 AM

Challenge Problems - Grade 4

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Abigail starts homework at 8:00 AM. They spend 94 minutes on math and 73 minutes on reading. What time do they finish?

Answer: 10:47 AM

Solution: Total time: $94 + 73 = 167$ minutes = 2 hours 47 minutes. Finish time: $8:00 + 2:47 = 10:47$ AM.

Challenge 2: Sofia starts homework at 10:00 AM. They spend 81 minutes on math and 33 minutes on reading. What time do they finish?

Answer: 11:54 AM

Solution: Total time: $81 + 33 = 114$ minutes = 1 hours 54 minutes. Finish time: $10:00 + 1:54 = 11:54$ AM.

Challenge 3: Liam starts homework at 8:00 AM. They spend 49 minutes on math and 54 minutes on reading. What time do they finish?

Answer: 9:43 AM

Solution: Total time: $49 + 54 = 103$ minutes = 1 hours 43 minutes. Finish time: $8:00 + 1:43 = 9:43$ AM.

Challenge 4: James starts homework at 8:00 AM. They spend 117 minutes on math and 46 minutes on reading. What time do they finish?

Answer: 10:43 AM

Solution: Total time: $117 + 46 = 163$ minutes = 2 hours 43 minutes. Finish time: $8:00 + 2:43 = 10:43$ AM.

Challenge 5: James starts homework at 10:00 AM. They spend 94 minutes on math and 85 minutes on reading. What time do they finish?

Answer: 12:59 AM

Solution: Total time: $94 + 85 = 179$ minutes = 2 hours 59 minutes. Finish time: $10:00 + 2:59 = 12:59$ AM.

Challenge 6: Sophia starts homework at 8:00 AM. They spend 95 minutes on math and 47 minutes on reading. What time do they finish?

Answer: 10:22 AM

Solution: Total time: $95 + 47 = 142$ minutes = 2 hours 22 minutes. Finish time: $8:00 + 2:22 = 10:22$ AM.

Challenge 7: Joseph starts homework at 8:00 AM. They spend 65 minutes on math and 82 minutes on reading. What time do they finish?

Answer: 10:27 AM

Solution: Total time: $65 + 82 = 147$ minutes = 2 hours 27 minutes. Finish time: $8:00 + 2:27 = 10:27$ AM.

Challenge 8: Liam starts homework at 9:00 AM. They spend 69 minutes on math and 68 minutes on reading. What time do they finish?

Answer: 11:17 AM

Solution: Total time: $69 + 68 = 137$ minutes = 2 hours 17 minutes. Finish time: $9:00 + 2:17 = 11:17$ AM.

Challenge 9: Amelia starts homework at 11:00 AM. They spend 89 minutes on math and 71 minutes on reading. What time do they finish?

Answer: 13:40 AM

Solution: Total time: $89 + 71 = 160$ minutes = 2 hours 40 minutes. Finish time: $11:00 + 2:40 = 13:40$ AM.

Challenge 10: Alexander starts homework at 11:00 AM. They spend 118 minutes on math and 82 minutes on reading. What time do they finish?

Answer: 14:20 AM

Solution: Total time: $118 + 82 = 200$ minutes = 3 hours 20 minutes. Finish time: $11:00 + 3:20 = 14:20$ AM.

Challenge 11: Alexander starts homework at 8:00 AM. They spend 96 minutes on math and 52 minutes on reading. What time do they finish?

Answer: 10:28 AM

Solution: Total time: $96 + 52 = 148$ minutes = 2 hours 28 minutes. Finish time: $8:00 + 2:28 = 10:28$ AM.

Challenge 12: James starts homework at 10:00 AM. They spend 67 minutes on math and 48 minutes on reading. What time do they finish?

Answer: 11:55 AM

Solution: Total time: $67 + 48 = 115$ minutes = 1 hours 55 minutes. Finish time: $10:00 + 1:55 = 11:55$ AM.

Challenge 13: Mason starts homework at 10:00 AM. They spend 91 minutes on math and 43 minutes on reading. What time do they finish?

Answer: 12:14 AM

Solution: Total time: $91 + 43 = 134$ minutes = 2 hours 14 minutes. Finish time: $10:00 + 2:14 = 12:14$ AM.

Challenge 14: Ava starts homework at 11:00 AM. They spend 107 minutes on math and 47 minutes on reading. What time do they finish?

Answer: 13:34 AM

Solution: Total time: $107 + 47 = 154$ minutes = 2 hours 34 minutes. Finish time: $11:00 + 2:34 = 13:34$ AM.

Challenge 15: James starts homework at 8:00 AM. They spend 48 minutes on math and 80 minutes on reading. What time do they finish?

Answer: 10:08 AM

Solution: Total time: $48 + 80 = 128$ minutes = 2 hours 8 minutes. Finish time: $8:00 + 2:08 = 10:08$ AM.

Grade 5 Problems

Problem 1: A bicycle costs \$74. The sales tax is 10%. What is the total price including tax?

Answer: 81

Explanation: Tax amount: $\$74 \times 10\% = \$74 \times 0.10 = \$7$. Total: $\$74 + \$7 = \$81$.

Problem 2: A bicycle costs \$44. The sales tax is 10%. What is the total price including tax?

Answer: 48

Explanation: Tax amount: $\$44 \times 10\% = \$44 \times 0.10 = \$4$. Total: $\$44 + \$4 = \$48$.

Problem 3: A bicycle costs \$49. The sales tax is 10%. What is the total price including tax?

Answer: 53

Explanation: Tax amount: $\$49 \times 10\% = \$49 \times 0.10 = \$4$. Total: $\$49 + \$4 = \$53$.

Problem 4: A bicycle costs \$47. The sales tax is 10%. What is the total price including tax?

Answer: 51

Explanation: Tax amount: $\$47 \times 10\% = \$47 \times 0.10 = \$4$. Total: $\$47 + \$4 = \$51$.

Problem 5: A bicycle costs \$99. The sales tax is 10%. What is the total price including tax?

Answer: 108

Explanation: Tax amount: $\$99 \times 10\% = \$99 \times 0.10 = \$9$. Total: $\$99 + \$9 = \$108$.

Problem 6: A bicycle costs \$79. The sales tax is 10%. What is the total price including tax?

Answer: 86

Explanation: Tax amount: $\$79 \times 10\% = \$79 \times 0.10 = \$7$. Total: $\$79 + \$7 = \$86$.

Problem 7: A bicycle costs \$41. The sales tax is 10%. What is the total price including tax?

Answer: 45

Explanation: Tax amount: $\$41 \times 10\% = \$41 \times 0.10 = \$4$. Total: $\$41 + \$4 = \$45$.

Problem 8: A bicycle costs \$45. The sales tax is 10%. What is the total price including tax?

Answer: 49

Explanation: Tax amount: $\$45 \times 10\% = \$45 \times 0.10 = \$4$. Total: $\$45 + \$4 = \$49$.

Problem 9: A bicycle costs \$89. The sales tax is 10%. What is the total price including tax?

Answer: 97

Explanation: Tax amount: $\$89 \times 10\% = \$89 \times 0.10 = \$8$. Total: $\$89 + \$8 = \$97$.

Problem 10: A bicycle costs \$47. The sales tax is 10%. What is the total price including tax?

Answer: 51

Explanation: Tax amount: $\$47 \times 10\% = \$47 \times 0.10 = \$4$. Total: $\$47 + \$4 = \$51$.

Problem 11: A bicycle costs \$79. The sales tax is 10%. What is the total price including tax?

Answer: 86

Explanation: Tax amount: $\$79 \times 10\% = \$79 \times 0.10 = \$7$. Total: $\$79 + \$7 = \$86$.

Problem 12: A bicycle costs \$77. The sales tax is 10%. What is the total price including tax?

Answer: 84

Explanation: Tax amount: $\$77 \times 10\% = \$77 \times 0.10 = \$7$. Total: $\$77 + \$7 = \$84$.

Problem 13: A bicycle costs \$89. The sales tax is 10%. What is the total price including tax?

Answer: 97

Explanation: Tax amount: $\$89 \times 10\% = \$89 \times 0.10 = \$8$. Total: $\$89 + \$8 = \$97$.

Problem 14: A bicycle costs \$86. The sales tax is 10%. What is the total price including tax?

Answer: 94

Explanation: Tax amount: $\$86 \times 10\% = \$86 \times 0.10 = \$8$. Total: $\$86 + \$8 = \$94$.

Problem 15: A bicycle costs \$50. The sales tax is 10%. What is the total price including tax?

Answer: 55

Explanation: Tax amount: $\$50 \times 10\% = \$50 \times 0.10 = \$5$. Total: $\$50 + \$5 = \$55$.

Problem 16: A bicycle costs \$45. The sales tax is 10%. What is the total price including tax?

Answer: 49

Explanation: Tax amount: $\$45 \times 10\% = \$45 \times 0.10 = \$4$. Total: $\$45 + \$4 = \$49$.

Problem 17: A bicycle costs \$72. The sales tax is 10%. What is the total price including tax?

Answer: 79

Explanation: Tax amount: $\$72 \times 10\% = \$72 \times 0.10 = \$7$. Total: $\$72 + \$7 = \$79$.

Problem 18: A bicycle costs \$88. The sales tax is 10%. What is the total price including tax?

Answer: 96

Explanation: Tax amount: $\$88 \times 10\% = \$88 \times 0.10 = \$8$. Total: $\$88 + \$8 = \$96$.

Problem 19: A bicycle costs \$87. The sales tax is 10%. What is the total price including tax?

Answer: 95

Explanation: Tax amount: $\$87 \times 10\% = \$87 \times 0.10 = \$8$. Total: $\$87 + \$8 = \$95$.

Problem 20: A bicycle costs \$49. The sales tax is 10%. What is the total price including tax?

Answer: 53

Explanation: Tax amount: $\$49 \times 10\% = \$49 \times 0.10 = \$4$. Total: $\$49 + \$4 = \$53$.

Problem 21: A bicycle costs \$77. The sales tax is 10%. What is the total price including tax?

Answer: 84

Explanation: Tax amount: $\$77 \times 10\% = \$77 \times 0.10 = \$7$. Total: $\$77 + \$7 = \$84$.

Problem 22: A bicycle costs \$42. The sales tax is 10%. What is the total price including tax?

Answer: 46

Explanation: Tax amount: $\$42 \times 10\% = \$42 \times 0.10 = \$4$. Total: $\$42 + \$4 = \$46$.

Problem 23: A bicycle costs \$52. The sales tax is 10%. What is the total price including tax?

Answer: 57

Explanation: Tax amount: $\$52 \times 10\% = \$52 \times 0.10 = \$5$. Total: $\$52 + \$5 = \$57$.

Problem 24: A bicycle costs \$79. The sales tax is 10%. What is the total price including tax?

Answer: 86

Explanation: Tax amount: $\$79 \times 10\% = \$79 \times 0.10 = \$7$. Total: $\$79 + \$7 = \$86$.

Problem 25: A bicycle costs \$79. The sales tax is 10%. What is the total price including tax?

Answer: 86

Explanation: Tax amount: $\$79 \times 10\% = \$79 \times 0.10 = \$7$. Total: $\$79 + \$7 = \$86$.

Problem 26: A bicycle costs \$82. The sales tax is 10%. What is the total price including tax?

Answer: 90

Explanation: Tax amount: $\$82 \times 10\% = \$82 \times 0.10 = \$8$. Total: $\$82 + \$8 = \$90$.

Problem 27: A bicycle costs \$93. The sales tax is 10%. What is the total price including tax?

Answer: 102

Explanation: Tax amount: $\$93 \times 10\% = \$93 \times 0.10 = \$9$. Total: $\$93 + \$9 = \$102$.

Problem 28: A bicycle costs \$86. The sales tax is 10%. What is the total price including tax?

Answer: 94

Explanation: Tax amount: $\$86 \times 10\% = \$86 \times 0.10 = \$8$. Total: $\$86 + \$8 = \$94$.

Problem 29: A bicycle costs \$43. The sales tax is 10%. What is the total price including tax?

Answer: 47

Explanation: Tax amount: $\$43 \times 10\% = \$43 \times 0.10 = \$4$. Total: $\$43 + \$4 = \$47$.

Problem 30: A bicycle costs \$64. The sales tax is 10%. What is the total price including tax?

Answer: 70

Explanation: Tax amount: $\$64 \times 10\% = \$64 \times 0.10 = \$6$. Total: $\$64 + \$6 = \$70$.

Problem 31: A bicycle costs \$58. The sales tax is 10%. What is the total price including tax?

Answer: 63

Explanation: Tax amount: $\$58 \times 10\% = \$58 \times 0.10 = \$5$. Total: $\$58 + \$5 = \$63$.

Problem 32: A bicycle costs \$57. The sales tax is 10%. What is the total price including tax?

Answer: 62

Explanation: Tax amount: $\$57 \times 10\% = \$57 \times 0.10 = \$5$. Total: $\$57 + \$5 = \$62$.

Problem 33: A bicycle costs \$63. The sales tax is 10%. What is the total price including tax?

Answer: 69

Explanation: Tax amount: $\$63 \times 10\% = \$63 \times 0.10 = \$6$. Total: $\$63 + \$6 = \$69$.

Problem 34: A bicycle costs \$54. The sales tax is 10%. What is the total price including tax?

Answer: 59

Explanation: Tax amount: $\$54 \times 10\% = \$54 \times 0.10 = \$5$. Total: $\$54 + \$5 = \$59$.

Problem 35: A bicycle costs \$61. The sales tax is 10%. What is the total price including tax?

Answer: 67

Explanation: Tax amount: $\$61 \times 10\% = \$61 \times 0.10 = \$6$. Total: $\$61 + \$6 = \$67$.

Problem 36: A bicycle costs \$59. The sales tax is 10%. What is the total price including tax?

Answer: 64

Explanation: Tax amount: $\$59 \times 10\% = \$59 \times 0.10 = \$5$. Total: $\$59 + \$5 = \$64$.

Problem 37: A bicycle costs \$90. The sales tax is 10%. What is the total price including tax?

Answer: 99

Explanation: Tax amount: $\$90 \times 10\% = \$90 \times 0.10 = \$9$. Total: $\$90 + \$9 = \$99$.

Problem 38: A bicycle costs \$42. The sales tax is 10%. What is the total price including tax?

Answer: 46

Explanation: Tax amount: $\$42 \times 10\% = \$42 \times 0.10 = \$4$. Total: $\$42 + \$4 = \$46$.

Problem 39: A bicycle costs \$87. The sales tax is 10%. What is the total price including tax?

Answer: 95

Explanation: Tax amount: $\$87 \times 10\% = \$87 \times 0.10 = \$8$. Total: $\$87 + \$8 = \$95$.

Problem 40: A bicycle costs \$99. The sales tax is 10%. What is the total price including tax?

Answer: 108

Explanation: Tax amount: $\$99 \times 10\% = \$99 \times 0.10 = \$9$. Total: $\$99 + \$9 = \$108$.

Problem 41: A bicycle costs \$71. The sales tax is 10%. What is the total price including tax?

Answer: 78

Explanation: Tax amount: $\$71 \times 10\% = \$71 \times 0.10 = \$7$. Total: $\$71 + \$7 = \$78$.

Problem 42: A bicycle costs \$80. The sales tax is 10%. What is the total price including tax?

Answer: 88

Explanation: Tax amount: $\$80 \times 10\% = \$80 \times 0.10 = \$8$. Total: $\$80 + \$8 = \$88$.

Problem 43: A bicycle costs \$54. The sales tax is 10%. What is the total price including tax?

Answer: 59

Explanation: Tax amount: $\$54 \times 10\% = \$54 \times 0.10 = \$5$. Total: $\$54 + \$5 = \$59$.

Problem 44: A bicycle costs \$54. The sales tax is 10%. What is the total price including tax?

Answer: 59

Explanation: Tax amount: $\$54 \times 10\% = \$54 \times 0.10 = \$5$. Total: $\$54 + \$5 = \$59$.

Problem 45: A bicycle costs \$91. The sales tax is 10%. What is the total price including tax?

Answer: 100

Explanation: Tax amount: $\$91 \times 10\% = \$91 \times 0.10 = \$9$. Total: $\$91 + \$9 = \$100$.

Practice Exercises - Grade 5

Exercise 1: A bicycle costs \$83. The sales tax is 10%. What is the total price including tax?

Exercise 2: A bicycle costs \$91. The sales tax is 10%. What is the total price including tax?

Exercise 3: A bicycle costs \$80. The sales tax is 10%. What is the total price including tax?

Exercise 4: A bicycle costs \$54. The sales tax is 10%. What is the total price including tax?

Exercise 5: A bicycle costs \$85. The sales tax is 10%. What is the total price including tax?

Exercise 6: A bicycle costs \$63. The sales tax is 10%. What is the total price including tax?

Exercise 7: A bicycle costs \$78. The sales tax is 10%. What is the total price including tax?

Exercise 8: A bicycle costs \$66. The sales tax is 10%. What is the total price including tax?

Exercise 9: A bicycle costs \$42. The sales tax is 10%. What is the total price including tax?

Exercise 10: A bicycle costs \$50. The sales tax is 10%. What is the total price including tax?

Exercise 11: A bicycle costs \$79. The sales tax is 10%. What is the total price including tax?

Exercise 12: A bicycle costs \$84. The sales tax is 10%. What is the total price including tax?

Exercise 13: A bicycle costs \$94. The sales tax is 10%. What is the total price including tax?

Exercise 14: A bicycle costs \$72. The sales tax is 10%. What is the total price including tax?

Exercise 15: A bicycle costs \$80. The sales tax is 10%. What is the total price including tax?

Exercise 16: A bicycle costs \$99. The sales tax is 10%. What is the total price including tax?

Exercise 17: A bicycle costs \$74. The sales tax is 10%. What is the total price including tax?

Exercise 18: A bicycle costs \$98. The sales tax is 10%. What is the total price including tax?

Exercise 19: A bicycle costs \$95. The sales tax is 10%. What is the total price including tax?

Exercise 20: A bicycle costs \$87. The sales tax is 10%. What is the total price including tax?

Practice Exercise Answers:

Exercise 1: 91

Exercise 2: 100

Exercise 3: 88

Exercise 4: 59

Exercise 5: 93

Exercise 6: 69

Exercise 7: 85

Exercise 8: 72

Exercise 9: 46

Exercise 10: 55

Exercise 11: 86

Exercise 12: 92

Exercise 13: 103

Exercise 14: 79

Exercise 15: 88

Exercise 16: 108

Exercise 17: 81

Exercise 18: 107

Exercise 19: 104

Exercise 20: 95

Challenge Problems - Grade 5

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: A bicycle costs \$48. The sales tax is 10%. What is the total price including tax?

Answer: 52

Solution: Tax amount: $\$48 \times 10\% = \$48 \times 0.10 = \$4$. Total: $\$48 + \$4 = \$52$.

Challenge 2: A bicycle costs \$100. The sales tax is 10%. What is the total price including tax?

Answer: 110

Solution: Tax amount: $\$100 \times 10\% = \$100 \times 0.10 = \$10$. Total: $\$100 + \$10 = \$110$.

Challenge 3: A bicycle costs \$82. The sales tax is 10%. What is the total price including tax?

Answer: 90

Solution: Tax amount: $\$82 \times 10\% = \$82 \times 0.10 = \$8$. Total: $\$82 + \$8 = \$90$.

Challenge 4: A bicycle costs \$95. The sales tax is 10%. What is the total price including tax?

Answer: 104

Solution: Tax amount: $\$95 \times 10\% = \$95 \times 0.10 = \$9$. Total: $\$95 + \$9 = \$104$.

Challenge 5: A bicycle costs \$79. The sales tax is 10%. What is the total price including tax?

Answer: 86

Solution: Tax amount: $\$79 \times 10\% = \$79 \times 0.10 = \$7$. Total: $\$79 + \$7 = \$86$.

Challenge 6: A bicycle costs \$74. The sales tax is 10%. What is the total price including tax?

Answer: 81

Solution: Tax amount: $\$74 \times 10\% = \$74 \times 0.10 = \$7$. Total: $\$74 + \$7 = \$81$.

Challenge 7: A bicycle costs \$63. The sales tax is 10%. What is the total price including tax?

Answer: 69

Solution: Tax amount: $\$63 \times 10\% = \$63 \times 0.10 = \$6$. Total: $\$63 + \$6 = \$69$.

Challenge 8: A bicycle costs \$50. The sales tax is 10%. What is the total price including tax?

Answer: 55

Solution: Tax amount: $\$50 \times 10\% = \$50 \times 0.10 = \$5$. Total: $\$50 + \$5 = \$55$.

Challenge 9: A bicycle costs \$99. The sales tax is 10%. What is the total price including tax?

Answer: 108

Solution: Tax amount: $\$99 \times 10\% = \$99 \times 0.10 = \$9$. Total: $\$99 + \$9 = \$108$.

Challenge 10: A bicycle costs \$40. The sales tax is 10%. What is the total price including tax?

Answer: 44

Solution: Tax amount: $\$40 \times 10\% = \$40 \times 0.10 = \$4$. Total: $\$40 + \$4 = \$44$.

Challenge 11: A bicycle costs \$43. The sales tax is 10%. What is the total price including tax?

Answer: 47

Solution: Tax amount: $\$43 \times 10\% = \$43 \times 0.10 = \$4$. Total: $\$43 + \$4 = \$47$.

Challenge 12: A bicycle costs \$89. The sales tax is 10%. What is the total price including tax?

Answer: 97

Solution: Tax amount: $\$89 \times 10\% = \$89 \times 0.10 = \$8$. Total: $\$89 + \$8 = \$97$.

Challenge 13: A bicycle costs \$67. The sales tax is 10%. What is the total price including tax?

Answer: 73

Solution: Tax amount: $\$67 \times 10\% = \$67 \times 0.10 = \$6$. Total: $\$67 + \$6 = \$73$.

Challenge 14: A bicycle costs \$94. The sales tax is 10%. What is the total price including tax?

Answer: 103

Solution: Tax amount: $\$94 \times 10\% = \$94 \times 0.10 = \$9$. Total: $\$94 + \$9 = \$103$.

Challenge 15: A bicycle costs \$85. The sales tax is 10%. What is the total price including tax?

Answer: 93

Solution: Tax amount: $\$85 \times 10\% = \$85 \times 0.10 = \$8$. Total: $\$85 + \$8 = \$93$.

Grade 6 Problems

Problem 1: David works 5 hours per day for 8 days, earning \$13 per hour. After paying \$23 in expenses, how much money does David have?

Answer: 497

Explanation: Total earned: $\$13 \times 5 \times 8 = \520 . After expenses: $\$520 - \$23 = \$497$.

Problem 2: Sofia works 7 hours per day for 10 days, earning \$9 per hour. After paying \$34 in expenses, how much money does Sofia have?

Answer: 596

Explanation: Total earned: $\$9 \times 7 \times 10 = \630 . After expenses: $\$630 - \$34 = \$596$.

Problem 3: Charlotte works 6 hours per day for 8 days, earning \$10 per hour. After paying \$39 in expenses, how much money does Charlotte have?

Answer: 441

Explanation: Total earned: $\$10 \times 6 \times 8 = \480 . After expenses: $\$480 - \$39 = \$441$.

Problem 4: Avery works 7 hours per day for 10 days, earning \$8 per hour. After paying \$20 in expenses, how much money does Avery have?

Answer: 540

Explanation: Total earned: $\$8 \times 7 \times 10 = \560 . After expenses: $\$560 - \$20 = \$540$.

Problem 5: David works 5 hours per day for 7 days, earning \$11 per hour. After paying \$34 in expenses, how much money does David have?

Answer: 351

Explanation: Total earned: $\$11 \times 5 \times 7 = \385 . After expenses: $\$385 - \$34 = \$351$.

Problem 6: Avery works 5 hours per day for 10 days, earning \$8 per hour. After paying \$35 in expenses, how much money does Avery have?

Answer: 365

Explanation: Total earned: $\$8 \times 5 \times 10 = \400 . After expenses: $\$400 - \$35 = \$365$.

Problem 7: Evelyn works 4 hours per day for 7 days, earning \$12 per hour. After paying \$37 in expenses, how much money does Evelyn have?

Answer: 299

Explanation: Total earned: $\$12 \times 4 \times 7 = \336 . After expenses: $\$336 - \$37 = \$299$.

Problem 8: Sophia works 4 hours per day for 10 days, earning \$11 per hour. After paying \$49 in expenses, how much money does Sophia have?

Answer: 391

Explanation: Total earned: $\$11 \times 4 \times 10 = \440 . After expenses: $\$440 - \$49 = \$391$.

Problem 9: Joseph works 8 hours per day for 5 days, earning \$8 per hour. After paying \$43 in expenses, how much money does Joseph have?

Answer: 277

Explanation: Total earned: $\$8 \times 8 \times 5 = \320 . After expenses: $\$320 - \$43 = \$277$.

Problem 10: Jack works 7 hours per day for 7 days, earning \$8 per hour. After paying \$41 in expenses, how much money does Jack have?

Answer: 351

Explanation: Total earned: $\$8 \times 7 \times 7 = \392 . After expenses: $\$392 - \$41 = \$351$.

Problem 11: Sofia works 4 hours per day for 8 days, earning \$14 per hour. After paying \$45 in expenses, how much money does Sofia have?

Answer: 403

Explanation: Total earned: $\$14 \times 4 \times 8 = \448 . After expenses: $\$448 - \$45 = \$403$.

Problem 12: Charlotte works 5 hours per day for 7 days, earning \$15 per hour. After paying \$40 in expenses, how much money does Charlotte have?

Answer: 485

Explanation: Total earned: $\$15 \times 5 \times 7 = \525 . After expenses: $\$525 - \$40 = \$485$.

Problem 13: Noah works 6 hours per day for 7 days, earning \$15 per hour. After paying \$44 in expenses, how much money does Noah have?

Answer: 586

Explanation: Total earned: $\$15 \times 6 \times 7 = \630 . After expenses: $\$630 - \$44 = \$586$.

Problem 14: Evelyn works 6 hours per day for 6 days, earning \$14 per hour. After paying \$26 in expenses, how much money does Evelyn have?

Answer: 478

Explanation: Total earned: $\$14 \times 6 \times 6 = \504 . After expenses: $\$504 - \$26 = \$478$.

Problem 15: Alexander works 8 hours per day for 9 days, earning \$10 per hour. After paying \$38 in expenses, how much money does Alexander have?

Answer: 682

Explanation: Total earned: $\$10 \times 8 \times 9 = \720 . After expenses: $\$720 - \$38 = \$682$.

Problem 16: Emily works 5 hours per day for 7 days, earning \$11 per hour. After paying \$45 in expenses, how much money does Emily have?

Answer: 340

Explanation: Total earned: $\$11 \times 5 \times 7 = \385 . After expenses: $\$385 - \$45 = \$340$.

Problem 17: Mason works 6 hours per day for 10 days, earning \$13 per hour. After paying \$27 in expenses, how much money does Mason have?

Answer: 753

Explanation: Total earned: $\$13 \times 6 \times 10 = \780 . After expenses: $\$780 - \$27 = \$753$.

Problem 18: Harper works 5 hours per day for 8 days, earning \$14 per hour. After paying \$33 in expenses, how much money does Harper have?

Answer: 527

Explanation: Total earned: $\$14 \times 5 \times 8 = \560 . After expenses: $\$560 - \$33 = \$527$.

Problem 19: Sebastian works 4 hours per day for 6 days, earning \$11 per hour. After paying \$45 in expenses, how much money does Sebastian have?

Answer: 219

Explanation: Total earned: $\$11 \times 4 \times 6 = \264 . After expenses: $\$264 - \$45 = \$219$.

Problem 20: Alexander works 4 hours per day for 7 days, earning \$8 per hour. After paying \$22 in expenses, how much money does Alexander have?

Answer: 202

Explanation: Total earned: $\$8 \times 4 \times 7 = \224 . After expenses: $\$224 - \$22 = \$202$.

Problem 21: Sebastian works 7 hours per day for 10 days, earning \$15 per hour. After paying \$42 in expenses, how much money does Sebastian have?

Answer: 1008

Explanation: Total earned: $\$15 \times 7 \times 10 = \1050 . After expenses: $\$1050 - \$42 = \$1008$.

Problem 22: James works 5 hours per day for 9 days, earning \$9 per hour. After paying \$50 in expenses, how much money does James have?

Answer: 355

Explanation: Total earned: $\$9 \times 5 \times 9 = \405 . After expenses: $\$405 - \$50 = \$355$.

Problem 23: Alexander works 7 hours per day for 8 days, earning \$12 per hour. After paying \$29 in expenses, how much money does Alexander have?

Answer: 643

Explanation: Total earned: $\$12 \times 7 \times 8 = \672 . After expenses: $\$672 - \$29 = \$643$.

Problem 24: David works 7 hours per day for 6 days, earning \$9 per hour. After paying \$24 in expenses, how much money does David have?

Answer: 354

Explanation: Total earned: $\$9 \times 7 \times 6 = \378 . After expenses: $\$378 - \$24 = \$354$.

Problem 25: Benjamin works 6 hours per day for 9 days, earning \$8 per hour. After paying \$27 in expenses, how much money does Benjamin have?

Answer: 405

Explanation: Total earned: $\$8 \times 6 \times 9 = \432 . After expenses: $\$432 - \$27 = \$405$.

Problem 26: David works 5 hours per day for 7 days, earning \$9 per hour. After paying \$41 in expenses, how much money does David have?

Answer: 274

Explanation: Total earned: $\$9 \times 5 \times 7 = \315 . After expenses: $\$315 - \$41 = \$274$.

Problem 27: Sofia works 8 hours per day for 6 days, earning \$8 per hour. After paying \$50 in expenses, how much money does Sofia have?

Answer: 334

Explanation: Total earned: $\$8 \times 8 \times 6 = \384 . After expenses: $\$384 - \$50 = \$334$.

Problem 28: Samuel works 5 hours per day for 6 days, earning \$9 per hour. After paying \$42 in expenses, how much money does Samuel have?

Answer: 228

Explanation: Total earned: $\$9 \times 5 \times 6 = \270 . After expenses: $\$270 - \$42 = \$228$.

Problem 29: Isabella works 5 hours per day for 7 days, earning \$12 per hour. After paying \$32 in expenses, how much money does Isabella have?

Answer: 388

Explanation: Total earned: $\$12 \times 5 \times 7 = \420 . After expenses: $\$420 - \$32 = \$388$.

Problem 30: Henry works 8 hours per day for 5 days, earning \$14 per hour. After paying \$48 in expenses, how much money does Henry have?

Answer: 512

Explanation: Total earned: $\$14 \times 8 \times 5 = \560 . After expenses: $\$560 - \$48 = \$512$.

Problem 31: Ava works 8 hours per day for 9 days, earning \$14 per hour. After paying \$21 in expenses, how much money does Ava have?

Answer: 987

Explanation: Total earned: $\$14 \times 8 \times 9 = \1008 . After expenses: $\$1008 - \$21 = \$987$.

Problem 32: Mason works 4 hours per day for 6 days, earning \$8 per hour. After paying \$31 in expenses, how much money does Mason have?

Answer: 161

Explanation: Total earned: $\$8 \times 4 \times 6 = \192 . After expenses: $\$192 - \$31 = \$161$.

Problem 33: Alexander works 6 hours per day for 8 days, earning \$8 per hour. After paying \$34 in expenses, how much money does Alexander have?

Answer: 350

Explanation: Total earned: $\$8 \times 6 \times 8 = \384 . After expenses: $\$384 - \$34 = \$350$.

Problem 34: Isabella works 8 hours per day for 10 days, earning \$13 per hour. After paying \$21 in expenses, how much money does Isabella have?

Answer: 1019

Explanation: Total earned: $\$13 \times 8 \times 10 = \1040 . After expenses: $\$1040 - \$21 = \$1019$.

Problem 35: Evelyn works 7 hours per day for 7 days, earning \$15 per hour. After paying \$44 in expenses, how much money does Evelyn have?

Answer: 691

Explanation: Total earned: $\$15 \times 7 \times 7 = \735 . After expenses: $\$735 - \$44 = \$691$.

Problem 36: James works 4 hours per day for 10 days, earning \$14 per hour. After paying \$43 in expenses, how much money does James have?

Answer: 517

Explanation: Total earned: $\$14 \times 4 \times 10 = \560 . After expenses: $\$560 - \$43 = \$517$.

Problem 37: Olivia works 7 hours per day for 5 days, earning \$15 per hour. After paying \$41 in expenses, how much money does Olivia have?

Answer: 484

Explanation: Total earned: $\$15 \times 7 \times 5 = \525 . After expenses: $\$525 - \$41 = \$484$.

Problem 38: Emily works 6 hours per day for 10 days, earning \$14 per hour. After paying \$29 in expenses, how much money does Emily have?

Answer: 811

Explanation: Total earned: $\$14 \times 6 \times 10 = \840 . After expenses: $\$840 - \$29 = \$811$.

Problem 39: Emily works 6 hours per day for 5 days, earning \$10 per hour. After paying \$44 in expenses, how much money does Emily have?

Answer: 256

Explanation: Total earned: $\$10 \times 6 \times 5 = \300 . After expenses: $\$300 - \$44 = \$256$.

Problem 40: Aiden works 7 hours per day for 7 days, earning \$11 per hour. After paying \$28 in expenses, how much money does Aiden have?

Answer: 511

Explanation: Total earned: $\$11 \times 7 \times 7 = \539 . After expenses: $\$539 - \$28 = \$511$.

Problem 41: Charlotte works 8 hours per day for 6 days, earning \$15 per hour. After paying \$30 in expenses, how much money does Charlotte have?

Answer: 690

Explanation: Total earned: $\$15 \times 8 \times 6 = \720 . After expenses: $\$720 - \$30 = \$690$.

Problem 42: Sebastian works 7 hours per day for 5 days, earning \$11 per hour. After paying \$23 in expenses, how much money does Sebastian have?

Answer: 362

Explanation: Total earned: $\$11 \times 7 \times 5 = \385 . After expenses: $\$385 - \$23 = \$362$.

Problem 43: Emily works 5 hours per day for 5 days, earning \$8 per hour. After paying \$28 in expenses, how much money does Emily have?

Answer: 172

Explanation: Total earned: $\$8 \times 5 \times 5 = \200 . After expenses: $\$200 - \$28 = \$172$.

Problem 44: Aiden works 7 hours per day for 6 days, earning \$12 per hour. After paying \$36 in expenses, how much money does Aiden have?

Answer: 468

Explanation: Total earned: $\$12 \times 7 \times 6 = \504 . After expenses: $\$504 - \$36 = \$468$.

Problem 45: David works 8 hours per day for 9 days, earning \$14 per hour. After paying \$44 in expenses, how much money does David have?

Answer: 964

Explanation: Total earned: $\$14 \times 8 \times 9 = \1008 . After expenses: $\$1008 - \$44 = \$964$.

Practice Exercises - Grade 6

Exercise 1: Harper works 5 hours per day for 5 days, earning \$13 per hour. After paying \$44 in expenses, how much money does Harper have?

Exercise 2: Sebastian works 7 hours per day for 5 days, earning \$11 per hour. After paying \$21 in expenses, how much money does Sebastian have?

Exercise 3: Evelyn works 7 hours per day for 9 days, earning \$14 per hour. After paying \$35 in expenses, how much money does Evelyn have?

Exercise 4: Avery works 7 hours per day for 10 days, earning \$12 per hour. After paying \$27 in expenses, how much money does Avery have?

Exercise 5: Mason works 5 hours per day for 9 days, earning \$12 per hour. After paying \$35 in expenses, how much money does Mason have?

Exercise 6: Amelia works 6 hours per day for 10 days, earning \$9 per hour. After paying \$38 in expenses, how much money does Amelia have?

Exercise 7: Jack works 6 hours per day for 10 days, earning \$15 per hour. After paying \$27 in expenses, how much money does Jack have?

Exercise 8: Joseph works 8 hours per day for 5 days, earning \$8 per hour. After paying \$28 in expenses, how much money does Joseph have?

Exercise 9: Lucas works 4 hours per day for 10 days, earning \$12 per hour. After paying \$41 in expenses, how much money does Lucas have?

Exercise 10: Liam works 5 hours per day for 8 days, earning \$14 per hour. After paying \$40 in expenses, how much money does Liam have?

Exercise 11: Lucas works 6 hours per day for 9 days, earning \$10 per hour. After paying \$24 in expenses, how much money does Lucas have?

Exercise 12: Mason works 4 hours per day for 8 days, earning \$15 per hour. After paying \$31 in expenses, how much money does Mason have?

Exercise 13: Henry works 5 hours per day for 8 days, earning \$14 per hour. After paying \$20 in expenses, how much money does Henry have?

Exercise 14: Abigail works 7 hours per day for 5 days, earning \$9 per hour. After paying \$38 in expenses, how much money does Abigail have?

Exercise 15: Mason works 4 hours per day for 9 days, earning \$11 per hour. After paying \$49 in expenses, how much money does Mason have?

Exercise 16: Sofia works 7 hours per day for 10 days, earning \$11 per hour. After paying \$42 in expenses, how much money does Sofia have?

Exercise 17: Ava works 5 hours per day for 7 days, earning \$13 per hour. After paying \$32 in expenses, how much money does Ava have?

Exercise 18: James works 8 hours per day for 7 days, earning \$12 per hour. After paying \$43 in expenses, how much money does James have?

Exercise 19: Aiden works 4 hours per day for 9 days, earning \$9 per hour. After paying \$47 in expenses, how much money does Aiden have?

Exercise 20: Alexander works 5 hours per day for 7 days, earning \$9 per hour. After paying \$32 in expenses, how much money does Alexander have?

Practice Exercise Answers:

Exercise 1: 281

Exercise 2: 364

Exercise 3: 847

Exercise 4: 813

Exercise 5: 505

Exercise 6: 502

Exercise 7: 873

Exercise 8: 292

Exercise 9: 439

Exercise 10: 520

Exercise 11: 516

Exercise 12: 449

Exercise 13: 540

Exercise 14: 277

Exercise 15: 347

Exercise 16: 728

Exercise 17: 423

Exercise 18: 629

Exercise 19: 277

Exercise 20: 283

Challenge Problems - Grade 6

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Ava works 7 hours per day for 9 days, earning \$13 per hour. After paying \$26 in expenses, how much money does Ava have?

Answer: 793

Solution: Total earned: $\$13 \times 7 \times 9 = \819 . After expenses: $\$819 - \$26 = \$793$.

Challenge 2: Noah works 4 hours per day for 10 days, earning \$14 per hour. After paying \$46 in expenses, how much money does Noah have?

Answer: 514

Solution: Total earned: $\$14 \times 4 \times 10 = \560 . After expenses: $\$560 - \$46 = \$514$.

Challenge 3: Charlotte works 5 hours per day for 5 days, earning \$10 per hour. After paying \$35 in expenses, how much money does Charlotte have?

Answer: 215

Solution: Total earned: $\$10 \times 5 \times 5 = \250 . After expenses: $\$250 - \$35 = \$215$.

Challenge 4: Amelia works 8 hours per day for 10 days, earning \$14 per hour. After paying \$24 in expenses, how much money does Amelia have?

Answer: 1096

Solution: Total earned: $\$14 \times 8 \times 10 = \1120 . After expenses: $\$1120 - \$24 = \$1096$.

Challenge 5: Sebastian works 7 hours per day for 8 days, earning \$13 per hour. After paying \$23 in expenses, how much money does Sebastian have?

Answer: 705

Solution: Total earned: $\$13 \times 7 \times 8 = \728 . After expenses: $\$728 - \$23 = \$705$.

Challenge 6: Sebastian works 4 hours per day for 10 days, earning \$12 per hour. After paying \$38 in expenses, how much money does Sebastian have?

Answer: 442

Solution: Total earned: $\$12 \times 4 \times 10 = \480 . After expenses: $\$480 - \$38 = \$442$.

Challenge 7: Mason works 5 hours per day for 10 days, earning \$10 per hour. After paying \$39 in expenses, how much money does Mason have?

Answer: 461

Solution: Total earned: $\$10 \times 5 \times 10 = \500 . After expenses: $\$500 - \$39 = \$461$.

Challenge 8: Sofia works 7 hours per day for 8 days, earning \$15 per hour. After paying \$49 in expenses, how much money does Sofia have?

Answer: 791

Solution: Total earned: $\$15 \times 7 \times 8 = \840 . After expenses: $\$840 - \$49 = \$791$.

Challenge 9: Mia works 7 hours per day for 5 days, earning \$8 per hour. After paying \$46 in expenses, how much money does Mia have?

Answer: 234

Solution: Total earned: $\$8 \times 7 \times 5 = \280 . After expenses: $\$280 - \$46 = \$234$.

Challenge 10: Sofia works 4 hours per day for 9 days, earning \$11 per hour. After paying \$44 in expenses, how much money does Sofia have?

Answer: 352

Solution: Total earned: $\$11 \times 4 \times 9 = \396 . After expenses: $\$396 - \$44 = \$352$.

Challenge 11: Noah works 4 hours per day for 5 days, earning \$11 per hour. After paying \$22 in expenses, how much money does Noah have?

Answer: 198

Solution: Total earned: $\$11 \times 4 \times 5 = \220 . After expenses: $\$220 - \$22 = \$198$.

Challenge 12: Avery works 8 hours per day for 8 days, earning \$8 per hour. After paying \$47 in expenses, how much money does Avery have?

Answer: 465

Solution: Total earned: $\$8 \times 8 \times 8 = \512 . After expenses: $\$512 - \$47 = \$465$.

Challenge 13: Mia works 8 hours per day for 7 days, earning \$8 per hour. After paying \$46 in expenses, how much money does Mia have?

Answer: 402

Solution: Total earned: $\$8 \times 8 \times 7 = \448 . After expenses: $\$448 - \$46 = \$402$.

Challenge 14: Ethan works 7 hours per day for 5 days, earning \$11 per hour. After paying \$35 in expenses, how much money does Ethan have?

Answer: 350

Solution: Total earned: $\$11 \times 7 \times 5 = \385 . After expenses: $\$385 - \$35 = \$350$.

Challenge 15: Benjamin works 4 hours per day for 7 days, earning \$14 per hour. After paying \$39 in expenses, how much money does Benjamin have?

Answer: 353

Solution: Total earned: $\$14 \times 4 \times 7 = \392 . After expenses: $\$392 - \$39 = \$353$.

Chapter 6: Measurement

Measurement is fundamental to daily life. Students will practice converting units, calculating with different measurement systems, and solving real-world problems involving length, weight, volume, and time.

Grade 1 Problems

Problem 1: Ethan has a pencil that is 8 inches long and another that is 5 inches long. How much longer is the longer pencil?

Answer: 3

Explanation: Difference: $|8 - 5| = 3$ inches.

Problem 2: Elizabeth has a pencil that is 3 inches long and another that is 11 inches long. How much longer is the longer pencil?

Answer: 8

Explanation: Difference: $|3 - 11| = 8$ inches.

Problem 3: Samuel has a pencil that is 7 inches long and another that is 12 inches long. How much longer is the longer pencil?

Answer: 5

Explanation: Difference: $|7 - 12| = 5$ inches.

Problem 4: Isabella has a pencil that is 9 inches long and another that is 12 inches long. How much longer is the longer pencil?

Answer: 3

Explanation: Difference: $|9 - 12| = 3$ inches.

Problem 5: Benjamin has a pencil that is 10 inches long and another that is 4 inches long. How much longer is the longer pencil?

Answer: 6

Explanation: Difference: $|10 - 4| = 6$ inches.

Problem 6: Emily has a pencil that is 7 inches long and another that is 4 inches long. How much longer is the longer pencil?

Answer: 3

Explanation: Difference: $|7 - 4| = 3$ inches.

Problem 7: Ethan has a pencil that is 3 inches long and another that is 7 inches long. How much longer is the longer pencil?

Answer: 4

Explanation: Difference: $|3 - 7| = 4$ inches.

Problem 8: Emma has a pencil that is 11 inches long and another that is 3 inches long. How much longer is the longer pencil?

Answer: 8

Explanation: Difference: $|11 - 3| = 8$ inches.

Problem 9: Mia has a pencil that is 3 inches long and another that is 7 inches long. How much longer is the longer pencil?

Answer: 4

Explanation: Difference: $|3 - 7| = 4$ inches.

Problem 10: Liam has a pencil that is 10 inches long and another that is 7 inches long. How much longer is the longer pencil?

Answer: 3

Explanation: Difference: $|10 - 7| = 3$ inches.

Problem 11: James has a pencil that is 11 inches long and another that is 10 inches long. How much longer is the longer pencil?

Answer: 1

Explanation: Difference: $|11 - 10| = 1$ inches.

Problem 12: Avery has a pencil that is 11 inches long and another that is 3 inches long. How much longer is the longer pencil?

Answer: 8

Explanation: Difference: $|11 - 3| = 8$ inches.

Problem 13: Emily has a pencil that is 8 inches long and another that is 10 inches long. How much longer is the longer pencil?

Answer: 2

Explanation: Difference: $|8 - 10| = 2$ inches.

Problem 14: Liam has a pencil that is 10 inches long and another that is 3 inches long. How much longer is the longer pencil?

Answer: 7

Explanation: Difference: $|10 - 3| = 7$ inches.

Problem 15: Mia has a pencil that is 8 inches long and another that is 11 inches long. How much longer is the longer pencil?

Answer: 3

Explanation: Difference: $|8 - 11| = 3$ inches.

Problem 16: Noah has a pencil that is 10 inches long and another that is 7 inches long. How much longer is the longer pencil?

Answer: 3

Explanation: Difference: $|10 - 7| = 3$ inches.

Problem 17: Elizabeth has a pencil that is 9 inches long and another that is 12 inches long. How much longer is the longer pencil?

Answer: 3

Explanation: Difference: $|9 - 12| = 3$ inches.

Problem 18: Avery has a pencil that is 6 inches long and another that is 3 inches long. How much longer is the longer pencil?

Answer: 3

Explanation: Difference: $|6 - 3| = 3$ inches.

Problem 19: Mason has a pencil that is 5 inches long and another that is 12 inches long. How much longer is the longer pencil?

Answer: 7

Explanation: Difference: $|5 - 12| = 7$ inches.

Problem 20: Samuel has a pencil that is 4 inches long and another that is 4 inches long. How much longer is the longer pencil?

Answer: 0

Explanation: Difference: $|4 - 4| = 0$ inches.

Problem 21: Mia has a pencil that is 6 inches long and another that is 11 inches long. How much longer is the longer pencil?

Answer: 5

Explanation: Difference: $|6 - 11| = 5$ inches.

Problem 22: Joseph has a pencil that is 11 inches long and another that is 3 inches long. How much longer is the longer pencil?

Answer: 8

Explanation: Difference: $|11 - 3| = 8$ inches.

Problem 23: Isabella has a pencil that is 4 inches long and another that is 11 inches long. How much longer is the longer pencil?

Answer: 7

Explanation: Difference: $|4 - 11| = 7$ inches.

Problem 24: Jack has a pencil that is 10 inches long and another that is 3 inches long. How much longer is the longer pencil?

Answer: 7

Explanation: Difference: $|10 - 3| = 7$ inches.

Problem 25: Elizabeth has a pencil that is 4 inches long and another that is 6 inches long. How much longer is the longer pencil?

Answer: 2

Explanation: Difference: $|4 - 6| = 2$ inches.

Problem 26: James has a pencil that is 3 inches long and another that is 12 inches long. How much longer is the longer pencil?

Answer: 9

Explanation: Difference: $|3 - 12| = 9$ inches.

Problem 27: Isabella has a pencil that is 8 inches long and another that is 6 inches long. How much longer is the longer pencil?

Answer: 2

Explanation: Difference: $|8 - 6| = 2$ inches.

Problem 28: Amelia has a pencil that is 12 inches long and another that is 6 inches long. How much longer is the longer pencil?

Answer: 6

Explanation: Difference: $|12 - 6| = 6$ inches.

Problem 29: Elizabeth has a pencil that is 6 inches long and another that is 7 inches long. How much longer is the longer pencil?

Answer: 1

Explanation: Difference: $|6 - 7| = 1$ inches.

Problem 30: David has a pencil that is 4 inches long and another that is 7 inches long. How much longer is the longer pencil?

Answer: 3

Explanation: Difference: $|4 - 7| = 3$ inches.

Problem 31: Liam has a pencil that is 11 inches long and another that is 4 inches long. How much longer is the longer pencil?

Answer: 7

Explanation: Difference: $|11 - 4| = 7$ inches.

Problem 32: Aiden has a pencil that is 11 inches long and another that is 11 inches long. How much longer is the longer pencil?

Answer: 0

Explanation: Difference: $|11 - 11| = 0$ inches.

Problem 33: Aiden has a pencil that is 10 inches long and another that is 8 inches long. How much longer is the longer pencil?

Answer: 2

Explanation: Difference: $|10 - 8| = 2$ inches.

Problem 34: Sebastian has a pencil that is 4 inches long and another that is 4 inches long. How much longer is the longer pencil?

Answer: 0

Explanation: Difference: $|4 - 4| = 0$ inches.

Problem 35: Samuel has a pencil that is 3 inches long and another that is 6 inches long. How much longer is the longer pencil?

Answer: 3

Explanation: Difference: $|3 - 6| = 3$ inches.

Problem 36: David has a pencil that is 8 inches long and another that is 8 inches long. How much longer is the longer pencil?

Answer: 0

Explanation: Difference: $|8 - 8| = 0$ inches.

Problem 37: Ava has a pencil that is 5 inches long and another that is 7 inches long. How much longer is the longer pencil?

Answer: 2

Explanation: Difference: $|5 - 7| = 2$ inches.

Problem 38: Mia has a pencil that is 10 inches long and another that is 12 inches long. How much longer is the longer pencil?

Answer: 2

Explanation: Difference: $|10 - 12| = 2$ inches.

Problem 39: Sofia has a pencil that is 10 inches long and another that is 11 inches long. How much longer is the longer pencil?

Answer: 1

Explanation: Difference: $|10 - 11| = 1$ inches.

Problem 40: Emily has a pencil that is 5 inches long and another that is 6 inches long. How much longer is the longer pencil?

Answer: 1

Explanation: Difference: $|5 - 6| = 1$ inches.

Problem 41: Joseph has a pencil that is 11 inches long and another that is 11 inches long. How much longer is the longer pencil?

Answer: 0

Explanation: Difference: $|11 - 11| = 0$ inches.

Problem 42: Avery has a pencil that is 11 inches long and another that is 8 inches long. How much longer is the longer pencil?

Answer: 3

Explanation: Difference: $|11 - 8| = 3$ inches.

Problem 43: Amelia has a pencil that is 3 inches long and another that is 7 inches long. How much longer is the longer pencil?

Answer: 4

Explanation: Difference: $|3 - 7| = 4$ inches.

Problem 44: Sophia has a pencil that is 7 inches long and another that is 11 inches long. How much longer is the longer pencil?

Answer: 4

Explanation: Difference: $|7 - 11| = 4$ inches.

Problem 45: Avery has a pencil that is 11 inches long and another that is 12 inches long. How much longer is the longer pencil?

Answer: 1

Explanation: Difference: $|11 - 12| = 1$ inches.

Practice Exercises - Grade 1

Exercise 1: Mason has a pencil that is 10 inches long and another that is 3 inches long. How much longer is the longer pencil?

Exercise 2: Benjamin has a pencil that is 6 inches long and another that is 10 inches long. How much longer is the longer pencil?

Exercise 3: Isabella has a pencil that is 9 inches long and another that is 12 inches long. How much longer is the longer pencil?

Exercise 4: Mia has a pencil that is 4 inches long and another that is 5 inches long. How much longer is the longer pencil?

Exercise 5: Joseph has a pencil that is 6 inches long and another that is 7 inches long. How much longer is the longer pencil?

Exercise 6: Evelyn has a pencil that is 12 inches long and another that is 8 inches long. How much longer is the longer pencil?

Exercise 7: Elizabeth has a pencil that is 5 inches long and another that is 4 inches long. How much longer is the longer pencil?

Exercise 8: Lucas has a pencil that is 12 inches long and another that is 6 inches long. How much longer is the longer pencil?

Exercise 9: Elizabeth has a pencil that is 10 inches long and another that is 4 inches long. How much longer is the longer pencil?

Exercise 10: Isabella has a pencil that is 12 inches long and another that is 10 inches long. How much longer is the longer pencil?

Exercise 11: Jack has a pencil that is 7 inches long and another that is 10 inches long. How much longer is the longer pencil?

Exercise 12: David has a pencil that is 8 inches long and another that is 5 inches long. How much longer is the longer pencil?

Exercise 13: Sophia has a pencil that is 5 inches long and another that is 12 inches long. How much longer is the longer pencil?

Exercise 14: Harper has a pencil that is 5 inches long and another that is 12 inches long. How much longer is the longer pencil?

Exercise 15: Elizabeth has a pencil that is 7 inches long and another that is 7 inches long. How much longer is the longer pencil?

Exercise 16: Alexander has a pencil that is 5 inches long and another that is 11 inches long. How much longer is the longer pencil?

Exercise 17: Mason has a pencil that is 11 inches long and another that is 9 inches long. How much longer is the longer pencil?

Exercise 18: Benjamin has a pencil that is 5 inches long and another that is 4 inches long. How much longer is the longer pencil?

Exercise 19: Mason has a pencil that is 6 inches long and another that is 3 inches long. How much longer is the longer pencil?

Exercise 20: James has a pencil that is 12 inches long and another that is 5 inches long. How much longer is the longer pencil?

Practice Exercise Answers:

Exercise 1: 7

Exercise 2: 4

Exercise 3: 3

Exercise 4: 1

Exercise 5: 1

Exercise 6: 4

Exercise 7: 1

Exercise 8: 6

Exercise 9: 6

Exercise 10: 2

Exercise 11: 3

Exercise 12: 3

Exercise 13: 7

Exercise 14: 7

Exercise 15: 0

Exercise 16: 6

Exercise 17: 2

Exercise 18: 1

Exercise 19: 3

Exercise 20: 7

Challenge Problems - Grade 1

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Samuel has a pencil that is 8 inches long and another that is 8 inches long. How much longer is the longer pencil?

Answer: 0

Solution: Difference: $|8 - 8| = 0$ inches.

Challenge 2: Joseph has a pencil that is 5 inches long and another that is 5 inches long. How much longer is the longer pencil?

Answer: 0

Solution: Difference: $|5 - 5| = 0$ inches.

Challenge 3: Amelia has a pencil that is 12 inches long and another that is 3 inches long. How much longer is the longer pencil?

Answer: 9

Solution: Difference: $|12 - 3| = 9$ inches.

Challenge 4: Elizabeth has a pencil that is 11 inches long and another that is 4 inches long. How much longer is the longer pencil?

Answer: 7

Solution: Difference: $|11 - 4| = 7$ inches.

Challenge 5: Ethan has a pencil that is 6 inches long and another that is 4 inches long. How much longer is the longer pencil?

Answer: 2

Solution: Difference: $|6 - 4| = 2$ inches.

Challenge 6: Samuel has a pencil that is 5 inches long and another that is 4 inches long. How much longer is the longer pencil?

Answer: 1

Solution: Difference: $|5 - 4| = 1$ inches.

Challenge 7: Alexander has a pencil that is 8 inches long and another that is 11 inches long. How much longer is the longer pencil?

Answer: 3

Solution: Difference: $|8 - 11| = 3$ inches.

Challenge 8: Avery has a pencil that is 8 inches long and another that is 6 inches long. How much longer is the longer pencil?

Answer: 2

Solution: Difference: $|8 - 6| = 2$ inches.

Challenge 9: Avery has a pencil that is 7 inches long and another that is 7 inches long. How much longer is the longer pencil?

Answer: 0

Solution: Difference: $|7 - 7| = 0$ inches.

Challenge 10: Sebastian has a pencil that is 4 inches long and another that is 3 inches long. How much longer is the longer pencil?

Answer: 1

Solution: Difference: $|4 - 3| = 1$ inches.

Challenge 11: Liam has a pencil that is 8 inches long and another that is 8 inches long. How much longer is the longer pencil?

Answer: 0

Solution: Difference: $|8 - 8| = 0$ inches.

Challenge 12: Henry has a pencil that is 9 inches long and another that is 8 inches long. How much longer is the longer pencil?

Answer: 1

Solution: Difference: $|9 - 8| = 1$ inches.

Challenge 13: Lucas has a pencil that is 9 inches long and another that is 6 inches long. How much longer is the longer pencil?

Answer: 3

Solution: Difference: $|9 - 6| = 3$ inches.

Challenge 14: Benjamin has a pencil that is 12 inches long and another that is 4 inches long. How much longer is the longer pencil?

Answer: 8

Solution: Difference: $|12 - 4| = 8$ inches.

Challenge 15: Evelyn has a pencil that is 11 inches long and another that is 12 inches long. How much longer is the longer pencil?

Answer: 1

Solution: Difference: $|11 - 12| = 1$ inches.

Grade 2 Problems

Problem 1: Charlotte measures a table that is 6 feet long. How many inches is this?

Answer: 72

Explanation: 1 foot = 12 inches. $6 \text{ feet} \times 12 = 72$ inches.

Problem 2: Benjamin measures a table that is 3 feet long. How many inches is this?

Answer: 36

Explanation: 1 foot = 12 inches. $3 \text{ feet} \times 12 = 36$ inches.

Problem 3: Alexander measures a table that is 6 feet long. How many inches is this?

Answer: 72

Explanation: 1 foot = 12 inches. $6 \text{ feet} \times 12 = 72 \text{ inches}$.

Problem 4: Ethan measures a table that is 6 feet long. How many inches is this?

Answer: 72

Explanation: 1 foot = 12 inches. $6 \text{ feet} \times 12 = 72 \text{ inches}$.

Problem 5: Sofia measures a table that is 2 feet long. How many inches is this?

Answer: 24

Explanation: 1 foot = 12 inches. $2 \text{ feet} \times 12 = 24 \text{ inches}$.

Problem 6: Aiden measures a table that is 3 feet long. How many inches is this?

Answer: 36

Explanation: 1 foot = 12 inches. $3 \text{ feet} \times 12 = 36 \text{ inches}$.

Problem 7: Noah measures a table that is 3 feet long. How many inches is this?

Answer: 36

Explanation: 1 foot = 12 inches. $3 \text{ feet} \times 12 = 36 \text{ inches}$.

Problem 8: Sebastian measures a table that is 2 feet long. How many inches is this?

Answer: 24

Explanation: 1 foot = 12 inches. $2 \text{ feet} \times 12 = 24 \text{ inches}$.

Problem 9: Liam measures a table that is 2 feet long. How many inches is this?

Answer: 24

Explanation: 1 foot = 12 inches. $2 \text{ feet} \times 12 = 24 \text{ inches}$.

Problem 10: Evelyn measures a table that is 3 feet long. How many inches is this?

Answer: 36

Explanation: 1 foot = 12 inches. $3 \text{ feet} \times 12 = 36 \text{ inches}$.

Problem 11: Mason measures a table that is 2 feet long. How many inches is this?

Answer: 24

Explanation: 1 foot = 12 inches. $2 \text{ feet} \times 12 = 24 \text{ inches}$.

Problem 12: Avery measures a table that is 6 feet long. How many inches is this?

Answer: 72

Explanation: 1 foot = 12 inches. $6 \text{ feet} \times 12 = 72 \text{ inches}$.

Problem 13: Avery measures a table that is 5 feet long. How many inches is this?

Answer: 60

Explanation: 1 foot = 12 inches. $5 \text{ feet} \times 12 = 60 \text{ inches}$.

Problem 14: Mason measures a table that is 4 feet long. How many inches is this?

Answer: 48

Explanation: 1 foot = 12 inches. $4 \text{ feet} \times 12 = 48 \text{ inches}$.

Problem 15: James measures a table that is 6 feet long. How many inches is this?

Answer: 72

Explanation: 1 foot = 12 inches. $6 \text{ feet} \times 12 = 72 \text{ inches}$.

Problem 16: Emma measures a table that is 2 feet long. How many inches is this?

Answer: 24

Explanation: 1 foot = 12 inches. $2 \text{ feet} \times 12 = 24 \text{ inches}$.

Problem 17: Ava measures a table that is 2 feet long. How many inches is this?

Answer: 24

Explanation: 1 foot = 12 inches. $2 \text{ feet} \times 12 = 24 \text{ inches}$.

Problem 18: Ethan measures a table that is 6 feet long. How many inches is this?

Answer: 72

Explanation: 1 foot = 12 inches. $6 \text{ feet} \times 12 = 72 \text{ inches}$.

Problem 19: Emma measures a table that is 5 feet long. How many inches is this?

Answer: 60

Explanation: 1 foot = 12 inches. 5 feet $\times$ 12 = 60 inches.

Problem 20: Sofia measures a table that is 3 feet long. How many inches is this?

Answer: 36

Explanation: 1 foot = 12 inches. 3 feet $\times$ 12 = 36 inches.

Problem 21: Isabella measures a table that is 5 feet long. How many inches is this?

Answer: 60

Explanation: 1 foot = 12 inches. 5 feet $\times$ 12 = 60 inches.

Problem 22: Emma measures a table that is 5 feet long. How many inches is this?

Answer: 60

Explanation: 1 foot = 12 inches. 5 feet $\times$ 12 = 60 inches.

Problem 23: Elizabeth measures a table that is 4 feet long. How many inches is this?

Answer: 48

Explanation: 1 foot = 12 inches. 4 feet $\times$ 12 = 48 inches.

Problem 24: Olivia measures a table that is 2 feet long. How many inches is this?

Answer: 24

Explanation: 1 foot = 12 inches. 2 feet $\times$ 12 = 24 inches.

Problem 25: Ethan measures a table that is 5 feet long. How many inches is this?

Answer: 60

Explanation: 1 foot = 12 inches. 5 feet $\times$ 12 = 60 inches.

Problem 26: Emily measures a table that is 2 feet long. How many inches is this?

Answer: 24

Explanation: 1 foot = 12 inches. 2 feet $\times$ 12 = 24 inches.

Problem 27: Harper measures a table that is 5 feet long. How many inches is this?

Answer: 60

Explanation: 1 foot = 12 inches. 5 feet $\times$ 12 = 60 inches.

Problem 28: Aiden measures a table that is 5 feet long. How many inches is this?

Answer: 60

Explanation: 1 foot = 12 inches. 5 feet $\times$ 12 = 60 inches.

Problem 29: Ethan measures a table that is 3 feet long. How many inches is this?

Answer: 36

Explanation: 1 foot = 12 inches. 3 feet $\times$ 12 = 36 inches.

Problem 30: Liam measures a table that is 6 feet long. How many inches is this?

Answer: 72

Explanation: 1 foot = 12 inches. 6 feet $\times$ 12 = 72 inches.

Problem 31: Isabella measures a table that is 6 feet long. How many inches is this?

Answer: 72

Explanation: 1 foot = 12 inches. 6 feet $\times$ 12 = 72 inches.

Problem 32: Noah measures a table that is 5 feet long. How many inches is this?

Answer: 60

Explanation: 1 foot = 12 inches. 5 feet $\times$ 12 = 60 inches.

Problem 33: Sofia measures a table that is 3 feet long. How many inches is this?

Answer: 36

Explanation: 1 foot = 12 inches. 3 feet $\times$ 12 = 36 inches.

Problem 34: Henry measures a table that is 5 feet long. How many inches is this?

Answer: 60

Explanation: 1 foot = 12 inches. 5 feet $\times$ 12 = 60 inches.

Problem 35: Sebastian measures a table that is 4 feet long. How many inches is this?

Answer: 48

Explanation: 1 foot = 12 inches. $4 \text{ feet} \times 12 = 48 \text{ inches}$.

Problem 36: Lucas measures a table that is 6 feet long. How many inches is this?

Answer: 72

Explanation: 1 foot = 12 inches. $6 \text{ feet} \times 12 = 72 \text{ inches}$.

Problem 37: Ethan measures a table that is 2 feet long. How many inches is this?

Answer: 24

Explanation: 1 foot = 12 inches. $2 \text{ feet} \times 12 = 24 \text{ inches}$.

Problem 38: Isabella measures a table that is 4 feet long. How many inches is this?

Answer: 48

Explanation: 1 foot = 12 inches. $4 \text{ feet} \times 12 = 48 \text{ inches}$.

Problem 39: David measures a table that is 5 feet long. How many inches is this?

Answer: 60

Explanation: 1 foot = 12 inches. $5 \text{ feet} \times 12 = 60 \text{ inches}$.

Problem 40: Mason measures a table that is 6 feet long. How many inches is this?

Answer: 72

Explanation: 1 foot = 12 inches. $6 \text{ feet} \times 12 = 72 \text{ inches}$.

Problem 41: Henry measures a table that is 4 feet long. How many inches is this?

Answer: 48

Explanation: 1 foot = 12 inches. $4 \text{ feet} \times 12 = 48 \text{ inches}$.

Problem 42: Emma measures a table that is 6 feet long. How many inches is this?

Answer: 72

Explanation: 1 foot = 12 inches. $6 \text{ feet} \times 12 = 72 \text{ inches}$.

Problem 43: Sophia measures a table that is 4 feet long. How many inches is this?

Answer: 48

Explanation: 1 foot = 12 inches. $4 \text{ feet} \times 12 = 48 \text{ inches}$.

Problem 44: Alexander measures a table that is 4 feet long. How many inches is this?

Answer: 48

Explanation: 1 foot = 12 inches. $4 \text{ feet} \times 12 = 48 \text{ inches}$.

Problem 45: Ava measures a table that is 6 feet long. How many inches is this?

Answer: 72

Explanation: 1 foot = 12 inches. $6 \text{ feet} \times 12 = 72 \text{ inches}$.

Practice Exercises - Grade 2

Exercise 1: Liam measures a table that is 4 feet long. How many inches is this?

Exercise 2: Emma measures a table that is 2 feet long. How many inches is this?

Exercise 3: Benjamin measures a table that is 2 feet long. How many inches is this?

Exercise 4: Joseph measures a table that is 5 feet long. How many inches is this?

Exercise 5: Sebastian measures a table that is 4 feet long. How many inches is this?

Exercise 6: Sebastian measures a table that is 3 feet long. How many inches is this?

Exercise 7: Noah measures a table that is 2 feet long. How many inches is this?

Exercise 8: Mia measures a table that is 3 feet long. How many inches is this?

Exercise 9: Emma measures a table that is 3 feet long. How many inches is this?

Exercise 10: Samuel measures a table that is 2 feet long. How many inches is this?

Exercise 11: Samuel measures a table that is 2 feet long. How many inches is this?

Exercise 12: Harper measures a table that is 5 feet long. How many inches is this?

Exercise 13: Ava measures a table that is 5 feet long. How many inches is this?

Exercise 14: Evelyn measures a table that is 6 feet long. How many inches is this?

Exercise 15: Amelia measures a table that is 2 feet long. How many inches is this?

Exercise 16: Lucas measures a table that is 6 feet long. How many inches is this?

Exercise 17: Sofia measures a table that is 2 feet long. How many inches is this?

Exercise 18: Mia measures a table that is 4 feet long. How many inches is this?

Exercise 19: David measures a table that is 2 feet long. How many inches is this?

Exercise 20: Samuel measures a table that is 2 feet long. How many inches is this?

Practice Exercise Answers:

Exercise 1: 48

Exercise 2: 24

Exercise 3: 24

Exercise 4: 60

Exercise 5: 48

Exercise 6: 36

Exercise 7: 24

Exercise 8: 36

Exercise 9: 36

Exercise 10: 24

Exercise 11: 24

Exercise 12: 60

Exercise 13: 60

Exercise 14: 72

Exercise 15: 24

Exercise 16: 72

Exercise 17: 24

Exercise 18: 48

Exercise 19: 24

Exercise 20: 24

Challenge Problems - Grade 2

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Jack measures a table that is 2 feet long. How many inches is this?

Answer: 24

Solution: 1 foot = 12 inches. $2 \text{ feet} \times 12 = 24 \text{ inches}$.

Challenge 2: Harper measures a table that is 5 feet long. How many inches is this?

Answer: 60

Solution: 1 foot = 12 inches. $5 \text{ feet} \times 12 = 60 \text{ inches}$.

Challenge 3: Jack measures a table that is 3 feet long. How many inches is this?

Answer: 36

Solution: 1 foot = 12 inches. $3 \text{ feet} \times 12 = 36 \text{ inches}$.

Challenge 4: Alexander measures a table that is 3 feet long. How many inches is this?

Answer: 36

Solution: 1 foot = 12 inches. $3 \text{ feet} \times 12 = 36 \text{ inches}$.

Challenge 5: Mia measures a table that is 6 feet long. How many inches is this?

Answer: 72

Solution: 1 foot = 12 inches. $6 \text{ feet} \times 12 = 72 \text{ inches}$.

Challenge 6: Jack measures a table that is 5 feet long. How many inches is this?

Answer: 60

Solution: 1 foot = 12 inches. $5 \text{ feet} \times 12 = 60 \text{ inches}$.

Challenge 7: Elizabeth measures a table that is 5 feet long. How many inches is this?

Answer: 60

Solution: 1 foot = 12 inches. $5 \text{ feet} \times 12 = 60 \text{ inches}$.

Challenge 8: Harper measures a table that is 4 feet long. How many inches is this?

Answer: 48

Solution: 1 foot = 12 inches. $4 \text{ feet} \times 12 = 48 \text{ inches}$.

Challenge 9: Noah measures a table that is 2 feet long. How many inches is this?

Answer: 24

Solution: 1 foot = 12 inches. $2 \text{ feet} \times 12 = 24 \text{ inches}$.

Challenge 10: Mason measures a table that is 6 feet long. How many inches is this?

Answer: 72

Solution: 1 foot = 12 inches. $6 \text{ feet} \times 12 = 72 \text{ inches}$.

Challenge 11: Sophia measures a table that is 5 feet long. How many inches is this?

Answer: 60

Solution: 1 foot = 12 inches. $5 \text{ feet} \times 12 = 60 \text{ inches}$.

Challenge 12: Benjamin measures a table that is 2 feet long. How many inches is this?

Answer: 24

Solution: 1 foot = 12 inches. $2 \text{ feet} \times 12 = 24 \text{ inches}$.

Challenge 13: Isabella measures a table that is 4 feet long. How many inches is this?

Answer: 48

Solution: 1 foot = 12 inches. $4 \text{ feet} \times 12 = 48 \text{ inches}$.

Challenge 14: Olivia measures a table that is 5 feet long. How many inches is this?

Answer: 60

Solution: 1 foot = 12 inches. $5 \text{ feet} \times 12 = 60 \text{ inches}$.

Challenge 15: Sofia measures a table that is 4 feet long. How many inches is this?

Answer: 48

Solution: 1 foot = 12 inches. $4 \text{ feet} \times 12 = 48 \text{ inches}$.

Grade 3 Problems

Problem 1: A bag of flour weighs 15 pounds. How many ounces is this?

Answer: 240

Explanation: 1 pound = 16 ounces. $15 \text{ pounds} \times 16 = 240 \text{ ounces}$.

Problem 2: A bag of flour weighs 13 pounds. How many ounces is this?

Answer: 208

Explanation: 1 pound = 16 ounces. $13 \text{ pounds} \times 16 = 208 \text{ ounces}$.

Problem 3: A bag of flour weighs 10 pounds. How many ounces is this?

Answer: 160

Explanation: 1 pound = 16 ounces. $10 \text{ pounds} \times 16 = 160 \text{ ounces}$.

Problem 4: A bag of flour weighs 16 pounds. How many ounces is this?

Answer: 256

Explanation: 1 pound = 16 ounces. $16 \text{ pounds} \times 16 = 256 \text{ ounces}$.

Problem 5: A bag of flour weighs 17 pounds. How many ounces is this?

Answer: 272

Explanation: 1 pound = 16 ounces. $17 \text{ pounds} \times 16 = 272 \text{ ounces}$.

Problem 6: A bag of flour weighs 14 pounds. How many ounces is this?

Answer: 224

Explanation: 1 pound = 16 ounces. $14 \text{ pounds} \times 16 = 224 \text{ ounces}$.

Problem 7: A bag of flour weighs 13 pounds. How many ounces is this?

Answer: 208

Explanation: 1 pound = 16 ounces. $13 \text{ pounds} \times 16 = 208 \text{ ounces}$.

Problem 8: A bag of flour weighs 6 pounds. How many ounces is this?

Answer: 96

Explanation: 1 pound = 16 ounces. $6 \text{ pounds} \times 16 = 96 \text{ ounces}$.

Problem 9: A bag of flour weighs 17 pounds. How many ounces is this?

Answer: 272

Explanation: 1 pound = 16 ounces. $17 \text{ pounds} \times 16 = 272 \text{ ounces}$.

Problem 10: A bag of flour weighs 15 pounds. How many ounces is this?

Answer: 240

Explanation: 1 pound = 16 ounces. $15 \text{ pounds} \times 16 = 240 \text{ ounces}$.

Problem 11: A bag of flour weighs 20 pounds. How many ounces is this?

Answer: 320

Explanation: 1 pound = 16 ounces. $20 \text{ pounds} \times 16 = 320 \text{ ounces}$.

Problem 12: A bag of flour weighs 5 pounds. How many ounces is this?

Answer: 80

Explanation: 1 pound = 16 ounces. $5 \text{ pounds} \times 16 = 80 \text{ ounces}$.

Problem 13: A bag of flour weighs 9 pounds. How many ounces is this?

Answer: 144

Explanation: 1 pound = 16 ounces. $9 \text{ pounds} \times 16 = 144 \text{ ounces}$.

Problem 14: A bag of flour weighs 6 pounds. How many ounces is this?

Answer: 96

Explanation: 1 pound = 16 ounces. $6 \text{ pounds} \times 16 = 96 \text{ ounces}$.

Problem 15: A bag of flour weighs 19 pounds. How many ounces is this?

Answer: 304

Explanation: 1 pound = 16 ounces. $19 \text{ pounds} \times 16 = 304 \text{ ounces}$.

Problem 16: A bag of flour weighs 9 pounds. How many ounces is this?

Answer: 144

Explanation: 1 pound = 16 ounces. $9 \text{ pounds} \times 16 = 144 \text{ ounces}$.

Problem 17: A bag of flour weighs 5 pounds. How many ounces is this?

Answer: 80

Explanation: 1 pound = 16 ounces. $5 \text{ pounds} \times 16 = 80 \text{ ounces}$.

Problem 18: A bag of flour weighs 18 pounds. How many ounces is this?

Answer: 288

Explanation: 1 pound = 16 ounces. $18 \text{ pounds} \times 16 = 288 \text{ ounces}$.

Problem 19: A bag of flour weighs 7 pounds. How many ounces is this?

Answer: 112

Explanation: 1 pound = 16 ounces. $7 \text{ pounds} \times 16 = 112 \text{ ounces}$.

Problem 20: A bag of flour weighs 18 pounds. How many ounces is this?

Answer: 288

Explanation: 1 pound = 16 ounces. $18 \text{ pounds} \times 16 = 288 \text{ ounces}$.

Problem 21: A bag of flour weighs 12 pounds. How many ounces is this?

Answer: 192

Explanation: 1 pound = 16 ounces. $12 \text{ pounds} \times 16 = 192 \text{ ounces}$.

Problem 22: A bag of flour weighs 10 pounds. How many ounces is this?

Answer: 160

Explanation: 1 pound = 16 ounces. $10 \text{ pounds} \times 16 = 160 \text{ ounces}$.

Problem 23: A bag of flour weighs 20 pounds. How many ounces is this?

Answer: 320

Explanation: 1 pound = 16 ounces. $20 \text{ pounds} \times 16 = 320 \text{ ounces}$.

Problem 24: A bag of flour weighs 19 pounds. How many ounces is this?

Answer: 304

Explanation: 1 pound = 16 ounces. $19 \text{ pounds} \times 16 = 304 \text{ ounces}$.

Problem 25: A bag of flour weighs 8 pounds. How many ounces is this?

Answer: 128

Explanation: 1 pound = 16 ounces. $8 \text{ pounds} \times 16 = 128 \text{ ounces}$.

Problem 26: A bag of flour weighs 18 pounds. How many ounces is this?

Answer: 288

Explanation: 1 pound = 16 ounces. $18 \text{ pounds} \times 16 = 288 \text{ ounces}$.

Problem 27: A bag of flour weighs 18 pounds. How many ounces is this?

Answer: 288

Explanation: 1 pound = 16 ounces. $18 \text{ pounds} \times 16 = 288 \text{ ounces}$.

Problem 28: A bag of flour weighs 5 pounds. How many ounces is this?

Answer: 80

Explanation: 1 pound = 16 ounces. $5 \text{ pounds} \times 16 = 80 \text{ ounces}$.

Problem 29: A bag of flour weighs 10 pounds. How many ounces is this?

Answer: 160

Explanation: 1 pound = 16 ounces. $10 \text{ pounds} \times 16 = 160 \text{ ounces}$.

Problem 30: A bag of flour weighs 17 pounds. How many ounces is this?

Answer: 272

Explanation: 1 pound = 16 ounces. $17 \text{ pounds} \times 16 = 272 \text{ ounces}$.

Problem 31: A bag of flour weighs 13 pounds. How many ounces is this?

Answer: 208

Explanation: 1 pound = 16 ounces. $13 \text{ pounds} \times 16 = 208 \text{ ounces}$.

Problem 32: A bag of flour weighs 18 pounds. How many ounces is this?

Answer: 288

Explanation: 1 pound = 16 ounces. $18 \text{ pounds} \times 16 = 288 \text{ ounces}$.

Problem 33: A bag of flour weighs 14 pounds. How many ounces is this?

Answer: 224

Explanation: 1 pound = 16 ounces. $14 \text{ pounds} \times 16 = 224 \text{ ounces}$.

Problem 34: A bag of flour weighs 5 pounds. How many ounces is this?

Answer: 80

Explanation: 1 pound = 16 ounces. $5 \text{ pounds} \times 16 = 80 \text{ ounces}$.

Problem 35: A bag of flour weighs 18 pounds. How many ounces is this?

Answer: 288

Explanation: 1 pound = 16 ounces. $18 \text{ pounds} \times 16 = 288 \text{ ounces}$.

Problem 36: A bag of flour weighs 10 pounds. How many ounces is this?

Answer: 160

Explanation: 1 pound = 16 ounces. $10 \text{ pounds} \times 16 = 160 \text{ ounces}$.

Problem 37: A bag of flour weighs 13 pounds. How many ounces is this?

Answer: 208

Explanation: 1 pound = 16 ounces. $13 \text{ pounds} \times 16 = 208 \text{ ounces}$.

Problem 38: A bag of flour weighs 19 pounds. How many ounces is this?

Answer: 304

Explanation: 1 pound = 16 ounces. $19 \text{ pounds} \times 16 = 304 \text{ ounces}$.

Problem 39: A bag of flour weighs 10 pounds. How many ounces is this?

Answer: 160

Explanation: 1 pound = 16 ounces. $10 \text{ pounds} \times 16 = 160 \text{ ounces}$.

Problem 40: A bag of flour weighs 9 pounds. How many ounces is this?

Answer: 144

Explanation: 1 pound = 16 ounces. $9 \text{ pounds} \times 16 = 144 \text{ ounces}$.

Problem 41: A bag of flour weighs 12 pounds. How many ounces is this?

Answer: 192

Explanation: 1 pound = 16 ounces. $12 \text{ pounds} \times 16 = 192 \text{ ounces}$.

Problem 42: A bag of flour weighs 10 pounds. How many ounces is this?

Answer: 160

Explanation: 1 pound = 16 ounces. $10 \text{ pounds} \times 16 = 160 \text{ ounces}$.

Problem 43: A bag of flour weighs 9 pounds. How many ounces is this?

Answer: 144

Explanation: 1 pound = 16 ounces. $9 \text{ pounds} \times 16 = 144 \text{ ounces}$.

Problem 44: A bag of flour weighs 9 pounds. How many ounces is this?

Answer: 144

Explanation: 1 pound = 16 ounces. $9 \text{ pounds} \times 16 = 144 \text{ ounces}$.

Problem 45: A bag of flour weighs 18 pounds. How many ounces is this?

Answer: 288

Explanation: 1 pound = 16 ounces. $18 \text{ pounds} \times 16 = 288 \text{ ounces}$.

Practice Exercises - Grade 3

Exercise 1: A bag of flour weighs 19 pounds. How many ounces is this?

Exercise 2: A bag of flour weighs 12 pounds. How many ounces is this?

Exercise 3: A bag of flour weighs 17 pounds. How many ounces is this?

Exercise 4: A bag of flour weighs 9 pounds. How many ounces is this?

Exercise 5: A bag of flour weighs 16 pounds. How many ounces is this?

Exercise 6: A bag of flour weighs 15 pounds. How many ounces is this?

Exercise 7: A bag of flour weighs 11 pounds. How many ounces is this?

Exercise 8: A bag of flour weighs 16 pounds. How many ounces is this?

Exercise 9: A bag of flour weighs 15 pounds. How many ounces is this?

Exercise 10: A bag of flour weighs 20 pounds. How many ounces is this?

Exercise 11: A bag of flour weighs 16 pounds. How many ounces is this?

Exercise 12: A bag of flour weighs 20 pounds. How many ounces is this?

Exercise 13: A bag of flour weighs 18 pounds. How many ounces is this?

Exercise 14: A bag of flour weighs 7 pounds. How many ounces is this?

Exercise 15: A bag of flour weighs 14 pounds. How many ounces is this?

Exercise 16: A bag of flour weighs 11 pounds. How many ounces is this?

Exercise 17: A bag of flour weighs 7 pounds. How many ounces is this?

Exercise 18: A bag of flour weighs 10 pounds. How many ounces is this?

Exercise 19: A bag of flour weighs 8 pounds. How many ounces is this?

Exercise 20: A bag of flour weighs 20 pounds. How many ounces is this?

Practice Exercise Answers:

Exercise 1: 304

Exercise 2: 192

Exercise 3: 272

Exercise 4: 144

Exercise 5: 256

Exercise 6: 240

Exercise 7: 176

Exercise 8: 256

Exercise 9: 240

Exercise 10: 320

Exercise 11: 256

Exercise 12: 320

Exercise 13: 288

Exercise 14: 112

Exercise 15: 224

Exercise 16: 176

Exercise 17: 112

Exercise 18: 160

Exercise 19: 128

Exercise 20: 320

Challenge Problems - Grade 3

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: A bag of flour weighs 9 pounds. How many ounces is this?

Answer: 144

Solution: 1 pound = 16 ounces. $9 \text{ pounds} \times 16 = 144 \text{ ounces}$.

Challenge 2: A bag of flour weighs 12 pounds. How many ounces is this?

Answer: 192

Solution: 1 pound = 16 ounces. $12 \text{ pounds} \times 16 = 192 \text{ ounces}$.

Challenge 3: A bag of flour weighs 10 pounds. How many ounces is this?

Answer: 160

Solution: 1 pound = 16 ounces. $10 \text{ pounds} \times 16 = 160 \text{ ounces}$.

Challenge 4: A bag of flour weighs 18 pounds. How many ounces is this?

Answer: 288

Solution: 1 pound = 16 ounces. $18 \text{ pounds} \times 16 = 288 \text{ ounces}$.

Challenge 5: A bag of flour weighs 19 pounds. How many ounces is this?

Answer: 304

Solution: 1 pound = 16 ounces. $19 \text{ pounds} \times 16 = 304 \text{ ounces}$.

Challenge 6: A bag of flour weighs 15 pounds. How many ounces is this?

Answer: 240

Solution: 1 pound = 16 ounces. $15 \text{ pounds} \times 16 = 240 \text{ ounces}$.

Challenge 7: A bag of flour weighs 13 pounds. How many ounces is this?

Answer: 208

Solution: 1 pound = 16 ounces. $13 \text{ pounds} \times 16 = 208 \text{ ounces}$.

Challenge 8: A bag of flour weighs 13 pounds. How many ounces is this?

Answer: 208

Solution: 1 pound = 16 ounces. $13 \text{ pounds} \times 16 = 208 \text{ ounces}$.

Challenge 9: A bag of flour weighs 8 pounds. How many ounces is this?

Answer: 128

Solution: 1 pound = 16 ounces. $8 \text{ pounds} \times 16 = 128 \text{ ounces}$.

Challenge 10: A bag of flour weighs 6 pounds. How many ounces is this?

Answer: 96

Solution: 1 pound = 16 ounces. $6 \text{ pounds} \times 16 = 96 \text{ ounces}$.

Challenge 11: A bag of flour weighs 11 pounds. How many ounces is this?

Answer: 176

Solution: 1 pound = 16 ounces. $11 \text{ pounds} \times 16 = 176 \text{ ounces}$.

Challenge 12: A bag of flour weighs 18 pounds. How many ounces is this?

Answer: 288

Solution: 1 pound = 16 ounces. $18 \text{ pounds} \times 16 = 288 \text{ ounces}$.

Challenge 13: A bag of flour weighs 19 pounds. How many ounces is this?

Answer: 304

Solution: 1 pound = 16 ounces. $19 \text{ pounds} \times 16 = 304 \text{ ounces}$.

Challenge 14: A bag of flour weighs 11 pounds. How many ounces is this?

Answer: 176

Solution: 1 pound = 16 ounces. $11 \text{ pounds} \times 16 = 176 \text{ ounces}$.

Challenge 15: A bag of flour weighs 12 pounds. How many ounces is this?

Answer: 192

Solution: 1 pound = 16 ounces. $12 \text{ pounds} \times 16 = 192 \text{ ounces}$.

Grade 4 Problems

Problem 1: Jack fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Explanation: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Problem 2: Jack fills a container with 4 gallons of water. How many quarts is this?

Answer: 16

Explanation: 1 gallon = 4 quarts. $4 \text{ gallons} \times 4 = 16 \text{ quarts}$.

Problem 3: Isabella fills a container with 7 gallons of water. How many quarts is this?

Answer: 28

Explanation: 1 gallon = 4 quarts. $7 \text{ gallons} \times 4 = 28 \text{ quarts}$.

Problem 4: Harper fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Explanation: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Problem 5: Harper fills a container with 5 gallons of water. How many quarts is this?

Answer: 20

Explanation: 1 gallon = 4 quarts. $5 \text{ gallons} \times 4 = 20 \text{ quarts}$.

Problem 6: Amelia fills a container with 4 gallons of water. How many quarts is this?

Answer: 16

Explanation: 1 gallon = 4 quarts. $4 \text{ gallons} \times 4 = 16 \text{ quarts}$.

Problem 7: Isabella fills a container with 2 gallons of water. How many quarts is this?

Answer: 8

Explanation: 1 gallon = 4 quarts. $2 \text{ gallons} \times 4 = 8 \text{ quarts}$.

Problem 8: Sebastian fills a container with 7 gallons of water. How many quarts is this?

Answer: 28

Explanation: 1 gallon = 4 quarts. $7 \text{ gallons} \times 4 = 28 \text{ quarts}$.

Problem 9: Emma fills a container with 5 gallons of water. How many quarts is this?

Answer: 20

Explanation: 1 gallon = 4 quarts. $5 \text{ gallons} \times 4 = 20 \text{ quarts}$.

Problem 10: Amelia fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Explanation: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Problem 11: Isabella fills a container with 6 gallons of water. How many quarts is this?

Answer: 24

Explanation: 1 gallon = 4 quarts. $6 \text{ gallons} \times 4 = 24 \text{ quarts}$.

Problem 12: Ava fills a container with 5 gallons of water. How many quarts is this?

Answer: 20

Explanation: 1 gallon = 4 quarts. $5 \text{ gallons} \times 4 = 20 \text{ quarts}$.

Problem 13: Evelyn fills a container with 2 gallons of water. How many quarts is this?

Answer: 8

Explanation: 1 gallon = 4 quarts. $2 \text{ gallons} \times 4 = 8 \text{ quarts}$.

Problem 14: Joseph fills a container with 7 gallons of water. How many quarts is this?

Answer: 28

Explanation: 1 gallon = 4 quarts. $7 \text{ gallons} \times 4 = 28 \text{ quarts}$.

Problem 15: Sofia fills a container with 6 gallons of water. How many quarts is this?

Answer: 24

Explanation: 1 gallon = 4 quarts. $6 \text{ gallons} \times 4 = 24 \text{ quarts}$.

Problem 16: Emma fills a container with 2 gallons of water. How many quarts is this?

Answer: 8

Explanation: 1 gallon = 4 quarts. $2 \text{ gallons} \times 4 = 8 \text{ quarts}$.

Problem 17: Charlotte fills a container with 6 gallons of water. How many quarts is this?

Answer: 24

Explanation: 1 gallon = 4 quarts. $6 \text{ gallons} \times 4 = 24 \text{ quarts}$.

Problem 18: Ethan fills a container with 5 gallons of water. How many quarts is this?

Answer: 20

Explanation: 1 gallon = 4 quarts. $5 \text{ gallons} \times 4 = 20 \text{ quarts}$.

Problem 19: Benjamin fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Explanation: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Problem 20: Samuel fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Explanation: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Problem 21: David fills a container with 7 gallons of water. How many quarts is this?

Answer: 28

Explanation: 1 gallon = 4 quarts. $7 \text{ gallons} \times 4 = 28 \text{ quarts}$.

Problem 22: Ava fills a container with 5 gallons of water. How many quarts is this?

Answer: 20

Explanation: 1 gallon = 4 quarts. $5 \text{ gallons} \times 4 = 20 \text{ quarts}$.

Problem 23: Elizabeth fills a container with 2 gallons of water. How many quarts is this?

Answer: 8

Explanation: 1 gallon = 4 quarts. $2 \text{ gallons} \times 4 = 8 \text{ quarts}$.

Problem 24: Avery fills a container with 3 gallons of water. How many quarts is this?

Answer: 12

Explanation: 1 gallon = 4 quarts. $3 \text{ gallons} \times 4 = 12 \text{ quarts}$.

Problem 25: Joseph fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Explanation: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Problem 26: Liam fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Explanation: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Problem 27: Olivia fills a container with 5 gallons of water. How many quarts is this?

Answer: 20

Explanation: 1 gallon = 4 quarts. $5 \text{ gallons} \times 4 = 20 \text{ quarts}$.

Problem 28: Henry fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Explanation: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Problem 29: James fills a container with 5 gallons of water. How many quarts is this?

Answer: 20

Explanation: 1 gallon = 4 quarts. $5 \text{ gallons} \times 4 = 20 \text{ quarts}$.

Problem 30: Samuel fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Explanation: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Problem 31: Henry fills a container with 6 gallons of water. How many quarts is this?

Answer: 24

Explanation: 1 gallon = 4 quarts. $6 \text{ gallons} \times 4 = 24 \text{ quarts}$.

Problem 32: Mason fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Explanation: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Problem 33: Alexander fills a container with 4 gallons of water. How many quarts is this?

Answer: 16

Explanation: 1 gallon = 4 quarts. $4 \text{ gallons} \times 4 = 16 \text{ quarts}$.

Problem 34: Mia fills a container with 2 gallons of water. How many quarts is this?

Answer: 8

Explanation: 1 gallon = 4 quarts. $2 \text{ gallons} \times 4 = 8 \text{ quarts}$.

Problem 35: Sophia fills a container with 3 gallons of water. How many quarts is this?

Answer: 12

Explanation: 1 gallon = 4 quarts. $3 \text{ gallons} \times 4 = 12 \text{ quarts}$.

Problem 36: Mason fills a container with 2 gallons of water. How many quarts is this?

Answer: 8

Explanation: 1 gallon = 4 quarts. $2 \text{ gallons} \times 4 = 8 \text{ quarts}$.

Problem 37: Evelyn fills a container with 5 gallons of water. How many quarts is this?

Answer: 20

Explanation: 1 gallon = 4 quarts. $5 \text{ gallons} \times 4 = 20 \text{ quarts}$.

Problem 38: Samuel fills a container with 5 gallons of water. How many quarts is this?

Answer: 20

Explanation: 1 gallon = 4 quarts. $5 \text{ gallons} \times 4 = 20 \text{ quarts}$.

Problem 39: Benjamin fills a container with 7 gallons of water. How many quarts is this?

Answer: 28

Explanation: 1 gallon = 4 quarts. $7 \text{ gallons} \times 4 = 28 \text{ quarts}$.

Problem 40: Alexander fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Explanation: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Problem 41: Sofia fills a container with 7 gallons of water. How many quarts is this?

Answer: 28

Explanation: 1 gallon = 4 quarts. $7 \text{ gallons} \times 4 = 28 \text{ quarts}$.

Problem 42: Mia fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Explanation: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Problem 43: Emily fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Explanation: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Problem 44: Isabella fills a container with 2 gallons of water. How many quarts is this?

Answer: 8

Explanation: 1 gallon = 4 quarts. $2 \text{ gallons} \times 4 = 8 \text{ quarts}$.

Problem 45: Sofia fills a container with 5 gallons of water. How many quarts is this?

Answer: 20

Explanation: 1 gallon = 4 quarts. $5 \text{ gallons} \times 4 = 20 \text{ quarts}$.

Practice Exercises - Grade 4

Exercise 1: Benjamin fills a container with 5 gallons of water. How many quarts is this?

Exercise 2: Henry fills a container with 6 gallons of water. How many quarts is this?

Exercise 3: Ava fills a container with 5 gallons of water. How many quarts is this?

Exercise 4: Harper fills a container with 2 gallons of water. How many quarts is this?

Exercise 5: Harper fills a container with 5 gallons of water. How many quarts is this?

Exercise 6: Henry fills a container with 5 gallons of water. How many quarts is this?

Exercise 7: Liam fills a container with 6 gallons of water. How many quarts is this?

Exercise 8: Sebastian fills a container with 5 gallons of water. How many quarts is this?

Exercise 9: Noah fills a container with 4 gallons of water. How many quarts is this?

Exercise 10: Benjamin fills a container with 4 gallons of water. How many quarts is this?

Exercise 11: Isabella fills a container with 6 gallons of water. How many quarts is this?

Exercise 12: Charlotte fills a container with 4 gallons of water. How many quarts is this?

Exercise 13: Abigail fills a container with 6 gallons of water. How many quarts is this?

Exercise 14: Aiden fills a container with 8 gallons of water. How many quarts is this?

Exercise 15: Samuel fills a container with 4 gallons of water. How many quarts is this?

Exercise 16: Isabella fills a container with 7 gallons of water. How many quarts is this?

Exercise 17: Sofia fills a container with 5 gallons of water. How many quarts is this?

Exercise 18: Aiden fills a container with 3 gallons of water. How many quarts is this?

Exercise 19: Alexander fills a container with 5 gallons of water. How many quarts is this?

Exercise 20: Lucas fills a container with 3 gallons of water. How many quarts is this?

Practice Exercise Answers:

Exercise 1: 20

Exercise 2: 24

Exercise 3: 20

Exercise 4: 8

Exercise 5: 20

Exercise 6: 20

Exercise 7: 24

Exercise 8: 20

Exercise 9: 16

Exercise 10: 16

Exercise 11: 24

Exercise 12: 16

Exercise 13: 24

Exercise 14: 32

Exercise 15: 16

Exercise 16: 28

Exercise 17: 20

Exercise 18: 12

Exercise 19: 20

Exercise 20: 12

Challenge Problems - Grade 4

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Mason fills a container with 3 gallons of water. How many quarts is this?

Answer: 12

Solution: 1 gallon = 4 quarts. $3 \text{ gallons} \times 4 = 12 \text{ quarts}$.

Challenge 2: Isabella fills a container with 4 gallons of water. How many quarts is this?

Answer: 16

Solution: 1 gallon = 4 quarts. $4 \text{ gallons} \times 4 = 16 \text{ quarts}$.

Challenge 3: David fills a container with 3 gallons of water. How many quarts is this?

Answer: 12

Solution: 1 gallon = 4 quarts. $3 \text{ gallons} \times 4 = 12 \text{ quarts}$.

Challenge 4: Alexander fills a container with 6 gallons of water. How many quarts is this?

Answer: 24

Solution: 1 gallon = 4 quarts. $6 \text{ gallons} \times 4 = 24 \text{ quarts}$.

Challenge 5: Henry fills a container with 7 gallons of water. How many quarts is this?

Answer: 28

Solution: 1 gallon = 4 quarts. $7 \text{ gallons} \times 4 = 28 \text{ quarts}$.

Challenge 6: Henry fills a container with 3 gallons of water. How many quarts is this?

Answer: 12

Solution: 1 gallon = 4 quarts. $3 \text{ gallons} \times 4 = 12 \text{ quarts}$.

Challenge 7: Joseph fills a container with 2 gallons of water. How many quarts is this?

Answer: 8

Solution: 1 gallon = 4 quarts. $2 \text{ gallons} \times 4 = 8 \text{ quarts}$.

Challenge 8: Samuel fills a container with 7 gallons of water. How many quarts is this?

Answer: 28

Solution: 1 gallon = 4 quarts. $7 \text{ gallons} \times 4 = 28 \text{ quarts}$.

Challenge 9: Samuel fills a container with 2 gallons of water. How many quarts is this?

Answer: 8

Solution: 1 gallon = 4 quarts. $2 \text{ gallons} \times 4 = 8 \text{ quarts}$.

Challenge 10: Elizabeth fills a container with 6 gallons of water. How many quarts is this?

Answer: 24

Solution: 1 gallon = 4 quarts. $6 \text{ gallons} \times 4 = 24 \text{ quarts}$.

Challenge 11: Ava fills a container with 3 gallons of water. How many quarts is this?

Answer: 12

Solution: 1 gallon = 4 quarts. $3 \text{ gallons} \times 4 = 12 \text{ quarts}$.

Challenge 12: Elizabeth fills a container with 3 gallons of water. How many quarts is this?

Answer: 12

Solution: 1 gallon = 4 quarts. $3 \text{ gallons} \times 4 = 12 \text{ quarts}$.

Challenge 13: Evelyn fills a container with 6 gallons of water. How many quarts is this?

Answer: 24

Solution: 1 gallon = 4 quarts. $6 \text{ gallons} \times 4 = 24 \text{ quarts}$.

Challenge 14: David fills a container with 8 gallons of water. How many quarts is this?

Answer: 32

Solution: 1 gallon = 4 quarts. $8 \text{ gallons} \times 4 = 32 \text{ quarts}$.

Challenge 15: Ethan fills a container with 3 gallons of water. How many quarts is this?

Answer: 12

Solution: 1 gallon = 4 quarts. $3 \text{ gallons} \times 4 = 12 \text{ quarts}$.

Grade 5 Problems

Problem 1: Joseph runs 9 kilometers. How many meters is this?

Answer: 9000

Explanation: 1 kilometer = 1000 meters. $9 \text{ km} \times 1000 = 9000 \text{ meters}$.

Problem 2: Olivia runs 7 kilometers. How many meters is this?

Answer: 7000

Explanation: 1 kilometer = 1000 meters. $7 \text{ km} \times 1000 = 7000 \text{ meters}$.

Problem 3: Lucas runs 4 kilometers. How many meters is this?

Answer: 4000

Explanation: 1 kilometer = 1000 meters. $4 \text{ km} \times 1000 = 4000 \text{ meters}$.

Problem 4: Liam runs 10 kilometers. How many meters is this?

Answer: 10000

Explanation: 1 kilometer = 1000 meters. $10 \text{ km} \times 1000 = 10000 \text{ meters}$.

Problem 5: Aiden runs 4 kilometers. How many meters is this?

Answer: 4000

Explanation: 1 kilometer = 1000 meters. $4 \text{ km} \times 1000 = 4000 \text{ meters}$.

Problem 6: Noah runs 3 kilometers. How many meters is this?

Answer: 3000

Explanation: 1 kilometer = 1000 meters. $3 \text{ km} \times 1000 = 3000 \text{ meters}$.

Problem 7: Sebastian runs 10 kilometers. How many meters is this?

Answer: 10000

Explanation: 1 kilometer = 1000 meters. $10 \text{ km} \times 1000 = 10000 \text{ meters}$.

Problem 8: Alexander runs 4 kilometers. How many meters is this?

Answer: 4000

Explanation: 1 kilometer = 1000 meters. $4 \text{ km} \times 1000 = 4000 \text{ meters}$.

Problem 9: Elizabeth runs 2 kilometers. How many meters is this?

Answer: 2000

Explanation: 1 kilometer = 1000 meters. $2 \text{ km} \times 1000 = 2000 \text{ meters}$.

Problem 10: Elizabeth runs 7 kilometers. How many meters is this?

Answer: 7000

Explanation: 1 kilometer = 1000 meters. $7 \text{ km} \times 1000 = 7000 \text{ meters}$.

Problem 11: Emma runs 5 kilometers. How many meters is this?

Answer: 5000

Explanation: 1 kilometer = 1000 meters. $5 \text{ km} \times 1000 = 5000 \text{ meters}$.

Problem 12: Emily runs 8 kilometers. How many meters is this?

Answer: 8000

Explanation: 1 kilometer = 1000 meters. $8 \text{ km} \times 1000 = 8000 \text{ meters}$.

Problem 13: Elizabeth runs 3 kilometers. How many meters is this?

Answer: 3000

Explanation: 1 kilometer = 1000 meters. $3 \text{ km} \times 1000 = 3000 \text{ meters}$.

Problem 14: James runs 8 kilometers. How many meters is this?

Answer: 8000

Explanation: 1 kilometer = 1000 meters. $8 \text{ km} \times 1000 = 8000 \text{ meters}$.

Problem 15: Charlotte runs 5 kilometers. How many meters is this?

Answer: 5000

Explanation: 1 kilometer = 1000 meters. $5 \text{ km} \times 1000 = 5000 \text{ meters}$.

Problem 16: Joseph runs 10 kilometers. How many meters is this?

Answer: 10000

Explanation: 1 kilometer = 1000 meters. $10 \text{ km} \times 1000 = 10000 \text{ meters}$.

Problem 17: Liam runs 2 kilometers. How many meters is this?

Answer: 2000

Explanation: 1 kilometer = 1000 meters. $2 \text{ km} \times 1000 = 2000 \text{ meters}$.

Problem 18: Sofia runs 6 kilometers. How many meters is this?

Answer: 6000

Explanation: 1 kilometer = 1000 meters. $6 \text{ km} \times 1000 = 6000 \text{ meters}$.

Problem 19: Mia runs 7 kilometers. How many meters is this?

Answer: 7000

Explanation: 1 kilometer = 1000 meters. $7 \text{ km} \times 1000 = 7000 \text{ meters}$.

Problem 20: Ethan runs 5 kilometers. How many meters is this?

Answer: 5000

Explanation: 1 kilometer = 1000 meters. $5 \text{ km} \times 1000 = 5000 \text{ meters}$.

Problem 21: Isabella runs 2 kilometers. How many meters is this?

Answer: 2000

Explanation: 1 kilometer = 1000 meters. $2 \text{ km} \times 1000 = 2000 \text{ meters}$.

Problem 22: Amelia runs 6 kilometers. How many meters is this?

Answer: 6000

Explanation: 1 kilometer = 1000 meters. $6 \text{ km} \times 1000 = 6000 \text{ meters}$.

Problem 23: James runs 3 kilometers. How many meters is this?

Answer: 3000

Explanation: 1 kilometer = 1000 meters. $3 \text{ km} \times 1000 = 3000 \text{ meters}$.

Problem 24: Olivia runs 5 kilometers. How many meters is this?

Answer: 5000

Explanation: 1 kilometer = 1000 meters. $5 \text{ km} \times 1000 = 5000 \text{ meters}$.

Problem 25: Olivia runs 5 kilometers. How many meters is this?

Answer: 5000

Explanation: 1 kilometer = 1000 meters. $5 \text{ km} \times 1000 = 5000 \text{ meters}$.

Problem 26: David runs 5 kilometers. How many meters is this?

Answer: 5000

Explanation: 1 kilometer = 1000 meters. $5 \text{ km} \times 1000 = 5000 \text{ meters}$.

Problem 27: David runs 3 kilometers. How many meters is this?

Answer: 3000

Explanation: 1 kilometer = 1000 meters. $3 \text{ km} \times 1000 = 3000 \text{ meters}$.

Problem 28: Abigail runs 6 kilometers. How many meters is this?

Answer: 6000

Explanation: 1 kilometer = 1000 meters. $6 \text{ km} \times 1000 = 6000 \text{ meters}$.

Problem 29: Harper runs 6 kilometers. How many meters is this?

Answer: 6000

Explanation: 1 kilometer = 1000 meters. $6 \text{ km} \times 1000 = 6000 \text{ meters}$.

Problem 30: Sofia runs 3 kilometers. How many meters is this?

Answer: 3000

Explanation: 1 kilometer = 1000 meters. $3 \text{ km} \times 1000 = 3000 \text{ meters}$.

Problem 31: Evelyn runs 2 kilometers. How many meters is this?

Answer: 2000

Explanation: 1 kilometer = 1000 meters. $2 \text{ km} \times 1000 = 2000 \text{ meters}$.

Problem 32: Emily runs 8 kilometers. How many meters is this?

Answer: 8000

Explanation: 1 kilometer = 1000 meters. $8 \text{ km} \times 1000 = 8000 \text{ meters}$.

Problem 33: David runs 7 kilometers. How many meters is this?

Answer: 7000

Explanation: 1 kilometer = 1000 meters. $7 \text{ km} \times 1000 = 7000 \text{ meters}$.

Problem 34: Evelyn runs 7 kilometers. How many meters is this?

Answer: 7000

Explanation: 1 kilometer = 1000 meters. $7 \text{ km} \times 1000 = 7000 \text{ meters}$.

Problem 35: Isabella runs 3 kilometers. How many meters is this?

Answer: 3000

Explanation: 1 kilometer = 1000 meters. $3 \text{ km} \times 1000 = 3000 \text{ meters}$.

Problem 36: Amelia runs 2 kilometers. How many meters is this?

Answer: 2000

Explanation: 1 kilometer = 1000 meters. $2 \text{ km} \times 1000 = 2000 \text{ meters}$.

Problem 37: Joseph runs 6 kilometers. How many meters is this?

Answer: 6000

Explanation: 1 kilometer = 1000 meters. $6 \text{ km} \times 1000 = 6000 \text{ meters}$.

Problem 38: Emma runs 6 kilometers. How many meters is this?

Answer: 6000

Explanation: 1 kilometer = 1000 meters. $6 \text{ km} \times 1000 = 6000 \text{ meters}$.

Problem 39: Mia runs 9 kilometers. How many meters is this?

Answer: 9000

Explanation: 1 kilometer = 1000 meters. $9 \text{ km} \times 1000 = 9000 \text{ meters}$.

Problem 40: Amelia runs 5 kilometers. How many meters is this?

Answer: 5000

Explanation: 1 kilometer = 1000 meters. $5 \text{ km} \times 1000 = 5000 \text{ meters}$.

Problem 41: Harper runs 8 kilometers. How many meters is this?

Answer: 8000

Explanation: 1 kilometer = 1000 meters. $8 \text{ km} \times 1000 = 8000 \text{ meters}$.

Problem 42: Jack runs 3 kilometers. How many meters is this?

Answer: 3000

Explanation: 1 kilometer = 1000 meters. $3 \text{ km} \times 1000 = 3000 \text{ meters}$.

Problem 43: Avery runs 6 kilometers. How many meters is this?

Answer: 6000

Explanation: 1 kilometer = 1000 meters. $6 \text{ km} \times 1000 = 6000 \text{ meters}$.

Problem 44: Charlotte runs 2 kilometers. How many meters is this?

Answer: 2000

Explanation: 1 kilometer = 1000 meters. $2 \text{ km} \times 1000 = 2000 \text{ meters}$.

Problem 45: Sebastian runs 5 kilometers. How many meters is this?

Answer: 5000

Explanation: 1 kilometer = 1000 meters. $5 \text{ km} \times 1000 = 5000 \text{ meters}$.

Practice Exercises - Grade 5

Exercise 1: Abigail runs 3 kilometers. How many meters is this?

Exercise 2: Jack runs 7 kilometers. How many meters is this?

Exercise 3: Lucas runs 5 kilometers. How many meters is this?

Exercise 4: Samuel runs 7 kilometers. How many meters is this?

Exercise 5: Amelia runs 3 kilometers. How many meters is this?

Exercise 6: Sofia runs 6 kilometers. How many meters is this?

Exercise 7: Isabella runs 4 kilometers. How many meters is this?

Exercise 8: Henry runs 2 kilometers. How many meters is this?

Exercise 9: Aiden runs 5 kilometers. How many meters is this?

Exercise 10: Isabella runs 4 kilometers. How many meters is this?

Exercise 11: Charlotte runs 8 kilometers. How many meters is this?

Exercise 12: James runs 3 kilometers. How many meters is this?

Exercise 13: Emily runs 4 kilometers. How many meters is this?

Exercise 14: Alexander runs 7 kilometers. How many meters is this?

Exercise 15: Jack runs 9 kilometers. How many meters is this?

Exercise 16: Noah runs 8 kilometers. How many meters is this?

Exercise 17: Avery runs 6 kilometers. How many meters is this?

Exercise 18: Amelia runs 9 kilometers. How many meters is this?

Exercise 19: Benjamin runs 2 kilometers. How many meters is this?

Exercise 20: Joseph runs 9 kilometers. How many meters is this?

Practice Exercise Answers:

Exercise 1: 3000

Exercise 2: 7000

Exercise 3: 5000

Exercise 4: 7000

Exercise 5: 3000

Exercise 6: 6000

Exercise 7: 4000

Exercise 8: 2000

Exercise 9: 5000

Exercise 10: 4000

Exercise 11: 8000

Exercise 12: 3000

Exercise 13: 4000

Exercise 14: 7000

Exercise 15: 9000

Exercise 16: 8000

Exercise 17: 6000

Exercise 18: 9000

Exercise 19: 2000

Exercise 20: 9000

Challenge Problems - Grade 5

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Emily runs 6 kilometers. How many meters is this?

Answer: 6000

Solution: 1 kilometer = 1000 meters. $6 \text{ km} \times 1000 = 6000 \text{ meters}$.

Challenge 2: Sofia runs 6 kilometers. How many meters is this?

Answer: 6000

Solution: 1 kilometer = 1000 meters. $6 \text{ km} \times 1000 = 6000$ meters.

Challenge 3: Sophia runs 7 kilometers. How many meters is this?

Answer: 7000

Solution: 1 kilometer = 1000 meters. $7 \text{ km} \times 1000 = 7000$ meters.

Challenge 4: Emma runs 6 kilometers. How many meters is this?

Answer: 6000

Solution: 1 kilometer = 1000 meters. $6 \text{ km} \times 1000 = 6000$ meters.

Challenge 5: Olivia runs 2 kilometers. How many meters is this?

Answer: 2000

Solution: 1 kilometer = 1000 meters. $2 \text{ km} \times 1000 = 2000$ meters.

Challenge 6: Elizabeth runs 9 kilometers. How many meters is this?

Answer: 9000

Solution: 1 kilometer = 1000 meters. $9 \text{ km} \times 1000 = 9000$ meters.

Challenge 7: Harper runs 8 kilometers. How many meters is this?

Answer: 8000

Solution: 1 kilometer = 1000 meters. $8 \text{ km} \times 1000 = 8000$ meters.

Challenge 8: Sofia runs 4 kilometers. How many meters is this?

Answer: 4000

Solution: 1 kilometer = 1000 meters. $4 \text{ km} \times 1000 = 4000$ meters.

Challenge 9: Ava runs 7 kilometers. How many meters is this?

Answer: 7000

Solution: 1 kilometer = 1000 meters. $7 \text{ km} \times 1000 = 7000$ meters.

Challenge 10: Isabella runs 3 kilometers. How many meters is this?

Answer: 3000

Solution: 1 kilometer = 1000 meters. $3 \text{ km} \times 1000 = 3000$ meters.

Challenge 11: Abigail runs 6 kilometers. How many meters is this?

Answer: 6000

Solution: 1 kilometer = 1000 meters. $6 \text{ km} \times 1000 = 6000$ meters.

Challenge 12: Emily runs 4 kilometers. How many meters is this?

Answer: 4000

Solution: 1 kilometer = 1000 meters. $4 \text{ km} \times 1000 = 4000$ meters.

Challenge 13: Aiden runs 5 kilometers. How many meters is this?

Answer: 5000

Solution: 1 kilometer = 1000 meters. $5 \text{ km} \times 1000 = 5000$ meters.

Challenge 14: Alexander runs 3 kilometers. How many meters is this?

Answer: 3000

Solution: 1 kilometer = 1000 meters. $3 \text{ km} \times 1000 = 3000$ meters.

Challenge 15: Evelyn runs 7 kilometers. How many meters is this?

Answer: 7000

Solution: 1 kilometer = 1000 meters. $7 \text{ km} \times 1000 = 7000$ meters.

Grade 6 Problems

Problem 1: Avery spends 3 hours and 11 minutes on a project. How many total minutes is this?

Answer: 191

Explanation: 3 hours = 180 minutes. Total: $180 + 11 = 191$ minutes.

Problem 2: Jack spends 2 hours and 0 minutes on a project. How many total minutes is this?

Answer: 120

Explanation: 2 hours = 120 minutes. Total: $120 + 0 = 120$ minutes.

Problem 3: Henry spends 5 hours and 21 minutes on a project. How many total minutes is this?

Answer: 321

Explanation: 5 hours = 300 minutes. Total: $300 + 21 = 321$ minutes.

Problem 4: Sofia spends 5 hours and 38 minutes on a project. How many total minutes is this?

Answer: 338

Explanation: 5 hours = 300 minutes. Total: $300 + 38 = 338$ minutes.

Problem 5: Jack spends 2 hours and 16 minutes on a project. How many total minutes is this?

Answer: 136

Explanation: 2 hours = 120 minutes. Total: $120 + 16 = 136$ minutes.

Problem 6: Emily spends 3 hours and 45 minutes on a project. How many total minutes is this?

Answer: 225

Explanation: 3 hours = 180 minutes. Total: $180 + 45 = 225$ minutes.

Problem 7: Emily spends 3 hours and 30 minutes on a project. How many total minutes is this?

Answer: 210

Explanation: 3 hours = 180 minutes. Total: $180 + 30 = 210$ minutes.

Problem 8: Evelyn spends 3 hours and 16 minutes on a project. How many total minutes is this?

Answer: 196

Explanation: 3 hours = 180 minutes. Total: $180 + 16 = 196$ minutes.

Problem 9: Harper spends 4 hours and 36 minutes on a project. How many total minutes is this?

Answer: 276

Explanation: 4 hours = 240 minutes. Total: $240 + 36 = 276$ minutes.

Problem 10: Charlotte spends 2 hours and 17 minutes on a project. How many total minutes is this?

Answer: 137

Explanation: 2 hours = 120 minutes. Total: $120 + 17 = 137$ minutes.

Problem 11: Sofia spends 5 hours and 6 minutes on a project. How many total minutes is this?

Answer: 306

Explanation: 5 hours = 300 minutes. Total: $300 + 6 = 306$ minutes.

Problem 12: David spends 3 hours and 7 minutes on a project. How many total minutes is this?

Answer: 187

Explanation: 3 hours = 180 minutes. Total: $180 + 7 = 187$ minutes.

Problem 13: Mia spends 5 hours and 6 minutes on a project. How many total minutes is this?

Answer: 306

Explanation: 5 hours = 300 minutes. Total: $300 + 6 = 306$ minutes.

Problem 14: Samuel spends 4 hours and 37 minutes on a project. How many total minutes is this?

Answer: 277

Explanation: 4 hours = 240 minutes. Total: $240 + 37 = 277$ minutes.

Problem 15: Evelyn spends 3 hours and 45 minutes on a project. How many total minutes is this?

Answer: 225

Explanation: 3 hours = 180 minutes. Total: $180 + 45 = 225$ minutes.

Problem 16: Alexander spends 5 hours and 31 minutes on a project. How many total minutes is this?

Answer: 331

Explanation: 5 hours = 300 minutes. Total: $300 + 31 = 331$ minutes.

Problem 17: Ethan spends 2 hours and 30 minutes on a project. How many total minutes is this?

Answer: 150

Explanation: 2 hours = 120 minutes. Total: $120 + 30 = 150$ minutes.

Problem 18: Emily spends 5 hours and 16 minutes on a project. How many total minutes is this?

Answer: 316

Explanation: 5 hours = 300 minutes. Total: $300 + 16 = 316$ minutes.

Problem 19: Jack spends 4 hours and 20 minutes on a project. How many total minutes is this?

Answer: 260

Explanation: 4 hours = 240 minutes. Total: $240 + 20 = 260$ minutes.

Problem 20: James spends 4 hours and 37 minutes on a project. How many total minutes is this?

Answer: 277

Explanation: 4 hours = 240 minutes. Total: $240 + 37 = 277$ minutes.

Problem 21: Liam spends 2 hours and 37 minutes on a project. How many total minutes is this?

Answer: 157

Explanation: 2 hours = 120 minutes. Total: $120 + 37 = 157$ minutes.

Problem 22: Jack spends 4 hours and 1 minutes on a project. How many total minutes is this?

Answer: 241

Explanation: 4 hours = 240 minutes. Total: $240 + 1 = 241$ minutes.

Problem 23: Elizabeth spends 4 hours and 26 minutes on a project. How many total minutes is this?

Answer: 266

Explanation: 4 hours = 240 minutes. Total: $240 + 26 = 266$ minutes.

Problem 24: Emily spends 5 hours and 4 minutes on a project. How many total minutes is this?

Answer: 304

Explanation: 5 hours = 300 minutes. Total: $300 + 4 = 304$ minutes.

Problem 25: Aiden spends 3 hours and 42 minutes on a project. How many total minutes is this?

Answer: 222

Explanation: 3 hours = 180 minutes. Total: $180 + 42 = 222$ minutes.

Problem 26: Noah spends 3 hours and 3 minutes on a project. How many total minutes is this?

Answer: 183

Explanation: 3 hours = 180 minutes. Total: $180 + 3 = 183$ minutes.

Problem 27: Lucas spends 2 hours and 19 minutes on a project. How many total minutes is this?

Answer: 139

Explanation: 2 hours = 120 minutes. Total: $120 + 19 = 139$ minutes.

Problem 28: Sofia spends 5 hours and 39 minutes on a project. How many total minutes is this?

Answer: 339

Explanation: 5 hours = 300 minutes. Total: $300 + 39 = 339$ minutes.

Problem 29: Aiden spends 5 hours and 2 minutes on a project. How many total minutes is this?

Answer: 302

Explanation: 5 hours = 300 minutes. Total: $300 + 2 = 302$ minutes.

Problem 30: Joseph spends 2 hours and 9 minutes on a project. How many total minutes is this?

Answer: 129

Explanation: 2 hours = 120 minutes. Total: $120 + 9 = 129$ minutes.

Problem 31: Lucas spends 3 hours and 6 minutes on a project. How many total minutes is this?

Answer: 186

Explanation: 3 hours = 180 minutes. Total: $180 + 6 = 186$ minutes.

Problem 32: Ava spends 4 hours and 23 minutes on a project. How many total minutes is this?

Answer: 263

Explanation: 4 hours = 240 minutes. Total: $240 + 23 = 263$ minutes.

Problem 33: Jack spends 3 hours and 26 minutes on a project. How many total minutes is this?

Answer: 206

Explanation: 3 hours = 180 minutes. Total: $180 + 26 = 206$ minutes.

Problem 34: Isabella spends 2 hours and 39 minutes on a project. How many total minutes is this?

Answer: 159

Explanation: 2 hours = 120 minutes. Total: $120 + 39 = 159$ minutes.

Problem 35: Aiden spends 3 hours and 1 minutes on a project. How many total minutes is this?

Answer: 181

Explanation: 3 hours = 180 minutes. Total: $180 + 1 = 181$ minutes.

Problem 36: Joseph spends 5 hours and 20 minutes on a project. How many total minutes is this?

Answer: 320

Explanation: 5 hours = 300 minutes. Total: $300 + 20 = 320$ minutes.

Problem 37: Henry spends 4 hours and 15 minutes on a project. How many total minutes is this?

Answer: 255

Explanation: 4 hours = 240 minutes. Total: $240 + 15 = 255$ minutes.

Problem 38: Aiden spends 3 hours and 29 minutes on a project. How many total minutes is this?

Answer: 209

Explanation: 3 hours = 180 minutes. Total: $180 + 29 = 209$ minutes.

Problem 39: Benjamin spends 4 hours and 33 minutes on a project. How many total minutes is this?

Answer: 273

Explanation: 4 hours = 240 minutes. Total: $240 + 33 = 273$ minutes.

Problem 40: Emma spends 4 hours and 13 minutes on a project. How many total minutes is this?

Answer: 253

Explanation: 4 hours = 240 minutes. Total: $240 + 13 = 253$ minutes.

Problem 41: Elizabeth spends 4 hours and 22 minutes on a project. How many total minutes is this?

Answer: 262

Explanation: 4 hours = 240 minutes. Total: $240 + 22 = 262$ minutes.

Problem 42: Jack spends 3 hours and 42 minutes on a project. How many total minutes is this?

Answer: 222

Explanation: 3 hours = 180 minutes. Total: $180 + 42 = 222$ minutes.

Problem 43: Alexander spends 2 hours and 19 minutes on a project. How many total minutes is this?

Answer: 139

Explanation: 2 hours = 120 minutes. Total: $120 + 19 = 139$ minutes.

Problem 44: David spends 4 hours and 21 minutes on a project. How many total minutes is this?

Answer: 261

Explanation: 4 hours = 240 minutes. Total: $240 + 21 = 261$ minutes.

Problem 45: Emily spends 5 hours and 9 minutes on a project. How many total minutes is this?

Answer: 309

Explanation: 5 hours = 300 minutes. Total: $300 + 9 = 309$ minutes.

Practice Exercises - Grade 6

Exercise 1: Harper spends 4 hours and 24 minutes on a project. How many total minutes is this?

Exercise 2: Joseph spends 3 hours and 12 minutes on a project. How many total minutes is this?

Exercise 3: Abigail spends 5 hours and 36 minutes on a project. How many total minutes is this?

Exercise 4: Jack spends 3 hours and 8 minutes on a project. How many total minutes is this?

Exercise 5: Liam spends 5 hours and 30 minutes on a project. How many total minutes is this?

Exercise 6: Lucas spends 3 hours and 2 minutes on a project. How many total minutes is this?

Exercise 7: James spends 4 hours and 22 minutes on a project. How many total minutes is this?

Exercise 8: Samuel spends 3 hours and 20 minutes on a project. How many total minutes is this?

Exercise 9: Benjamin spends 4 hours and 11 minutes on a project. How many total minutes is this?

Exercise 10: Emily spends 4 hours and 26 minutes on a project. How many total minutes is this?

Exercise 11: Avery spends 3 hours and 7 minutes on a project. How many total minutes is this?

Exercise 12: Harper spends 3 hours and 32 minutes on a project. How many total minutes is this?

Exercise 13: David spends 2 hours and 17 minutes on a project. How many total minutes is this?

Exercise 14: Elizabeth spends 5 hours and 19 minutes on a project. How many total minutes is this?

Exercise 15: Liam spends 3 hours and 28 minutes on a project. How many total minutes is this?

Exercise 16: Alexander spends 4 hours and 35 minutes on a project. How many total minutes is this?

Exercise 17: Lucas spends 4 hours and 29 minutes on a project. How many total minutes is this?

Exercise 18: Henry spends 3 hours and 4 minutes on a project. How many total minutes is this?

Exercise 19: Jack spends 4 hours and 10 minutes on a project. How many total minutes is this?

Exercise 20: Sebastian spends 3 hours and 23 minutes on a project. How many total minutes is this?

Practice Exercise Answers:

Exercise 1: 264

Exercise 2: 192

Exercise 3: 336

Exercise 4: 188

Exercise 5: 330

Exercise 6: 182

Exercise 7: 262

Exercise 8: 200

Exercise 9: 251

Exercise 10: 266

Exercise 11: 187

Exercise 12: 212

Exercise 13: 137

Exercise 14: 319

Exercise 15: 208

Exercise 16: 275

Exercise 17: 269

Exercise 18: 184

Exercise 19: 250

Exercise 20: 203

Challenge Problems - Grade 6

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Jack spends 3 hours and 27 minutes on a project. How many total minutes is this?

Answer: 207

Solution: 3 hours = 180 minutes. Total: $180 + 27 = 207$ minutes.

Challenge 2: Henry spends 2 hours and 45 minutes on a project. How many total minutes is this?

Answer: 165

Solution: 2 hours = 120 minutes. Total: $120 + 45 = 165$ minutes.

Challenge 3: Sofia spends 5 hours and 9 minutes on a project. How many total minutes is this?

Answer: 309

Solution: 5 hours = 300 minutes. Total: $300 + 9 = 309$ minutes.

Challenge 4: Mason spends 5 hours and 45 minutes on a project. How many total minutes is this?

Answer: 345

Solution: 5 hours = 300 minutes. Total: $300 + 45 = 345$ minutes.

Challenge 5: Henry spends 3 hours and 44 minutes on a project. How many total minutes is this?

Answer: 224

Solution: 3 hours = 180 minutes. Total: $180 + 44 = 224$ minutes.

Challenge 6: Isabella spends 5 hours and 35 minutes on a project. How many total minutes is this?

Answer: 335

Solution: 5 hours = 300 minutes. Total: $300 + 35 = 335$ minutes.

Challenge 7: Harper spends 4 hours and 17 minutes on a project. How many total minutes is this?

Answer: 257

Solution: 4 hours = 240 minutes. Total: $240 + 17 = 257$ minutes.

Challenge 8: Mason spends 3 hours and 5 minutes on a project. How many total minutes is this?

Answer: 185

Solution: 3 hours = 180 minutes. Total: $180 + 5 = 185$ minutes.

Challenge 9: Abigail spends 2 hours and 19 minutes on a project. How many total minutes is this?

Answer: 139

Solution: 2 hours = 120 minutes. Total: $120 + 19 = 139$ minutes.

Challenge 10: Noah spends 2 hours and 2 minutes on a project. How many total minutes is this?

Answer: 122

Solution: 2 hours = 120 minutes. Total: $120 + 2 = 122$ minutes.

Challenge 11: Avery spends 5 hours and 17 minutes on a project. How many total minutes is this?

Answer: 317

Solution: 5 hours = 300 minutes. Total: $300 + 17 = 317$ minutes.

Challenge 12: Aiden spends 4 hours and 12 minutes on a project. How many total minutes is this?

Answer: 252

Solution: 4 hours = 240 minutes. Total: $240 + 12 = 252$ minutes.

Challenge 13: Sophia spends 2 hours and 40 minutes on a project. How many total minutes is this?

Answer: 160

Solution: 2 hours = 120 minutes. Total: $120 + 40 = 160$ minutes.

Challenge 14: Evelyn spends 4 hours and 7 minutes on a project. How many total minutes is this?

Answer: 247

Solution: 4 hours = 240 minutes. Total: $240 + 7 = 247$ minutes.

Challenge 15: Aiden spends 3 hours and 39 minutes on a project. How many total minutes is this?

Answer: 219

Solution: 3 hours = 180 minutes. Total: $180 + 39 = 219$ minutes.

Chapter 7: Data and Graphs

Data analysis and graphing help us make sense of information. This chapter teaches students to collect, organize, interpret, and present data using various graphs and statistical measures.

Grade 1 Problems

Problem 1: Isabella counts 9 red flowers and 5 blue flowers. How many flowers in total?

Answer: 14

Explanation: Total flowers: $9 + 5 = 14$ flowers.

Problem 2: Olivia counts 3 red flowers and 7 blue flowers. How many flowers in total?

Answer: 10

Explanation: Total flowers: $3 + 7 = 10$ flowers.

Problem 3: Mason counts 8 red flowers and 3 blue flowers. How many flowers in total?

Answer: 11

Explanation: Total flowers: $8 + 3 = 11$ flowers.

Problem 4: Joseph counts 8 red flowers and 5 blue flowers. How many flowers in total?

Answer: 13

Explanation: Total flowers: $8 + 5 = 13$ flowers.

Problem 5: David counts 5 red flowers and 8 blue flowers. How many flowers in total?

Answer: 13

Explanation: Total flowers: $5 + 8 = 13$ flowers.

Problem 6: Joseph counts 4 red flowers and 9 blue flowers. How many flowers in total?

Answer: 13

Explanation: Total flowers: $4 + 9 = 13$ flowers.

Problem 7: David counts 10 red flowers and 4 blue flowers. How many flowers in total?

Answer: 14

Explanation: Total flowers: $10 + 4 = 14$ flowers.

Problem 8: Elizabeth counts 8 red flowers and 9 blue flowers. How many flowers in total?

Answer: 17

Explanation: Total flowers: $8 + 9 = 17$ flowers.

Problem 9: Henry counts 3 red flowers and 6 blue flowers. How many flowers in total?

Answer: 9

Explanation: Total flowers: $3 + 6 = 9$ flowers.

Problem 10: Liam counts 9 red flowers and 8 blue flowers. How many flowers in total?

Answer: 17

Explanation: Total flowers: $9 + 8 = 17$ flowers.

Problem 11: Benjamin counts 3 red flowers and 7 blue flowers. How many flowers in total?

Answer: 10

Explanation: Total flowers: $3 + 7 = 10$ flowers.

Problem 12: Emma counts 4 red flowers and 5 blue flowers. How many flowers in total?

Answer: 9

Explanation: Total flowers: $4 + 5 = 9$ flowers.

Problem 13: Sophia counts 9 red flowers and 8 blue flowers. How many flowers in total?

Answer: 17

Explanation: Total flowers: $9 + 8 = 17$ flowers.

Problem 14: Alexander counts 3 red flowers and 3 blue flowers. How many flowers in total?

Answer: 6

Explanation: Total flowers: $3 + 3 = 6$ flowers.

Problem 15: Emma counts 7 red flowers and 3 blue flowers. How many flowers in total?

Answer: 10

Explanation: Total flowers: $7 + 3 = 10$ flowers.

Problem 16: Amelia counts 10 red flowers and 6 blue flowers. How many flowers in total?

Answer: 16

Explanation: Total flowers: $10 + 6 = 16$ flowers.

Problem 17: Isabella counts 9 red flowers and 9 blue flowers. How many flowers in total?

Answer: 18

Explanation: Total flowers: $9 + 9 = 18$ flowers.

Problem 18: Isabella counts 4 red flowers and 7 blue flowers. How many flowers in total?

Answer: 11

Explanation: Total flowers: $4 + 7 = 11$ flowers.

Problem 19: Elizabeth counts 10 red flowers and 6 blue flowers. How many flowers in total?

Answer: 16

Explanation: Total flowers: $10 + 6 = 16$ flowers.

Problem 20: Charlotte counts 5 red flowers and 6 blue flowers. How many flowers in total?

Answer: 11

Explanation: Total flowers: $5 + 6 = 11$ flowers.

Problem 21: Amelia counts 8 red flowers and 4 blue flowers. How many flowers in total?

Answer: 12

Explanation: Total flowers: $8 + 4 = 12$ flowers.

Problem 22: Amelia counts 8 red flowers and 3 blue flowers. How many flowers in total?

Answer: 11

Explanation: Total flowers: $8 + 3 = 11$ flowers.

Problem 23: Mia counts 4 red flowers and 10 blue flowers. How many flowers in total?

Answer: 14

Explanation: Total flowers: $4 + 10 = 14$ flowers.

Problem 24: Benjamin counts 9 red flowers and 10 blue flowers. How many flowers in total?

Answer: 19

Explanation: Total flowers: $9 + 10 = 19$ flowers.

Problem 25: Alexander counts 3 red flowers and 3 blue flowers. How many flowers in total?

Answer: 6

Explanation: Total flowers: $3 + 3 = 6$ flowers.

Problem 26: Liam counts 10 red flowers and 3 blue flowers. How many flowers in total?

Answer: 13

Explanation: Total flowers: $10 + 3 = 13$ flowers.

Problem 27: Jack counts 4 red flowers and 4 blue flowers. How many flowers in total?

Answer: 8

Explanation: Total flowers: $4 + 4 = 8$ flowers.

Problem 28: Alexander counts 7 red flowers and 9 blue flowers. How many flowers in total?

Answer: 16

Explanation: Total flowers: $7 + 9 = 16$ flowers.

Problem 29: Aiden counts 3 red flowers and 5 blue flowers. How many flowers in total?

Answer: 8

Explanation: Total flowers: $3 + 5 = 8$ flowers.

Problem 30: Lucas counts 8 red flowers and 10 blue flowers. How many flowers in total?

Answer: 18

Explanation: Total flowers: $8 + 10 = 18$ flowers.

Problem 31: Isabella counts 10 red flowers and 8 blue flowers. How many flowers in total?

Answer: 18

Explanation: Total flowers: $10 + 8 = 18$ flowers.

Problem 32: Amelia counts 8 red flowers and 6 blue flowers. How many flowers in total?

Answer: 14

Explanation: Total flowers: $8 + 6 = 14$ flowers.

Problem 33: Aiden counts 9 red flowers and 7 blue flowers. How many flowers in total?

Answer: 16

Explanation: Total flowers: $9 + 7 = 16$ flowers.

Problem 34: Sebastian counts 7 red flowers and 5 blue flowers. How many flowers in total?

Answer: 12

Explanation: Total flowers: $7 + 5 = 12$ flowers.

Problem 35: Avery counts 5 red flowers and 10 blue flowers. How many flowers in total?

Answer: 15

Explanation: Total flowers: $5 + 10 = 15$ flowers.

Problem 36: Emily counts 6 red flowers and 5 blue flowers. How many flowers in total?

Answer: 11

Explanation: Total flowers: $6 + 5 = 11$ flowers.

Problem 37: Isabella counts 4 red flowers and 5 blue flowers. How many flowers in total?

Answer: 9

Explanation: Total flowers: $4 + 5 = 9$ flowers.

Problem 38: Elizabeth counts 7 red flowers and 9 blue flowers. How many flowers in total?

Answer: 16

Explanation: Total flowers: $7 + 9 = 16$ flowers.

Problem 39: Abigail counts 5 red flowers and 10 blue flowers. How many flowers in total?

Answer: 15

Explanation: Total flowers: $5 + 10 = 15$ flowers.

Problem 40: Avery counts 7 red flowers and 8 blue flowers. How many flowers in total?

Answer: 15

Explanation: Total flowers: $7 + 8 = 15$ flowers.

Problem 41: Aiden counts 4 red flowers and 4 blue flowers. How many flowers in total?

Answer: 8

Explanation: Total flowers: $4 + 4 = 8$ flowers.

Problem 42: Benjamin counts 10 red flowers and 3 blue flowers. How many flowers in total?

Answer: 13

Explanation: Total flowers: $10 + 3 = 13$ flowers.

Problem 43: Ava counts 9 red flowers and 4 blue flowers. How many flowers in total?

Answer: 13

Explanation: Total flowers: $9 + 4 = 13$ flowers.

Problem 44: Sofia counts 7 red flowers and 7 blue flowers. How many flowers in total?

Answer: 14

Explanation: Total flowers: $7 + 7 = 14$ flowers.

Problem 45: Benjamin counts 10 red flowers and 9 blue flowers. How many flowers in total?

Answer: 19

Explanation: Total flowers: $10 + 9 = 19$ flowers.

Practice Exercises - Grade 1

Exercise 1: Isabella counts 8 red flowers and 10 blue flowers. How many flowers in total?

Exercise 2: Lucas counts 10 red flowers and 8 blue flowers. How many flowers in total?

Exercise 3: Benjamin counts 4 red flowers and 10 blue flowers. How many flowers in total?

Exercise 4: Sofia counts 10 red flowers and 8 blue flowers. How many flowers in total?

Exercise 5: Charlotte counts 3 red flowers and 10 blue flowers. How many flowers in total?

Exercise 6: Benjamin counts 6 red flowers and 10 blue flowers. How many flowers in total?

Exercise 7: Mia counts 10 red flowers and 6 blue flowers. How many flowers in total?

Exercise 8: Liam counts 3 red flowers and 7 blue flowers. How many flowers in total?

Exercise 9: Avery counts 10 red flowers and 6 blue flowers. How many flowers in total?

Exercise 10: Avery counts 10 red flowers and 9 blue flowers. How many flowers in total?

Exercise 11: Mia counts 10 red flowers and 6 blue flowers. How many flowers in total?

Exercise 12: Jack counts 5 red flowers and 6 blue flowers. How many flowers in total?

Exercise 13: Liam counts 3 red flowers and 4 blue flowers. How many flowers in total?

Exercise 14: Elizabeth counts 7 red flowers and 3 blue flowers. How many flowers in total?

Exercise 15: Jack counts 3 red flowers and 9 blue flowers. How many flowers in total?

Exercise 16: James counts 7 red flowers and 3 blue flowers. How many flowers in total?

Exercise 17: Evelyn counts 3 red flowers and 7 blue flowers. How many flowers in total?

Exercise 18: Amelia counts 10 red flowers and 5 blue flowers. How many flowers in total?

Exercise 19: Charlotte counts 4 red flowers and 4 blue flowers. How many flowers in total?

Exercise 20: Mason counts 9 red flowers and 9 blue flowers. How many flowers in total?

Practice Exercise Answers:

Exercise 1: 18

Exercise 2: 18

Exercise 3: 14

Exercise 4: 18

Exercise 5: 13

Exercise 6: 16

Exercise 7: 16

Exercise 8: 10

Exercise 9: 16

Exercise 10: 19

Exercise 11: 16

Exercise 12: 11

Exercise 13: 7

Exercise 14: 10

Exercise 15: 12

Exercise 16: 10

Exercise 17: 10

Exercise 18: 15

Exercise 19: 8

Exercise 20: 18

Challenge Problems - Grade 1

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Emma counts 7 red flowers and 10 blue flowers. How many flowers in total?

Answer: 17

Solution: Total flowers: $7 + 10 = 17$ flowers.

Challenge 2: Emma counts 3 red flowers and 3 blue flowers. How many flowers in total?

Answer: 6

Solution: Total flowers: $3 + 3 = 6$ flowers.

Challenge 3: Evelyn counts 8 red flowers and 5 blue flowers. How many flowers in total?

Answer: 13

Solution: Total flowers: $8 + 5 = 13$ flowers.

Challenge 4: Mason counts 8 red flowers and 8 blue flowers. How many flowers in total?

Answer: 16

Solution: Total flowers: $8 + 8 = 16$ flowers.

Challenge 5: Sebastian counts 5 red flowers and 4 blue flowers. How many flowers in total?

Answer: 9

Solution: Total flowers: $5 + 4 = 9$ flowers.

Challenge 6: David counts 4 red flowers and 8 blue flowers. How many flowers in total?

Answer: 12

Solution: Total flowers: $4 + 8 = 12$ flowers.

Challenge 7: Ethan counts 7 red flowers and 7 blue flowers. How many flowers in total?

Answer: 14

Solution: Total flowers: $7 + 7 = 14$ flowers.

Challenge 8: Ava counts 7 red flowers and 6 blue flowers. How many flowers in total?

Answer: 13

Solution: Total flowers: $7 + 6 = 13$ flowers.

Challenge 9: Emma counts 4 red flowers and 4 blue flowers. How many flowers in total?

Answer: 8

Solution: Total flowers: $4 + 4 = 8$ flowers.

Challenge 10: James counts 10 red flowers and 9 blue flowers. How many flowers in total?

Answer: 19

Solution: Total flowers: $10 + 9 = 19$ flowers.

Challenge 11: Evelyn counts 5 red flowers and 10 blue flowers. How many flowers in total?

Answer: 15

Solution: Total flowers: $5 + 10 = 15$ flowers.

Challenge 12: Amelia counts 9 red flowers and 7 blue flowers. How many flowers in total?

Answer: 16

Solution: Total flowers: $9 + 7 = 16$ flowers.

Challenge 13: Olivia counts 3 red flowers and 5 blue flowers. How many flowers in total?

Answer: 8

Solution: Total flowers: $3 + 5 = 8$ flowers.

Challenge 14: Abigail counts 8 red flowers and 8 blue flowers. How many flowers in total?

Answer: 16

Solution: Total flowers: $8 + 8 = 16$ flowers.

Challenge 15: Elizabeth counts 3 red flowers and 7 blue flowers. How many flowers in total?

Answer: 10

Solution: Total flowers: $3 + 7 = 10$ flowers.

Grade 2 Problems

Problem 1: A class has 7 students who prefer cats and 4 who prefer dogs. How many more students prefer cats?

Answer: 3

Explanation: Difference: $|7 - 4| = 3$ students.

Problem 2: A class has 8 students who prefer cats and 7 who prefer dogs. How many more students prefer cats?

Answer: 1

Explanation: Difference: $|8 - 7| = 1$ students.

Problem 3: A class has 12 students who prefer cats and 11 who prefer dogs. How many more students prefer cats?

Answer: 1

Explanation: Difference: $|12 - 11| = 1$ students.

Problem 4: A class has 4 students who prefer cats and 5 who prefer dogs. How many more students prefer dogs?

Answer: 1

Explanation: Difference: $|4 - 5| = 1$ students.

Problem 5: A class has 7 students who prefer cats and 6 who prefer dogs. How many more students prefer cats?

Answer: 1

Explanation: Difference: $|7 - 6| = 1$ students.

Problem 6: A class has 5 students who prefer cats and 9 who prefer dogs. How many more students prefer dogs?

Answer: 4

Explanation: Difference: $|5 - 9| = 4$ students.

Problem 7: A class has 11 students who prefer cats and 7 who prefer dogs. How many more students prefer cats?

Answer: 4

Explanation: Difference: $|11 - 7| = 4$ students.

Problem 8: A class has 11 students who prefer cats and 10 who prefer dogs. How many more students prefer cats?

Answer: 1

Explanation: Difference: $|11 - 10| = 1$ students.

Problem 9: A class has 9 students who prefer cats and 6 who prefer dogs. How many more students prefer cats?

Answer: 3

Explanation: Difference: $|9 - 6| = 3$ students.

Problem 10: A class has 10 students who prefer cats and 10 who prefer dogs. How many more students prefer dogs?

Answer: 0

Explanation: Difference: $|10 - 10| = 0$ students.

Problem 11: A class has 5 students who prefer cats and 11 who prefer dogs. How many more students prefer dogs?

Answer: 6

Explanation: Difference: $|5 - 11| = 6$ students.

Problem 12: A class has 7 students who prefer cats and 4 who prefer dogs. How many more students prefer cats?

Answer: 3

Explanation: Difference: $|7 - 4| = 3$ students.

Problem 13: A class has 4 students who prefer cats and 5 who prefer dogs. How many more students prefer dogs?

Answer: 1

Explanation: Difference: $|4 - 5| = 1$ students.

Problem 14: A class has 6 students who prefer cats and 6 who prefer dogs. How many more students prefer dogs?

Answer: 0

Explanation: Difference: $|6 - 6| = 0$ students.

Problem 15: A class has 10 students who prefer cats and 12 who prefer dogs. How many more students prefer dogs?

Answer: 2

Explanation: Difference: $|10 - 12| = 2$ students.

Problem 16: A class has 11 students who prefer cats and 8 who prefer dogs. How many more students prefer cats?

Answer: 3

Explanation: Difference: $|11 - 8| = 3$ students.

Problem 17: A class has 5 students who prefer cats and 4 who prefer dogs. How many more students prefer cats?

Answer: 1

Explanation: Difference: $|5 - 4| = 1$ students.

Problem 18: A class has 4 students who prefer cats and 5 who prefer dogs. How many more students prefer dogs?

Answer: 1

Explanation: Difference: $|4 - 5| = 1$ students.

Problem 19: A class has 5 students who prefer cats and 8 who prefer dogs. How many more students prefer dogs?

Answer: 3

Explanation: Difference: $|5 - 8| = 3$ students.

Problem 20: A class has 11 students who prefer cats and 10 who prefer dogs. How many more students prefer cats?

Answer: 1

Explanation: Difference: $|11 - 10| = 1$ students.

Problem 21: A class has 8 students who prefer cats and 12 who prefer dogs. How many more students prefer dogs?

Answer: 4

Explanation: Difference: $|8 - 12| = 4$ students.

Problem 22: A class has 4 students who prefer cats and 6 who prefer dogs. How many more students prefer dogs?

Answer: 2

Explanation: Difference: $|4 - 6| = 2$ students.

Problem 23: A class has 7 students who prefer cats and 4 who prefer dogs. How many more students prefer cats?

Answer: 3

Explanation: Difference: $|7 - 4| = 3$ students.

Problem 24: A class has 7 students who prefer cats and 11 who prefer dogs. How many more students prefer dogs?

Answer: 4

Explanation: Difference: $|7 - 11| = 4$ students.

Problem 25: A class has 9 students who prefer cats and 6 who prefer dogs. How many more students prefer cats?

Answer: 3

Explanation: Difference: $|9 - 6| = 3$ students.

Problem 26: A class has 11 students who prefer cats and 6 who prefer dogs. How many more students prefer cats?

Answer: 5

Explanation: Difference: $|11 - 6| = 5$ students.

Problem 27: A class has 9 students who prefer cats and 9 who prefer dogs. How many more students prefer dogs?

Answer: 0

Explanation: Difference: $|9 - 9| = 0$ students.

Problem 28: A class has 5 students who prefer cats and 4 who prefer dogs. How many more students prefer cats?

Answer: 1

Explanation: Difference: $|5 - 4| = 1$ students.

Problem 29: A class has 4 students who prefer cats and 11 who prefer dogs. How many more students prefer dogs?

Answer: 7

Explanation: Difference: $|4 - 11| = 7$ students.

Problem 30: A class has 10 students who prefer cats and 10 who prefer dogs. How many more students prefer dogs?

Answer: 0

Explanation: Difference: $|10 - 10| = 0$ students.

Problem 31: A class has 9 students who prefer cats and 4 who prefer dogs. How many more students prefer cats?

Answer: 5

Explanation: Difference: $|9 - 4| = 5$ students.

Problem 32: A class has 11 students who prefer cats and 7 who prefer dogs. How many more students prefer cats?

Answer: 4

Explanation: Difference: $|11 - 7| = 4$ students.

Problem 33: A class has 8 students who prefer cats and 4 who prefer dogs. How many more students prefer cats?

Answer: 4

Explanation: Difference: $|8 - 4| = 4$ students.

Problem 34: A class has 7 students who prefer cats and 6 who prefer dogs. How many more students prefer cats?

Answer: 1

Explanation: Difference: $|7 - 6| = 1$ students.

Problem 35: A class has 5 students who prefer cats and 8 who prefer dogs. How many more students prefer dogs?

Answer: 3

Explanation: Difference: $|5 - 8| = 3$ students.

Problem 36: A class has 11 students who prefer cats and 9 who prefer dogs. How many more students prefer cats?

Answer: 2

Explanation: Difference: $|11 - 9| = 2$ students.

Problem 37: A class has 8 students who prefer cats and 9 who prefer dogs. How many more students prefer dogs?

Answer: 1

Explanation: Difference: $|8 - 9| = 1$ students.

Problem 38: A class has 5 students who prefer cats and 8 who prefer dogs. How many more students prefer dogs?

Answer: 3

Explanation: Difference: $|5 - 8| = 3$ students.

Problem 39: A class has 8 students who prefer cats and 12 who prefer dogs. How many more students prefer dogs?

Answer: 4

Explanation: Difference: $|8 - 12| = 4$ students.

Problem 40: A class has 9 students who prefer cats and 5 who prefer dogs. How many more students prefer cats?

Answer: 4

Explanation: Difference: $|9 - 5| = 4$ students.

Problem 41: A class has 6 students who prefer cats and 12 who prefer dogs. How many more students prefer dogs?

Answer: 6

Explanation: Difference: $|6 - 12| = 6$ students.

Problem 42: A class has 12 students who prefer cats and 5 who prefer dogs. How many more students prefer cats?

Answer: 7

Explanation: Difference: $|12 - 5| = 7$ students.

Problem 43: A class has 5 students who prefer cats and 8 who prefer dogs. How many more students prefer dogs?

Answer: 3

Explanation: Difference: $|5 - 8| = 3$ students.

Problem 44: A class has 8 students who prefer cats and 12 who prefer dogs. How many more students prefer dogs?

Answer: 4

Explanation: Difference: $|8 - 12| = 4$ students.

Problem 45: A class has 4 students who prefer cats and 5 who prefer dogs. How many more students prefer dogs?

Answer: 1

Explanation: Difference: $|4 - 5| = 1$ students.

Practice Exercises - Grade 2

Exercise 1: A class has 7 students who prefer cats and 7 who prefer dogs. How many more students prefer dogs?

Exercise 2: A class has 4 students who prefer cats and 9 who prefer dogs. How many more students prefer dogs?

Exercise 3: A class has 11 students who prefer cats and 7 who prefer dogs. How many more students prefer cats?

Exercise 4: A class has 9 students who prefer cats and 6 who prefer dogs. How many more students prefer cats?

Exercise 5: A class has 6 students who prefer cats and 11 who prefer dogs. How many more students prefer dogs?

Exercise 6: A class has 11 students who prefer cats and 4 who prefer dogs. How many more students prefer cats?

Exercise 7: A class has 4 students who prefer cats and 9 who prefer dogs. How many more students prefer dogs?

Exercise 8: A class has 12 students who prefer cats and 7 who prefer dogs. How many more students prefer cats?

Exercise 9: A class has 9 students who prefer cats and 9 who prefer dogs. How many more students prefer dogs?

Exercise 10: A class has 11 students who prefer cats and 9 who prefer dogs. How many more students prefer cats?

Exercise 11: A class has 9 students who prefer cats and 4 who prefer dogs. How many more students prefer cats?

Exercise 12: A class has 9 students who prefer cats and 11 who prefer dogs. How many more students prefer dogs?

Exercise 13: A class has 6 students who prefer cats and 10 who prefer dogs. How many more students prefer dogs?

Exercise 14: A class has 12 students who prefer cats and 12 who prefer dogs. How many more students prefer dogs?

Exercise 15: A class has 8 students who prefer cats and 6 who prefer dogs. How many more students prefer cats?

Exercise 16: A class has 8 students who prefer cats and 11 who prefer dogs. How many more students prefer dogs?

Exercise 17: A class has 5 students who prefer cats and 10 who prefer dogs. How many more students prefer dogs?

Exercise 18: A class has 8 students who prefer cats and 8 who prefer dogs. How many more students prefer dogs?

Exercise 19: A class has 9 students who prefer cats and 8 who prefer dogs. How many more students prefer cats?

Exercise 20: A class has 6 students who prefer cats and 7 who prefer dogs. How many more students prefer dogs?

Practice Exercise Answers:

Exercise 1: 0

Exercise 2: 5

Exercise 3: 4

Exercise 4: 3

Exercise 5: 5

Exercise 6: 7

Exercise 7: 5

Exercise 8: 5

Exercise 9: 0

Exercise 10: 2

Exercise 11: 5

Exercise 12: 2

Exercise 13: 4

Exercise 14: 0

Exercise 15: 2

Exercise 16: 3

Exercise 17: 5

Exercise 18: 0

Exercise 19: 1

Exercise 20: 1

Challenge Problems - Grade 2

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: A class has 10 students who prefer cats and 11 who prefer dogs. How many more students prefer dogs?

Answer: 1

Solution: Difference: $|10 - 11| = 1$ students.

Challenge 2: A class has 10 students who prefer cats and 4 who prefer dogs. How many more students prefer cats?

Answer: 6

Solution: Difference: $|10 - 4| = 6$ students.

Challenge 3: A class has 5 students who prefer cats and 6 who prefer dogs. How many more students prefer dogs?

Answer: 1

Solution: Difference: $|5 - 6| = 1$ students.

Challenge 4: A class has 9 students who prefer cats and 8 who prefer dogs. How many more students prefer cats?

Answer: 1

Solution: Difference: $|9 - 8| = 1$ students.

Challenge 5: A class has 10 students who prefer cats and 10 who prefer dogs. How many more students prefer dogs?

Answer: 0

Solution: Difference: $|10 - 10| = 0$ students.

Challenge 6: A class has 7 students who prefer cats and 4 who prefer dogs. How many more students prefer cats?

Answer: 3

Solution: Difference: $|7 - 4| = 3$ students.

Challenge 7: A class has 5 students who prefer cats and 4 who prefer dogs. How many more students prefer cats?

Answer: 1

Solution: Difference: $|5 - 4| = 1$ students.

Challenge 8: A class has 9 students who prefer cats and 10 who prefer dogs. How many more students prefer dogs?

Answer: 1

Solution: Difference: $|9 - 10| = 1$ students.

Challenge 9: A class has 5 students who prefer cats and 12 who prefer dogs. How many more students prefer dogs?

Answer: 7

Solution: Difference: $|5 - 12| = 7$ students.

Challenge 10: A class has 5 students who prefer cats and 9 who prefer dogs. How many more students prefer dogs?

Answer: 4

Solution: Difference: $|5 - 9| = 4$ students.

Challenge 11: A class has 10 students who prefer cats and 5 who prefer dogs. How many more students prefer cats?

Answer: 5

Solution: Difference: $|10 - 5| = 5$ students.

Challenge 12: A class has 6 students who prefer cats and 4 who prefer dogs. How many more students prefer cats?

Answer: 2

Solution: Difference: $|6 - 4| = 2$ students.

Challenge 13: A class has 11 students who prefer cats and 7 who prefer dogs. How many more students prefer cats?

Answer: 4

Solution: Difference: $|11 - 7| = 4$ students.

Challenge 14: A class has 12 students who prefer cats and 12 who prefer dogs. How many more students prefer dogs?

Answer: 0

Solution: Difference: $|12 - 12| = 0$ students.

Challenge 15: A class has 11 students who prefer cats and 9 who prefer dogs. How many more students prefer cats?

Answer: 2

Solution: Difference: $|11 - 9| = 2$ students.

Grade 3 Problems

Problem 1: David scored 88, 66, 94, and 76 on four tests. What is the average score?

Answer: 81

Explanation: Average = $(88 + 66 + 94 + 76) \div 4 = 324 \div 4 = 81$.

Problem 2: Henry scored 89, 72, 72, and 98 on four tests. What is the average score?

Answer: 82

Explanation: Average = $(89 + 72 + 72 + 98) \div 4 = 331 \div 4 = 82$.

Problem 3: Aiden scored 99, 70, 82, and 89 on four tests. What is the average score?

Answer: 85

Explanation: Average = $(99 + 70 + 82 + 89) \div 4 = 340 \div 4 = 85$.

Problem 4: Alexander scored 80, 90, 73, and 62 on four tests. What is the average score?

Answer: 76

Explanation: Average = $(80 + 90 + 73 + 62) \div 4 = 305 \div 4 = 76$.

Problem 5: Aiden scored 76, 61, 100, and 90 on four tests. What is the average score?

Answer: 81

Explanation: Average = $(76 + 61 + 100 + 90) \div 4 = 327 \div 4 = 81$.

Problem 6: Abigail scored 73, 86, 93, and 62 on four tests. What is the average score?

Answer: 78

Explanation: Average = $(73 + 86 + 93 + 62) \div 4 = 314 \div 4 = 78$.

Problem 7: Ava scored 92, 83, 98, and 99 on four tests. What is the average score?

Answer: 93

Explanation: Average = $(92 + 83 + 98 + 99) \div 4 = 372 \div 4 = 93$.

Problem 8: Liam scored 60, 78, 85, and 66 on four tests. What is the average score?

Answer: 72

Explanation: Average = $(60 + 78 + 85 + 66) \div 4 = 289 \div 4 = 72$.

Problem 9: James scored 76, 90, 78, and 95 on four tests. What is the average score?

Answer: 84

Explanation: Average = $(76 + 90 + 78 + 95) \div 4 = 339 \div 4 = 84$.

Problem 10: Abigail scored 83, 76, 69, and 79 on four tests. What is the average score?

Answer: 76

Explanation: Average = $(83 + 76 + 69 + 79) \div 4 = 307 \div 4 = 76$.

Problem 11: Evelyn scored 71, 86, 89, and 77 on four tests. What is the average score?

Answer: 80

Explanation: Average = $(71 + 86 + 89 + 77) \div 4 = 323 \div 4 = 80$.

Problem 12: Amelia scored 61, 76, 73, and 77 on four tests. What is the average score?

Answer: 71

Explanation: Average = $(61 + 76 + 73 + 77) \div 4 = 287 \div 4 = 71$.

Problem 13: Mason scored 82, 90, 67, and 81 on four tests. What is the average score?

Answer: 80

Explanation: Average = $(82 + 90 + 67 + 81) \div 4 = 320 \div 4 = 80$.

Problem 14: Harper scored 94, 60, 60, and 78 on four tests. What is the average score?

Answer: 73

Explanation: Average = $(94 + 60 + 60 + 78) \div 4 = 292 \div 4 = 73$.

Problem 15: Emily scored 73, 69, 84, and 67 on four tests. What is the average score?

Answer: 73

Explanation: Average = $(73 + 69 + 84 + 67) \div 4 = 293 \div 4 = 73$.

Problem 16: Sofia scored 91, 80, 83, and 72 on four tests. What is the average score?

Answer: 81

Explanation: Average = $(91 + 80 + 83 + 72) \div 4 = 326 \div 4 = 81$.

Problem 17: Amelia scored 85, 89, 82, and 70 on four tests. What is the average score?

Answer: 81

Explanation: Average = $(85 + 89 + 82 + 70) \div 4 = 326 \div 4 = 81$.

Problem 18: Charlotte scored 93, 96, 96, and 65 on four tests. What is the average score?

Answer: 87

Explanation: Average = $(93 + 96 + 96 + 65) \div 4 = 350 \div 4 = 87$.

Problem 19: James scored 90, 87, 83, and 98 on four tests. What is the average score?

Answer: 89

Explanation: Average = $(90 + 87 + 83 + 98) \div 4 = 358 \div 4 = 89$.

Problem 20: Amelia scored 88, 80, 85, and 60 on four tests. What is the average score?

Answer: 78

Explanation: Average = $(88 + 80 + 85 + 60) \div 4 = 313 \div 4 = 78$.

Problem 21: Sophia scored 81, 95, 97, and 77 on four tests. What is the average score?

Answer: 87

Explanation: Average = $(81 + 95 + 97 + 77) \div 4 = 350 \div 4 = 87$.

Problem 22: Abigail scored 68, 100, 63, and 96 on four tests. What is the average score?

Answer: 81

Explanation: Average = $(68 + 100 + 63 + 96) \div 4 = 327 \div 4 = 81$.

Problem 23: Elizabeth scored 65, 100, 90, and 86 on four tests. What is the average score?

Answer: 85

Explanation: Average = $(65 + 100 + 90 + 86) \div 4 = 341 \div 4 = 85$.

Problem 24: Mia scored 82, 90, 63, and 97 on four tests. What is the average score?

Answer: 83

Explanation: Average = $(82 + 90 + 63 + 97) \div 4 = 332 \div 4 = 83$.

Problem 25: Evelyn scored 95, 83, 81, and 64 on four tests. What is the average score?

Answer: 80

Explanation: Average = $(95 + 83 + 81 + 64) \div 4 = 323 \div 4 = 80$.

Problem 26: Henry scored 65, 88, 62, and 60 on four tests. What is the average score?

Answer: 68

Explanation: Average = $(65 + 88 + 62 + 60) \div 4 = 275 \div 4 = 68$.

Problem 27: Sophia scored 60, 82, 66, and 100 on four tests. What is the average score?

Answer: 77

Explanation: Average = $(60 + 82 + 66 + 100) \div 4 = 308 \div 4 = 77$.

Problem 28: Joseph scored 94, 97, 78, and 96 on four tests. What is the average score?

Answer: 91

Explanation: Average = $(94 + 97 + 78 + 96) \div 4 = 365 \div 4 = 91$.

Problem 29: Emily scored 93, 84, 96, and 75 on four tests. What is the average score?

Answer: 87

Explanation: Average = $(93 + 84 + 96 + 75) \div 4 = 348 \div 4 = 87$.

Problem 30: Abigail scored 78, 69, 76, and 81 on four tests. What is the average score?

Answer: 76

Explanation: Average = $(78 + 69 + 76 + 81) \div 4 = 304 \div 4 = 76$.

Problem 31: Ethan scored 80, 76, 68, and 86 on four tests. What is the average score?

Answer: 77

Explanation: Average = $(80 + 76 + 68 + 86) \div 4 = 310 \div 4 = 77$.

Problem 32: Lucas scored 100, 61, 90, and 62 on four tests. What is the average score?

Answer: 78

Explanation: Average = $(100 + 61 + 90 + 62) \div 4 = 313 \div 4 = 78$.

Problem 33: Emma scored 76, 86, 78, and 87 on four tests. What is the average score?

Answer: 81

Explanation: Average = $(76 + 86 + 78 + 87) \div 4 = 327 \div 4 = 81$.

Problem 34: Harper scored 74, 63, 87, and 82 on four tests. What is the average score?

Answer: 76

Explanation: Average = $(74 + 63 + 87 + 82) \div 4 = 306 \div 4 = 76$.

Problem 35: Alexander scored 88, 99, 85, and 64 on four tests. What is the average score?

Answer: 84

Explanation: Average = $(88 + 99 + 85 + 64) \div 4 = 336 \div 4 = 84$.

Problem 36: Elizabeth scored 85, 90, 70, and 86 on four tests. What is the average score?

Answer: 82

Explanation: Average = $(85 + 90 + 70 + 86) \div 4 = 331 \div 4 = 82$.

Problem 37: Jack scored 93, 98, 89, and 69 on four tests. What is the average score?

Answer: 87

Explanation: Average = $(93 + 98 + 89 + 69) \div 4 = 349 \div 4 = 87$.

Problem 38: Harper scored 65, 61, 100, and 69 on four tests. What is the average score?

Answer: 73

Explanation: Average = $(65 + 61 + 100 + 69) \div 4 = 295 \div 4 = 73$.

Problem 39: Lucas scored 79, 67, 61, and 75 on four tests. What is the average score?

Answer: 70

Explanation: Average = $(79 + 67 + 61 + 75) \div 4 = 282 \div 4 = 70$.

Problem 40: Elizabeth scored 89, 74, 81, and 97 on four tests. What is the average score?

Answer: 85

Explanation: Average = $(89 + 74 + 81 + 97) \div 4 = 341 \div 4 = 85$.

Problem 41: Emily scored 72, 79, 95, and 70 on four tests. What is the average score?

Answer: 79

Explanation: Average = $(72 + 79 + 95 + 70) \div 4 = 316 \div 4 = 79$.

Problem 42: Noah scored 78, 69, 80, and 97 on four tests. What is the average score?

Answer: 81

Explanation: Average = $(78 + 69 + 80 + 97) \div 4 = 324 \div 4 = 81$.

Problem 43: Amelia scored 82, 83, 84, and 83 on four tests. What is the average score?

Answer: 83

Explanation: Average = $(82 + 83 + 84 + 83) \div 4 = 332 \div 4 = 83$.

Problem 44: Olivia scored 94, 97, 82, and 77 on four tests. What is the average score?

Answer: 87

Explanation: Average = $(94 + 97 + 82 + 77) \div 4 = 350 \div 4 = 87$.

Problem 45: Benjamin scored 62, 65, 70, and 88 on four tests. What is the average score?

Answer: 71

Explanation: Average = $(62 + 65 + 70 + 88) \div 4 = 285 \div 4 = 71$.

Practice Exercises - Grade 3

Exercise 1: Aiden scored 85, 76, 71, and 64 on four tests. What is the average score?

Exercise 2: Charlotte scored 88, 66, 93, and 76 on four tests. What is the average score?

Exercise 3: Evelyn scored 78, 91, 82, and 80 on four tests. What is the average score?

Exercise 4: Abigail scored 87, 98, 63, and 89 on four tests. What is the average score?

Exercise 5: Olivia scored 86, 76, 71, and 63 on four tests. What is the average score?

Exercise 6: Samuel scored 75, 97, 62, and 79 on four tests. What is the average score?

Exercise 7: Ava scored 84, 94, 63, and 61 on four tests. What is the average score?

Exercise 8: Abigail scored 97, 84, 87, and 78 on four tests. What is the average score?

Exercise 9: Abigail scored 93, 62, 83, and 96 on four tests. What is the average score?

Exercise 10: Ava scored 88, 64, 95, and 96 on four tests. What is the average score?

Exercise 11: Alexander scored 68, 65, 87, and 63 on four tests. What is the average score?

Exercise 12: Evelyn scored 71, 61, 94, and 91 on four tests. What is the average score?

Exercise 13: Noah scored 75, 62, 60, and 70 on four tests. What is the average score?

Exercise 14: Emma scored 93, 100, 74, and 71 on four tests. What is the average score?

Exercise 15: Joseph scored 60, 85, 85, and 79 on four tests. What is the average score?

Exercise 16: Sophia scored 71, 95, 84, and 63 on four tests. What is the average score?

Exercise 17: Henry scored 90, 96, 69, and 99 on four tests. What is the average score?

Exercise 18: Avery scored 95, 97, 85, and 71 on four tests. What is the average score?

Exercise 19: Mason scored 67, 68, 91, and 80 on four tests. What is the average score?

Exercise 20: Emma scored 90, 81, 92, and 70 on four tests. What is the average score?

Practice Exercise Answers:

Exercise 1: 74

Exercise 2: 80

Exercise 3: 82

Exercise 4: 84

Exercise 5: 74

Exercise 6: 78

Exercise 7: 75

Exercise 8: 86

Exercise 9: 83

Exercise 10: 85

Exercise 11: 70

Exercise 12: 79

Exercise 13: 66

Exercise 14: 84

Exercise 15: 77

Exercise 16: 78

Exercise 17: 88

Exercise 18: 87

Exercise 19: 76

Exercise 20: 83

Challenge Problems - Grade 3

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: David scored 93, 90, 85, and 87 on four tests. What is the average score?

Answer: 88

Solution: Average = $(93 + 90 + 85 + 87) \div 4 = 355 \div 4 = 88$.

Challenge 2: Noah scored 91, 80, 91, and 84 on four tests. What is the average score?

Answer: 86

Solution: Average = $(91 + 80 + 91 + 84) \div 4 = 346 \div 4 = 86$.

Challenge 3: Liam scored 81, 73, 100, and 87 on four tests. What is the average score?

Answer: 85

Solution: Average = $(81 + 73 + 100 + 87) \div 4 = 341 \div 4 = 85$.

Challenge 4: Abigail scored 67, 73, 98, and 75 on four tests. What is the average score?

Answer: 78

Solution: Average = $(67 + 73 + 98 + 75) \div 4 = 313 \div 4 = 78$.

Challenge 5: Abigail scored 100, 60, 86, and 69 on four tests. What is the average score?

Answer: 78

Solution: Average = $(100 + 60 + 86 + 69) \div 4 = 315 \div 4 = 78$.

Challenge 6: Ethan scored 95, 81, 77, and 69 on four tests. What is the average score?

Answer: 80

Solution: Average = $(95 + 81 + 77 + 69) \div 4 = 322 \div 4 = 80$.

Challenge 7: Joseph scored 88, 91, 68, and 95 on four tests. What is the average score?

Answer: 85

Solution: Average = $(88 + 91 + 68 + 95) \div 4 = 342 \div 4 = 85$.

Challenge 8: Ethan scored 82, 79, 71, and 98 on four tests. What is the average score?

Answer: 82

Solution: Average = $(82 + 79 + 71 + 98) \div 4 = 330 \div 4 = 82$.

Challenge 9: Emily scored 83, 88, 91, and 82 on four tests. What is the average score?

Answer: 86

Solution: Average = $(83 + 88 + 91 + 82) \div 4 = 344 \div 4 = 86$.

Challenge 10: Samuel scored 86, 71, 94, and 97 on four tests. What is the average score?

Answer: 87

Solution: Average = $(86 + 71 + 94 + 97) \div 4 = 348 \div 4 = 87$.

Challenge 11: Emily scored 96, 88, 95, and 60 on four tests. What is the average score?

Answer: 84

Solution: Average = $(96 + 88 + 95 + 60) \div 4 = 339 \div 4 = 84$.

Challenge 12: Liam scored 89, 77, 75, and 70 on four tests. What is the average score?

Answer: 77

Solution: Average = $(89 + 77 + 75 + 70) \div 4 = 311 \div 4 = 77$.

Challenge 13: Sofia scored 69, 83, 100, and 78 on four tests. What is the average score?

Answer: 82

Solution: Average = $(69 + 83 + 100 + 78) \div 4 = 330 \div 4 = 82$.

Challenge 14: Joseph scored 90, 62, 75, and 99 on four tests. What is the average score?

Answer: 81

Solution: Average = $(90 + 62 + 75 + 99) \div 4 = 326 \div 4 = 81$.

Challenge 15: Mia scored 61, 81, 87, and 81 on four tests. What is the average score?

Answer: 77

Solution: Average = $(61 + 81 + 87 + 81) \div 4 = 310 \div 4 = 77$.

Grade 4 Problems

Problem 1: Students measured heights: 10, 8, 2, 4, 9, 9, 6, 9 inches. What is the mode (most common value)?

Answer: 9

Explanation: The value 9 appears most frequently in the data.

Problem 2: Students measured heights: 1, 10, 1, 1, 2, 6, 9, 2 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Problem 3: Students measured heights: 3, 6, 1, 7, 7, 3, 4, 1 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Problem 4: Students measured heights: 2, 2, 10, 7, 1, 1, 9, 7 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Problem 5: Students measured heights: 2, 10, 5, 4, 3, 2, 8, 5 inches. What is the mode (most common value)?

Answer: 2

Explanation: The value 2 appears most frequently in the data.

Problem 6: Students measured heights: 6, 8, 8, 5, 10, 4, 3, 9 inches. What is the mode (most common value)?

Answer: 8

Explanation: The value 8 appears most frequently in the data.

Problem 7: Students measured heights: 2, 1, 10, 7, 7, 6, 3, 10 inches. What is the mode (most common value)?

Answer: 7

Explanation: The value 7 appears most frequently in the data.

Problem 8: Students measured heights: 3, 9, 9, 1, 9, 10, 3, 7 inches. What is the mode (most common value)?

Answer: 9

Explanation: The value 9 appears most frequently in the data.

Problem 9: Students measured heights: 4, 1, 6, 10, 10, 7, 3, 8 inches. What is the mode (most common value)?

Answer: 10

Explanation: The value 10 appears most frequently in the data.

Problem 10: Students measured heights: 8, 2, 1, 3, 7, 1, 1, 6 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Problem 11: Students measured heights: 3, 7, 4, 1, 1, 2, 8, 5 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Problem 12: Students measured heights: 9, 1, 6, 8, 3, 1, 4, 6 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Problem 13: Students measured heights: 7, 4, 8, 4, 1, 4, 1, 8 inches. What is the mode (most common value)?

Answer: 4

Explanation: The value 4 appears most frequently in the data.

Problem 14: Students measured heights: 3, 1, 5, 8, 3, 3, 5, 8 inches. What is the mode (most common value)?

Answer: 3

Explanation: The value 3 appears most frequently in the data.

Problem 15: Students measured heights: 7, 8, 8, 5, 8, 6, 3, 6 inches. What is the mode (most common value)?

Answer: 8

Explanation: The value 8 appears most frequently in the data.

Problem 16: Students measured heights: 4, 6, 8, 10, 2, 9, 7, 3 inches. What is the mode (most common value)?

Answer: 2

Explanation: The value 2 appears most frequently in the data.

Problem 17: Students measured heights: 6, 9, 4, 3, 8, 10, 10, 9 inches. What is the mode (most common value)?

Answer: 9

Explanation: The value 9 appears most frequently in the data.

Problem 18: Students measured heights: 5, 8, 1, 2, 6, 6, 2, 8 inches. What is the mode (most common value)?

Answer: 2

Explanation: The value 2 appears most frequently in the data.

Problem 19: Students measured heights: 1, 6, 1, 1, 2, 3, 1, 2 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Problem 20: Students measured heights: 3, 3, 2, 4, 7, 5, 8, 6 inches. What is the mode (most common value)?

Answer: 3

Explanation: The value 3 appears most frequently in the data.

Problem 21: Students measured heights: 4, 2, 2, 5, 1, 9, 6, 10 inches. What is the mode (most common value)?

Answer: 2

Explanation: The value 2 appears most frequently in the data.

Problem 22: Students measured heights: 1, 2, 7, 5, 2, 9, 10, 5 inches. What is the mode (most common value)?

Answer: 2

Explanation: The value 2 appears most frequently in the data.

Problem 23: Students measured heights: 3, 5, 3, 5, 10, 2, 1, 7 inches. What is the mode (most common value)?

Answer: 3

Explanation: The value 3 appears most frequently in the data.

Problem 24: Students measured heights: 3, 8, 5, 7, 7, 2, 4, 3 inches. What is the mode (most common value)?

Answer: 3

Explanation: The value 3 appears most frequently in the data.

Problem 25: Students measured heights: 3, 9, 2, 10, 8, 9, 6, 2 inches. What is the mode (most common value)?

Answer: 2

Explanation: The value 2 appears most frequently in the data.

Problem 26: Students measured heights: 6, 4, 3, 9, 6, 10, 8, 8 inches. What is the mode (most common value)?

Answer: 6

Explanation: The value 6 appears most frequently in the data.

Problem 27: Students measured heights: 9, 6, 1, 3, 5, 5, 7, 7 inches. What is the mode (most common value)?

Answer: 5

Explanation: The value 5 appears most frequently in the data.

Problem 28: Students measured heights: 6, 7, 7, 9, 7, 7, 5, 6 inches. What is the mode (most common value)?

Answer: 7

Explanation: The value 7 appears most frequently in the data.

Problem 29: Students measured heights: 3, 10, 4, 8, 2, 2, 2, 7 inches. What is the mode (most common value)?

Answer: 2

Explanation: The value 2 appears most frequently in the data.

Problem 30: Students measured heights: 10, 10, 8, 8, 10, 6, 5, 6 inches. What is the mode (most common value)?

Answer: 10

Explanation: The value 10 appears most frequently in the data.

Problem 31: Students measured heights: 2, 1, 5, 9, 9, 1, 8, 8 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Problem 32: Students measured heights: 10, 3, 3, 1, 10, 6, 9, 2 inches. What is the mode (most common value)?

Answer: 3

Explanation: The value 3 appears most frequently in the data.

Problem 33: Students measured heights: 10, 10, 9, 1, 1, 6, 9, 3 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Problem 34: Students measured heights: 10, 4, 5, 3, 10, 4, 9, 5 inches. What is the mode (most common value)?

Answer: 4

Explanation: The value 4 appears most frequently in the data.

Problem 35: Students measured heights: 8, 4, 6, 4, 5, 3, 10, 5 inches. What is the mode (most common value)?

Answer: 4

Explanation: The value 4 appears most frequently in the data.

Problem 36: Students measured heights: 10, 7, 10, 1, 5, 8, 7, 7 inches. What is the mode (most common value)?

Answer: 7

Explanation: The value 7 appears most frequently in the data.

Problem 37: Students measured heights: 1, 6, 8, 8, 1, 1, 7, 2 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Problem 38: Students measured heights: 8, 10, 4, 8, 5, 9, 9, 8 inches. What is the mode (most common value)?

Answer: 8

Explanation: The value 8 appears most frequently in the data.

Problem 39: Students measured heights: 5, 10, 5, 7, 3, 6, 2, 9 inches. What is the mode (most common value)?

Answer: 5

Explanation: The value 5 appears most frequently in the data.

Problem 40: Students measured heights: 9, 8, 8, 6, 6, 1, 1, 3 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Problem 41: Students measured heights: 4, 10, 9, 5, 10, 6, 4, 9 inches. What is the mode (most common value)?

Answer: 4

Explanation: The value 4 appears most frequently in the data.

Problem 42: Students measured heights: 1, 3, 7, 6, 5, 9, 10, 4 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Problem 43: Students measured heights: 6, 7, 5, 1, 10, 2, 6, 2 inches. What is the mode (most common value)?

Answer: 2

Explanation: The value 2 appears most frequently in the data.

Problem 44: Students measured heights: 2, 9, 5, 2, 1, 5, 1, 3 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Problem 45: Students measured heights: 4, 10, 2, 3, 2, 5, 1, 1 inches. What is the mode (most common value)?

Answer: 1

Explanation: The value 1 appears most frequently in the data.

Practice Exercises - Grade 4

Exercise 1: Students measured heights: 3, 1, 9, 8, 2, 4, 2, 8 inches. What is the mode (most common value)?

Exercise 2: Students measured heights: 3, 10, 8, 8, 7, 1, 5, 1 inches. What is the mode (most common value)?

Exercise 3: Students measured heights: 10, 2, 7, 6, 2, 7, 10, 1 inches. What is the mode (most common value)?

Exercise 4: Students measured heights: 3, 2, 9, 10, 2, 5, 6, 2 inches. What is the mode (most common value)?

Exercise 5: Students measured heights: 7, 8, 6, 2, 5, 9, 6, 5 inches. What is the mode (most common value)?

Exercise 6: Students measured heights: 5, 7, 2, 7, 4, 4, 9, 9 inches. What is the mode (most common value)?

Exercise 7: Students measured heights: 6, 10, 9, 8, 8, 8, 6, 5 inches. What is the mode (most common value)?

Exercise 8: Students measured heights: 1, 10, 5, 10, 9, 6, 5, 8 inches. What is the mode (most common value)?

Exercise 9: Students measured heights: 2, 3, 5, 5, 5, 1, 4, 5 inches. What is the mode (most common value)?

Exercise 10: Students measured heights: 7, 2, 6, 10, 2, 2, 1, 2 inches. What is the mode (most common value)?

Exercise 11: Students measured heights: 9, 7, 8, 3, 3, 7, 8, 3 inches. What is the mode (most common value)?

Exercise 12: Students measured heights: 3, 5, 8, 1, 5, 7, 6, 9 inches. What is the mode (most common value)?

Exercise 13: Students measured heights: 5, 8, 2, 2, 1, 10, 7, 5 inches. What is the mode (most common value)?

Exercise 14: Students measured heights: 9, 6, 1, 4, 6, 3, 9, 2 inches. What is the mode (most common value)?

Exercise 15: Students measured heights: 9, 4, 7, 6, 7, 8, 1, 3 inches. What is the mode (most common value)?

Exercise 16: Students measured heights: 1, 5, 8, 8, 4, 5, 6, 8 inches. What is the mode (most common value)?

Exercise 17: Students measured heights: 9, 2, 7, 4, 1, 8, 7, 4 inches. What is the mode (most common value)?

Exercise 18: Students measured heights: 6, 10, 10, 9, 8, 10, 10, 8 inches. What is the mode (most common value)?

Exercise 19: Students measured heights: 7, 9, 6, 5, 3, 3, 1, 3 inches. What is the mode (most common value)?

Exercise 20: Students measured heights: 6, 2, 10, 7, 2, 6, 1, 3 inches. What is the mode (most common value)?

Practice Exercise Answers:

Exercise 1: 2

Exercise 2: 1

Exercise 3: 2

Exercise 4: 2

Exercise 5: 5

Exercise 6: 4

Exercise 7: 8

Exercise 8: 5

Exercise 9: 5

Exercise 10: 2

Exercise 11: 3

Exercise 12: 5

Exercise 13: 2

Exercise 14: 6

Exercise 15: 7

Exercise 16: 8

Exercise 17: 4

Exercise 18: 10

Exercise 19: 3

Exercise 20: 2

Challenge Problems - Grade 4

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Students measured heights: 4, 1, 5, 4, 8, 9, 8, 7 inches. What is the mode (most common value)?

Answer: 4

Solution: The value 4 appears most frequently in the data.

Challenge 2: Students measured heights: 2, 7, 10, 2, 10, 5, 5, 6 inches. What is the mode (most common value)?

Answer: 2

Solution: The value 2 appears most frequently in the data.

Challenge 3: Students measured heights: 6, 7, 1, 3, 5, 1, 6, 10 inches. What is the mode (most common value)?

Answer: 1

Solution: The value 1 appears most frequently in the data.

Challenge 4: Students measured heights: 4, 5, 7, 2, 10, 3, 5, 2 inches. What is the mode (most common value)?

Answer: 2

Solution: The value 2 appears most frequently in the data.

Challenge 5: Students measured heights: 10, 8, 2, 3, 8, 3, 9, 10 inches. What is the mode (most common value)?

Answer: 3

Solution: The value 3 appears most frequently in the data.

Challenge 6: Students measured heights: 9, 8, 8, 9, 9, 5, 4, 1 inches. What is the mode (most common value)?

Answer: 9

Solution: The value 9 appears most frequently in the data.

Challenge 7: Students measured heights: 2, 2, 2, 3, 5, 9, 8, 7 inches. What is the mode (most common value)?

Answer: 2

Solution: The value 2 appears most frequently in the data.

Challenge 8: Students measured heights: 4, 2, 8, 10, 8, 8, 9, 8 inches. What is the mode (most common value)?

Answer: 8

Solution: The value 8 appears most frequently in the data.

Challenge 9: Students measured heights: 5, 6, 1, 5, 1, 2, 4, 3 inches. What is the mode (most common value)?

Answer: 1

Solution: The value 1 appears most frequently in the data.

Challenge 10: Students measured heights: 10, 4, 3, 1, 6, 7, 6, 6 inches. What is the mode (most common value)?

Answer: 6

Solution: The value 6 appears most frequently in the data.

Challenge 11: Students measured heights: 3, 1, 4, 3, 7, 10, 1, 6 inches. What is the mode (most common value)?

Answer: 1

Solution: The value 1 appears most frequently in the data.

Challenge 12: Students measured heights: 4, 10, 8, 10, 6, 5, 4, 5 inches. What is the mode (most common value)?

Answer: 4

Solution: The value 4 appears most frequently in the data.

Challenge 13: Students measured heights: 2, 2, 10, 9, 4, 4, 8, 10 inches. What is the mode (most common value)?

Answer: 2

Solution: The value 2 appears most frequently in the data.

Challenge 14: Students measured heights: 6, 2, 7, 2, 8, 2, 6, 6 inches. What is the mode (most common value)?

Answer: 2

Solution: The value 2 appears most frequently in the data.

Challenge 15: Students measured heights: 3, 9, 6, 7, 9, 9, 6, 8 inches. What is the mode (most common value)?

Answer: 9

Solution: The value 9 appears most frequently in the data.

Grade 5 Problems

Problem 1: Five students scored: 23, 31, 35, 50, 50 points. What is the median score?

Answer: 35

Explanation: Ordered data: 23, 31, 35, 50, 50. The median (middle value) is 35.

Problem 2: Five students scored: 18, 32, 38, 49, 50 points. What is the median score?

Answer: 38

Explanation: Ordered data: 18, 32, 38, 49, 50. The median (middle value) is 38.

Problem 3: Five students scored: 14, 22, 33, 34, 47 points. What is the median score?

Answer: 33

Explanation: Ordered data: 14, 22, 33, 34, 47. The median (middle value) is 33.

Problem 4: Five students scored: 22, 29, 31, 35, 50 points. What is the median score?

Answer: 31

Explanation: Ordered data: 22, 29, 31, 35, 50. The median (middle value) is 31.

Problem 5: Five students scored: 16, 24, 26, 41, 47 points. What is the median score?

Answer: 26

Explanation: Ordered data: 16, 24, 26, 41, 47. The median (middle value) is 26.

Problem 6: Five students scored: 11, 13, 26, 29, 47 points. What is the median score?

Answer: 26

Explanation: Ordered data: 11, 13, 26, 29, 47. The median (middle value) is 26.

Problem 7: Five students scored: 15, 16, 18, 34, 37 points. What is the median score?

Answer: 18

Explanation: Ordered data: 15, 16, 18, 34, 37. The median (middle value) is 18.

Problem 8: Five students scored: 15, 24, 24, 42, 44 points. What is the median score?

Answer: 24

Explanation: Ordered data: 15, 24, 24, 42, 44. The median (middle value) is 24.

Problem 9: Five students scored: 12, 17, 28, 30, 40 points. What is the median score?

Answer: 28

Explanation: Ordered data: 12, 17, 28, 30, 40. The median (middle value) is 28.

Problem 10: Five students scored: 11, 12, 18, 30, 34 points. What is the median score?

Answer: 18

Explanation: Ordered data: 11, 12, 18, 30, 34. The median (middle value) is 18.

Problem 11: Five students scored: 15, 19, 31, 33, 47 points. What is the median score?

Answer: 31

Explanation: Ordered data: 15, 19, 31, 33, 47. The median (middle value) is 31.

Problem 12: Five students scored: 17, 23, 25, 37, 40 points. What is the median score?

Answer: 25

Explanation: Ordered data: 17, 23, 25, 37, 40. The median (middle value) is 25.

Problem 13: Five students scored: 21, 27, 29, 36, 45 points. What is the median score?

Answer: 29

Explanation: Ordered data: 21, 27, 29, 36, 45. The median (middle value) is 29.

Problem 14: Five students scored: 10, 37, 46, 47, 50 points. What is the median score?

Answer: 46

Explanation: Ordered data: 10, 37, 46, 47, 50. The median (middle value) is 46.

Problem 15: Five students scored: 14, 26, 36, 49, 49 points. What is the median score?

Answer: 36

Explanation: Ordered data: 14, 26, 36, 49, 49. The median (middle value) is 36.

Problem 16: Five students scored: 14, 19, 30, 41, 43 points. What is the median score?

Answer: 30

Explanation: Ordered data: 14, 19, 30, 41, 43. The median (middle value) is 30.

Problem 17: Five students scored: 12, 15, 19, 47, 49 points. What is the median score?

Answer: 19

Explanation: Ordered data: 12, 15, 19, 47, 49. The median (middle value) is 19.

Problem 18: Five students scored: 12, 16, 24, 24, 33 points. What is the median score?

Answer: 24

Explanation: Ordered data: 12, 16, 24, 24, 33. The median (middle value) is 24.

Problem 19: Five students scored: 15, 28, 29, 29, 35 points. What is the median score?

Answer: 29

Explanation: Ordered data: 15, 28, 29, 29, 35. The median (middle value) is 29.

Problem 20: Five students scored: 11, 15, 16, 43, 46 points. What is the median score?

Answer: 16

Explanation: Ordered data: 11, 15, 16, 43, 46. The median (middle value) is 16.

Problem 21: Five students scored: 23, 28, 39, 41, 50 points. What is the median score?

Answer: 39

Explanation: Ordered data: 23, 28, 39, 41, 50. The median (middle value) is 39.

Problem 22: Five students scored: 22, 33, 39, 41, 44 points. What is the median score?

Answer: 39

Explanation: Ordered data: 22, 33, 39, 41, 44. The median (middle value) is 39.

Problem 23: Five students scored: 18, 19, 34, 44, 49 points. What is the median score?

Answer: 34

Explanation: Ordered data: 18, 19, 34, 44, 49. The median (middle value) is 34.

Problem 24: Five students scored: 11, 25, 39, 45, 50 points. What is the median score?

Answer: 39

Explanation: Ordered data: 11, 25, 39, 45, 50. The median (middle value) is 39.

Problem 25: Five students scored: 21, 29, 33, 35, 43 points. What is the median score?

Answer: 33

Explanation: Ordered data: 21, 29, 33, 35, 43. The median (middle value) is 33.

Problem 26: Five students scored: 10, 32, 34, 39, 48 points. What is the median score?

Answer: 34

Explanation: Ordered data: 10, 32, 34, 39, 48. The median (middle value) is 34.

Problem 27: Five students scored: 11, 19, 37, 42, 43 points. What is the median score?

Answer: 37

Explanation: Ordered data: 11, 19, 37, 42, 43. The median (middle value) is 37.

Problem 28: Five students scored: 19, 31, 34, 34, 49 points. What is the median score?

Answer: 34

Explanation: Ordered data: 19, 31, 34, 34, 49. The median (middle value) is 34.

Problem 29: Five students scored: 14, 19, 33, 45, 48 points. What is the median score?

Answer: 33

Explanation: Ordered data: 14, 19, 33, 45, 48. The median (middle value) is 33.

Problem 30: Five students scored: 10, 13, 26, 26, 32 points. What is the median score?

Answer: 26

Explanation: Ordered data: 10, 13, 26, 26, 32. The median (middle value) is 26.

Problem 31: Five students scored: 12, 33, 34, 39, 50 points. What is the median score?

Answer: 34

Explanation: Ordered data: 12, 33, 34, 39, 50. The median (middle value) is 34.

Problem 32: Five students scored: 18, 24, 25, 45, 50 points. What is the median score?

Answer: 25

Explanation: Ordered data: 18, 24, 25, 45, 50. The median (middle value) is 25.

Problem 33: Five students scored: 13, 18, 22, 35, 48 points. What is the median score?

Answer: 22

Explanation: Ordered data: 13, 18, 22, 35, 48. The median (middle value) is 22.

Problem 34: Five students scored: 13, 23, 25, 29, 39 points. What is the median score?

Answer: 25

Explanation: Ordered data: 13, 23, 25, 29, 39. The median (middle value) is 25.

Problem 35: Five students scored: 21, 31, 44, 48, 50 points. What is the median score?

Answer: 44

Explanation: Ordered data: 21, 31, 44, 48, 50. The median (middle value) is 44.

Problem 36: Five students scored: 12, 14, 25, 38, 41 points. What is the median score?

Answer: 25

Explanation: Ordered data: 12, 14, 25, 38, 41. The median (middle value) is 25.

Problem 37: Five students scored: 21, 31, 37, 38, 40 points. What is the median score?

Answer: 37

Explanation: Ordered data: 21, 31, 37, 38, 40. The median (middle value) is 37.

Problem 38: Five students scored: 18, 19, 24, 34, 38 points. What is the median score?

Answer: 24

Explanation: Ordered data: 18, 19, 24, 34, 38. The median (middle value) is 24.

Problem 39: Five students scored: 17, 24, 33, 40, 48 points. What is the median score?

Answer: 33

Explanation: Ordered data: 17, 24, 33, 40, 48. The median (middle value) is 33.

Problem 40: Five students scored: 10, 21, 21, 26, 43 points. What is the median score?

Answer: 21

Explanation: Ordered data: 10, 21, 21, 26, 43. The median (middle value) is 21.

Problem 41: Five students scored: 14, 21, 34, 38, 49 points. What is the median score?

Answer: 34

Explanation: Ordered data: 14, 21, 34, 38, 49. The median (middle value) is 34.

Problem 42: Five students scored: 21, 36, 38, 41, 45 points. What is the median score?

Answer: 38

Explanation: Ordered data: 21, 36, 38, 41, 45. The median (middle value) is 38.

Problem 43: Five students scored: 19, 25, 28, 48, 49 points. What is the median score?

Answer: 28

Explanation: Ordered data: 19, 25, 28, 48, 49. The median (middle value) is 28.

Problem 44: Five students scored: 13, 19, 43, 47, 48 points. What is the median score?

Answer: 43

Explanation: Ordered data: 13, 19, 43, 47, 48. The median (middle value) is 43.

Problem 45: Five students scored: 13, 29, 31, 45, 47 points. What is the median score?

Answer: 31

Explanation: Ordered data: 13, 29, 31, 45, 47. The median (middle value) is 31.

Practice Exercises - Grade 5

Exercise 1: Five students scored: 35, 40, 47, 49, 49 points. What is the median score?

Exercise 2: Five students scored: 13, 18, 28, 37, 44 points. What is the median score?

Exercise 3: Five students scored: 11, 18, 18, 34, 48 points. What is the median score?

Exercise 4: Five students scored: 15, 21, 31, 35, 42 points. What is the median score?

Exercise 5: Five students scored: 17, 22, 38, 42, 44 points. What is the median score?

Exercise 6: Five students scored: 17, 22, 42, 45, 50 points. What is the median score?

Exercise 7: Five students scored: 11, 17, 21, 28, 45 points. What is the median score?

Exercise 8: Five students scored: 14, 17, 20, 26, 38 points. What is the median score?

Exercise 9: Five students scored: 23, 23, 27, 30, 42 points. What is the median score?

Exercise 10: Five students scored: 10, 15, 18, 22, 33 points. What is the median score?

Exercise 11: Five students scored: 19, 23, 26, 30, 48 points. What is the median score?

Exercise 12: Five students scored: 16, 23, 48, 50, 50 points. What is the median score?

Exercise 13: Five students scored: 11, 13, 25, 36, 50 points. What is the median score?

Exercise 14: Five students scored: 21, 28, 37, 44, 49 points. What is the median score?

Exercise 15: Five students scored: 13, 23, 23, 32, 36 points. What is the median score?

Exercise 16: Five students scored: 11, 13, 13, 16, 47 points. What is the median score?

Exercise 17: Five students scored: 11, 18, 19, 22, 48 points. What is the median score?

Exercise 18: Five students scored: 15, 15, 40, 44, 50 points. What is the median score?

Exercise 19: Five students scored: 17, 24, 27, 30, 36 points. What is the median score?

Exercise 20: Five students scored: 12, 14, 18, 31, 32 points. What is the median score?

Practice Exercise Answers:

Exercise 1: 47

Exercise 2: 28

Exercise 3: 18

Exercise 4: 31

Exercise 5: 38

Exercise 6: 42

Exercise 7: 21

Exercise 8: 20

Exercise 9: 27

Exercise 10: 18

Exercise 11: 26

Exercise 12: 48

Exercise 13: 25

Exercise 14: 37

Exercise 15: 23

Exercise 16: 13

Exercise 17: 19

Exercise 18: 40

Exercise 19: 27

Exercise 20: 18

Challenge Problems - Grade 5

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Five students scored: 10, 13, 18, 22, 27 points. What is the median score?

Answer: 18

Solution: Ordered data: 10, 13, 18, 22, 27. The median (middle value) is 18.

Challenge 2: Five students scored: 15, 19, 29, 38, 42 points. What is the median score?

Answer: 29

Solution: Ordered data: 15, 19, 29, 38, 42. The median (middle value) is 29.

Challenge 3: Five students scored: 12, 23, 36, 44, 44 points. What is the median score?

Answer: 36

Solution: Ordered data: 12, 23, 36, 44, 44. The median (middle value) is 36.

Challenge 4: Five students scored: 13, 25, 43, 48, 50 points. What is the median score?

Answer: 43

Solution: Ordered data: 13, 25, 43, 48, 50. The median (middle value) is 43.

Challenge 5: Five students scored: 23, 31, 37, 37, 39 points. What is the median score?

Answer: 37

Solution: Ordered data: 23, 31, 37, 37, 39. The median (middle value) is 37.

Challenge 6: Five students scored: 26, 30, 31, 34, 49 points. What is the median score?

Answer: 31

Solution: Ordered data: 26, 30, 31, 34, 49. The median (middle value) is 31.

Challenge 7: Five students scored: 18, 19, 28, 30, 42 points. What is the median score?

Answer: 28

Solution: Ordered data: 18, 19, 28, 30, 42. The median (middle value) is 28.

Challenge 8: Five students scored: 14, 18, 27, 30, 34 points. What is the median score?

Answer: 27

Solution: Ordered data: 14, 18, 27, 30, 34. The median (middle value) is 27.

Challenge 9: Five students scored: 37, 40, 41, 46, 47 points. What is the median score?

Answer: 41

Solution: Ordered data: 37, 40, 41, 46, 47. The median (middle value) is 41.

Challenge 10: Five students scored: 24, 26, 30, 33, 41 points. What is the median score?

Answer: 30

Solution: Ordered data: 24, 26, 30, 33, 41. The median (middle value) is 30.

Challenge 11: Five students scored: 15, 20, 21, 22, 34 points. What is the median score?

Answer: 21

Solution: Ordered data: 15, 20, 21, 22, 34. The median (middle value) is 21.

Challenge 12: Five students scored: 17, 23, 35, 42, 45 points. What is the median score?

Answer: 35

Solution: Ordered data: 17, 23, 35, 42, 45. The median (middle value) is 35.

Challenge 13: Five students scored: 15, 23, 37, 44, 48 points. What is the median score?

Answer: 37

Solution: Ordered data: 15, 23, 37, 44, 48. The median (middle value) is 37.

Challenge 14: Five students scored: 14, 27, 36, 38, 49 points. What is the median score?

Answer: 36

Solution: Ordered data: 14, 27, 36, 38, 49. The median (middle value) is 36.

Challenge 15: Five students scored: 20, 31, 34, 35, 47 points. What is the median score?

Answer: 34

Solution: Ordered data: 20, 31, 34, 35, 47. The median (middle value) is 34.

Grade 6 Problems

Problem 1: Test scores were: 33, 37, 38, 43, 50, 57, 70. What is the range?

Answer: 37

Explanation: Range = highest - lowest = $70 - 33 = 37$.

Problem 2: Test scores were: 26, 27, 43, 61, 67, 68, 78. What is the range?

Answer: 52

Explanation: Range = highest - lowest = $78 - 26 = 52$.

Problem 3: Test scores were: 23, 24, 32, 42, 50, 58, 68. What is the range?

Answer: 45

Explanation: Range = highest - lowest = $68 - 23 = 45$.

Problem 4: Test scores were: 32, 35, 48, 48, 50, 72, 78. What is the range?

Answer: 46

Explanation: Range = highest - lowest = $78 - 32 = 46$.

Problem 5: Test scores were: 21, 31, 44, 49, 56, 61, 78. What is the range?

Answer: 57

Explanation: Range = highest - lowest = $78 - 21 = 57$.

Problem 6: Test scores were: 23, 23, 31, 43, 48, 59, 64. What is the range?

Answer: 41

Explanation: Range = highest - lowest = $64 - 23 = 41$.

Problem 7: Test scores were: 22, 27, 33, 37, 48, 57, 71. What is the range?

Answer: 49

Explanation: Range = highest - lowest = $71 - 22 = 49$.

Problem 8: Test scores were: 26, 36, 36, 45, 45, 46, 61. What is the range?

Answer: 35

Explanation: Range = highest - lowest = $61 - 26 = 35$.

Problem 9: Test scores were: 32, 45, 56, 57, 68, 75, 79. What is the range?

Answer: 47

Explanation: Range = highest - lowest = $79 - 32 = 47$.

Problem 10: Test scores were: 31, 32, 50, 55, 69, 74, 76. What is the range?

Answer: 45

Explanation: Range = highest - lowest = $76 - 31 = 45$.

Problem 11: Test scores were: 22, 34, 51, 52, 54, 65, 72. What is the range?

Answer: 50

Explanation: Range = highest - lowest = $72 - 22 = 50$.

Problem 12: Test scores were: 20, 31, 46, 55, 55, 70, 75. What is the range?

Answer: 55

Explanation: Range = highest - lowest = $75 - 20 = 55$.

Problem 13: Test scores were: 28, 42, 51, 61, 63, 73, 75. What is the range?

Answer: 47

Explanation: Range = highest - lowest = $75 - 28 = 47$.

Problem 14: Test scores were: 39, 55, 59, 61, 66, 70, 72. What is the range?

Answer: 33

Explanation: Range = highest - lowest = $72 - 39 = 33$.

Problem 15: Test scores were: 21, 39, 48, 52, 54, 64, 77. What is the range?

Answer: 56

Explanation: Range = highest - lowest = $77 - 21 = 56$.

Problem 16: Test scores were: 26, 31, 33, 44, 56, 61, 77. What is the range?

Answer: 51

Explanation: Range = highest - lowest = $77 - 26 = 51$.

Problem 17: Test scores were: 31, 37, 37, 59, 65, 66, 74. What is the range?

Answer: 43

Explanation: Range = highest - lowest = $74 - 31 = 43$.

Problem 18: Test scores were: 23, 25, 42, 45, 55, 56, 60. What is the range?

Answer: 37

Explanation: Range = highest - lowest = $60 - 23 = 37$.

Problem 19: Test scores were: 40, 44, 46, 48, 59, 64, 74. What is the range?

Answer: 34

Explanation: Range = highest - lowest = $74 - 40 = 34$.

Problem 20: Test scores were: 25, 41, 43, 45, 53, 63, 66. What is the range?

Answer: 41

Explanation: Range = highest - lowest = $66 - 25 = 41$.

Problem 21: Test scores were: 24, 34, 39, 42, 51, 60, 77. What is the range?

Answer: 53

Explanation: Range = highest - lowest = $77 - 24 = 53$.

Problem 22: Test scores were: 20, 28, 44, 46, 53, 67, 79. What is the range?

Answer: 59

Explanation: Range = highest - lowest = $79 - 20 = 59$.

Problem 23: Test scores were: 28, 35, 37, 37, 53, 71, 77. What is the range?

Answer: 49

Explanation: Range = highest - lowest = $77 - 28 = 49$.

Problem 24: Test scores were: 22, 27, 31, 48, 60, 65, 66. What is the range?

Answer: 44

Explanation: Range = highest - lowest = $66 - 22 = 44$.

Problem 25: Test scores were: 20, 21, 31, 51, 52, 53, 80. What is the range?

Answer: 60

Explanation: Range = highest - lowest = $80 - 20 = 60$.

Problem 26: Test scores were: 27, 35, 36, 37, 58, 67, 69. What is the range?

Answer: 42

Explanation: Range = highest - lowest = $69 - 27 = 42$.

Problem 27: Test scores were: 21, 28, 29, 41, 45, 76, 76. What is the range?

Answer: 55

Explanation: Range = highest - lowest = $76 - 21 = 55$.

Problem 28: Test scores were: 27, 32, 38, 44, 62, 72, 80. What is the range?

Answer: 53

Explanation: Range = highest - lowest = $80 - 27 = 53$.

Problem 29: Test scores were: 30, 38, 45, 52, 53, 58, 75. What is the range?

Answer: 45

Explanation: Range = highest - lowest = $75 - 30 = 45$.

Problem 30: Test scores were: 21, 45, 50, 53, 55, 55, 61. What is the range?

Answer: 40

Explanation: Range = highest - lowest = $61 - 21 = 40$.

Problem 31: Test scores were: 31, 41, 46, 47, 53, 53, 63. What is the range?

Answer: 32

Explanation: Range = highest - lowest = $63 - 31 = 32$.

Problem 32: Test scores were: 26, 38, 47, 50, 57, 60, 70. What is the range?

Answer: 44

Explanation: Range = highest - lowest = $70 - 26 = 44$.

Problem 33: Test scores were: 28, 33, 45, 50, 54, 56, 68. What is the range?

Answer: 40

Explanation: Range = highest - lowest = $68 - 28 = 40$.

Problem 34: Test scores were: 20, 25, 25, 39, 58, 63, 66. What is the range?

Answer: 46

Explanation: Range = highest - lowest = $66 - 20 = 46$.

Problem 35: Test scores were: 20, 44, 44, 48, 57, 72, 80. What is the range?

Answer: 60

Explanation: Range = highest - lowest = $80 - 20 = 60$.

Problem 36: Test scores were: 31, 31, 34, 48, 58, 60, 77. What is the range?

Answer: 46

Explanation: Range = highest - lowest = $77 - 31 = 46$.

Problem 37: Test scores were: 31, 36, 39, 46, 49, 71, 76. What is the range?

Answer: 45

Explanation: Range = highest - lowest = $76 - 31 = 45$.

Problem 38: Test scores were: 22, 31, 33, 48, 50, 64, 68. What is the range?

Answer: 46

Explanation: Range = highest - lowest = $68 - 22 = 46$.

Problem 39: Test scores were: 31, 44, 54, 55, 63, 64, 72. What is the range?

Answer: 41

Explanation: Range = highest - lowest = $72 - 31 = 41$.

Problem 40: Test scores were: 26, 36, 42, 42, 54, 65, 79. What is the range?

Answer: 53

Explanation: Range = highest - lowest = $79 - 26 = 53$.

Problem 41: Test scores were: 21, 32, 42, 55, 56, 61, 76. What is the range?

Answer: 55

Explanation: Range = highest - lowest = $76 - 21 = 55$.

Problem 42: Test scores were: 24, 30, 34, 41, 50, 53, 67. What is the range?

Answer: 43

Explanation: Range = highest - lowest = $67 - 24 = 43$.

Problem 43: Test scores were: 23, 24, 48, 50, 60, 74, 74. What is the range?

Answer: 51

Explanation: Range = highest - lowest = $74 - 23 = 51$.

Problem 44: Test scores were: 26, 39, 47, 49, 60, 69, 80. What is the range?

Answer: 54

Explanation: Range = highest - lowest = $80 - 26 = 54$.

Problem 45: Test scores were: 26, 29, 34, 43, 52, 58, 80. What is the range?

Answer: 54

Explanation: Range = highest - lowest = $80 - 26 = 54$.

Practice Exercises - Grade 6

Exercise 1: Test scores were: 30, 47, 51, 54, 65, 70, 72. What is the range?

Exercise 2: Test scores were: 20, 28, 41, 53, 73, 74, 79. What is the range?

Exercise 3: Test scores were: 20, 25, 33, 41, 47, 64, 76. What is the range?

Exercise 4: Test scores were: 22, 26, 47, 49, 53, 69, 72. What is the range?

Exercise 5: Test scores were: 23, 27, 31, 43, 55, 69, 72. What is the range?

Exercise 6: Test scores were: 20, 29, 30, 38, 40, 40, 62. What is the range?

Exercise 7: Test scores were: 24, 29, 40, 55, 57, 65, 71. What is the range?

Exercise 8: Test scores were: 24, 24, 41, 64, 67, 76, 80. What is the range?

Exercise 9: Test scores were: 31, 38, 41, 43, 57, 66, 73. What is the range?

Exercise 10: Test scores were: 23, 29, 38, 40, 55, 69, 78. What is the range?

Exercise 11: Test scores were: 32, 33, 35, 40, 42, 43, 73. What is the range?

Exercise 12: Test scores were: 26, 29, 30, 31, 49, 76, 77. What is the range?

Exercise 13: Test scores were: 20, 32, 32, 34, 37, 65, 78. What is the range?

Exercise 14: Test scores were: 24, 32, 34, 43, 45, 70, 73. What is the range?

Exercise 15: Test scores were: 27, 33, 36, 42, 57, 63, 63. What is the range?

Exercise 16: Test scores were: 21, 21, 27, 35, 40, 64, 70. What is the range?

Exercise 17: Test scores were: 20, 23, 28, 43, 44, 75, 78. What is the range?

Exercise 18: Test scores were: 28, 35, 53, 57, 60, 67, 73. What is the range?

Exercise 19: Test scores were: 25, 32, 33, 36, 58, 63, 76. What is the range?

Exercise 20: Test scores were: 20, 23, 40, 40, 45, 68, 75. What is the range?

Practice Exercise Answers:

Exercise 1: 42

Exercise 2: 59

Exercise 3: 56

Exercise 4: 50

Exercise 5: 49

Exercise 6: 42

Exercise 7: 47

Exercise 8: 56

Exercise 9: 42

Exercise 10: 55

Exercise 11: 41

Exercise 12: 51

Exercise 13: 58

Exercise 14: 49

Exercise 15: 36

Exercise 16: 49

Exercise 17: 58

Exercise 18: 45

Exercise 19: 51

Exercise 20: 55

Challenge Problems - Grade 6

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Test scores were: 26, 27, 48, 54, 55, 61, 74. What is the range?

Answer: 48

Solution: Range = highest - lowest = $74 - 26 = 48$.

Challenge 2: Test scores were: 26, 32, 46, 48, 50, 53, 62. What is the range?

Answer: 36

Solution: Range = highest - lowest = $62 - 26 = 36$.

Challenge 3: Test scores were: 26, 41, 43, 54, 59, 73, 74. What is the range?

Answer: 48

Solution: Range = highest - lowest = $74 - 26 = 48$.

Challenge 4: Test scores were: 22, 46, 49, 56, 62, 63, 77. What is the range?

Answer: 55

Solution: Range = highest - lowest = $77 - 22 = 55$.

Challenge 5: Test scores were: 23, 23, 33, 39, 65, 73, 76. What is the range?

Answer: 53

Solution: Range = highest - lowest = $76 - 23 = 53$.

Challenge 6: Test scores were: 20, 22, 22, 24, 43, 69, 75. What is the range?

Answer: 55

Solution: Range = highest - lowest = $75 - 20 = 55$.

Challenge 7: Test scores were: 29, 32, 63, 66, 71, 74, 80. What is the range?

Answer: 51

Solution: Range = highest - lowest = $80 - 29 = 51$.

Challenge 8: Test scores were: 33, 39, 41, 54, 61, 66, 66. What is the range?

Answer: 33

Solution: Range = highest - lowest = $66 - 33 = 33$.

Challenge 9: Test scores were: 20, 23, 31, 35, 41, 46, 61. What is the range?

Answer: 41

Solution: Range = highest - lowest = $61 - 20 = 41$.

Challenge 10: Test scores were: 24, 38, 55, 56, 59, 63, 69. What is the range?

Answer: 45

Solution: Range = highest - lowest = $69 - 24 = 45$.

Challenge 11: Test scores were: 20, 25, 26, 35, 44, 50, 66. What is the range?

Answer: 46

Solution: Range = highest - lowest = $66 - 20 = 46$.

Challenge 12: Test scores were: 24, 44, 46, 55, 57, 65, 66. What is the range?

Answer: 42

Solution: Range = highest - lowest = $66 - 24 = 42$.

Challenge 13: Test scores were: 37, 41, 42, 53, 68, 70, 72. What is the range?

Answer: 35

Solution: Range = highest - lowest = $72 - 37 = 35$.

Challenge 14: Test scores were: 39, 40, 40, 62, 64, 73, 79. What is the range?

Answer: 40

Solution: Range = highest - lowest = $79 - 39 = 40$.

Challenge 15: Test scores were: 36, 52, 58, 64, 64, 70, 74. What is the range?

Answer: 38

Solution: Range = highest - lowest = $74 - 36 = 38$.

Chapter 8: Mathematical Vocabulary and Examples

Mathematical vocabulary is the language of mathematics. This chapter provides clear definitions, examples, and applications of key mathematical terms that students encounter throughout their learning journey.

Grade 1 Problems

Problem 1: Term: SUM

Answer: The answer when you add numbers together

Explanation: Example: $5 + 3 = 8$ (8 is the sum)

Problem 2: Term: DIFFERENCE

Answer: The answer when you subtract numbers

Explanation: Example: $9 - 4 = 5$ (5 is the difference)

Problem 3: Term: EQUAL

Answer: Having the same value

Explanation: Example: $3 + 2 = 5$ means both sides are equal

Problem 4: Term: GREATER THAN

Answer: Larger in value

Explanation: Example: $7 > 3$ means 7 is greater than 3

Problem 5: Term: LESS THAN

Answer: Smaller in value

Explanation: Example: $2 < 6$ means 2 is less than 6

Problem 6: Term: ADDEND

Answer: A number being added

Explanation: Example: In $4 + 5 = 9$, both 4 and 5 are addends

Problem 7: Term: COUNT

Answer: To find the total number

Explanation: Example: Count the apples: 1, 2, 3, 4, 5

Problem 8: Term: COMPARE

Answer: To see which is more or less

Explanation: Example: Compare 8 and 5: 8 is more

Problem 9: Term: ONES

Answer: The rightmost digit in a number

Explanation: Example: In 47, the 7 is in the ones place

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Answer: The second digit from the right

Explanation: Example: In 47, the 4 is in the tens place

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Answer: A larger amount

Explanation: Example: 10 is more than 7

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Answer: A smaller amount

Explanation: Example: 3 is fewer than 8

Problem 13: Term: PAIR

Answer: Two items together

Explanation: Example: A pair of shoes means 2 shoes

Problem 14: Term: HALF

Answer: One of two equal parts

Explanation: Example: Half of 8 is 4

Problem 15: Term: DOUBLE

Answer: Two times a number

Explanation: Example: Double 5 is 10

Problem 16: Term: NUMBER LINE

Answer: A line showing numbers in order

Explanation: Example: 0, 1, 2, 3, 4, 5... on a line

Problem 17: Term: PATTERN

Answer: Something that repeats

Explanation: Example: 2, 4, 6, 8 is a pattern of even numbers

Problem 18: Term: SHAPE

Answer: The form of an object

Explanation: Example: Circle, square, triangle are shapes

Problem 19: Term: CORNER

Answer: Where two sides meet

Explanation: Example: A square has 4 corners

Problem 20: Term: SIDE

Answer: An edge of a shape

Explanation: Example: A triangle has 3 sides

Problem 21: Term: WHOLE

Answer: All of something

Explanation: Example: The whole pie, not just a slice

Problem 22: Term: PART

Answer: A piece of something

Explanation: Example: I ate part of the cookie

Problem 23: Term: GROUP

Answer: Things put together

Explanation: Example: A group of 5 pencils

Problem 24: Term: ORDER

Answer: Arrangement from smallest to largest

Explanation: Example: Put in order: 2, 5, 7, 9

Problem 25: Term: TALLY

Answer: A mark to count

Explanation: Example: Make tally marks: |||| means 4

Problem 26: Term: BEFORE

Answer: Coming earlier

Explanation: Example: 5 comes before 6

Problem 27: Term: AFTER

Answer: Coming later

Explanation: Example: 8 comes after 7

Problem 28: Term: BETWEEN

Answer: In the middle of two numbers

Explanation: Example: 5 is between 4 and 6

Problem 29: Term: SUM

Answer: The answer when you add numbers together

Explanation: Example: $5 + 3 = 8$ (8 is the sum)

Problem 30: Term: DIFFERENCE

Answer: The answer when you subtract numbers

Explanation: Example: $9 - 4 = 5$ (5 is the difference)

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Answer: The second digit from the right

Explanation: Example: In 47, the 4 is in the tens place

Problem 39: Term: MORE

Answer: A larger amount

Explanation: Example: 10 is more than 7

Problem 40: Term: FEWER

Answer: A smaller amount

Explanation: Example: 3 is fewer than 8

Problem 41: Term: PAIR

Answer: Two items together

Explanation: Example: A pair of shoes means 2 shoes

Problem 42: Term: HALF

Answer: One of two equal parts

Explanation: Example: Half of 8 is 4

Problem 43: Term: DOUBLE

Answer: Two times a number

Explanation: Example: Double 5 is 10

Problem 44: Term: NUMBER LINE

Answer: A line showing numbers in order

Explanation: Example: 0, 1, 2, 3, 4, 5... on a line

Problem 45: Term: PATTERN

Answer: Something that repeats

Explanation: Example: 2, 4, 6, 8 is a pattern of even numbers

Practice Exercises - Grade 1

Exercise 1: Term: SUM

Exercise 2: Term: DIFFERENCE

Exercise 3: Term: EQUAL

Exercise 4: Term: GREATER THAN

Exercise 5: Term: LESS THAN

Exercise 6: Term: ADDEND

Exercise 7: Term: COUNT

Exercise 8: Term: COMPARE

Exercise 9: Term: ONES

Exercise 10: Term: TENS

Exercise 11: Term: MORE

Exercise 12: Term: FEWER

Exercise 13: Term: PAIR

Exercise 14: Term: HALF

Exercise 15: Term: DOUBLE

Exercise 16: Term: NUMBER LINE

Exercise 17: Term: PATTERN

Exercise 18: Term: SHAPE

Exercise 19: Term: CORNER

Exercise 20: Term: SIDE

Practice Exercise Answers:

Exercise 1: The answer when you add numbers together

Exercise 2: The answer when you subtract numbers

Exercise 3: Having the same value

Exercise 4: Larger in value

Exercise 5: Smaller in value

Exercise 6: A number being added

Exercise 7: To find the total number

Exercise 8: To see which is more or less

Exercise 9: The rightmost digit in a number

Exercise 10: The second digit from the right

Exercise 11: A larger amount

Exercise 12: A smaller amount

Exercise 13: Two items together

Exercise 14: One of two equal parts

Exercise 15: Two times a number

Exercise 16: A line showing numbers in order

Exercise 17: Something that repeats

Exercise 18: The form of an object

Exercise 19: Where two sides meet

Exercise 20: An edge of a shape

Challenge Problems - Grade 1

These advanced problems require multi-step thinking and deeper understanding.

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Solution: Example: $5 + 3 = 8$ (8 is the sum)

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Answer: The answer when you subtract numbers

Solution: Example: $9 - 4 = 5$ (5 is the difference)

Challenge 3: Term: EQUAL

Answer: Having the same value

Solution: Example: $3 + 2 = 5$ means both sides are equal

Challenge 4: Term: GREATER THAN

Answer: Larger in value

Solution: Example: $7 > 3$ means 7 is greater than 3

Challenge 5: Term: LESS THAN

Answer: Smaller in value

Solution: Example: $2 < 6$ means 2 is less than 6

Challenge 6: Term: ADDEND

Answer: A number being added

Solution: Example: In $4 + 5 = 9$, both 4 and 5 are addends

Challenge 7: Term: COUNT

Answer: To find the total number

Solution: Example: Count the apples: 1, 2, 3, 4, 5

Challenge 8: Term: COMPARE

Answer: To see which is more or less

Solution: Example: Compare 8 and 5: 8 is more

Challenge 9: Term: ONES

Answer: The rightmost digit in a number

Solution: Example: In 47, the 7 is in the ones place

Challenge 10: Term: TENS

Answer: The second digit from the right

Solution: Example: In 47, the 4 is in the tens place

Challenge 11: Term: MORE

Answer: A larger amount

Solution: Example: 10 is more than 7

Challenge 12: Term: FEWER

Answer: A smaller amount

Solution: Example: 3 is fewer than 8

Challenge 13: Term: PAIR

Answer: Two items together

Solution: Example: A pair of shoes means 2 shoes

Challenge 14: Term: HALF

Answer: One of two equal parts

Solution: Example: Half of 8 is 4

Challenge 15: Term: DOUBLE

Answer: Two times a number

Solution: Example: Double 5 is 10

Grade 2 Problems

Problem 1: Term: PRODUCT

Answer: The answer when you multiply numbers

Explanation: Example: $4 \times 3 = 12$ (12 is the product)

Problem 2: Term: EVEN NUMBER

Answer: A number divisible by 2

Explanation: Example: 2, 4, 6, 8, 10 are even numbers

Problem 3: Term: ODD NUMBER

Answer: A number not divisible by 2

Explanation: Example: 1, 3, 5, 7, 9 are odd numbers

Problem 4: Term: DIGIT

Answer: A single symbol used to make numbers

Explanation: Example: The number 47 has two digits: 4 and 7

Problem 5: Term: PLACE VALUE

Answer: The value of where a digit is in a number

Explanation: Example: In 25, the 2 is in the tens place

Problem 6: Term: REGROUP

Answer: To exchange 10 ones for 1 ten (or similar)

Explanation: Example: $16 + 8$: regroup 14 ones as 1 ten 4 ones

Problem 7: Term: ARRAY

Answer: Objects arranged in rows and columns

Explanation: Example: 3 rows of 4 = $3 \times 4 = 12$

Problem 8: Term: SKIP COUNT

Answer: Counting by a number other than 1

Explanation: Example: Skip count by 5: 5, 10, 15, 20

Problem 9: Term: HUNDREDS

Answer: The third digit from the right

Explanation: Example: In 347, the 3 is in the hundreds place

Problem 10: Term: ESTIMATE

Answer: To make a reasonable guess

Explanation: Example: Estimate $48 + 33$: about $50 + 30 = 80$

Problem 11: Term: ROUND

Answer: To change to the nearest ten or hundred

Explanation: Example: Round 47 to nearest 10: 50

Problem 12: Term: EQUATION

Answer: A math sentence with an equals sign

Explanation: Example: $7 + 5 = 12$ is an equation

Problem 13: Term: NUMBER SENTENCE

Answer: A complete math statement

Explanation: Example: $9 - 3 = 6$ is a number sentence

Problem 14: Term: MENTAL MATH

Answer: Solving in your head without paper

Explanation: Example: $20 + 15 = 35$ (done mentally)

Problem 15: Term: FACT FAMILY

Answer: Related addition and subtraction facts

Explanation: Example: $3+4=7$, $4+3=7$, $7-3=4$, $7-4=3$

Problem 16: Term: MEASUREMENT

Answer: Finding the size of something

Explanation: Example: Measure the pencil: 6 inches

Problem 17: Term: LENGTH

Answer: How long something is

Explanation: Example: The length of the table is 3 feet

Problem 18: Term: WIDTH

Answer: How wide something is

Explanation: Example: The width of the book is 8 inches

Problem 19: Term: CENTIMETER

Answer: A metric unit of length

Explanation: Example: A paperclip is about 3 centimeters

Problem 20: Term: METER

Answer: A metric unit equal to 100 centimeters

Explanation: Example: A doorway is about 2 meters tall

Problem 21: Term: INCH

Answer: A unit of length

Explanation: Example: A quarter is about 1 inch across

Problem 22: Term: FOOT

Answer: A unit equal to 12 inches

Explanation: Example: A ruler is 1 foot long

Problem 23: Term: TIME

Answer: When something happens

Explanation: Example: What time is it? 3:30

Problem 24: Term: HOUR

Answer: A unit of time (60 minutes)

Explanation: Example: School lasts 6 hours

Problem 25: Term: MINUTE

Answer: A unit of time (60 seconds)

Explanation: Example: The test takes 20 minutes

Problem 26: Term: MONEY

Answer: Coins and bills used to buy things

Explanation: Example: A dollar equals 100 cents

Problem 27: Term: CENT

Answer: A penny, 1/100 of a dollar

Explanation: Example: 1 cent is written as \$0.01

Problem 28: Term: DOLLAR

Answer: 100 cents

Explanation: Example: 1 dollar = 100 cents = \$1.00

Problem 29: Term: PRODUCT

Answer: The answer when you multiply numbers

Explanation: Example: $4 \times 3 = 12$ (12 is the product)

Problem 30: Term: EVEN NUMBER

Answer: A number divisible by 2

Explanation: Example: 2, 4, 6, 8, 10 are even numbers

Problem 31: Term: ODD NUMBER

Answer: A number not divisible by 2

Explanation: Example: 1, 3, 5, 7, 9 are odd numbers

Problem 32: Term: DIGIT

Answer: A single symbol used to make numbers

Explanation: Example: The number 47 has two digits: 4 and 7

Problem 33: Term: PLACE VALUE

Answer: The value of where a digit is in a number

Explanation: Example: In 25, the 2 is in the tens place

Problem 34: Term: REGROUP

Answer: To exchange 10 ones for 1 ten (or similar)

Explanation: Example: $16 + 8$: regroup 14 ones as 1 ten 4 ones

Problem 35: Term: ARRAY

Answer: Objects arranged in rows and columns

Explanation: Example: 3 rows of 4 = $3 \times 4 = 12$

Problem 36: Term: SKIP COUNT

Answer: Counting by a number other than 1

Explanation: Example: Skip count by 5: 5, 10, 15, 20

Problem 37: Term: HUNDREDS

Answer: The third digit from the right

Explanation: Example: In 347, the 3 is in the hundreds place

Problem 38: Term: ESTIMATE

Answer: To make a reasonable guess

Explanation: Example: Estimate $48 + 33$: about $50 + 30 = 80$

Problem 39: Term: ROUND

Answer: To change to the nearest ten or hundred

Explanation: Example: Round 47 to nearest 10: 50

Problem 40: Term: EQUATION

Answer: A math sentence with an equals sign

Explanation: Example: $7 + 5 = 12$ is an equation

Problem 41: Term: NUMBER SENTENCE

Answer: A complete math statement

Explanation: Example: $9 - 3 = 6$ is a number sentence

Problem 42: Term: MENTAL MATH

Answer: Solving in your head without paper

Explanation: Example: $20 + 15 = 35$ (done mentally)

Problem 43: Term: FACT FAMILY

Answer: Related addition and subtraction facts

Explanation: Example: $3+4=7$, $4+3=7$, $7-3=4$, $7-4=3$

Problem 44: Term: MEASUREMENT

Answer: Finding the size of something

Explanation: Example: Measure the pencil: 6 inches

Problem 45: Term: LENGTH

Answer: How long something is

Explanation: Example: The length of the table is 3 feet

Practice Exercises - Grade 2

Exercise 1: Term: PRODUCT

Exercise 2: Term: EVEN NUMBER

Exercise 3: Term: ODD NUMBER

Exercise 4: Term: DIGIT

Exercise 5: Term: PLACE VALUE

Exercise 6: Term: REGROUP

Exercise 7: Term: ARRAY

Exercise 8: Term: SKIP COUNT

Exercise 9: Term: HUNDREDS

Exercise 10: Term: ESTIMATE

Exercise 11: Term: ROUND

Exercise 12: Term: EQUATION

Exercise 13: Term: NUMBER SENTENCE

Exercise 14: Term: MENTAL MATH

Exercise 15: Term: FACT FAMILY

Exercise 16: Term: MEASUREMENT

Exercise 17: Term: LENGTH

Exercise 18: Term: WIDTH

Exercise 19: Term: CENTIMETER

Exercise 20: Term: METER

Practice Exercise Answers:

Exercise 1: The answer when you multiply numbers

Exercise 2: A number divisible by 2

Exercise 3: A number not divisible by 2

Exercise 4: A single symbol used to make numbers

Exercise 5: The value of where a digit is in a number

Exercise 6: To exchange 10 ones for 1 ten (or similar)

Exercise 7: Objects arranged in rows and columns

Exercise 8: Counting by a number other than 1

Exercise 9: The third digit from the right

Exercise 10: To make a reasonable guess

Exercise 11: To change to the nearest ten or hundred

Exercise 12: A math sentence with an equals sign

Exercise 13: A complete math statement

Exercise 14: Solving in your head without paper

Exercise 15: Related addition and subtraction facts

Exercise 16: Finding the size of something

Exercise 17: How long something is

Exercise 18: How wide something is

Exercise 19: A metric unit of length

Exercise 20: A metric unit equal to 100 centimeters

Challenge Problems - Grade 2

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Term: PRODUCT

Answer: The answer when you multiply numbers

Solution: Example: $4 \times 3 = 12$ (12 is the product)

Challenge 2: Term: EVEN NUMBER

Answer: A number divisible by 2

Solution: Example: 2, 4, 6, 8, 10 are even numbers

Challenge 3: Term: ODD NUMBER

Answer: A number not divisible by 2

Solution: Example: 1, 3, 5, 7, 9 are odd numbers

Challenge 4: Term: DIGIT

Answer: A single symbol used to make numbers

Solution: Example: The number 47 has two digits: 4 and 7

Challenge 5: Term: PLACE VALUE

Answer: The value of where a digit is in a number

Solution: Example: In 25, the 2 is in the tens place

Challenge 6: Term: REGROUP

Answer: To exchange 10 ones for 1 ten (or similar)

Solution: Example: $16 + 8$: regroup 14 ones as 1 ten 4 ones

Challenge 7: Term: ARRAY

Answer: Objects arranged in rows and columns

Solution: Example: 3 rows of 4 = $3 \times 4 = 12$

Challenge 8: Term: SKIP COUNT

Answer: Counting by a number other than 1

Solution: Example: Skip count by 5: 5, 10, 15, 20

Challenge 9: Term: HUNDREDS

Answer: The third digit from the right

Solution: Example: In 347, the 3 is in the hundreds place

Challenge 10: Term: ESTIMATE

Answer: To make a reasonable guess

Solution: Example: Estimate $48 + 33$: about $50 + 30 = 80$

Challenge 11: Term: ROUND

Answer: To change to the nearest ten or hundred

Solution: Example: Round 47 to nearest 10: 50

Challenge 12: Term: EQUATION

Answer: A math sentence with an equals sign

Solution: Example: $7 + 5 = 12$ is an equation

Challenge 13: Term: NUMBER SENTENCE

Answer: A complete math statement

Solution: Example: $9 - 3 = 6$ is a number sentence

Challenge 14: Term: MENTAL MATH

Answer: Solving in your head without paper

Solution: Example: $20 + 15 = 35$ (done mentally)

Challenge 15: Term: FACT FAMILY

Answer: Related addition and subtraction facts

Solution: Example: $3+4=7$, $4+3=7$, $7-3=4$, $7-4=3$

Grade 3 Problems

Problem 1: Term: QUOTIENT

Answer: The answer when you divide numbers

Explanation: Example: $12 \div 3 = 4$ (4 is the quotient)

Problem 2: Term: REMAINDER

Answer: What is left after division

Explanation: Example: $13 \div 4 = 3 \text{ R}1$ (1 is the remainder)

Problem 3: Term: FRACTION

Answer: Part of a whole

Explanation: Example: $1/2$ means one part out of two equal parts

Problem 4: Term: NUMERATOR

Answer: The top number in a fraction

Explanation: Example: In $3/4$, the numerator is 3

Problem 5: Term: DENOMINATOR

Answer: The bottom number in a fraction

Explanation: Example: In $3/4$, the denominator is 4

Problem 6: Term: EQUIVALENT

Answer: Having the same value

Explanation: Example: $\frac{1}{2} = \frac{2}{4} = \frac{3}{6}$ are equivalent fractions

Problem 7: Term: COMPARE FRACTIONS

Answer: Deciding which fraction is larger

Explanation: Example: $\frac{3}{4} > \frac{1}{2}$

Problem 8: Term: UNIT FRACTION

Answer: A fraction with numerator of 1

Explanation: Example: $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$ are unit fractions

Problem 9: Term: MIXED NUMBER

Answer: A whole number and a fraction

Explanation: Example: $2\frac{1}{2}$ means 2 and one-half

Problem 10: Term: AREA

Answer: Space inside a 2D shape

Explanation: Example: A 3×4 rectangle has area of 12 square units

Problem 11: Term: PERIMETER

Answer: Distance around a shape

Explanation: Example: Square with side 5: perimeter = 20

Problem 12: Term: SQUARE UNIT

Answer: A 1×1 square used to measure area

Explanation: Example: Area is measured in square inches, etc.

Problem 13: Term: CAPACITY

Answer: How much a container holds

Explanation: Example: A pitcher has a capacity of 2 liters

Problem 14: Term: MASS

Answer: How much matter is in an object

Explanation: Example: The mass of the book is 500 grams

Problem 15: Term: WEIGHT

Answer: How heavy something is

Explanation: Example: The weight of the dog is 30 pounds

Problem 16: Term: GRAM

Answer: Metric unit of mass

Explanation: Example: A paperclip has mass of about 1 gram

Problem 17: Term: KILOGRAM

Answer: 1000 grams

Explanation: Example: A textbook might have mass of 1 kilogram

Problem 18: Term: LITER

Answer: Metric unit of capacity

Explanation: Example: A large bottle holds 2 liters

Problem 19: Term: MILLILITER

Answer: 1/1000 of a liter

Explanation: Example: A small medicine dose is 5 milliliters

Problem 20: Term: POLYGON

Answer: A closed shape with straight sides

Explanation: Example: Triangle, square, pentagon are polygons

Problem 21: Term: VERTEX

Answer: A corner point where sides meet

Explanation: Example: A triangle has 3 vertices

Problem 22: Term: EDGE

Answer: A line segment on a shape

Explanation: Example: A cube has 12 edges

Problem 23: Term: FACE

Answer: A flat surface on a 3D shape

Explanation: Example: A cube has 6 faces

Problem 24: Term: ANGLE

Answer: The space between two lines meeting at a point

Explanation: Example: The corner of a square is a right angle

Problem 25: Term: RIGHT ANGLE

Answer: A 90-degree angle (like a square corner)

Explanation: Example: The letter L shows a right angle

Problem 26: Term: PARALLEL

Answer: Lines that never meet

Explanation: Example: Railroad tracks are parallel

Problem 27: Term: PERPENDICULAR

Answer: Lines that meet at right angles

Explanation: Example: The letter T shows perpendicular lines

Problem 28: Term: SYMMETRY

Answer: When both sides match

Explanation: Example: A butterfly has symmetry

Problem 29: Term: QUOTIENT

Answer: The answer when you divide numbers

Explanation: Example: $12 \div 3 = 4$ (4 is the quotient)

Problem 30: Term: REMAINDER

Answer: What is left after division

Explanation: Example: $13 \div 4 = 3 \text{ R}1$ (1 is the remainder)

Problem 31: Term: FRACTION

Answer: Part of a whole

Explanation: Example: $1/2$ means one part out of two equal parts

Problem 32: Term: NUMERATOR

Answer: The top number in a fraction

Explanation: Example: In $3/4$, the numerator is 3

Problem 33: Term: DENOMINATOR

Answer: The bottom number in a fraction

Explanation: Example: In $3/4$, the denominator is 4

Problem 34: Term: EQUIVALENT

Answer: Having the same value

Explanation: Example: $1/2 = 2/4 = 3/6$ are equivalent fractions

Problem 35: Term: COMPARE FRACTIONS

Answer: Deciding which fraction is larger

Explanation: Example: $3/4 > 1/2$

Problem 36: Term: UNIT FRACTION

Answer: A fraction with numerator of 1

Explanation: Example: $1/3$, $1/4$, $1/5$ are unit fractions

Problem 37: Term: MIXED NUMBER

Answer: A whole number and a fraction

Explanation: Example: $2 \frac{1}{2}$ means 2 and one-half

Problem 38: Term: AREA

Answer: Space inside a 2D shape

Explanation: Example: A 3×4 rectangle has area of 12 square units

Problem 39: Term: PERIMETER

Answer: Distance around a shape

Explanation: Example: Square with side 5: perimeter = 20

Problem 40: Term: SQUARE UNIT

Answer: A 1×1 square used to measure area

Explanation: Example: Area is measured in square inches, etc.

Problem 41: Term: CAPACITY

Answer: How much a container holds

Explanation: Example: A pitcher has a capacity of 2 liters

Problem 42: Term: MASS

Answer: How much matter is in an object

Explanation: Example: The mass of the book is 500 grams

Problem 43: Term: WEIGHT

Answer: How heavy something is

Explanation: Example: The weight of the dog is 30 pounds

Problem 44: Term: GRAM

Answer: Metric unit of mass

Explanation: Example: A paperclip has mass of about 1 gram

Problem 45: Term: KILOGRAM

Answer: 1000 grams

Explanation: Example: A textbook might have mass of 1 kilogram

Practice Exercises - Grade 3

Exercise 1: Term: QUOTIENT

Exercise 2: Term: REMAINDER

Exercise 3: Term: FRACTION

Exercise 4: Term: NUMERATOR

Exercise 5: Term: DENOMINATOR

Exercise 6: Term: EQUIVALENT

Exercise 7: Term: COMPARE FRACTIONS

Exercise 8: Term: UNIT FRACTION

Exercise 9: Term: MIXED NUMBER

Exercise 10: Term: AREA

Exercise 11: Term: PERIMETER

Exercise 12: Term: SQUARE UNIT

Exercise 13: Term: CAPACITY

Exercise 14: Term: MASS

Exercise 15: Term: WEIGHT

Exercise 16: Term: GRAM

Exercise 17: Term: KILOGRAM

Exercise 18: Term: LITER

Exercise 19: Term: MILLILITER

Exercise 20: Term: POLYGON

Practice Exercise Answers:

Exercise 1: The answer when you divide numbers

Exercise 2: What is left after division

Exercise 3: Part of a whole

Exercise 4: The top number in a fraction

Exercise 5: The bottom number in a fraction

Exercise 6: Having the same value

Exercise 7: Deciding which fraction is larger

Exercise 8: A fraction with numerator of 1

Exercise 9: A whole number and a fraction

Exercise 10: Space inside a 2D shape

Exercise 11: Distance around a shape

Exercise 12: A 1×1 square used to measure area

Exercise 13: How much a container holds

Exercise 14: How much matter is in an object

Exercise 15: How heavy something is

Exercise 16: Metric unit of mass

Exercise 17: 1000 grams

Exercise 18: Metric unit of capacity

Exercise 19: 1/1000 of a liter

Exercise 20: A closed shape with straight sides

Challenge Problems - Grade 3

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Term: QUOTIENT

Answer: The answer when you divide numbers

Solution: Example: $12 \div 3 = 4$ (4 is the quotient)

Challenge 2: Term: REMAINDER

Answer: What is left after division

Solution: Example: $13 \div 4 = 3 \text{ R}1$ (1 is the remainder)

Challenge 3: Term: FRACTION

Answer: Part of a whole

Solution: Example: $1/2$ means one part out of two equal parts

Challenge 4: Term: NUMERATOR

Answer: The top number in a fraction

Solution: Example: In $3/4$, the numerator is 3

Challenge 5: Term: DENOMINATOR

Answer: The bottom number in a fraction

Solution: Example: In $3/4$, the denominator is 4

Challenge 6: Term: EQUIVALENT

Answer: Having the same value

Solution: Example: $1/2 = 2/4 = 3/6$ are equivalent fractions

Challenge 7: Term: COMPARE FRACTIONS

Answer: Deciding which fraction is larger

Solution: Example: $3/4 > 1/2$

Challenge 8: Term: UNIT FRACTION

Answer: A fraction with numerator of 1

Solution: Example: $1/3$, $1/4$, $1/5$ are unit fractions

Challenge 9: Term: MIXED NUMBER

Answer: A whole number and a fraction

Solution: Example: $2 \frac{1}{2}$ means 2 and one-half

Challenge 10: Term: AREA

Answer: Space inside a 2D shape

Solution: Example: A 3×4 rectangle has area of 12 square units

Challenge 11: Term: PERIMETER

Answer: Distance around a shape

Solution: Example: Square with side 5: perimeter = 20

Challenge 12: Term: SQUARE UNIT

Answer: A 1×1 square used to measure area

Solution: Example: Area is measured in square inches, etc.

Challenge 13: Term: CAPACITY

Answer: How much a container holds

Solution: Example: A pitcher has a capacity of 2 liters

Challenge 14: Term: MASS

Answer: How much matter is in an object

Solution: Example: The mass of the book is 500 grams

Challenge 15: Term: WEIGHT

Answer: How heavy something is

Solution: Example: The weight of the dog is 30 pounds

Grade 4 Problems

Problem 1: Term: FACTOR

Answer: A number that divides evenly into another

Explanation: Example: Factors of 12: 1, 2, 3, 4, 6, 12

Problem 2: Term: MULTIPLE

Answer: The product of a number and another integer

Explanation: Example: Multiples of 5: 5, 10, 15, 20...

Problem 3: Term: PRIME NUMBER

Answer: A number with only 1 and itself as factors

Explanation: Example: 2, 3, 5, 7, 11 are prime

Problem 4: Term: COMPOSITE NUMBER

Answer: A number with more than two factors

Explanation: Example: 4, 6, 8, 9, 10 are composite

Problem 5: Term: PERIMETER

Answer: The distance around a shape

Explanation: Example: Rectangle 5×3 : perimeter = $5+3+5+3 = 16$

Problem 6: Term: COMMON FACTOR

Answer: A factor shared by two or more numbers

Explanation: Example: Common factors of 12 and 18: 1, 2, 3, 6

Problem 7: Term: GREATEST COMMON FACTOR

Answer: The largest common factor (GCF)

Explanation: Example: GCF of 12 and 18 is 6

Problem 8: Term: COMMON MULTIPLE

Answer: A multiple shared by two or more numbers

Explanation: Example: Common multiples of 3 and 4: 12, 24, 36

Problem 9: Term: LEAST COMMON MULTIPLE

Answer: The smallest common multiple (LCM)

Explanation: Example: LCM of 3 and 4 is 12

Problem 10: Term: DIVISIBLE

Answer: Can be divided evenly with no remainder

Explanation: Example: 20 is divisible by 4 ($20 \div 4 = 5$)

Problem 11: Term: DIVISIBILITY RULE

Answer: A shortcut to check if a number divides evenly

Explanation: Example: A number is divisible by 2 if it ends in 0,2,4,6,8

Problem 12: Term: SIMPLIFY

Answer: To reduce a fraction to lowest terms

Explanation: Example: Simplify $6/8$ to $3/4$

Problem 13: Term: IMPROPER FRACTION

Answer: Numerator is larger than denominator

Explanation: Example: $7/4$ is an improper fraction

Problem 14: Term: CONVERT

Answer: To change from one form to another

Explanation: Example: Convert $5/4$ to $1 \frac{1}{4}$

Problem 15: Term: DECIMAL

Answer: A number with a decimal point

Explanation: Example: $0.5 = 1/2$, $0.75 = 3/4$

Problem 16: Term: TENTH

Answer: One part out of ten

Explanation: Example: 0.1 or $1/10$

Problem 17: Term: HUNDREDTH

Answer: One part out of one hundred

Explanation: Example: 0.01 or $1/100$

Problem 18: Term: DECIMAL POINT

Answer: The dot separating whole and decimal parts

Explanation: Example: In 3.25, the point separates 3 and 25

Problem 19: Term: PLACE VALUE (DECIMALS)

Answer: Value based on position after decimal

Explanation: Example: In 1.47, 4 is in tenths, 7 in hundredths

Problem 20: Term: LINE GRAPH

Answer: A graph using points connected by lines

Explanation: Example: Shows temperature change over time

Problem 21: Term: BAR GRAPH

Answer: A graph using bars to show data

Explanation: Example: Compares quantities using bars

Problem 22: Term: PICTOGRAPH

Answer: A graph using pictures to represent data

Explanation: Example: Each picture = 5 students

Problem 23: Term: DATA

Answer: Information collected

Explanation: Example: Survey data: favorite colors of students

Problem 24: Term: SURVEY

Answer: Asking questions to collect data

Explanation: Example: Survey: What is your favorite sport?

Problem 25: Term: SCALE

Answer: The numbers on a graph axis

Explanation: Example: The scale goes from 0 to 100 by 10s

Problem 26: Term: INTERVAL

Answer: The space between marks on a scale

Explanation: Example: The interval is 5 (0, 5, 10, 15...)

Problem 27: Term: PROTRACTOR

Answer: A tool to measure angles

Explanation: Example: Use a protractor to measure a 45° angle

Problem 28: Term: COMPASS

Answer: A tool to draw circles

Explanation: Example: Use a compass to draw a perfect circle

Problem 29: Term: FACTOR

Answer: A number that divides evenly into another

Explanation: Example: Factors of 12: 1, 2, 3, 4, 6, 12

Problem 30: Term: MULTIPLE

Answer: The product of a number and another integer

Explanation: Example: Multiples of 5: 5, 10, 15, 20...

Problem 31: Term: PRIME NUMBER

Answer: A number with only 1 and itself as factors

Explanation: Example: 2, 3, 5, 7, 11 are prime

Problem 32: Term: COMPOSITE NUMBER

Answer: A number with more than two factors

Explanation: Example: 4, 6, 8, 9, 10 are composite

Problem 33: Term: PERIMETER

Answer: The distance around a shape

Explanation: Example: Rectangle 5×3 : perimeter = $5+3+5+3 = 16$

Problem 34: Term: COMMON FACTOR

Answer: A factor shared by two or more numbers

Explanation: Example: Common factors of 12 and 18: 1, 2, 3, 6

Problem 35: Term: GREATEST COMMON FACTOR

Answer: The largest common factor (GCF)

Explanation: Example: GCF of 12 and 18 is 6

Problem 36: Term: COMMON MULTIPLE

Answer: A multiple shared by two or more numbers

Explanation: Example: Common multiples of 3 and 4: 12, 24, 36

Problem 37: Term: LEAST COMMON MULTIPLE

Answer: The smallest common multiple (LCM)

Explanation: Example: LCM of 3 and 4 is 12

Problem 38: Term: DIVISIBLE

Answer: Can be divided evenly with no remainder

Explanation: Example: 20 is divisible by 4 ($20 \div 4 = 5$)

Problem 39: Term: DIVISIBILITY RULE

Answer: A shortcut to check if a number divides evenly

Explanation: Example: A number is divisible by 2 if it ends in 0,2,4,6,8

Problem 40: Term: SIMPLIFY

Answer: To reduce a fraction to lowest terms

Explanation: Example: Simplify $\frac{6}{8}$ to $\frac{3}{4}$

Problem 41: Term: IMPROPER FRACTION

Answer: Numerator is larger than denominator

Explanation: Example: $\frac{7}{4}$ is an improper fraction

Problem 42: Term: CONVERT

Answer: To change from one form to another

Explanation: Example: Convert $\frac{5}{4}$ to $1\frac{1}{4}$

Problem 43: Term: DECIMAL

Answer: A number with a decimal point

Explanation: Example: $0.5 = \frac{1}{2}$, $0.75 = \frac{3}{4}$

Problem 44: Term: TENTH

Answer: One part out of ten

Explanation: Example: 0.1 or $\frac{1}{10}$

Problem 45: Term: HUNDREDTH

Answer: One part out of one hundred

Explanation: Example: 0.01 or $\frac{1}{100}$

Practice Exercises - Grade 4

Exercise 1: Term: FACTOR

Exercise 2: Term: MULTIPLE

Exercise 3: Term: PRIME NUMBER

Exercise 4: Term: COMPOSITE NUMBER

Exercise 5: Term: PERIMETER

Exercise 6: Term: COMMON FACTOR

Exercise 7: Term: GREATEST COMMON FACTOR

Exercise 8: Term: COMMON MULTIPLE

Exercise 9: Term: LEAST COMMON MULTIPLE

Exercise 10: Term: DIVISIBLE

Exercise 11: Term: DIVISIBILITY RULE

Exercise 12: Term: SIMPLIFY

Exercise 13: Term: IMPROPER FRACTION

Exercise 14: Term: CONVERT

Exercise 15: Term: DECIMAL

Exercise 16: Term: TENTH

Exercise 17: Term: HUNDREDTH

Exercise 18: Term: DECIMAL POINT

Exercise 19: Term: PLACE VALUE (DECIMALS)

Exercise 20: Term: LINE GRAPH

Practice Exercise Answers:

Exercise 1: A number that divides evenly into another

Exercise 2: The product of a number and another integer

Exercise 3: A number with only 1 and itself as factors

Exercise 4: A number with more than two factors

Exercise 5: The distance around a shape

Exercise 6: A factor shared by two or more numbers

Exercise 7: The largest common factor (GCF)

Exercise 8: A multiple shared by two or more numbers

Exercise 9: The smallest common multiple (LCM)

Exercise 10: Can be divided evenly with no remainder

Exercise 11: A shortcut to check if a number divides evenly

Exercise 12: To reduce a fraction to lowest terms

Exercise 13: Numerator is larger than denominator

Exercise 14: To change from one form to another

Exercise 15: A number with a decimal point

Exercise 16: One part out of ten

Exercise 17: One part out of one hundred

Exercise 18: The dot separating whole and decimal parts

Exercise 19: Value based on position after decimal

Exercise 20: A graph using points connected by lines

Challenge Problems - Grade 4

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Term: FACTOR

Answer: A number that divides evenly into another

Solution: Example: Factors of 12: 1, 2, 3, 4, 6, 12

Challenge 2: Term: MULTIPLE

Answer: The product of a number and another integer

Solution: Example: Multiples of 5: 5, 10, 15, 20...

Challenge 3: Term: PRIME NUMBER

Answer: A number with only 1 and itself as factors

Solution: Example: 2, 3, 5, 7, 11 are prime

Challenge 4: Term: COMPOSITE NUMBER

Answer: A number with more than two factors

Solution: Example: 4, 6, 8, 9, 10 are composite

Challenge 5: Term: PERIMETER

Answer: The distance around a shape

Solution: Example: Rectangle 5×3: perimeter = $5+3+5+3 = 16$

Challenge 6: Term: COMMON FACTOR

Answer: A factor shared by two or more numbers

Solution: Example: Common factors of 12 and 18: 1, 2, 3, 6

Challenge 7: Term: GREATEST COMMON FACTOR

Answer: The largest common factor (GCF)

Solution: Example: GCF of 12 and 18 is 6

Challenge 8: Term: COMMON MULTIPLE

Answer: A multiple shared by two or more numbers

Solution: Example: Common multiples of 3 and 4: 12, 24, 36

Challenge 9: Term: LEAST COMMON MULTIPLE

Answer: The smallest common multiple (LCM)

Solution: Example: LCM of 3 and 4 is 12

Challenge 10: Term: DIVISIBLE

Answer: Can be divided evenly with no remainder

Solution: Example: 20 is divisible by 4 ($20 \div 4 = 5$)

Challenge 11: Term: DIVISIBILITY RULE

Answer: A shortcut to check if a number divides evenly

Solution: Example: A number is divisible by 2 if it ends in 0,2,4,6,8

Challenge 12: Term: SIMPLIFY

Answer: To reduce a fraction to lowest terms

Solution: Example: Simplify $\frac{6}{8}$ to $\frac{3}{4}$

Challenge 13: Term: IMPROPER FRACTION

Answer: Numerator is larger than denominator

Solution: Example: $\frac{7}{4}$ is an improper fraction

Challenge 14: Term: CONVERT

Answer: To change from one form to another

Solution: Example: Convert $\frac{5}{4}$ to $1\frac{1}{4}$

Challenge 15: Term: DECIMAL

Answer: A number with a decimal point

Solution: Example: $0.5 = \frac{1}{2}$, $0.75 = \frac{3}{4}$

Grade 5 Problems

Problem 1: Term: DECIMAL

Answer: A number with a decimal point

Explanation: Example: 3.5, 0.75, 12.25 are decimals

Problem 2: Term: PERCENT

Answer: Per hundred

Explanation: Example: 50% means 50 out of 100

Problem 3: Term: AREA

Answer: The space inside a shape

Explanation: Example: Rectangle 4×6 : area = 24 square units

Problem 4: Term: VOLUME

Answer: Space inside a 3D shape

Explanation: Example: Box $3 \times 4 \times 5$: volume = 60 cubic units

Problem 5: Term: VARIABLE

Answer: A letter representing a number

Explanation: Example: In $x + 5 = 12$, x is the variable

Problem 6: Term: EXPRESSION

Answer: Numbers and operations without equals

Explanation: Example: $3x + 7$ is an expression

Problem 7: Term: EVALUATE

Answer: To find the value of an expression

Explanation: Example: Evaluate $2x + 3$ when $x = 5$: $2(5) + 3 = 13$

Problem 8: Term: COEFFICIENT

Answer: The number multiplying a variable

Explanation: Example: In $5x$, the coefficient is 5

Problem 9: Term: CONSTANT

Answer: A number that doesn't change

Explanation: Example: In $x + 7$, the constant is 7

Problem 10: Term: ORDER OF OPERATIONS

Answer: Rules for calculating (PEMDAS)

Explanation: Example: Do parentheses, then exponents, multiply/divide, add/subtract

Problem 11: Term: PARENTHESES

Answer: Grouping symbols ()

Explanation: Example: Calculate $(3 + 5) \times 2 = 8 \times 2 = 16$

Problem 12: Term: CUBIC UNIT

Answer: A $1 \times 1 \times 1$ cube for measuring volume

Explanation: Example: Volume in cubic inches, cubic feet, etc.

Problem 13: Term: RECTANGULAR PRISM

Answer: A box shape with 6 rectangular faces

Explanation: Example: A shoebox is a rectangular prism

Problem 14: Term: COORDINATE PLANE

Answer: A grid with x and y axes

Explanation: Example: Points like (3, 4) on the plane

Problem 15: Term: X-AXIS

Answer: The horizontal number line

Explanation: Example: The x-axis goes left and right

Problem 16: Term: Y-AXIS

Answer: The vertical number line

Explanation: Example: The y-axis goes up and down

Problem 17: Term: ORIGIN

Answer: The point (0, 0) where axes meet

Explanation: Example: The origin is the center of the plane

Problem 18: Term: ORDERED PAIR

Answer: Two numbers showing a location (x, y)

Explanation: Example: (2, 5) means 2 right, 5 up

Problem 19: Term: COORDINATE

Answer: One of the numbers in an ordered pair

Explanation: Example: In (3, 7), the coordinates are 3 and 7

Problem 20: Term: MEAN

Answer: The average of a set of numbers

Explanation: Example: Mean of 4, 6, 8: $(4+6+8) \div 3 = 6$

Problem 21: Term: MEDIAN

Answer: The middle number when ordered

Explanation: Example: Median of 3, 5, 9: the median is 5

Problem 22: Term: MODE

Answer: The most frequent number

Explanation: Example: Mode of 2, 3, 3, 5: the mode is 3

Problem 23: Term: RANGE

Answer: Difference between highest and lowest

Explanation: Example: Range of 2, 5, 9: $9 - 2 = 7$

Problem 24: Term: OUTLIER

Answer: A value far from the others

Explanation: Example: In 5, 6, 7, 25: the 25 is an outlier

Problem 25: Term: STEM-AND-LEAF PLOT

Answer: A way to organize data by place value

Explanation: Example: Shows tens in stem, ones in leaf

Problem 26: Term: LINE PLOT

Answer: Data shown with X marks above a number line

Explanation: Example: Each X represents one data point

Problem 27: Term: DISTRIBUTIVE PROPERTY

Answer: Multiply outside number by each inside

Explanation: Example: $3(x + 4) = 3x + 12$

Problem 28: Term: COMMUTATIVE PROPERTY

Answer: Order doesn't matter in addition/multiplication

Explanation: Example: $3 + 5 = 5 + 3$

Problem 29: Term: DECIMAL

Answer: A number with a decimal point

Explanation: Example: 3.5, 0.75, 12.25 are decimals

Problem 30: Term: PERCENT

Answer: Per hundred

Explanation: Example: 50% means 50 out of 100

Problem 31: Term: AREA

Answer: The space inside a shape

Explanation: Example: Rectangle 4×6 : area = 24 square units

Problem 32: Term: VOLUME

Answer: Space inside a 3D shape

Explanation: Example: Box $3 \times 4 \times 5$: volume = 60 cubic units

Problem 33: Term: VARIABLE

Answer: A letter representing a number

Explanation: Example: In $x + 5 = 12$, x is the variable

Problem 34: Term: EXPRESSION

Answer: Numbers and operations without equals

Explanation: Example: $3x + 7$ is an expression

Problem 35: Term: EVALUATE

Answer: To find the value of an expression

Explanation: Example: Evaluate $2x + 3$ when $x = 5$: $2(5) + 3 = 13$

Problem 36: Term: COEFFICIENT

Answer: The number multiplying a variable

Explanation: Example: In $5x$, the coefficient is 5

Problem 37: Term: CONSTANT

Answer: A number that doesn't change

Explanation: Example: In $x + 7$, the constant is 7

Problem 38: Term: ORDER OF OPERATIONS

Answer: Rules for calculating (PEMDAS)

Explanation: Example: Do parentheses, then exponents, multiply/divide, add/subtract

Problem 39: Term: PARENTHESES

Answer: Grouping symbols ()

Explanation: Example: Calculate $(3 + 5) \times 2 = 8 \times 2 = 16$

Problem 40: Term: CUBIC UNIT

Answer: A 1×1×1 cube for measuring volume

Explanation: Example: Volume in cubic inches, cubic feet, etc.

Problem 41: Term: RECTANGULAR PRISM

Answer: A box shape with 6 rectangular faces

Explanation: Example: A shoebox is a rectangular prism

Problem 42: Term: COORDINATE PLANE

Answer: A grid with x and y axes

Explanation: Example: Points like (3, 4) on the plane

Problem 43: Term: X-AXIS

Answer: The horizontal number line

Explanation: Example: The x-axis goes left and right

Problem 44: Term: Y-AXIS

Answer: The vertical number line

Explanation: Example: The y-axis goes up and down

Problem 45: Term: ORIGIN

Answer: The point (0, 0) where axes meet

Explanation: Example: The origin is the center of the plane

Practice Exercises - Grade 5

Exercise 1: Term: DECIMAL

Exercise 2: Term: PERCENT

Exercise 3: Term: AREA

Exercise 4: Term: VOLUME

Exercise 5: Term: VARIABLE

Exercise 6: Term: EXPRESSION

Exercise 7: Term: EVALUATE

Exercise 8: Term: COEFFICIENT

Exercise 9: Term: CONSTANT

Exercise 10: Term: ORDER OF OPERATIONS

Exercise 11: Term: PARENTHESES

Exercise 12: Term: CUBIC UNIT

Exercise 13: Term: RECTANGULAR PRISM

Exercise 14: Term: COORDINATE PLANE

Exercise 15: Term: X-AXIS

Exercise 16: Term: Y-AXIS

Exercise 17: Term: ORIGIN

Exercise 18: Term: ORDERED PAIR

Exercise 19: Term: COORDINATE

Exercise 20: Term: MEAN

Practice Exercise Answers:

Exercise 1: A number with a decimal point

Exercise 2: Per hundred

Exercise 3: The space inside a shape

Exercise 4: Space inside a 3D shape

Exercise 5: A letter representing a number

Exercise 6: Numbers and operations without equals

Exercise 7: To find the value of an expression

Exercise 8: The number multiplying a variable

Exercise 9: A number that doesn't change

Exercise 10: Rules for calculating (PEMDAS)

Exercise 11: Grouping symbols ()

Exercise 12: A $1 \times 1 \times 1$ cube for measuring volume

Exercise 13: A box shape with 6 rectangular faces

Exercise 14: A grid with x and y axes

Exercise 15: The horizontal number line

Exercise 16: The vertical number line

Exercise 17: The point (0, 0) where axes meet

Exercise 18: Two numbers showing a location (x, y)

Exercise 19: One of the numbers in an ordered pair

Exercise 20: The average of a set of numbers

Challenge Problems - Grade 5

These advanced problems require multi-step thinking and deeper understanding.

Challenge 1: Term: DECIMAL

Answer: A number with a decimal point

Solution: Example: 3.5, 0.75, 12.25 are decimals

Challenge 2: Term: PERCENT

Answer: Per hundred

Solution: Example: 50% means 50 out of 100

Challenge 3: Term: AREA

Answer: The space inside a shape

Solution: Example: Rectangle 4×6 : area = 24 square units

Challenge 4: Term: VOLUME

Answer: Space inside a 3D shape

Solution: Example: Box $3 \times 4 \times 5$: volume = 60 cubic units

Challenge 5: Term: VARIABLE

Answer: A letter representing a number

Solution: Example: In $x + 5 = 12$, x is the variable

Challenge 6: Term: EXPRESSION

Answer: Numbers and operations without equals

Solution: Example: $3x + 7$ is an expression

Challenge 7: Term: EVALUATE

Answer: To find the value of an expression

Solution: Example: Evaluate $2x + 3$ when $x = 5$: $2(5) + 3 = 13$

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Solution: Example: Points like (3, 4) on the plane

Challenge 15: Term: X-AXIS

Answer: The horizontal number line

Solution: Example: The x-axis goes left and right

Grade 6 Problems

Problem 1: Term: RATIO

Answer: A comparison of two quantities

Explanation: Example: 3:4 means 3 to 4

Problem 2: Term: PROPORTION

Answer: Two equal ratios

Explanation: Example: $\frac{2}{3} = \frac{4}{6}$ is a proportion

Problem 3: Term: INTEGER

Answer: Whole numbers including negatives

Explanation: Example: ...-2, -1, 0, 1, 2...

Problem 4: Term: ABSOLUTE VALUE

Answer: Distance from zero

Explanation: Example: $|-5| = 5$ and $|5| = 5$

Problem 5: Term: EXPONENT

Answer: How many times to multiply a number

Explanation: Example: $2^3 = 2 \times 2 \times 2 = 8$

Problem 6: Term: BASE

Answer: The number being multiplied in a power

Explanation: Example: In 5^3 , the base is 5

Problem 7: Term: POWER

Answer: A number written with an exponent

Explanation: Example: 10^2 is a power of 10

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Answer: Raised to the power of 2

Explanation: Example: 4 squared = $4^2 = 16$

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Explanation: Example: 25 is a perfect square (5×5)

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Answer: A ratio comparing different units

Explanation: Example: 60 miles per hour is a rate

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Answer: A rate with denominator of 1

Explanation: Example: \$3 per pound

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Answer: A number expressed as a percent

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Answer: A math sentence with $<$, $>$, $\leq$, or $\geq$

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Answer: A value that makes an equation true

Explanation: Example: $x = 7$ is the solution to $x + 3 = 10$

Problem 18: Term: SOLVE

Answer: To find the value of the variable

Explanation: Example: Solve for x : $2x = 14$, $x = 7$

Problem 19: Term: ISOLATE

Answer: To get the variable alone on one side

Explanation: Example: Isolate x in $x + 5 = 12$ by subtracting 5

Problem 20: Term: INVERSE OPERATION

Answer: The opposite operation

Explanation: Example: Addition and subtraction are inverse operations

Problem 21: Term: RECIPROCAL

Answer: A fraction flipped upside down

Explanation: Example: Reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$

Problem 22: Term: GCF

Answer: Greatest Common Factor

Explanation: Example: GCF of 12 and 18 is 6

Problem 23: Term: LCM

Answer: Least Common Multiple

Explanation: Example: LCM of 4 and 6 is 12

Problem 24: Term: CIRCLE

Answer: A round shape, all points equidistant from center

Explanation: Example: A wheel is a circle

Problem 25: Term: RADIUS

Answer: Distance from center to edge of circle

Explanation: Example: If radius is 5, diameter is 10

Problem 26: Term: DIAMETER

Answer: Distance across a circle through center

Explanation: Example: Diameter is twice the radius

Problem 27: Term: CIRCUMFERENCE

Answer: Distance around a circle

Explanation: Example: $C = \pi d$ or $C = 2\pi r$

Problem 28: Term: PI (II)

Answer: The ratio of circumference to diameter

Explanation: Example: $\pi \approx 3.14$ or $22/7$

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Exercise 3: Term: INTEGER

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Solution: Example: $|-5| = 5$ and $|5| = 5$

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